

- 4.52** Rod AD is acted upon by a vertical force $\mathbf{P}$ at end A and by two equal and opposite horizontal forces of magnitude Q at points B and C . Neglecting the weight of the rod, express the angle θ corresponding to the equilibrium position in terms of P and Q .

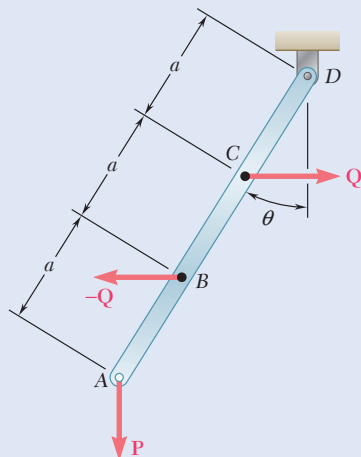


Fig. P4.52

- 4.53** A slender rod AB with a weight of W is attached to blocks A and B that move freely in the guides shown. The blocks are connected by an elastic cord that passes over a pulley at C . (a) Express the tension in the cord in terms of W and θ . (b) Determine the value of θ for which the tension in the cord is equal to $3W$.

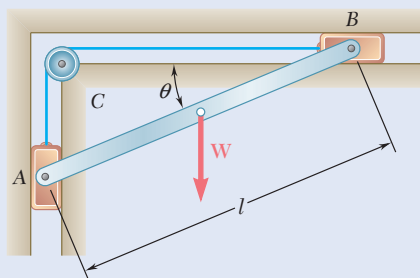


Fig. P4.53

- 4.54 and 4.55** A vertical load $\mathbf{P}$ is applied at end B of rod BC . (a) Neglecting the weight of the rod, express the angle θ corresponding to the equilibrium position in terms of P , l , and the counterweight W . (b) Determine the value of θ corresponding to equilibrium if $P = 2W$.

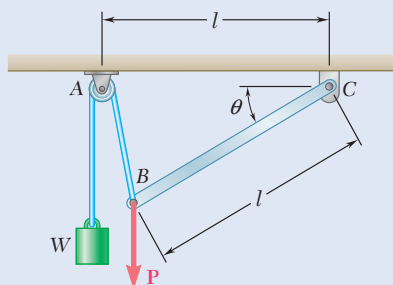


Fig. P4.54

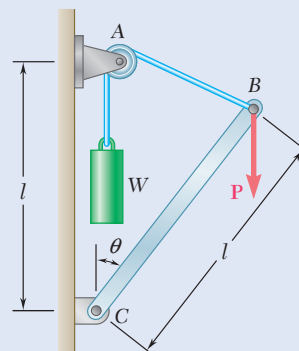


Fig. P4.55

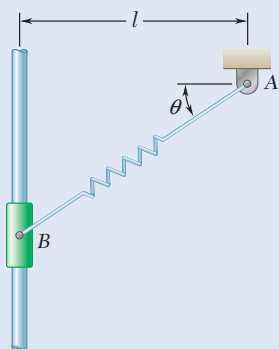


Fig. P4.56

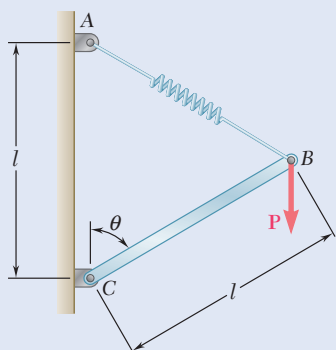


Fig. P4.58

4.56 A collar B with a weight of W can move freely along the vertical rod shown. The constant of the spring is k , and the spring is unstretched when $\theta = 0$. (a) Derive an equation in θ , W , k , and l that must be satisfied when the collar is in equilibrium. (b) Knowing that $W = 300$ N, $l = 500$ mm, and $k = 800$ N/m, determine the value of θ corresponding to equilibrium.

4.57 Solve Sample Prob. 4.5, assuming that the spring is unstretched when $\theta = 90^\circ$.

4.58 A vertical load P is applied at end B of rod BC . The constant of the spring is k , and the spring is unstretched when $\theta = 60^\circ$. (a) Neglecting the weight of the rod, express the angle θ corresponding to the equilibrium position in terms of P , k , and l . (b) Determine the value of θ corresponding to equilibrium if $P = \frac{1}{4}kl$.

4.59 Eight identical 500×750 -mm rectangular plates, each of mass $m = 40$ kg, are held in a vertical plane as shown. All connections consist of frictionless pins, rollers, or short links. In each case, determine whether (a) the plate is completely, partially, or improperly constrained, (b) the reactions are statically determinate or indeterminate, (c) the equilibrium of the plate is maintained in the position shown. Also, wherever possible, compute the reactions.

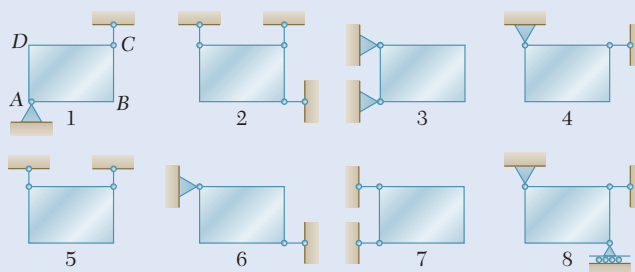


Fig. P4.59

4.60 The bracket ABC can be supported in the eight different ways shown. All connections consist of smooth pins, rollers, or short links. For each case, answer the questions listed in Prob. 4.59, and, wherever possible, compute the reactions, assuming that the magnitude of the force P is 100 lb.

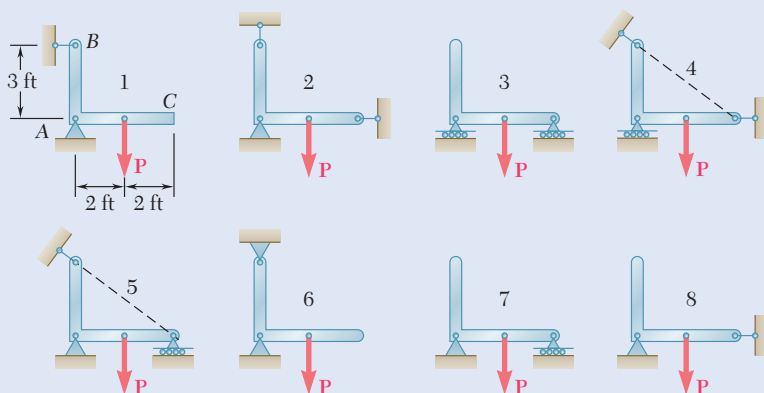


Fig. P4.60

4.2 TWO SPECIAL CASES

In practice, some simple cases of equilibrium occur quite often, either as part of a more complicated analysis or as the complete models of a situation. By understanding the characteristics of these cases, you can often simplify the overall analysis.

4.2A Equilibrium of a Two-Force Body

A particular case of equilibrium of considerable interest in practical applications is that of a rigid body subjected to two forces. Such a body is commonly called a **two-force body**. We show here that, **if a two-force body is in equilibrium, the two forces must have the same magnitude, the same line of action, and opposite sense.**

Consider a corner plate subjected to two forces $\mathbf{F}_1$ and $\mathbf{F}_2$ acting at A and B , respectively (Fig. 4.8a). If the plate is in equilibrium, the sum of the moments of $\mathbf{F}_1$ and $\mathbf{F}_2$ about any axis must be zero. First, we sum moments about A . Since the moment of $\mathbf{F}_1$ is obviously zero, the moment of $\mathbf{F}_2$ also must be zero and the line of action of $\mathbf{F}_2$ must pass through A (Fig. 4.8b). Similarly, summing moments about B , we can show that the line of action of $\mathbf{F}_1$ must pass through B (Fig. 4.8c). Therefore, both forces have the same line of action (line AB). You can see from either of the equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$ that they must also have the same magnitude but opposite sense.

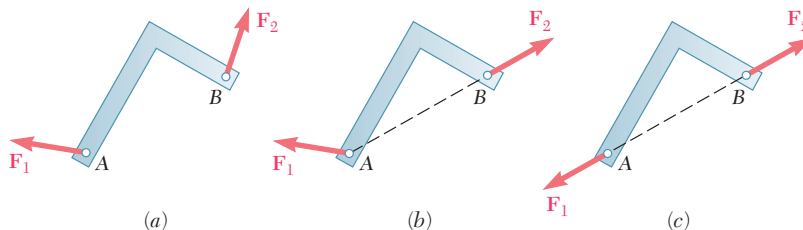


Fig. 4.8 A two-force body in equilibrium. (a) Forces act at two points of the body; (b) summing moments about point A shows that the line of action of $\mathbf{F}_2$ must pass through A ; (c) summing moments about point B shows that the line of action of $\mathbf{F}_1$ must pass through B .

If several forces act at two points A and B , the forces acting at A can be replaced by their resultant $\mathbf{F}_1$, and those acting at B can be replaced by their resultant $\mathbf{F}_2$. Thus, a two-force body can be more generally defined as **a rigid body subjected to forces acting at only two points**. The resultants $\mathbf{F}_1$ and $\mathbf{F}_2$ then must have the same line of action, the same magnitude, and opposite sense (Fig. 4.8).

Later, in the study of structures, frames, and machines, you will see how the recognition of two-force bodies simplifies the solution of certain problems.

4.2B Equilibrium of a Three-Force Body

Another case of equilibrium that is of great practical interest is that of a **three-force body**, i.e., a rigid body subjected to three forces or, more generally, a **rigid body subjected to forces acting at only three points**. Consider a rigid body subjected to a system of forces that can be reduced to three forces $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$ acting at A , B , and C , respectively (Fig. 4.9a). We show that if the body is in equilibrium, **the lines of action of the three forces must be either concurrent or parallel**.

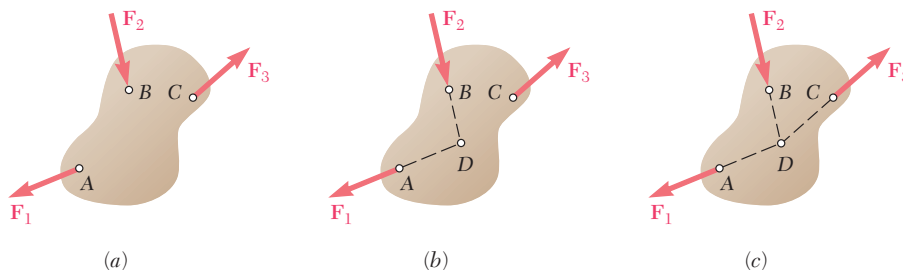


Fig. 4.9 A three-force body in equilibrium. (a–c) Demonstration that the lines of action of the three forces must be either concurrent or parallel.

Since the rigid body is in equilibrium, the sum of the moments of $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$ about any axis must be zero. Assuming that the lines of action of $\mathbf{F}_1$ and $\mathbf{F}_2$ intersect and denoting their point of intersection by D , we sum moments about D (Fig. 4.9b). Because the moments of $\mathbf{F}_1$ and $\mathbf{F}_2$ about D are zero, the moment of $\mathbf{F}_3$ about D also must be zero, and the line of action of $\mathbf{F}_3$ must pass through D (Fig. 4.9c). Therefore, the three lines of action are concurrent. The only exception occurs when none of the lines intersect; in this case, the lines of action are parallel.

Although problems concerning three-force bodies can be solved by the general methods of Sec. 4.1, we can use the property just established to solve these problems either graphically or mathematically using simple trigonometric or geometric relations (see Sample Problem 4.6).

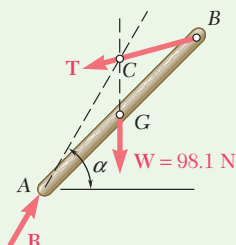
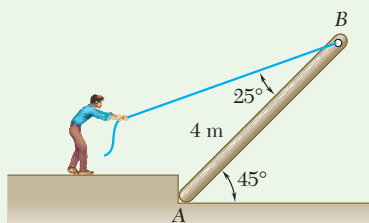


Fig. 1 Free-body diagram of joist.

Sample Problem 4.6

A man raises a 10-kg joist with a length of 4 m by pulling on a rope. Find the tension T in the rope and the reaction at A .

STRATEGY: The joist is acted upon by three forces: its weight $\mathbf{W}$, the force $\mathbf{T}$ exerted by the rope, and the reaction $\mathbf{R}$ of the ground at A . Therefore, it is a three-force body, and you can compute the forces by using a force triangle.

MODELING: First note that

$$W = mg = (10 \text{ kg})(9.81 \text{ m/s}^2) = 98.1 \text{ N}$$

Since the joist is a three-force body, the forces acting on it must be concurrent. The reaction $\mathbf{R}$ therefore must pass through the point of intersection C of the lines of action of the weight $\mathbf{W}$ and the tension force $\mathbf{T}$, as shown in the free-body diagram (Fig. 1). You can use this fact to determine the angle α that $\mathbf{R}$ forms with the horizontal.

ANALYSIS: Draw the vertical line BF through B and the horizontal line CD through C (Fig. 2). Then

$$AF = BF = (AB) \cos 45^\circ = (4 \text{ m}) \cos 45^\circ = 2.828 \text{ m}$$

$$CD = EF = AE = \frac{1}{2}(AF) = 1.414 \text{ m}$$

$$BD = (CD) \cot (45^\circ + 25^\circ) = (1.414 \text{ m}) \tan 20^\circ = 0.515 \text{ m}$$

$$CE = DF = BF - BD = 2.828 \text{ m} - 0.515 \text{ m} = 2.313 \text{ m}$$

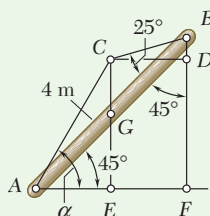


Fig. 2 Geometry analysis of the lines of action for the three forces acting on joist, concurrent at point C .

From these calculations, you can determine the angle α as

$$\tan \alpha = \frac{CE}{AE} = \frac{2.313 \text{ m}}{1.414 \text{ m}} = 1.636$$

$$\alpha = 58.6^\circ \quad \blacktriangleleft$$

You now know the directions of all the forces acting on the joist.

Force Triangle. Draw a force triangle as shown (Fig. 3) with its interior angles computed from the known directions of the forces. You can then use the law of sines to find the unknown forces.

$$\frac{T}{\sin 31.4^\circ} = \frac{R}{\sin 110^\circ} = \frac{98.1 \text{ N}}{\sin 38.6^\circ}$$

$$T = 81.9 \text{ N} \quad \blacktriangleleft$$

$$R = 147.8 \text{ N} \quad \blacktriangleleft 58.6^\circ$$

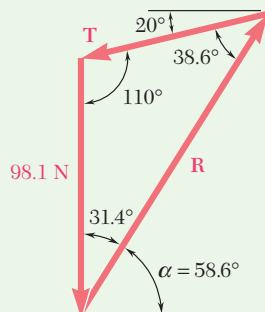


Fig. 3 Force triangle.

REFLECT and THINK: In practice, three-force members occur often, so learning this method of analysis is useful in many situations.

SOLVING PROBLEMS ON YOUR OWN

This section covered two particular cases of equilibrium of a rigid body.

1. A two-force body is subjected to forces at only two points. The resultants of the forces acting at each of these points must have the *same magnitude, the same line of action, and opposite sense*. This property allows you to simplify the solutions of some problems by replacing the two unknown components of a reaction by a single force of unknown magnitude but of *known direction*.

2. A three-force body is subjected to forces at only three points. The resultants of the forces acting at each of these points must be *concurrent or parallel*. To solve a problem involving a three-force body with concurrent forces, draw the free-body diagram showing that these three forces pass through the same point. You may be able to complete the solution by using simple geometry, such as a force triangle and the law of sines [see Sample Prob. 4.6].

This method for solving problems involving three-force bodies is not difficult to understand, but in practice, it can be difficult to sketch the necessary geometric constructions. If you encounter difficulty, first draw a reasonably large free-body diagram and then seek a relation between known or easily calculated lengths and a dimension that involves an unknown. Sample Prob. 4.6 illustrates this technique, where we used the easily calculated dimensions AE and CE to determine the angle α .

Problems

- 4.61** A 500-lb cylindrical tank, 8 ft in diameter, is to be raised over a 2-ft obstruction. A cable is wrapped around the tank and pulled horizontally as shown. Knowing that the corner of the obstruction at A is rough, find the required tension in the cable and the reaction at A .

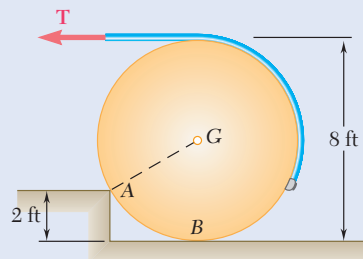


Fig. P4.61

- 4.62** Determine the reactions at A and B when $a = 180$ mm.
- 4.63** For the bracket and loading shown, determine the range of values of the distance a for which the magnitude of the reaction at B does not exceed 600 N.
- 4.64** The spanner shown is used to rotate a shaft. A pin fits in a hole at A , while a flat, frictionless surface rests against the shaft at B . If a 60-lb force P is exerted on the spanner at D , find the reactions at A and B .

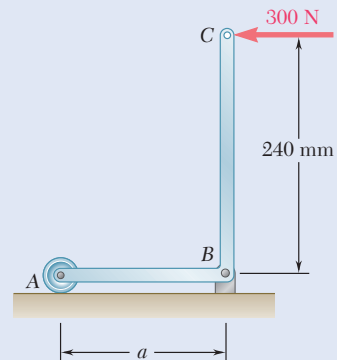


Fig. P4.62 and P4.63

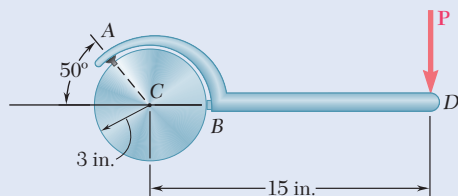


Fig. P4.64

- 4.65** Determine the reactions at B and C when $a = 30$ mm.

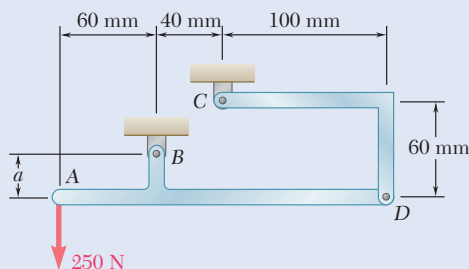


Fig. P4.65

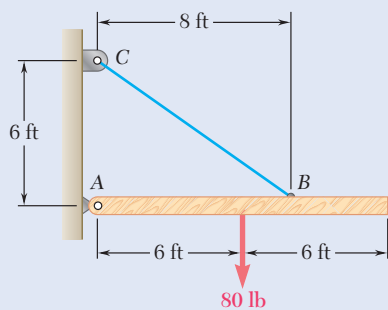


Fig. P4.66

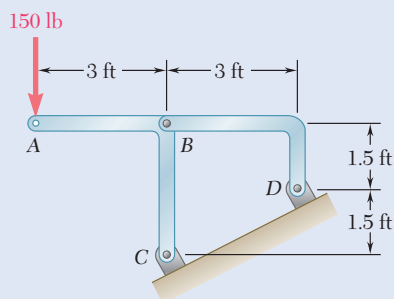


Fig. P4.68

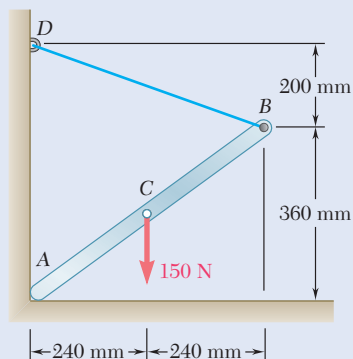


Fig. P4.70

4.66 A 12-ft wooden beam weighing 80 lb is supported by a pin and bracket at A and by cable BC. Find the reaction at A and the tension in the cable.

4.67 Determine the reactions at B and D when $b = 60$ mm.

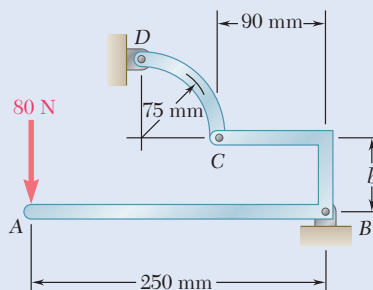


Fig. P4.67

4.68 For the frame and loading shown, determine the reactions at C and D.

4.69 A 50-kg crate is attached to the trolley-beam system shown. Knowing that $a = 1.5$ m, determine (a) the tension in cable CD, (b) the reaction at B.

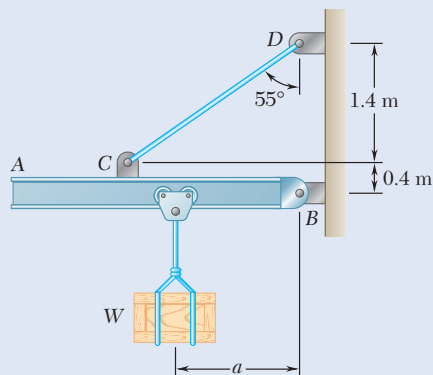


Fig. P4.69

4.70 One end of rod AB rests in the corner A and the other end is attached to cord BD. If the rod supports a 150-N load at its midpoint C, find the reaction at A and the tension in the cord.

4.71 For the boom and loading shown, determine (a) the tension in cord BD, (b) the reaction at C.

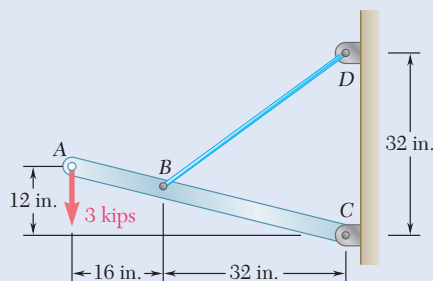


Fig. P4.71

- 4.72** A 40-lb roller of 8-in. diameter, which is to be used on a tile floor, is resting directly on the subflooring as shown. Knowing that the thickness of each tile is 0.3 in., determine the force $\mathbf{P}$ required to move the roller onto the tiles if the roller is (a) pushed to the left, (b) pulled to the right.

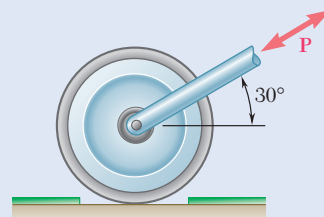


Fig. P4.72

- 4.73** A T-shaped bracket supports a 300-N load as shown. Determine the reactions at A and C when $\alpha = 45^\circ$.

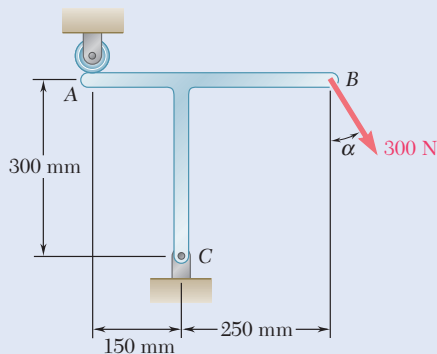


Fig. P4.73 and P4.74

- 4.74** A T-shaped bracket supports a 300-N load as shown. Determine the reactions at A and C when $\alpha = 60^\circ$.

- 4.75** Rod AB is supported by a pin and bracket at A and rests against a frictionless peg at C. Determine the reactions at A and C when a 170-N vertical force is applied at B.

- 4.76** Solve Prob. 4.75, assuming that the 170-N force applied at B is horizontal and directed to the left.

- 4.77** Member ABC is supported by a pin and bracket at B and by an inextensible cord attached at A and C and passing over a frictionless pulley at D. The tension may be assumed to be the same in portions AD and CD of the cord. For the loading shown and neglecting the size of the pulley, determine the tension in the cord and the reaction at B.

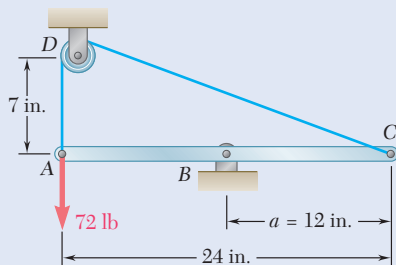


Fig. P4.77

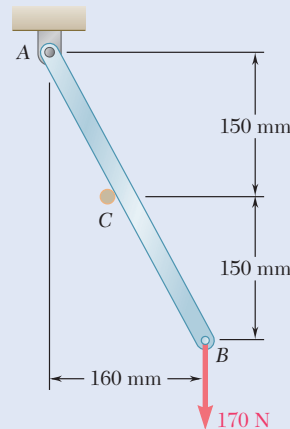


Fig. P4.75

- 4.78** Using the method of Sec. 4.2B, solve Prob. 4.22.

- 4.79** Knowing that $\theta = 30^\circ$, determine the reaction (a) at B, (b) at C.

- 4.80** Knowing that $\theta = 60^\circ$, determine the reaction (a) at B, (b) at C.

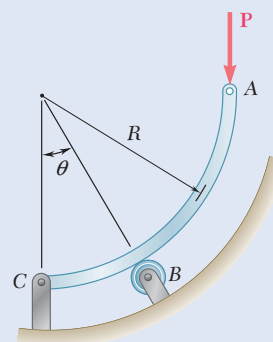


Fig. P4.79 and P4.80

4.81 Determine the reactions at A and B when $\beta = 50^\circ$.

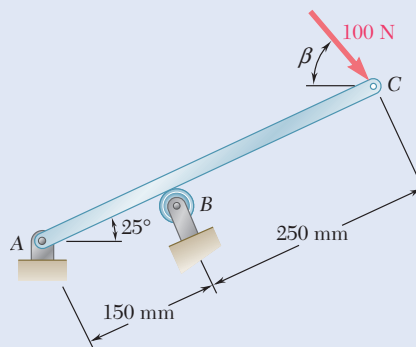


Fig. P4.81 and P4.82

4.82 Determine the reactions at A and B when $\beta = 80^\circ$.

4.83 Rod AB is bent into the shape of an arc of circle and is lodged between two pegs D and E . It supports a load P at end B . Neglecting friction and the weight of the rod, determine the distance c corresponding to equilibrium when $a = 20$ mm and $R = 100$ mm.

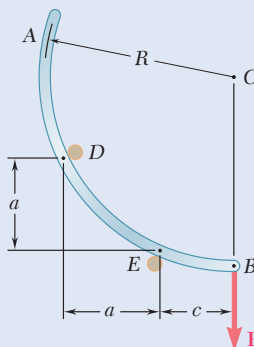


Fig. P4.83

4.84 A slender rod of length L is attached to collars that can slide freely along the guides shown. Knowing that the rod is in equilibrium, derive an expression for angle θ in terms of angle β .

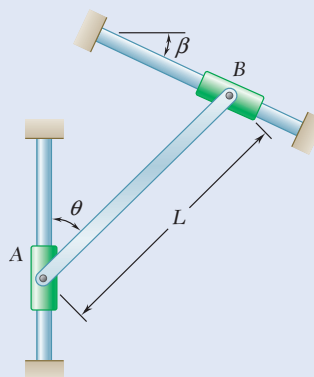


Fig. P4.84 and P4.85

4.85 An 8-kg slender rod of length L is attached to collars that can slide freely along the guides shown. Knowing that the rod is in equilibrium and that $\beta = 30^\circ$, determine (a) the angle θ that the rod forms with the vertical, (b) the reactions at A and B .

4.86 A uniform rod AB of length $2R$ rests inside a hemispherical bowl of radius R as shown. Neglecting friction, determine the angle θ corresponding to equilibrium.

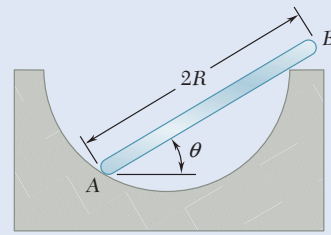


Fig. P4.86

4.87 A slender rod BC with a length of L and weight W is held by two cables as shown. Knowing that cable AB is horizontal and that the rod forms an angle of 40° with the horizontal, determine (a) the angle θ that cable CD forms with the horizontal, (b) the tension in each cable.

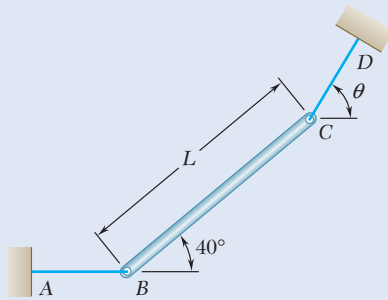


Fig. P4.87

4.88 A thin ring with a mass of 2 kg and radius $r = 140$ mm is held against a frictionless wall by a 125-mm string AB . Determine (a) the distance d , (b) the tension in the string, (c) the reaction at C .

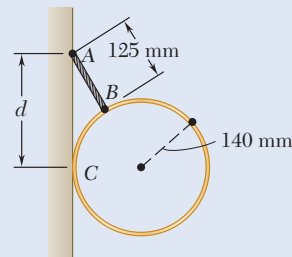


Fig. P4.88

4.89 A slender rod with a length of L and weight W is attached to a collar at A and is fitted with a small wheel at B . Knowing that the wheel rolls freely along a cylindrical surface of radius R , and neglecting friction, derive an equation in θ , L , and R that must be satisfied when the rod is in equilibrium.

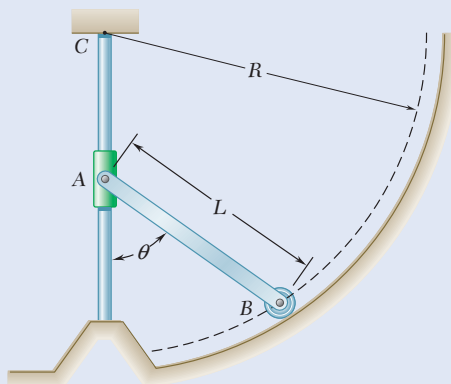


Fig. P4.89

4.90 Knowing that for the rod of Prob. 4.89, $L = 15$ in., $R = 20$ in., and $W = 10$ lb, determine (a) the angle θ corresponding to equilibrium, (b) the reactions at A and B .

4.3 EQUILIBRIUM IN THREE DIMENSIONS

The most general situation of rigid-body equilibrium occurs in three dimensions. The approach to modeling and analyzing these situations is the same as in two dimensions: Draw a free-body diagram and then write and solve the equilibrium equations. However, you now have more equations and more variables to deal with. In addition, reactions at supports and connections can be more varied, having as many as three force components and three couples acting at one support. As you will see in the Sample Problems, you need to visualize clearly in three dimensions and recall the vector analysis from Chapters 2 and 3.

4.3A Rigid-Body Equilibrium in Three Dimensions

We saw in Sec. 4.1 that six scalar equations are required to express the conditions for the equilibrium of a rigid body in the general three-dimensional case:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma F_z = 0 \quad (4.2)$$

$$\Sigma M_x = 0 \quad \Sigma M_y = 0 \quad \Sigma M_z = 0 \quad (4.3)$$

We can solve these equations for no more than *six unknowns*, which generally represent reactions at supports or connections.

In most problems, we can obtain the scalar equations (4.2) and (4.3) more conveniently if we first write the conditions for the equilibrium of the rigid body considered in vector form:

$$\Sigma \mathbf{F} = 0 \quad \Sigma \mathbf{M}_O = \Sigma (\mathbf{r} \times \mathbf{F}) = 0 \quad (4.1)$$

Then we can express the forces $\mathbf{F}$ and position vectors $\mathbf{r}$ in terms of scalar components and unit vectors. This enables us to compute all vector products either by direct calculation or by means of determinants (see Sec. 3.1F). Note that we can eliminate as many as three unknown reaction components from these computations through a judicious choice of the point O . By equating to zero the coefficients of the unit vectors in each of the two relations in Eq. (4.1), we obtain the desired scalar equations.[†]

Some equilibrium problems and their associated free-body diagrams might involve individual couples $\mathbf{M}_i$ either as applied loads or as support reactions. In such situations, you can accommodate these couples by expressing the second part of Eq. (4.1) as

$$\Sigma \mathbf{M}_O = \Sigma (\mathbf{r} \times \mathbf{F}) + \Sigma \mathbf{M}_i = 0 \quad (4.1')$$

4.3B Reactions for a Three-Dimensional Structure

The reactions on a three-dimensional structure range from a single force of known direction exerted by a frictionless surface to a force-couple system

[†]In some problems, it may be convenient to eliminate from the solution the reactions at two points A and B by writing the equilibrium equation $\Sigma M_{AB} = 0$. This involves determining the moments of the forces about the axis AB joining points A and B (see Sample Prob. 4.10).

exerted by a fixed support. Consequently, in problems involving the equilibrium of a three-dimensional structure, between one and six unknowns may be associated with the reaction at each support or connection.

Figure 4.10 shows various types of supports and connections with their corresponding reactions. A simple way of determining the type of reaction corresponding to a given support or connection and the number of unknowns involved is to find which of the six fundamental motions (translation in the x , y , and z directions and rotation about the x , y , and z axes) are allowed and which motions are prevented. The number of motions prevented equals the number of reactions.

Ball supports, frictionless surfaces, and cables, for example, prevent translation in one direction only and thus exert a single force whose line of action is known. Therefore, each of these supports involves one unknown—namely, the magnitude of the reaction. Rollers on rough surfaces and wheels on rails prevent translation in two directions; the corresponding reactions consist of two unknown force components. Rough surfaces in direct contact and ball-and-socket supports prevent translation in three directions while still allowing rotation; these supports involve three unknown force components.

Some supports and connections can prevent rotation as well as translation; the corresponding reactions include couples as well as forces. For example, the reaction at a fixed support, which prevents any motion (rotation as well as translation) consists of three unknown forces and three unknown couples. A universal joint, which is designed to allow rotation about two axes, exerts a reaction consisting of three unknown force components and one unknown couple.

Other supports and connections are primarily intended to prevent translation; their design, however, is such that they also prevent some rotations. The corresponding reactions consist essentially of force components but *may* also include couples. One group of supports of this type includes hinges and bearings designed to support radial loads only (for example, journal bearings or roller bearings). The corresponding reactions consist of two force components but may also include two couples. Another group includes pin-and-bracket supports, hinges, and bearings designed to support an axial thrust as well as a radial load (for example, ball bearings). The corresponding reactions consist of three force components but may include two couples. However, these supports do not exert any appreciable couples under normal conditions of use. Therefore, *only* force components should be included in their analysis *unless* it is clear that couples are necessary to maintain the equilibrium of the rigid body or unless the support is known to have been specifically designed to exert a couple (see Probs. 4.119 through 4.122).

If the reactions involve more than six unknowns, you have more unknowns than equations, and some of the reactions are **statically indeterminate**. If the reactions involve fewer than six unknowns, you have more equations than unknowns, and some of the equations of equilibrium cannot be satisfied under general loading conditions. In this case, the rigid body is only **partially constrained**. Under the particular loading conditions corresponding to a given problem, however, the extra equations often reduce to trivial identities, such as $0 = 0$, and can be disregarded; although only partially constrained, the rigid body remains in equilibrium (see Sample Probs. 4.7 and 4.8). Even with six or more unknowns, it is possible that some equations of equilibrium are not satisfied. This can occur when the reactions associated with the given supports either are parallel or intersect the same line; the rigid body is then **improperly constrained**.



Photo 4.3 Universal joints, seen on the drive shafts of rear-wheel-drive cars and trucks, allow rotational motion to be transferred between two noncollinear shafts.

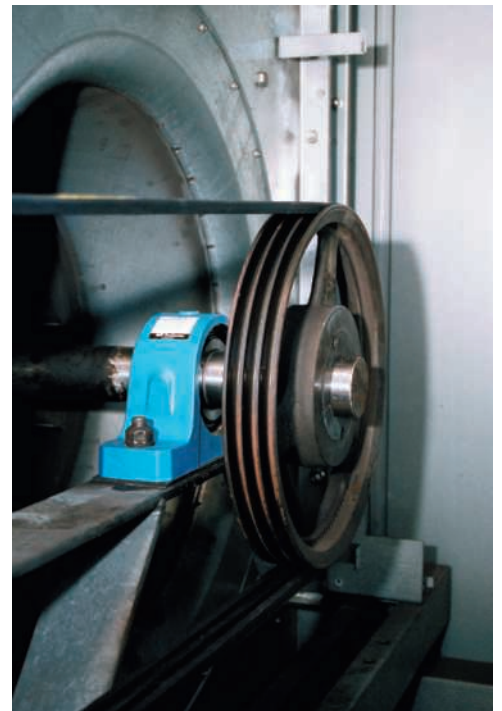


Photo 4.4 This pillow block bearing supports the shaft of a fan used in an industrial facility.

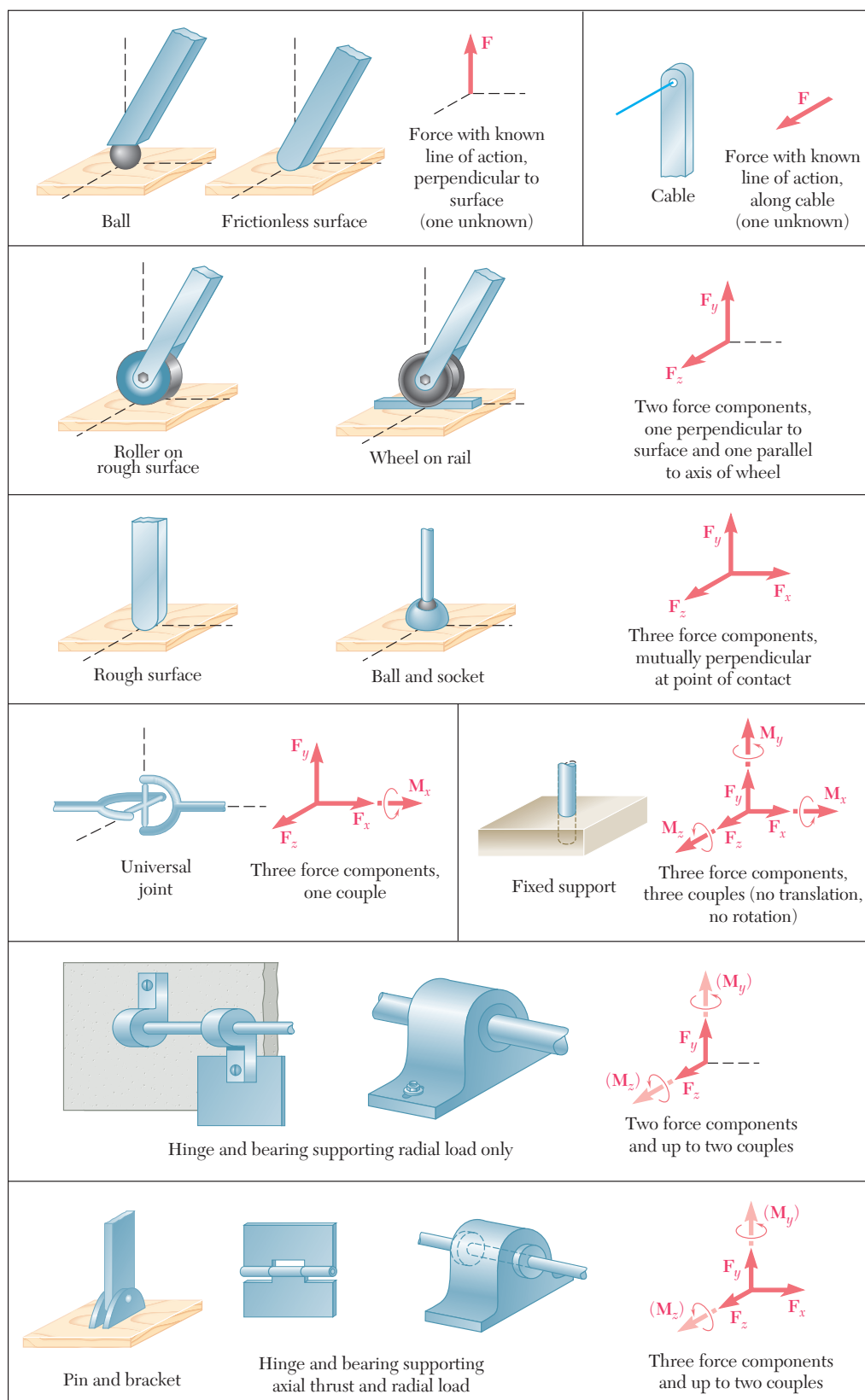


Fig. 4.10 Reactions at supports and connections in three dimensions.

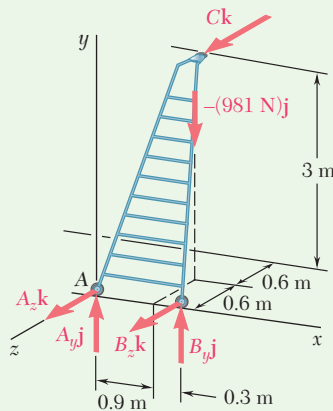
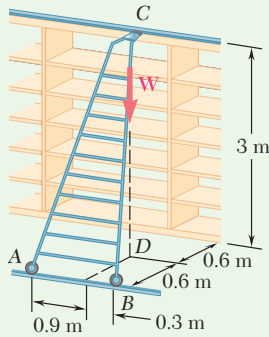


Fig. 1 Free-body diagram of ladder.

Sample Problem 4.7

A 20-kg ladder used to reach high shelves in a storeroom is supported by two flanged wheels *A* and *B* mounted on a rail and by a flangeless wheel *C* resting against a rail fixed to the wall. An 80-kg man stands on the ladder and leans to the right. The line of action of the combined weight **W** of the man and ladder intersects the floor at point *D*. Determine the reactions at *A*, *B*, and *C*.

STRATEGY: Draw a free-body diagram of the ladder, then write and solve the equilibrium equations in three dimensions.

MODELING:

Free-Body Diagram. The combined weight of the man and ladder is

$$\mathbf{W} = -mg\mathbf{j} = -(80 \text{ kg} + 20 \text{ kg})(9.81 \text{ m/s}^2)\mathbf{j} = -(981 \text{ N})\mathbf{j}$$

You have five unknown reaction components: two at each flanged wheel and one at the flangeless wheel (Fig. 1). The ladder is thus only partially constrained; it is free to roll along the rails. It is, however, in equilibrium under the given load because the equation $\Sigma F_x = 0$ is satisfied.

ANALYSIS:

Equilibrium Equations. The forces acting on the ladder form a system equivalent to zero:

$$\begin{aligned} \Sigma \mathbf{F} = 0: \quad & A_y\mathbf{j} + A_z\mathbf{k} + B_y\mathbf{j} + B_z\mathbf{k} - (981 \text{ N})\mathbf{j} + C\mathbf{k} = 0 \\ & (A_y + B_y - 981 \text{ N})\mathbf{j} + (A_z + B_z + C)\mathbf{k} = 0 \end{aligned} \quad (1)$$

$$\begin{aligned} \Sigma \mathbf{M}_A = \Sigma(\mathbf{r} \times \mathbf{F}) = 0: \quad & 1.2\mathbf{i} \times (B_y\mathbf{j} + B_z\mathbf{k}) + (0.9\mathbf{i} - 0.6\mathbf{k}) \times (-981\mathbf{j}) \\ & + (0.6\mathbf{i} + 3\mathbf{j} - 1.2\mathbf{k}) \times C\mathbf{k} = 0 \end{aligned}$$

Computing the vector products gives you[†]

$$\begin{aligned} 1.2B_y\mathbf{k} - 1.2B_z\mathbf{j} - 882.9\mathbf{k} - 588.6\mathbf{i} - 0.6C\mathbf{j} + 3C\mathbf{i} &= 0 \\ (3C - 588.6)\mathbf{i} - (1.2B_z + 0.6C)\mathbf{j} + (1.2B_y - 882.9)\mathbf{k} &= 0 \end{aligned} \quad (2)$$

Setting the coefficients of **i**, **j**, and **k** equal to zero in Eq. (2) produces the following three scalar equations, which state that the sum of the moments about each coordinate axis must be zero:

$$\begin{aligned} 3C - 588.6 &= 0 & C &= +196.2 \text{ N} \\ 1.2B_z + 0.6C &= 0 & B_z &= -98.1 \text{ N} \\ 1.2B_y - 882.9 &= 0 & B_y &= +736 \text{ N} \end{aligned}$$

The reactions at *B* and *C* are therefore

$$\mathbf{B} = +(736 \text{ N})\mathbf{j} - (98.1 \text{ N})\mathbf{k} \quad \mathbf{C} = +(196.2 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

[†]The moments in this sample problem, as well as in Sample Probs. 4.8 and 4.9, also can be expressed as determinants (see Sample Prob. 3.10).

Setting the coefficients of $\mathbf{j}$ and $\mathbf{k}$ equal to zero in Eq. (1), you obtain two scalar equations stating that the sums of the components in the y and z directions are zero. Substitute the values above for B_y , B_z , and C to get

$$\begin{array}{lll} A_y + B_y - 981 = 0 & A_y + 736 - 981 = 0 & A_y = +245 \text{ N} \\ A_z + B_z + C = 0 & A_z - 98.1 + 196.2 = 0 & A_z = -98.1 \text{ N} \end{array}$$

Therefore, the reaction at A is

$$\mathbf{A} = +(245 \text{ N})\mathbf{j} - (98.1 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

REFLECT and THINK: You summed moments about A as part of the analysis. As a check, you could now use these results and demonstrate that the sum of moments about any other point, such as point B , is also zero.

Sample Problem 4.8

A 5×8 -ft sign of uniform density weighs 270 lb and is supported by a ball-and-socket joint at A and by two cables. Determine the tension in each cable and the reaction at A .

STRATEGY: Draw a free-body diagram of the sign, and express the unknown cable tensions as Cartesian vectors. Then determine the cable tensions and the reaction at A by writing and solving the equilibrium equations.

MODELING:

Free-Body Diagram. The forces acting on the sign are its weight $\mathbf{W} = -(270 \text{ lb})\mathbf{j}$ and the reactions at A , B , and E (Fig. 1). The reaction at A is a force of unknown direction represented by three unknown components. Since the directions of the forces exerted by the cables are known, these forces involve only one unknown each: specifically, the magnitudes T_{BD} and T_{EC} . The total of five unknowns means that the sign is partially constrained. It can rotate freely about the x axis; it is, however, in equilibrium under the given loading, since the equation $\sum M_x = 0$ is satisfied.

ANALYSIS: You can express the components of the forces $\mathbf{T}_{BD}$ and $\mathbf{T}_{EC}$ in terms of the unknown magnitudes T_{BD} and T_{EC} as follows:

$$\begin{aligned} \overrightarrow{BD} &= -(8 \text{ ft})\mathbf{i} + (4 \text{ ft})\mathbf{j} - (8 \text{ ft})\mathbf{k} & BD &= 12 \text{ ft} \\ \overrightarrow{EC} &= -(6 \text{ ft})\mathbf{i} + (3 \text{ ft})\mathbf{j} + (2 \text{ ft})\mathbf{k} & EC &= 7 \text{ ft} \\ \mathbf{T}_{BD} &= T_{BD} \left(\frac{\overrightarrow{BD}}{BD} \right) = T_{BD} \left(-\frac{2}{3}\mathbf{i} + \frac{1}{3}\mathbf{j} - \frac{2}{3}\mathbf{k} \right) \\ \mathbf{T}_{EC} &= T_{EC} \left(\frac{\overrightarrow{EC}}{EC} \right) = T_{EC} \left(-\frac{6}{7}\mathbf{i} + \frac{3}{7}\mathbf{j} - \frac{2}{7}\mathbf{k} \right) \end{aligned}$$

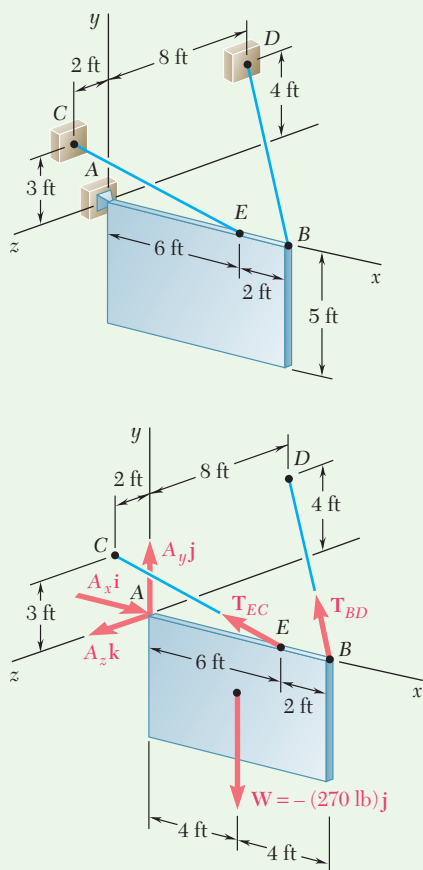


Fig. 1 Free-body diagram of sign.

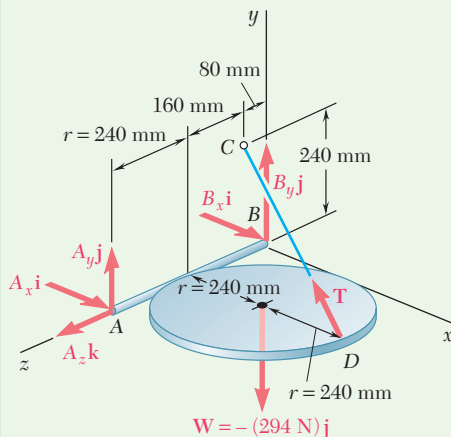


Fig. 1 Free-body diagram of pipe cover.

STRATEGY: Draw a free-body diagram with the coordinate axes shown (Fig. 1) and express the unknown cable tension as a Cartesian vector. Then apply the equilibrium equations to determine this tension and the support reactions.

MODELING:

Free-Body Diagram. The forces acting on the free body include its weight, which is

$$\mathbf{W} = -mg\mathbf{j} = -(30 \text{ kg})(9.81 \text{ m/s}^2)\mathbf{j} = -(294 \text{ N})\mathbf{j}$$

The reactions involve six unknowns: the magnitude of the force $\mathbf{T}$ exerted by the cable, three force components at hinge A, and two at hinge B. Express the components of $\mathbf{T}$ in terms of the unknown magnitude T by resolving the vector $\overrightarrow{DC}$ into rectangular components:

$$\overrightarrow{DC} = -(480 \text{ mm})\mathbf{i} + (240 \text{ mm})\mathbf{j} - (160 \text{ mm})\mathbf{k} \quad DC = 560 \text{ mm}$$

$$\mathbf{T} = T \frac{\overrightarrow{DC}}{DC} = -\frac{6}{7}T\mathbf{i} + \frac{3}{7}T\mathbf{j} - \frac{2}{7}T\mathbf{k}$$

ANALYSIS:

Equilibrium Equations. The forces acting on the pipe cover form a system equivalent to zero. Thus,

$$\begin{aligned} \Sigma \mathbf{F} = 0: \quad & A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k} + B_x\mathbf{i} + B_y\mathbf{j} + \mathbf{T} - (294 \text{ N})\mathbf{j} = 0 \\ & (A_x + B_x - \frac{6}{7}T)\mathbf{i} + (A_y + B_y + \frac{3}{7}T - 294 \text{ N})\mathbf{j} + (A_z - \frac{2}{7}T)\mathbf{k} = 0 \end{aligned} \quad (1)$$

$$\Sigma \mathbf{M}_B = \Sigma(\mathbf{r} \times \mathbf{F}) = 0:$$

$$\begin{aligned} 2r\mathbf{k} \times (A_x\mathbf{i} + A_y\mathbf{j} + A_z\mathbf{k}) \\ + (2r\mathbf{i} + r\mathbf{k}) \times (-\frac{6}{7}T\mathbf{i} + \frac{3}{7}T\mathbf{j} - \frac{2}{7}T\mathbf{k}) \\ + (r\mathbf{i} + r\mathbf{k}) \times (-294 \text{ N})\mathbf{j} = 0 \\ (-2A_y - \frac{3}{7}T + 294 \text{ N})r\mathbf{i} + (2A_x - \frac{2}{7}T)r\mathbf{j} + (\frac{6}{7}T - 294 \text{ N})r\mathbf{k} = 0 \end{aligned} \quad (2)$$

Setting the coefficients of the unit vectors equal to zero in Eq. (2) gives three scalar equations, which yield

$$A_x = +49.0 \text{ N} \quad A_y = +73.5 \text{ N} \quad T = 343 \text{ N} \quad \blacktriangleleft$$

Setting the coefficients of the unit vectors equal to zero in Eq. (1) produces three more scalar equations. After substituting the values of T , A_x , and A_y into these equations, you obtain

$$A_z = +98.0 \text{ N} \quad B_x = +245 \text{ N} \quad B_y = +73.5 \text{ N}$$

The reactions at A and B are therefore

$$\mathbf{A} = +(49.0 \text{ N})\mathbf{i} + (73.5 \text{ N})\mathbf{j} + (98.0 \text{ N})\mathbf{k} \quad \blacktriangleleft$$

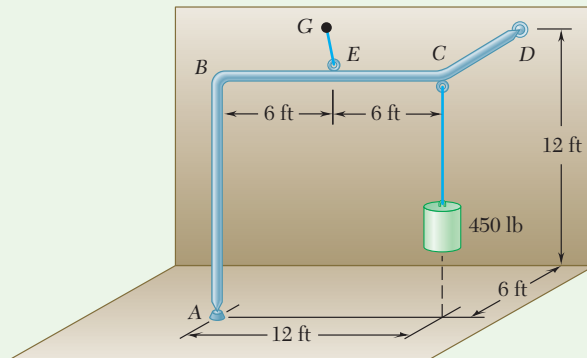
$$\mathbf{B} = +(245 \text{ N})\mathbf{i} + (73.5 \text{ N})\mathbf{j} \quad \blacktriangleleft$$

REFLECT and THINK: As a check, you can determine the tension in the cable using a scalar analysis. Assigning signs by the right-hand rule (rhr), we have

$$(+\text{rhr}) \quad \Sigma M_z = 0: \quad \frac{3}{7}T(0.48 \text{ m}) - (294 \text{ N})(0.24 \text{ m}) = 0 \quad T = 343 \text{ N} \quad \blacktriangleleft$$

Sample Problem 4.10

A 450-lb load hangs from the corner C of a rigid piece of pipe $ABCD$ that has been bent as shown. The pipe is supported by ball-and-socket joints A and D , which are fastened, respectively, to the floor and to a vertical wall, and by a cable attached at the midpoint E of the portion BC of the pipe and at a point G on the wall. Determine (a) where G should be located if the tension in the cable is to be minimum, (b) the corresponding minimum value of the tension.



STRATEGY: Draw the free-body diagram of the pipe showing the reactions at A and D . Isolate the unknown tension $\mathbf{T}$ and the known weight $\mathbf{W}$ by summing moments about the diagonal line AD , and compute values from the equilibrium equations.

MODELING and ANALYSIS:

Free-Body Diagram. The free-body diagram of the pipe includes the load $\mathbf{W} = (-450 \text{ lb})\mathbf{j}$, the reactions at A and D , and the force $\mathbf{T}$ exerted by the cable (Fig. 1). To eliminate the reactions at A and D from the computations, take the sum of the moments of the forces about the line AD and set it equal to zero. Denote the unit vector along AD by $\boldsymbol{\lambda}$, which enables you to write

$$\Sigma M_{AD} = 0: \quad \boldsymbol{\lambda} \cdot (\overrightarrow{AE} \times \mathbf{T}) + \boldsymbol{\lambda} \cdot (\overrightarrow{AC} \times \mathbf{W}) = 0 \quad (1)$$

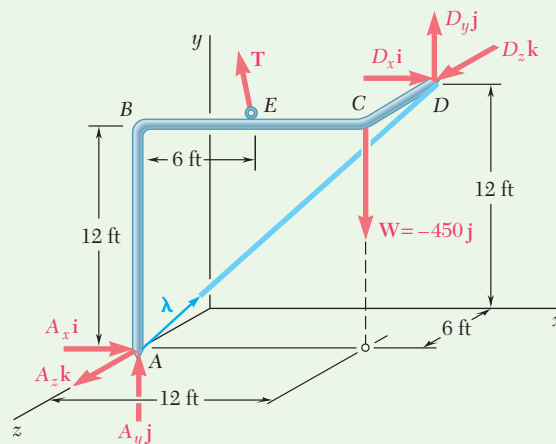


Fig. 1 Free-body diagram of pipe.

You can compute the second term in Eq. (1) as follows:

$$\begin{aligned}\overrightarrow{AC} \times \mathbf{W} &= (12\mathbf{i} + 12\mathbf{j}) \times (-450\mathbf{j}) = -5400\mathbf{k} \\ \boldsymbol{\lambda} &= \frac{\overrightarrow{AD}}{AD} = \frac{12\mathbf{i} + 12\mathbf{j} - 6\mathbf{k}}{18} = \frac{2}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{1}{3}\mathbf{k} \\ \boldsymbol{\lambda} \cdot (\overrightarrow{AC} \times \mathbf{W}) &= \left(\frac{2}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{1}{3}\mathbf{k}\right) \cdot (-5400\mathbf{k}) = +1800\end{aligned}$$

Substituting this value into Eq. (1) gives

$$\boldsymbol{\lambda} \cdot (\overrightarrow{AE} \times \mathbf{T}) = -1800 \text{ lb}\cdot\text{ft} \quad (2)$$

Minimum Value of Tension. Recalling the commutative property for mixed triple products, you can rewrite Eq. (2) in the form

$$\mathbf{T} \cdot (\boldsymbol{\lambda} \times \overrightarrow{AE}) = -1800 \text{ lb}\cdot\text{ft} \quad (3)$$

This shows that the projection of $\mathbf{T}$ on the vector $\boldsymbol{\lambda} \times \overrightarrow{AE}$ is a constant. It follows that $\mathbf{T}$ is minimum when it is parallel to the vector

$$\boldsymbol{\lambda} \times \overrightarrow{AE} = \left(\frac{2}{3}\mathbf{i} + \frac{2}{3}\mathbf{j} - \frac{1}{3}\mathbf{k}\right) \times (6\mathbf{i} + 12\mathbf{j}) = 4\mathbf{i} - 2\mathbf{j} + 4\mathbf{k}$$

The corresponding unit vector is $\frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}$, which gives

$$\mathbf{T}_{\min} = T\left(\frac{2}{3}\mathbf{i} - \frac{1}{3}\mathbf{j} + \frac{2}{3}\mathbf{k}\right) \quad (4)$$

Substituting for $\mathbf{T}$ and $\boldsymbol{\lambda} \times \overrightarrow{AE}$ in Eq. (3) and computing the dot products yields $6T = -1800$ and, thus, $T = -300$. Carrying this value into Eq. (4) gives you

$$\mathbf{T}_{\min} = -200\mathbf{i} + 100\mathbf{j} - 200\mathbf{k} \quad T_{\min} = 300 \text{ lb} \quad \blacktriangleleft$$

Location of G. Since the vector $\overrightarrow{EG}$ and the force $\mathbf{T}_{\min}$ have the same direction, their components must be proportional. Denoting the coordinates of G by x , y , and 0 (Fig. 2), you get

$$\frac{x - 6}{-200} = \frac{y - 12}{+100} = \frac{0 - 6}{-200} \quad x = 0 \quad y = 15 \text{ ft} \quad \blacktriangleleft$$

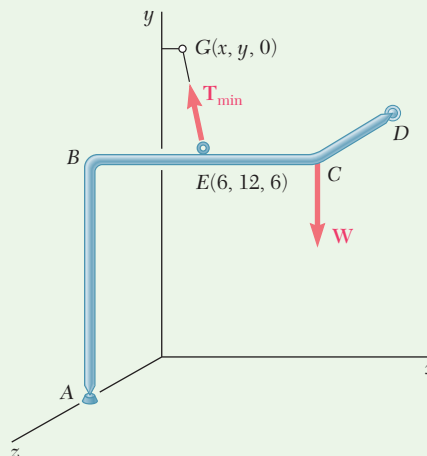


Fig. 2 Location of point G for minimum tension in cable.

REFLECT and THINK: Sometimes you have to rely on the vector analysis presented in Chapters 2 and 3 as much as on the conditions for equilibrium described in this chapter.

SOLVING PROBLEMS ON YOUR OWN

In this section, you considered the equilibrium of a *three-dimensional body*. It is again most important that you draw a complete *free-body diagram* as the first step of your solution.

1. Pay particular attention to the reactions at the supports as you draw the free-body diagram. The number of unknowns at a support can range from one to six (Fig. 4.10). To decide whether an unknown reaction or reaction component exists at a support, ask yourself whether the support prevents motion of the body in a certain direction or about a certain axis.

a. If motion is prevented in a certain direction, include in your free-body diagram an unknown *reaction* or *reaction component* that acts in the *same direction*.

b. If a support prevents rotation about a certain axis, include in your free-body diagram a *couple* of unknown magnitude that acts about the *same axis*.

2. The external forces acting on a three-dimensional body form a system equivalent to zero. Writing $\Sigma \mathbf{F} = 0$ and $\Sigma \mathbf{M}_A = 0$ about an appropriate point A and setting the coefficients of $\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$ in both equations equal to zero provides you with six scalar equations. In general, these equations contain six unknowns and may be solved for these unknowns.

3. After completing your free-body diagram, you may want to seek equations involving as few unknowns as possible. The following strategies may help you.

a. By summing moments about a ball-and-socket support or a hinge, you obtain equations from which three unknown reaction components have been eliminated [Sample Probs. 4.8 and 4.9].

b. If you can draw an axis through the points of application of all but one of the unknown reactions, summing moments about that axis will yield an equation in a single unknown [Sample Prob. 4.10].

4. After drawing your free-body diagram, you may find that one of the following situations exists.

a. The reactions involve fewer than six unknowns. The body is partially constrained and motion of the body is possible. However, you may be able to determine the reactions for a given loading condition [Sample Prob. 4.7].

b. The reactions involve more than six unknowns. The reactions are statically indeterminate. Although you may be able to calculate one or two reactions, you cannot determine all of them [Sample Prob. 4.10].

c. The reactions are parallel or intersect the same line. The body is improperly constrained, and motion can occur under a general loading condition.

Problems

FREE-BODY PRACTICE PROBLEMS

- 4.F5** Two tape spools are attached to an axle supported by bearings at A and D . The radius of spool B is 1.5 in. and the radius of spool C is 2 in. Knowing that $T_B = 20$ lb and that the system rotates at a constant rate, draw the free-body diagram needed to determine the reactions at A and D . Assume that the bearing at A does not exert any axial thrust and neglect the weights of the spools and axle.

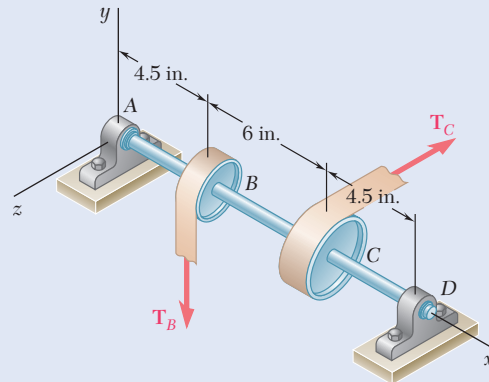


Fig. P4.F5

- 4.F6** A 12-m pole supports a horizontal cable CD and is held by a ball and socket at A and two cables BE and BF . Knowing that the tension in cable CD is 14 kN and assuming that CD is parallel to the x axis ($\phi = 0$), draw the free-body diagram needed to determine the tension in cables BE and BF and the reaction at A .
- 4.F7** A 20-kg cover for a roof opening is hinged at corners A and B . The roof forms an angle of 30° with the horizontal, and the cover is maintained in a horizontal position by the brace CE . Draw the free-body diagram needed to determine the magnitude of the force exerted by the brace and the reactions at the hinges. Assume that the hinge at A does not exert any axial thrust.

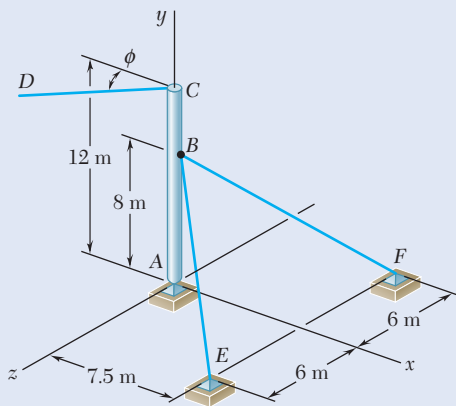


Fig. P4.F6

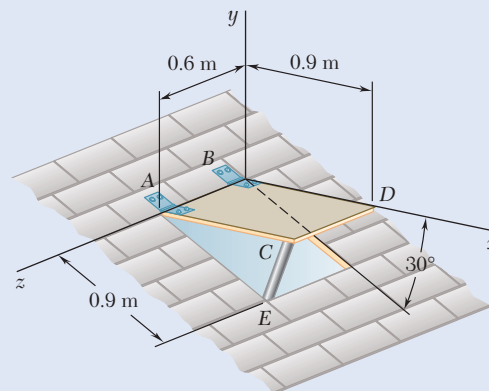


Fig. P4.F7

END-OF-SECTION PROBLEMS

4.91 Two transmission belts pass over a double-sheaved pulley that is attached to an axle supported by bearings at A and D . The radius of the inner sheave is 125 mm and the radius of the outer sheave is 250 mm. Knowing that when the system is at rest, the tension is 90 N in both portions of belt B and 150 N in both portions of belt C , determine the reactions at A and D . Assume that the bearing at D does not exert any axial thrust.

4.92 Solve Prob. 4.91, assuming that the pulley rotates at a constant rate and that $T_B = 104$ N, $T'_B = 84$ N, and $T_C = 175$ N.

4.93 A small winch is used to raise a 120-lb load. Find (a) the magnitude of the vertical force $\mathbf{P}$ that should be applied at C to maintain equilibrium in the position shown, (b) the reactions at A and B , assuming that the bearing at B does not exert any axial thrust.

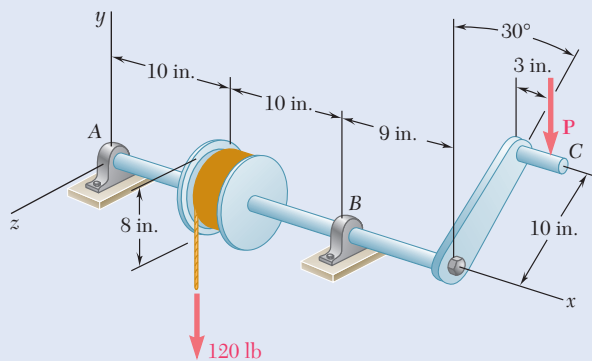


Fig. P4.93

4.94 A 4×8 -ft sheet of plywood weighing 34 lb has been temporarily placed among three pipe supports. The lower edge of the sheet rests on small collars at A and B and its upper edge leans against pipe C . Neglecting friction at all surfaces, determine the reactions at A , B , and C .

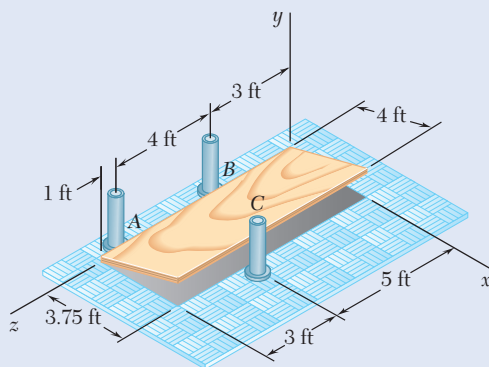


Fig. P4.94

4.95 A 250×400 -mm plate of mass 12 kg and a 300-mm-diameter pulley are welded to axle AC that is supported by bearings at A and B . For $\beta = 30^\circ$, determine (a) the tension in the cable, (b) the reactions at A and B . Assume that the bearing at B does not exert any axial thrust.

4.96 Solve Prob. 4.95 for $\beta = 60^\circ$.

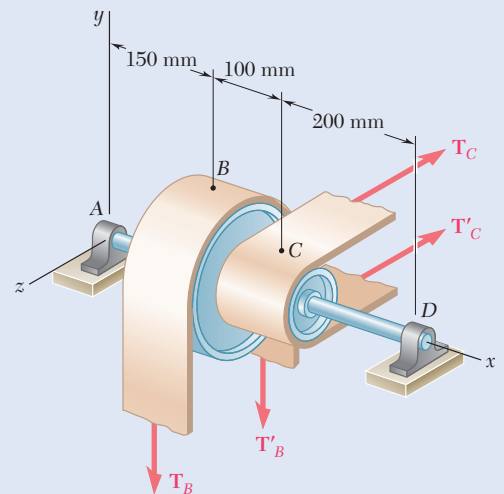


Fig. P4.91

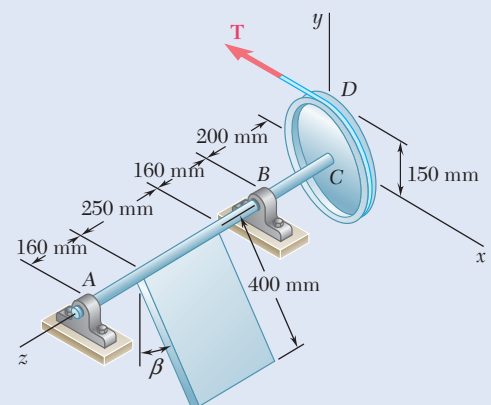


Fig. P4.95

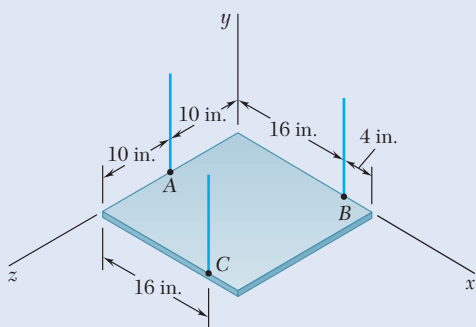


Fig. P4.97 and P4.98

4.97 The 20×20 -in. square plate shown weighs 56 lb and is supported by three vertical wires. Determine the tension in each wire.

4.98 The 20×20 -in. square plate shown weighs 56 lb and is supported by three vertical wires. Determine the weight and location of the lightest block that should be placed on the plate if the tensions in the three wires are to be equal.

4.99 An opening in a floor is covered by a 1×1.2 -m sheet of plywood with a mass of 18 kg. The sheet is hinged at A and B and is maintained in a position slightly above the floor by a small block C. Determine the vertical component of the reaction (a) at A, (b) at B, (c) at C.

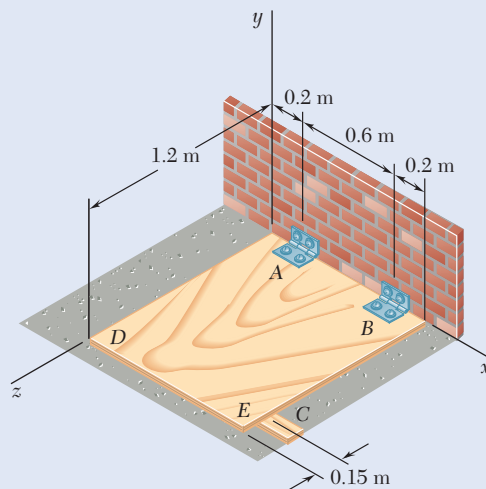


Fig. P4.99

4.100 Solve Prob. 4.99, assuming that the small block C is moved and placed under edge DE at a point 0.15 m from corner E.

4.101 Two steel pipes AB and BC, each having a mass per unit length of 8 kg/m, are welded together at B and supported by three vertical wires. Knowing that $a = 0.4$ m, determine the tension in each wire.

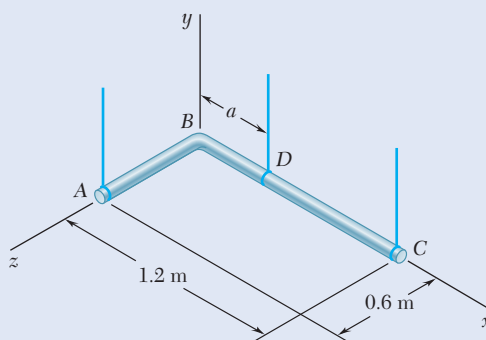


Fig. P4.101

4.102 For the pipe assembly of Prob. 4.101, determine (a) the largest permissible value of a if the assembly is not to tip, (b) the corresponding tension in each wire.

4.103 The 24-lb square plate shown is supported by three vertical wires. Determine (a) the tension in each wire when $a = 10$ in., (b) the value of a for which the tension in each wire is 8 lb.

4.104 The table shown weighs 30 lb and has a diameter of 4 ft. It is supported by three legs equally spaced around the edge. A vertical load $\mathbf{P}$ with a magnitude of 100 lb is applied to the top of the table at D . Determine the maximum value of a if the table is not to tip over. Show, on a sketch, the area of the table over which $\mathbf{P}$ can act without tipping the table.

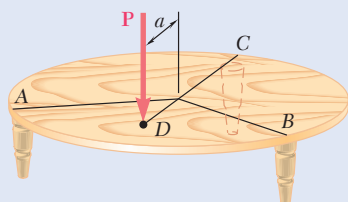


Fig. P4.104

4.105 A 10-ft boom is acted upon by the 840-lb force shown. Determine the tension in each cable and the reaction at the ball-and-socket joint at A .

4.106 The 6-m pole ABC is acted upon by a 455-N force as shown. The pole is held by a ball-and-socket joint at A and by two cables BD and BE . For $a = 3$ m, determine the tension in each cable and the reaction at A .

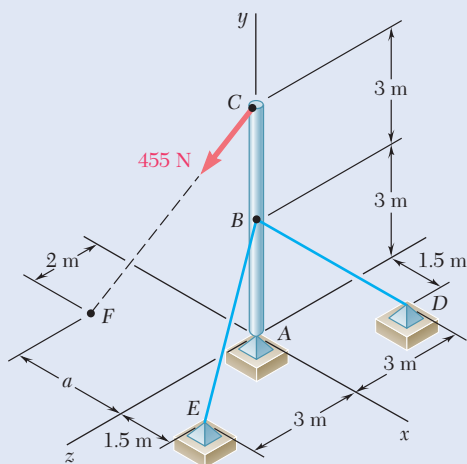


Fig. P4.106

4.107 Solve Prob. 4.106 for $a = 1.5$ m.

4.108 A 2.4-m boom is held by a ball-and-socket joint at C and by two cables AD and AE . Determine the tension in each cable and the reaction at C .

4.109 Solve Prob. 4.108, assuming that the 3.6-kN load is applied at point A .

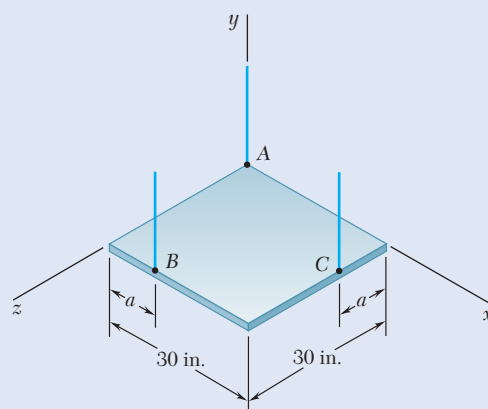


Fig. P4.103

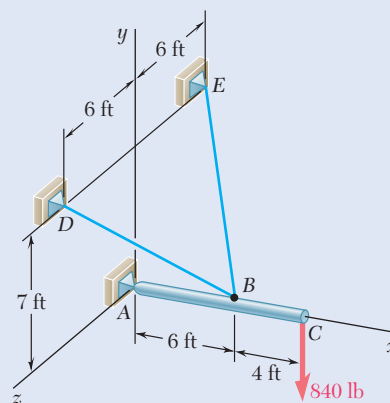


Fig. P4.105

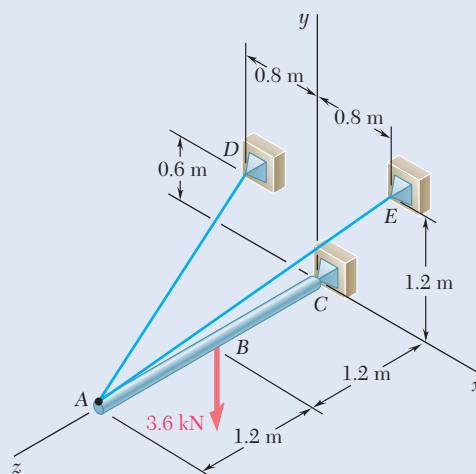


Fig. P4.108

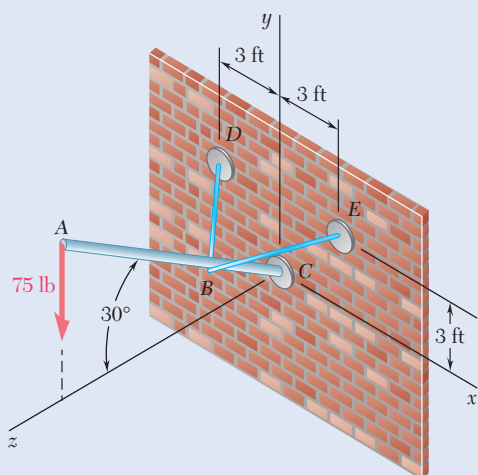


Fig. P4.110

4.110 The 10-ft flagpole AC forms an angle of 30° with the z axis. It is held by a ball-and-socket joint at C and by two thin braces BD and BE . Knowing that the distance BC is 3 ft, determine the tension in each brace and the reaction at C .

4.111 A 48-in. boom is held by a ball-and-socket joint at C and by two cables BF and DAE ; cable DAE passes around a frictionless pulley at A . For the loading shown, determine the tension in each cable and the reaction at C .

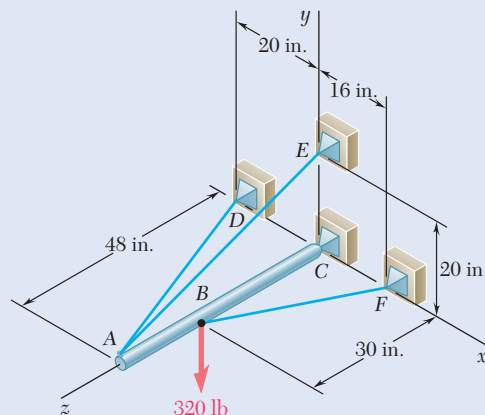


Fig. P4.111

4.112 Solve Prob. 4.111, assuming that the 320-lb load is applied at A .

4.113 A 10-kg storm window measuring 900×1500 mm is held by hinges at A and B . In the position shown, it is held away from the side of the house by a 600-mm stick CD . Assuming that the hinge at A does not exert any axial thrust, determine the magnitude of the force exerted by the stick and the components of the reactions at A and B .

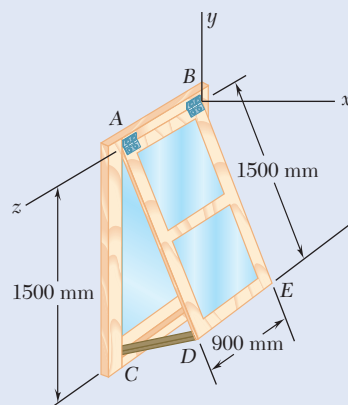


Fig. P4.113

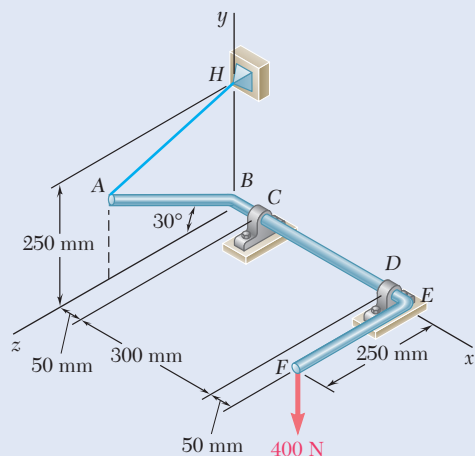


Fig. P4.114

4.114 The bent rod $ABEF$ is supported by bearings at C and D and by wire AH . Knowing that portion AB of the rod is 250 mm long, determine (a) the tension in wire AH , (b) the reactions at C and D . Assume that the bearing at D does not exert any axial thrust.

- 4.115** The horizontal platform $ABCD$ weighs 60 lb and supports a 240-lb load at its center. The platform is normally held in position by hinges at A and B and by braces CE and DE . If brace DE is removed, determine the reactions at the hinges and the force exerted by the remaining brace CE . The hinge at A does not exert any axial thrust.

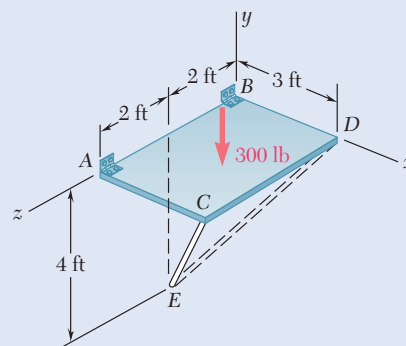


Fig. P4.115

- 4.116** The lid of a roof scuttle weighs 75 lb. It is hinged at corners A and B and maintained in the desired position by a rod CD pivoted at C . A pin at end D of the rod fits into one of several holes drilled in the edge of the lid. For $\alpha = 50^\circ$, determine (a) the magnitude of the force exerted by rod CD , (b) the reactions at the hinges. Assume that the hinge at B does not exert any axial thrust.

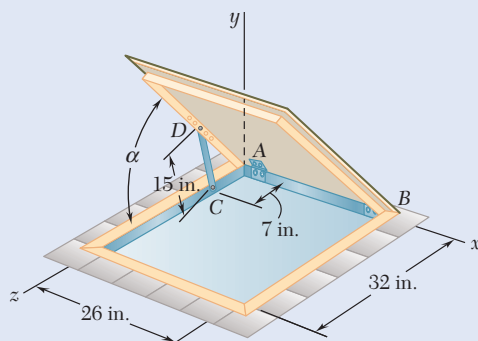


Fig. P4.116

- 4.117** A 100-kg uniform rectangular plate is supported in the position shown by hinges A and B and by cable DCE that passes over a frictionless hook at C . Assuming that the tension is the same in both parts of the cable, determine (a) the tension in the cable, (b) the reactions at A and B . Assume that the hinge at B does not exert any axial thrust.

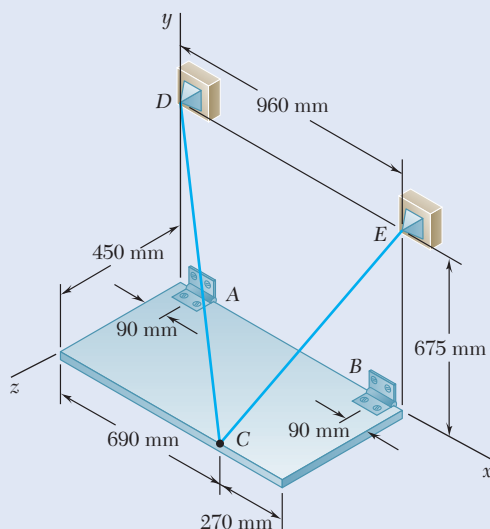


Fig. P4.117

- 4.118** Solve Prob. 4.117, assuming that cable DCE is replaced by a cable attached to point E and hook C .

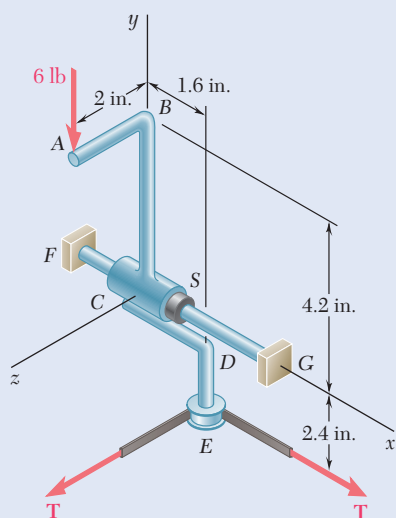


Fig. P4.121

4.119 Solve Prob. 4.113, assuming that the hinge at *A* has been removed and that the hinge at *B* can exert couples about axes parallel to the *x* and *y* axes.

4.120 Solve Prob. 4.115, assuming that the hinge at *B* has been removed and that the hinge at *A* can exert an axial thrust, as well as couples about axes parallel to the *x* and *y* axes.

4.121 The assembly shown is used to control the tension *T* in a tape that passes around a frictionless spool at *E*. Collar *C* is welded to rods *ABC* and *CDE*. It can rotate about shaft *FG* but its motion along the shaft is prevented by a washer *S*. For the loading shown, determine (a) the tension *T* in the tape, (b) the reaction at *C*.

4.122 The assembly shown is welded to collar *A* that fits on the vertical pin shown. The pin can exert couples about the *x* and *z* axes but does not prevent motion about or along the *y* axis. For the loading shown, determine the tension in each cable and the reaction at *A*.

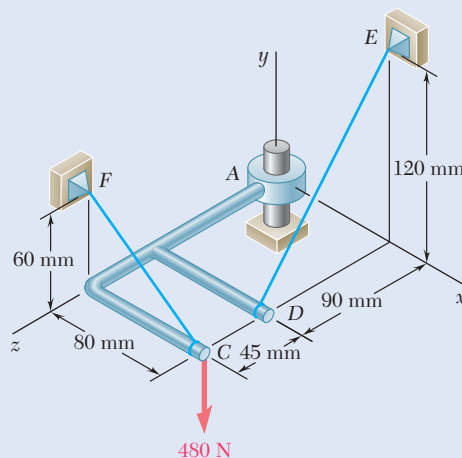


Fig. P4.122

4.123 The rigid L-shaped member *ABC* is supported by a ball-and-socket joint at *A* and by three cables. If a 1.8-kN load is applied at *F*, determine the tension in each cable.

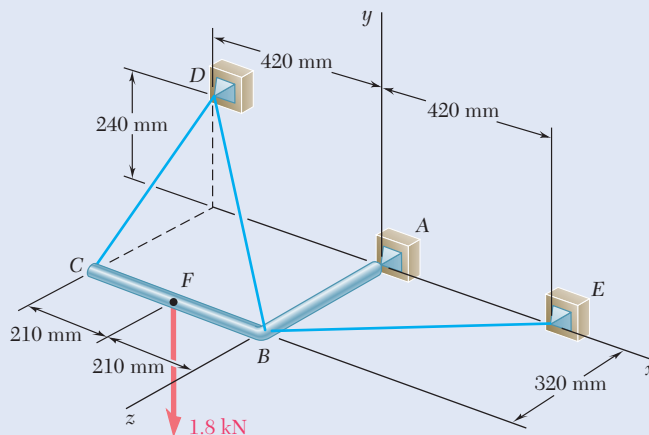


Fig. P4.123

4.124 Solve Prob. 4.123, assuming that the 1.8-kN load is applied at C .

4.125 The rigid L-shaped member ABF is supported by a ball-and-socket joint at A and by three cables. For the loading shown, determine the tension in each cable and the reaction at A .

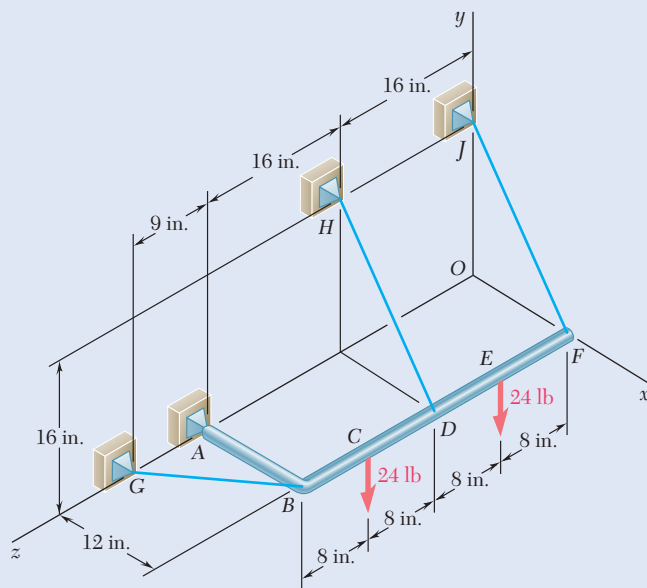


Fig. P4.125

4.126 Solve Prob. 4.125, assuming that the load at C has been removed.

4.127 Three rods are welded together to form a “corner” that is supported by three eyebolts. Neglecting friction, determine the reactions at A , B , and C when $P = 240$ lb, $a = 12$ in., $b = 8$ in., and $c = 10$ in.

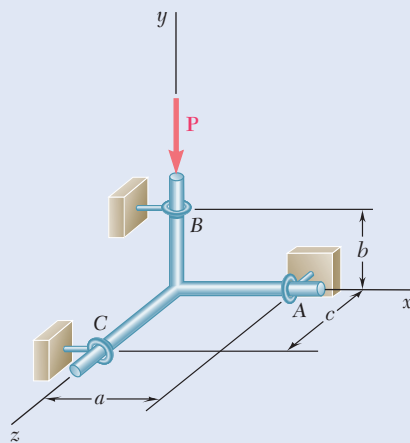


Fig. P4.127

4.128 Solve Prob. 4.127, assuming that the force $\mathbf{P}$ is removed and is replaced by a couple $\mathbf{M} = + (600 \text{ lb}\cdot\text{in.})\mathbf{j}$ acting at B .

- 4.129** Frame $ABCD$ is supported by a ball-and-socket joint at A and by three cables. For $a = 150$ mm, determine the tension in each cable and the reaction at A .

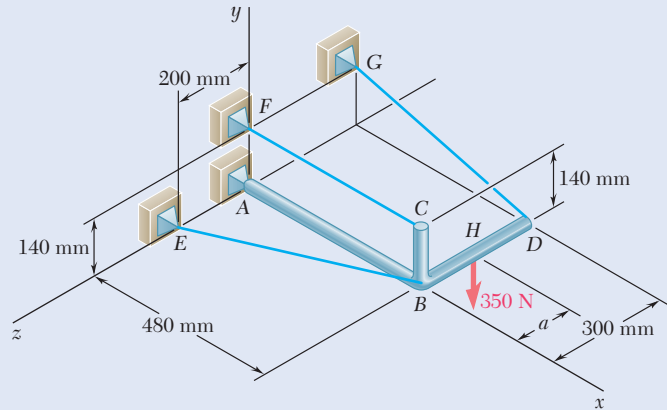


Fig. P4.129 and P4.130

- 4.130** Frame $ABCD$ is supported by a ball-and-socket joint at A and by three cables. Knowing that the 350-N load is applied at D ($a = 300$ mm), determine the tension in each cable and the reaction at A .

- 4.131** The assembly shown consists of an 80-mm rod AF that is welded to a cross frame consisting of four 200-mm arms. The assembly is supported by a ball-and-socket joint at F and by three short links, each of which forms an angle of 45° with the vertical. For the loading shown, determine (a) the tension in each link, (b) the reaction at F .

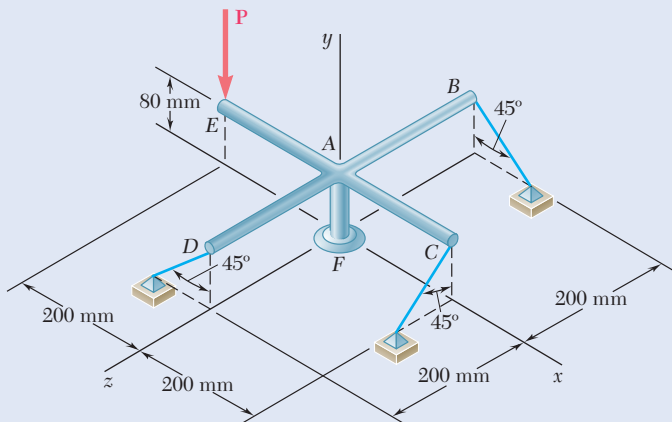


Fig. P4.131

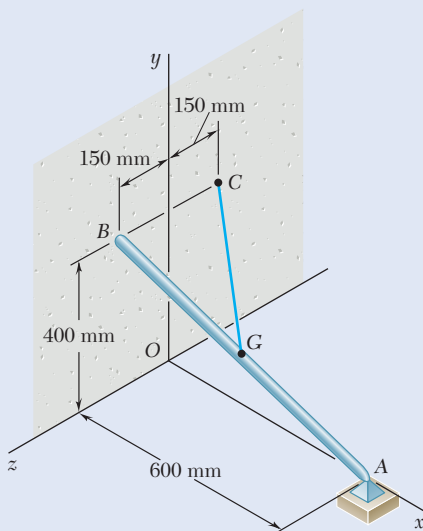


Fig. P4.132

- 4.132** The uniform 10-kg rod AB is supported by a ball-and-socket joint at A and by the cord CG that is attached to the midpoint G of the rod. Knowing that the rod leans against a frictionless vertical wall at B , determine (a) the tension in the cord, (b) the reactions at A and B .

- 4.133** The frame ACD is supported by ball-and-socket joints at A and D and by a cable that passes through a ring at B and is attached to hooks at G and H . Knowing that the frame supports at point C a load of magnitude $P = 268$ N, determine the tension in the cable.

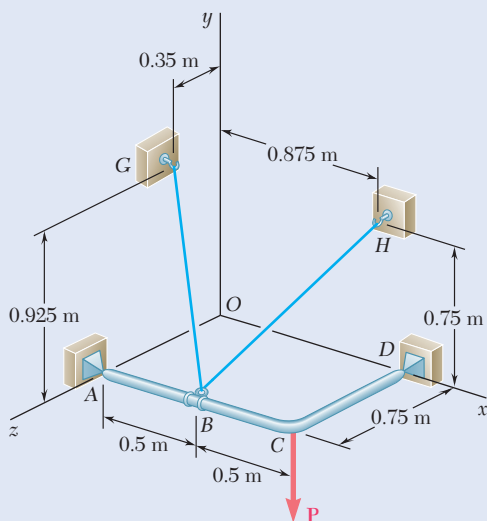


Fig. P4.133

- 4.134** Solve Prob. 4.133, assuming that cable GBH is replaced by a cable GB attached at G and B .

- 4.135** The bent rod $ABDE$ is supported by ball-and-socket joints at A and E and by the cable DF . If a 60-lb load is applied at C as shown, determine the tension in the cable.

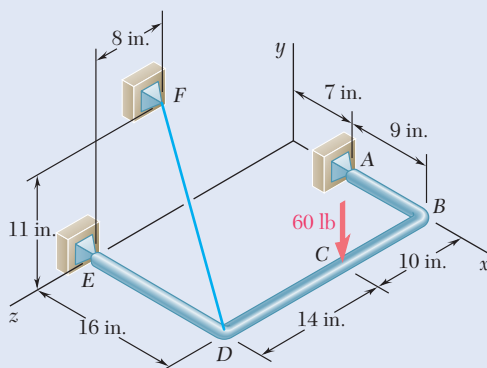


Fig. P4.135

- 4.136** Solve Prob. 4.135, assuming that cable DF is replaced by a cable connecting B and F .

- 4.137** Two rectangular plates are welded together to form the assembly shown. The assembly is supported by ball-and-socket joints at B and D and by a ball on a horizontal surface at C . For the loading shown, determine the reaction at C .

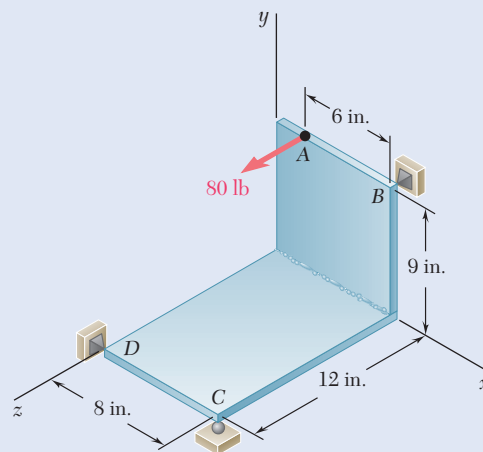


Fig. P4.137

- 4.138** The pipe $ACDE$ is supported by ball-and-socket joints at A and E and by the wire DF . Determine the tension in the wire when a 640-N load is applied at B as shown.

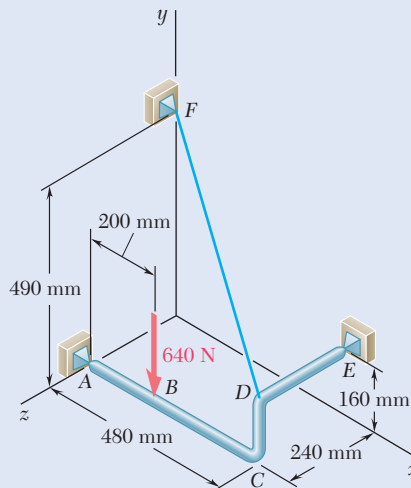


Fig. P4.138

- 4.139** Solve Prob. 4.138, assuming that wire DF is replaced by a wire connecting C and F .
- 4.140** Two 2×4 -ft plywood panels, each with a weight of 12 lb, are nailed together as shown. The panels are supported by ball-and-socket joints at A and F and by the wire BH . Determine (a) the location of H in the xy plane if the tension in the wire is to be minimum, (b) the corresponding minimum tension.

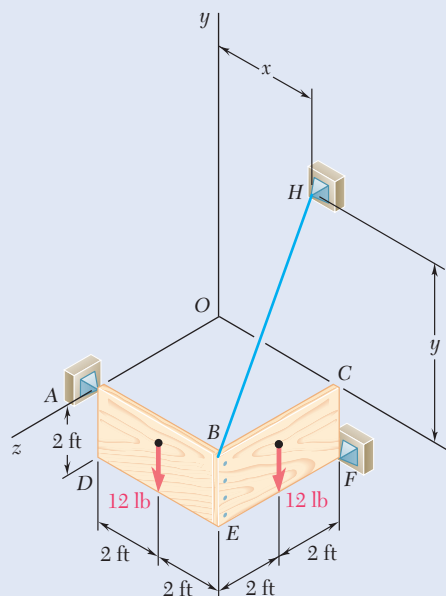


Fig. P4.140

- 4.141** Solve Prob. 4.140, subject to the restriction that H must lie on the y axis.

Review and Summary

Equilibrium Equations

This chapter was devoted to the study of the **equilibrium of rigid bodies**, i.e., to the situation when the external forces acting on a rigid body *form a system equivalent to zero* [Introduction]. We then have

$$\Sigma \mathbf{F} = 0 \quad \Sigma \mathbf{M}_O = \Sigma (\mathbf{r} \times \mathbf{F}) = 0 \quad (4.1)$$

Resolving each force and each moment into its rectangular components, we can express the necessary and sufficient conditions for the equilibrium of a rigid body with the following six scalar equations:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma F_z = 0 \quad (4.2)$$

$$\Sigma M_x = 0 \quad \Sigma M_y = 0 \quad \Sigma M_z = 0 \quad (4.3)$$

We can use these equations to determine unknown forces applied to the rigid body or unknown reactions exerted by its supports.

Free-Body Diagram

When solving a problem involving the equilibrium of a rigid body, it is essential to consider *all* of the forces acting on the body. Therefore, the first step in the solution of the problem should be to draw a **free-body diagram** showing the body under consideration and all of the unknown as well as known forces acting on it.

Equilibrium of a Two-Dimensional Structure

In the first part of this chapter, we considered the **equilibrium of a two-dimensional structure**; i.e., we assumed that the structure considered and the forces applied to it were contained in the same plane. We saw that each of the reactions exerted on the structure by its supports could involve one, two, or three unknowns, depending upon the type of support [Sec. 4.1A].

In the case of a two-dimensional structure, the equations given previously reduce to *three equilibrium equations*:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma M_A = 0 \quad (4.5)$$

where A is an arbitrary point in the plane of the structure [Sec. 4.1B]. We can use these equations to solve for three unknowns. Although the three equilibrium equations (4.5) cannot be *augmented* with additional equations, any of them can be *replaced* by another equation. Therefore, we can write alternative sets of equilibrium equations, such as

$$\Sigma F_x = 0 \quad \Sigma M_A = 0 \quad \Sigma M_B = 0 \quad (4.6)$$

where point B is chosen in such a way that the line AB is not parallel to the y axis, or

$$\Sigma M_A = 0 \quad \Sigma M_B = 0 \quad \Sigma M_C = 0 \quad (4.7)$$

where the points A , B , and C do not lie in a straight line.

Static Indeterminacy, Partial Constraints, Improper Constraints

Since any set of equilibrium equations can be solved for only three unknowns, the reactions at the supports of a rigid two-dimensional structure cannot be

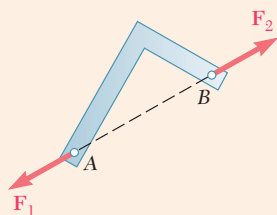


Fig. 4.11

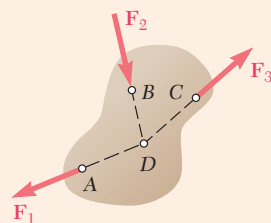


Fig. 4.12

completely determined if they involve *more than three unknowns*; they are said to be *statically indeterminate* [Sec. 4.1C]. On the other hand, if the reactions involve *fewer than three unknowns*, equilibrium is not maintained under general loading conditions; the structure is said to be *partially constrained*. The fact that the reactions involve exactly three unknowns is no guarantee that you can solve the equilibrium equations for all three unknowns. If the supports are arranged in such a way that the reactions are *either concurrent or parallel*, the reactions are statically indeterminate, and the structure is said to be *improperly constrained*.

Two-Force Body, Three-Force Body

We gave special attention in Sec. 4.2 to two particular cases of equilibrium of a rigid body. We defined a **two-force body** as a rigid body subjected to forces at only two points, and we showed that the resultants $\mathbf{F}_1$ and $\mathbf{F}_2$ of these forces must have the *same magnitude, the same line of action, and opposite sense* (Fig. 4.11), which is a property that simplifies the solution of certain problems in later chapters. We defined a **three-force body** as a rigid body subjected to forces at only three points, and we demonstrated that the resultants $\mathbf{F}_1$, $\mathbf{F}_2$, and $\mathbf{F}_3$ of these forces must be *either concurrent* (Fig. 4.12) *or parallel*. This property provides us with an alternative approach to the solution of problems involving a three-force body [Sample Prob. 4.6].

Equilibrium of a Three-Dimensional Body

In the second part of this chapter, we considered the *equilibrium of a three-dimensional body*. We saw that each of the reactions exerted on the body by its supports could involve between one and six unknowns, depending upon the type of support [Sec. 4.3A].

In the general case of the equilibrium of a three-dimensional body, all six of the scalar equilibrium equations (4.2) and (4.3) should be used and solved for *six unknowns* [Sec. 4.3B]. In most problems, however, we can obtain these equations more conveniently if we start from

$$\Sigma \mathbf{F} = 0 \quad \Sigma \mathbf{M}_O = \Sigma (\mathbf{r} \times \mathbf{F}) = 0 \quad (4.1)$$

and then express the forces $\mathbf{F}$ and position vectors $\mathbf{r}$ in terms of scalar components and unit vectors. We can compute the vector products either directly or by means of determinants, and obtain the desired scalar equations by equating to zero the coefficients of the unit vectors [Sample Probs. 4.7 through 4.9].

We noted that we may eliminate as many as three unknown reaction components from the computation of $\Sigma \mathbf{M}_O$ in the second of the relations (4.1) through a judicious choice of point O . Also, we can eliminate the reactions at two points A and B from the solution of some problems by writing the equation $\Sigma M_{AB} = 0$, which involves the computation of the moments of the forces about an axis AB joining points A and B [Sample Prob. 4.10].

We observed that when a body is subjected to individual couples $\mathbf{M}_i$, either as applied loads or as support reactions, we can include these couples by expressing the second part of Eq. (4.1) as

$$\Sigma \mathbf{M}_O = \Sigma (\mathbf{r} \times \mathbf{F}) + \Sigma \mathbf{M}_i = 0 \quad (4.1')$$

If the reactions involve more than six unknowns, some of the reactions are *statically indeterminate*; if they involve fewer than six unknowns, the rigid body is only *partially constrained*. Even with six or more unknowns, the rigid body is *improperly constrained* if the reactions associated with the given supports are either parallel or intersect the same line.

Review Problems

4.142 A 3200-lb forklift truck is used to lift a 1700-lb crate. Determine the reaction at each of the two (a) front wheels A , (b) rear wheels B .

4.143 The lever BCD is hinged at C and attached to a control rod at B . If $P = 100$ lb, determine (a) the tension in rod AB , (b) the reaction at C .

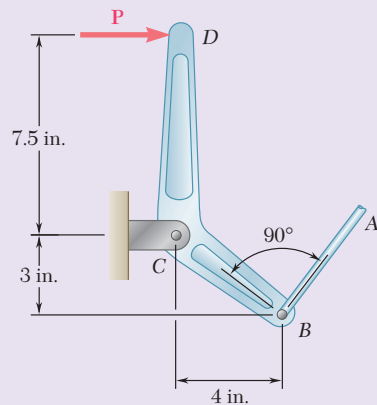


Fig. P4.143

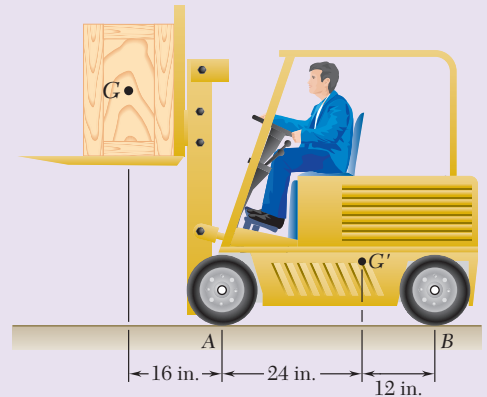


Fig. P4.142

4.144 Determine the reactions at A and B when (a) $h = 0$, (b) $h = 200$ mm.

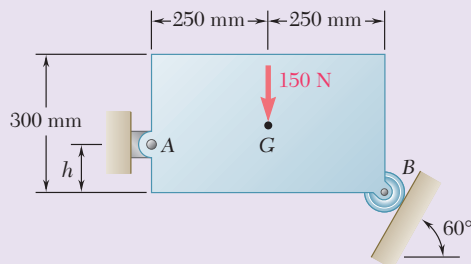


Fig. P4.144

4.145 Neglecting friction and the radius of the pulley, determine (a) the tension in cable ADB , (b) the reaction at C .

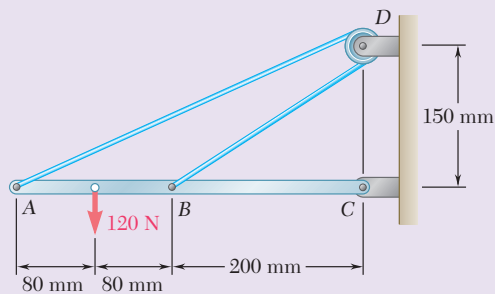


Fig. P4.145

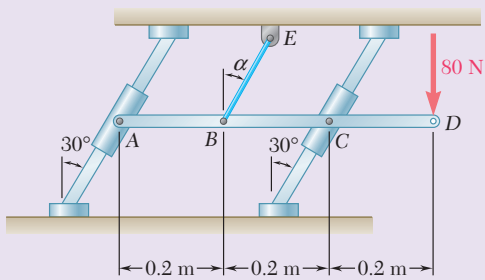


Fig. P4.146

4.146 Bar AD is attached at A and C to collars that can move freely on the rods shown. If the cord BE is vertical ($\alpha = 0$), determine the tension in the cord and the reactions at A and C .

4.147 A slender rod AB , with a weight of W , is attached to blocks A and B that move freely in the guides shown. The constant of the spring is k , and the spring is unstretched when $\theta = 0$. (a) Neglecting the weight of the blocks, derive an equation in W , k , l , and θ that must be satisfied when the rod is in equilibrium. (b) Determine the value of θ when $W = 75$ lb, $l = 30$ in., and $k = 3$ lb/in.

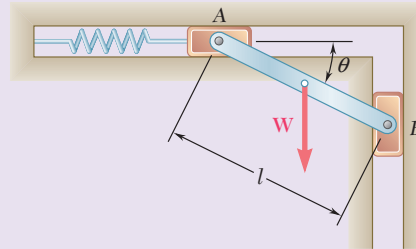


Fig. P4.147

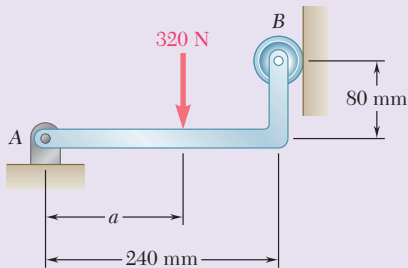


Fig. P4.148

4.148 Determine the reactions at A and B when $a = 150$ mm.

4.149 For the frame and loading shown, determine the reactions at A and C .

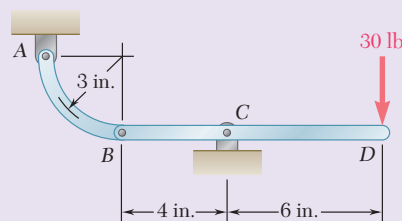


Fig. P4.149

4.150 A 200-mm lever and a 240-mm-diameter pulley are welded to the axle BE that is supported by bearings at C and D . If a 720-N vertical load is applied at A when the lever is horizontal, determine (a) the tension in the cord, (b) the reactions at C and D . Assume that the bearing at D does not exert any axial thrust.

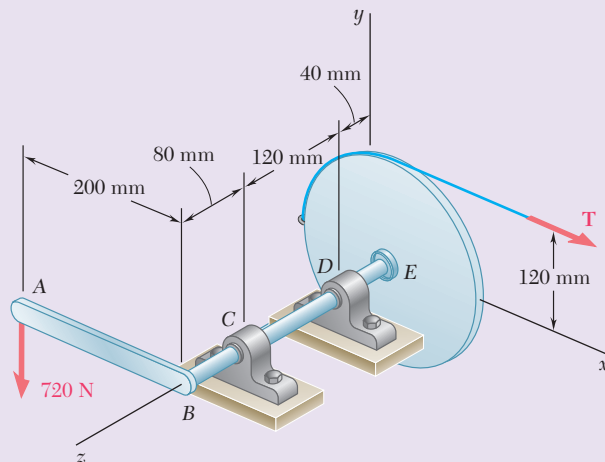


Fig. P4.150

4.151 The 45-lb square plate shown is supported by three vertical wires. Determine the tension in each wire.

4.152 The rectangular plate shown weighs 75 lb and is held in the position shown by hinges at A and B and by cable EF . Assuming that the hinge at B does not exert any axial thrust, determine (a) the tension in the cable, (b) the reactions at A and B .

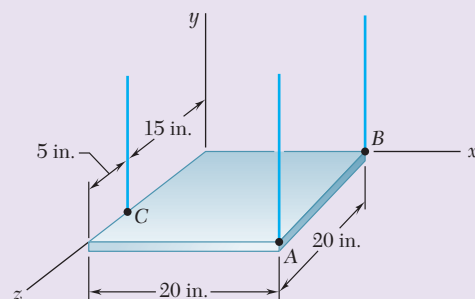


Fig. P4.151

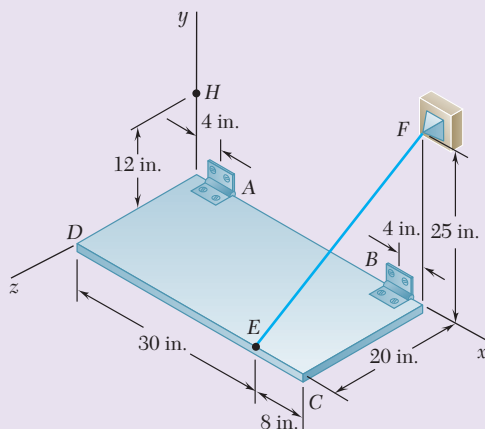


Fig. P4.152

4.153 A force P is applied to a bent rod ABC , which may be supported in four different ways as shown. In each case, if possible, determine the reactions at the supports.

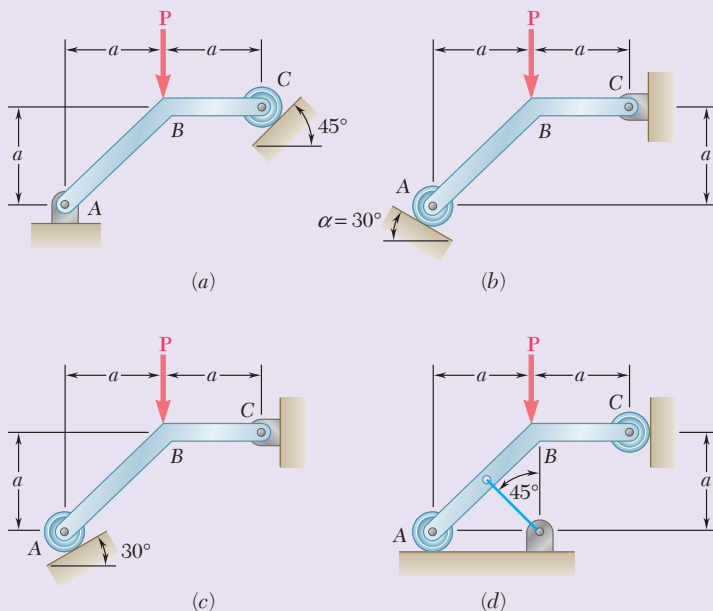


Fig. P4.153



5

Distributed Forces: Centroids and Centers of Gravity

Loads on dams include three types of distributed forces: the weights of its constituent elements, the pressure forces exerted by the water on its submerged face, and the pressure forces exerted by the ground on its base.

Introduction

5.1 PLANAR CENTERS OF GRAVITY AND CENTROIDS

- 5.1A Center of Gravity of a Two-Dimensional Body
- 5.1B Centroids of Areas and Lines
- 5.1C First Moments of Areas and Lines
- 5.1D Composite Plates and Wires

5.2 FURTHER CONSIDERATIONS OF CENTROIDS

- 5.2A Determination of Centroids by Integration
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5.3 ADDITIONAL APPLICATIONS OF CENTROIDS

- 5.3A Distributed Loads on Beams
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5.4 CENTERS OF GRAVITY AND CENTROIDS OF VOLUMES

- 5.4A Three-Dimensional Centers of Gravity and Centroids
- 5.4B Composite Bodies
- 5.4C Determination of Centroids of Volumes by Integration



Photo 5.1 The precise balancing of the components of a mobile requires an understanding of centers of gravity and centroids, the main topics of this chapter.

Objectives

- **Describe** the centers of gravity of two and three-dimensional bodies.
- **Define** the centroids of lines, areas, and volumes.
- **Consider** the first moments of lines and areas, and examine their properties.
- **Determine** centroids of composite lines, areas, and volumes by summation methods.
- **Determine** centroids of composite lines, areas, and volumes by integration.
- **Apply** the theorems of Pappus-Guldinus to analyze surfaces and bodies of revolution.
- **Analyze** distributed loads on beams and forces on submerged surfaces.

Introduction

We have assumed so far that we could represent the attraction exerted by the earth on a rigid body by a single force $\mathbf{W}$. This force, called the force due to gravity or the weight of the body, is applied at the **center of gravity** of the body (Sec. 3.1A). Actually, the earth exerts a force on each of the particles forming the body, so we should represent the attraction of the earth on a rigid body by a large number of small forces distributed over the entire body. You will see in this chapter, however, that all of these small forces can be replaced by a single equivalent force $\mathbf{W}$. You will also see how to determine the center of gravity—i.e., the point of application of the resultant $\mathbf{W}$ —for bodies of various shapes.

In the first part of this chapter, we study two-dimensional bodies, such as flat plates and wires contained in a given plane. We introduce two concepts closely associated with determining the center of gravity of a plate or a wire: the **centroid** of an area or a line and the **first moment** of an area or a line with respect to a given axis. Computing the area of a surface of revolution or the volume of a body of revolution is directly related to determining the centroid of the line or area used to generate that surface or body of revolution (theorems of Pappus-Guldinus). Also, as we show in Sec. 5.3, the determination of the centroid of an area simplifies the analysis of beams subjected to distributed loads and the computation of the forces exerted on submerged rectangular surfaces, such as hydraulic gates and portions of dams.

In the last part of this chapter, you will see how to determine the center of gravity of a three-dimensional body as well as how to calculate the centroid of a volume and the first moments of that volume with respect to the coordinate planes.

5.1 PLANAR CENTERS OF GRAVITY AND CENTROIDS

In Chapter 4, we showed how the locations of the lines of action of forces affects the replacement of a system of forces with an equivalent system of forces and couples. In this section, we extend this idea to show how a distributed system of forces (in particular, the elements of an object's weight) can be replaced by a single resultant force acting at a specific point on an object. The specific point is called the object's center of gravity.

5.1A Center of Gravity of a Two-Dimensional Body

Let us first consider a flat horizontal plate (Fig. 5.1). We can divide the plate into n small elements. We denote the coordinates of the first element by x_1 and y_1 , those of the second element by x_2 and y_2 , etc. The forces exerted by the earth on the elements of the plate are denoted, respectively, by $\Delta W_1, \Delta W_2, \dots, \Delta W_n$. These forces or weights are directed toward the center of the earth; however, for all practical purposes, we can assume them to be parallel. Their resultant is therefore a single force in the same direction. The magnitude W of this force is obtained by adding the magnitudes of the elemental weights.

$$\Sigma F_z: \quad W = \Delta W_1 + \Delta W_2 + \dots + \Delta W_n$$

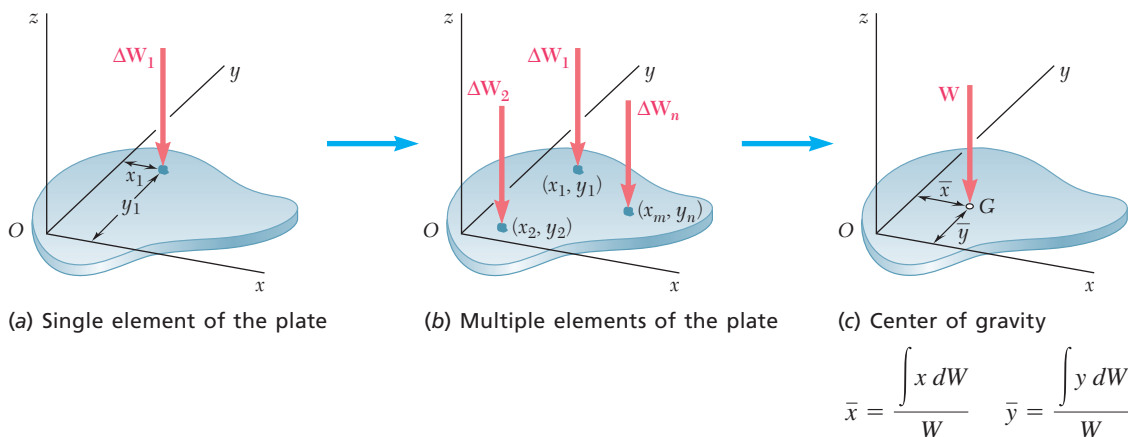


Fig. 5.1 The center of gravity of a plate is the point where the resultant weight of the plate acts. It is the weighted average of all the elements of weight that make up the plate.

To obtain the coordinates $\bar{x}$ and $\bar{y}$ of point G where the resultant $\mathbf{W}$ should be applied, we note that the moments of $\mathbf{W}$ about the y and x axes are equal to the sum of the corresponding moments of the elemental weights:

$$\begin{aligned} \Sigma M_y: \quad \bar{x}W &= x_1\Delta W_1 + x_2\Delta W_2 + \dots + x_n\Delta W_n \\ \Sigma M_x: \quad \bar{y}W &= y_1\Delta W_1 + y_2\Delta W_2 + \dots + y_n\Delta W_n \end{aligned} \quad (5.1)$$

Solving these equations for $\bar{x}$ and $\bar{y}$ gives us

$$\begin{aligned} \bar{x} &= \frac{x_1\Delta W_1 + x_2\Delta W_2 + \dots + x_n\Delta W_n}{W} \\ \bar{y} &= \frac{y_1\Delta W_1 + y_2\Delta W_2 + \dots + y_n\Delta W_n}{W} \end{aligned}$$

We could use these equations in this form to find the center of gravity of a collection of n objects, each with a weight of W_i .

If we now increase the number of elements into which we divide the plate and simultaneously decrease the size of each element, in the limit of infinitely many elements of infinitesimal size, we obtain the expressions

Weight, center of gravity of a flat plate

$$W = \int dW \quad \bar{x}W = \int x dW \quad \bar{y}W = \int y dW \quad (5.2)$$

Or, solving for $\bar{x}$ and $\bar{y}$, we have

$$W = \int dW \quad \bar{x} = \frac{\int x dW}{W} \quad \bar{y} = \frac{\int y dW}{W} \quad (5.2')$$

These equations define the weight W and the coordinates $\bar{x}$ and $\bar{y}$ of the **center of gravity** G of a flat plate. The same equations can be derived for a wire lying in the xy plane (Fig. 5.2). Note that the center of gravity of a wire is usually not located on the wire.

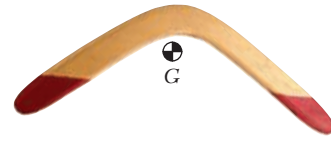


Photo 5.2 The center of gravity of a boomerang is not located on the object itself.

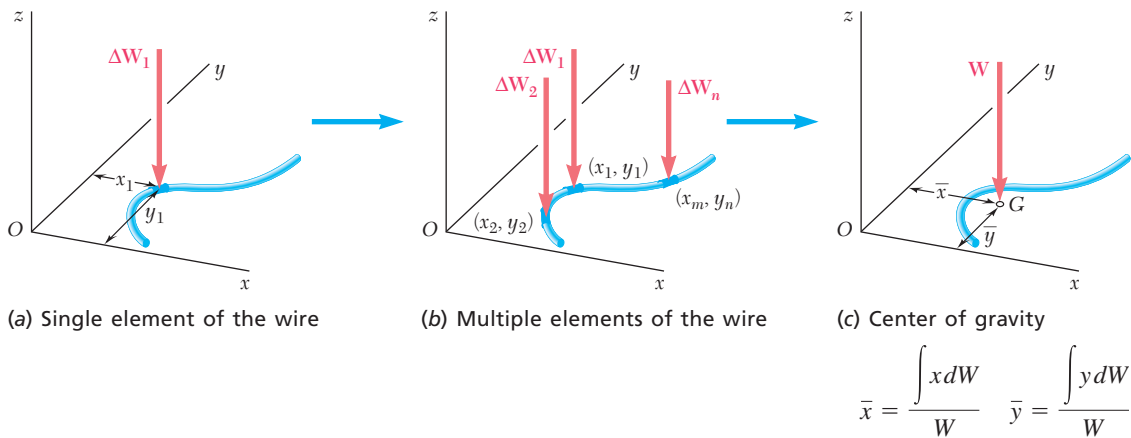


Fig. 5.2 The center of gravity of a wire is the point where the resultant weight of the wire acts. The center of gravity may not actually be located on the wire.

5.1B Centroids of Areas and Lines

In the case of a flat homogeneous plate of uniform thickness, we can express the magnitude ΔW of the weight of an element of the plate as

$$\Delta W = \gamma t \Delta A$$

where γ = specific weight (weight per unit volume) of the material

t = thickness of the plate

ΔA = area of the element

Similarly, we can express the magnitude W of the weight of the entire plate as

$$W = \gamma t A$$

where A is the total area of the plate.

If U.S. customary units are used, the specific weight γ should be expressed in lb/ft^3 , the thickness t in feet, and the areas ΔA and A in square feet. Then ΔW and W are expressed in pounds. If SI units are used, γ should be expressed in N/m^3 , t in meters, and the areas ΔA and A in square meters; the weights ΔW and W are then expressed in newtons.[†]

Substituting for ΔW and W in the moment equations (5.1) and dividing throughout by γt , we obtain

$$\begin{aligned}\Sigma M_y: \quad \bar{x}A &= x_1 \Delta A_1 + x_2 \Delta A_2 + \cdots + x_n \Delta A_n \\ \Sigma M_x: \quad \bar{y}A &= y_1 \Delta A_1 + y_2 \Delta A_2 + \cdots + y_n \Delta A_n\end{aligned}$$

If we increase the number of elements into which the area A is divided and simultaneously decrease the size of each element, in the limit we obtain

Centroid of an area A

$$\bar{x}A = \int x dA \quad \bar{y}A = \int y dA \quad (5.3)$$

Or, solving for $\bar{x}$ and $\bar{y}$, we obtain

$$\bar{x} = \frac{\int x dA}{A} \quad \bar{y} = \frac{\int y dA}{A} \quad (5.3')$$

These equations define the coordinates $\bar{x}$ and $\bar{y}$ of the center of gravity of a homogeneous plate. The point whose coordinates are $\bar{x}$ and $\bar{y}$ is also known as the **centroid C of the area A** of the plate (Fig. 5.3). If the plate is not homogeneous, you cannot use these equations to determine the center of gravity of the plate; they still define, however, the centroid of the area.

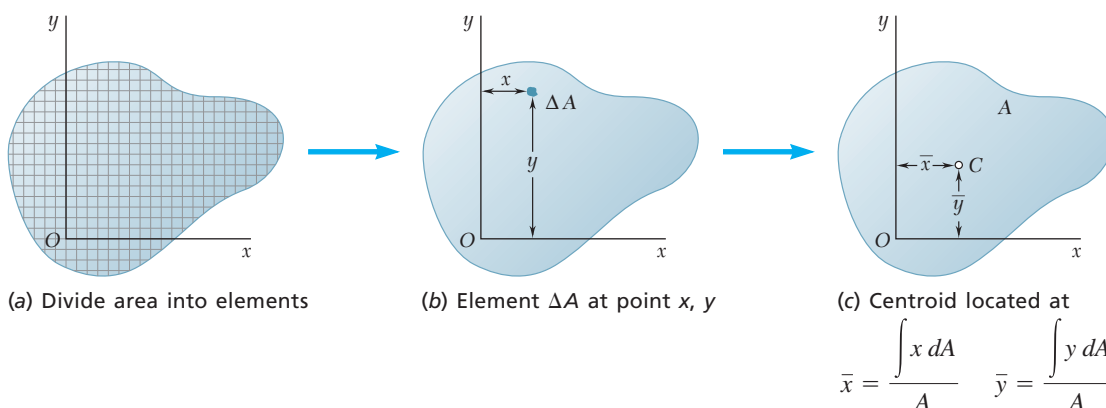


Fig. 5.3 The centroid of an area is the point where a homogeneous plate of uniform thickness would balance.

[†]We should note that in the SI system of units, a given material is generally characterized by its density ρ (mass per unit volume) rather than by its specific weight γ . You can obtain the specific weight of the material from the relation

$$\gamma = \rho g$$

where $g = 9.81 \text{ m/s}^2$. Note that since ρ is expressed in kg/m^3 , the units of γ are $(\text{kg/m}^3)(\text{m/s}^2)$, or N/m^3 .

In the case of a homogeneous wire of uniform cross section, we can express the magnitude ΔW of the weight of an element of wire as

$$\Delta W = \gamma a \Delta L$$

where γ = specific weight of the material
 a = cross-sectional area of the wire
 ΔL = length of the element

The center of gravity of the wire then coincides with the **centroid C of the line L** defining the shape of the wire (Fig. 5.4). We can obtain the coordinates $\bar{x}$ and $\bar{y}$ of the centroid of line L from the equations

Centroid of a line L

$$\bar{x}L = \int x dL \quad \bar{y}L = \int y dL \quad (5.4)$$

Solving for $\bar{x}$ and $\bar{y}$ gives us

$$\bar{x} = \frac{\int x dL}{L} \quad \bar{y} = \frac{\int y dL}{L} \quad (5.4')$$

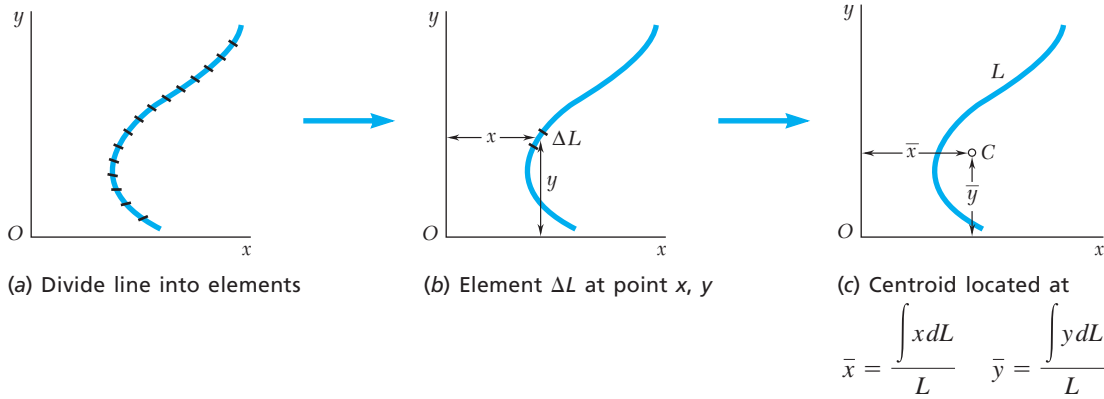


Fig. 5.4 The centroid of a line is the point where a homogeneous wire of uniform cross section would balance.

5.1C First Moments of Areas and Lines

The integral $\int x dA$ in Eqs. (5.3) is known as the **first moment of the area A with respect to the y axis** and is denoted by Q_y . Similarly, the integral $\int y dA$ defines the **first moment of A with respect to the x axis** and is denoted by Q_x . That is,

First moments of area A

$$Q_y = \int x dA \quad Q_x = \int y dA \quad (5.5)$$

Comparing Eqs. (5.3) with Eqs. (5.5), we note that we can express the first moments of the area A as the products of the area and the coordinates of its centroid:

$$Q_y = \bar{x}A \quad Q_x = \bar{y}A \quad (5.6)$$

It follows from Eqs. (5.6) that we can obtain the coordinates of the centroid of an area by dividing the first moments of that area by the area itself. The first moments of the area are also useful in mechanics of materials for determining the shearing stresses in beams under transverse loadings. Finally, we observe from Eqs. (5.6) that, if the centroid of an area is located on a coordinate axis, the first moment of the area with respect to that axis is zero. Conversely, if the first moment of an area with respect to a coordinate axis is zero, the centroid of the area is located on that axis.

We can use equations similar to Eqs. (5.5) and (5.6) to define the first moments of a line with respect to the coordinate axes and to express these moments as the products of the length L of the line and the coordinates $\bar{x}$ and $\bar{y}$ of its centroid.

An area A is said to be **symmetric with respect to an axis BB'** if for every point P of the area there exists a point P' of the same area such that the line PP' is perpendicular to BB' and is divided into two equal parts by that axis (Fig. 5.5a). The axis BB' is called an **axis of symmetry**. A line L is said to be symmetric with respect to an axis BB' if it satisfies similar conditions. When an area A or a line L possesses an axis of symmetry BB' , its first moment with respect to BB' is zero, and its centroid is located on that axis. For example, note that, for the area A of Fig. 5.5b, which is symmetric with respect to the y axis, every element of area dA

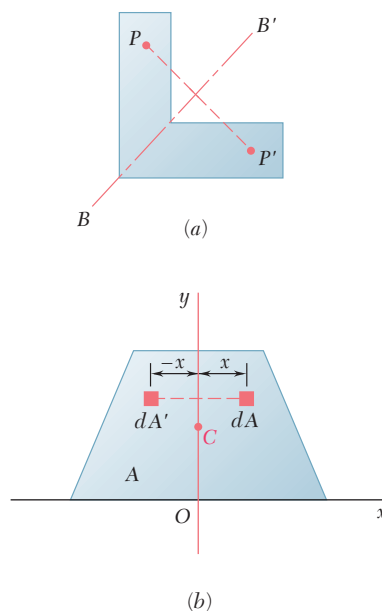


Fig. 5.5 Symmetry about an axis. (a) The area is symmetric about the axis BB' . (b) The centroid of the area is located on the axis of symmetry.

with abscissa x corresponds to an element dA' of equal area and with abscissa $-x$. It follows that the integral in the first of Eqs. (5.5) is zero and, thus, that $Q_y = 0$. It also follows from the first of the relations in Eq. (5.3) that $\bar{x} = 0$. Thus, if an area A or a line L possesses an axis of symmetry, its centroid C is located on that axis.

We further note that if an area or line possesses two axes of symmetry, its centroid C must be located at the intersection of the two axes (Fig. 5.6). This property enables us to determine immediately the centroids of areas such as circles, ellipses, squares, rectangles, equilateral triangles, or other symmetric figures, as well as the centroids of lines in the shape of the circumference of a circle, the perimeter of a square, etc.

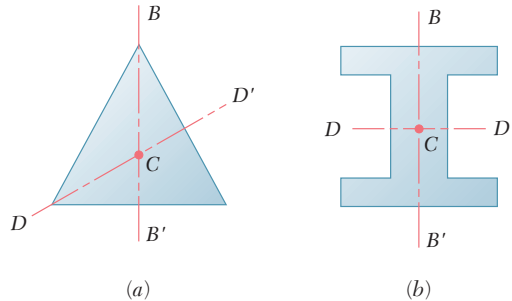


Fig. 5.6 If an area has two axes of symmetry, the centroid is located at their intersection. (a) An area with two axes of symmetry but no center of symmetry; (b) an area with two axes of symmetry and a center of symmetry.

We say that an area A is **symmetric with respect to a center** O if, for every element of area dA of coordinates x and y , there exists an element dA' of equal area with coordinates $-x$ and $-y$ (Fig. 5.7). It then follows that the integrals in Eqs. (5.5) are both zero and that $Q_x = Q_y = 0$. It also follows from Eqs. (5.3) that $\bar{x} = \bar{y} = 0$; that is, that the centroid of the area coincides with its center of symmetry O . Similarly, if a line possesses a center of symmetry O , the centroid of the line coincides with the center O .

Note that a figure possessing a center of symmetry does not necessarily possess an axis of symmetry (Fig. 5.7), whereas a figure possessing two axes of symmetry does not necessarily possess a center of symmetry (Fig. 5.6a). However, if a figure possesses two axes of symmetry at right angles to each other, the point of intersection of these axes is a center of symmetry (Fig. 5.6b).

Determining the centroids of unsymmetrical areas and lines and of areas and lines possessing only one axis of symmetry will be discussed in the next section. Centroids of common shapes of areas and lines are shown in Fig. 5.8A and B.

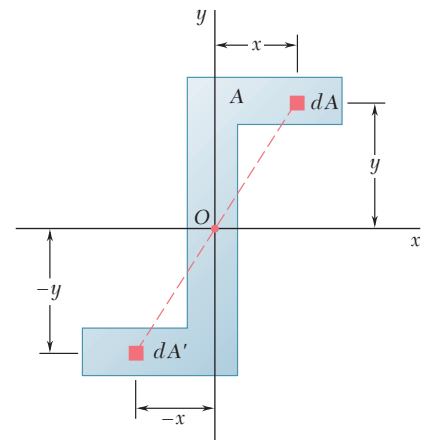


Fig. 5.7 An area may have a center of symmetry but no axis of symmetry.

5.1D Composite Plates and Wires

In many instances, we can divide a flat plate into rectangles, triangles, or the other common shapes shown in Fig. 5.8A. We can determine the abscissa $\bar{X}$ of the plate's center of gravity G from the abscissas $\bar{x}_1, \bar{x}_2, \dots, \bar{x}_n$ of the centers of gravity of the various parts. To do this, we equate the moment of the weight of the whole plate about the y axis to the sum of

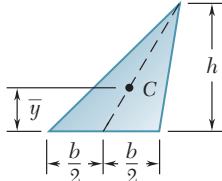
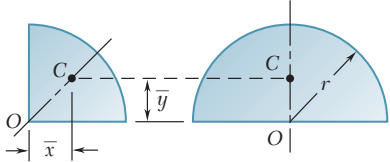
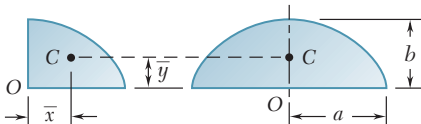
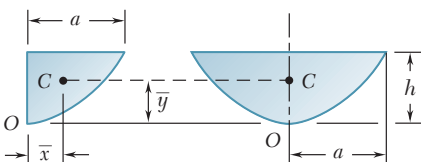
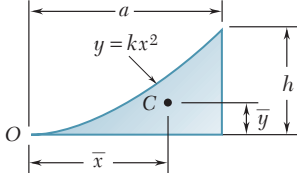
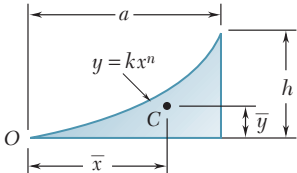
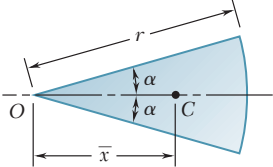
Shape		$\bar{x}$	$\bar{y}$	Area
Triangular area			$\frac{h}{3}$	$\frac{bh}{2}$
Quarter-circular area		$\frac{4r}{3\pi}$	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{4}$
Semicircular area		0	$\frac{4r}{3\pi}$	$\frac{\pi r^2}{2}$
Quarter-elliptical area		$\frac{4a}{3\pi}$	$\frac{4b}{3\pi}$	$\frac{\pi ab}{4}$
Semielliptical area		0	$\frac{4b}{3\pi}$	$\frac{\pi ab}{2}$
Semiparabolic area		$\frac{3a}{8}$	$\frac{3h}{5}$	$\frac{2ah}{3}$
Parabolic area		0	$\frac{3h}{5}$	$\frac{4ah}{3}$
Parabolic spandrel		$\frac{3a}{4}$	$\frac{3h}{10}$	$\frac{ah}{3}$
General spandrel		$\frac{n+1}{n+2}a$	$\frac{n+1}{4n+2}h$	$\frac{ah}{n+1}$
Circular sector		$\frac{2r \sin \alpha}{3\alpha}$	0	αr^2

Fig. 5.8A Centroids of common shapes of areas.

Shape		$\bar{x}$	$\bar{y}$	Length
Quarter-circular arc		$\frac{2r}{\pi}$	$\frac{2r}{\pi}$	$\frac{\pi r}{2}$
Semicircular arc		0	$\frac{2r}{\pi}$	πr
Arc of circle		$\frac{r \sin \alpha}{\alpha}$	0	$2\alpha r$

Fig. 5.8B Centroids of common shapes of lines.

the moments of the weights of the various parts about the same axis (Fig. 5.9). We can obtain the ordinate $\bar{Y}$ of the center of gravity of the plate in a similar way by equating moments about the x axis. Mathematically, we have

$$\Sigma M_y: \bar{X}(W_1 + W_2 + \cdots + W_n) = \bar{x}_1 W_1 + \bar{x}_2 W_2 + \cdots + \bar{x}_n W_n$$

$$\Sigma M_x: \bar{Y}(W_1 + W_2 + \cdots + W_n) = \bar{y}_1 W_1 + \bar{y}_2 W_2 + \cdots + \bar{y}_n W_n$$

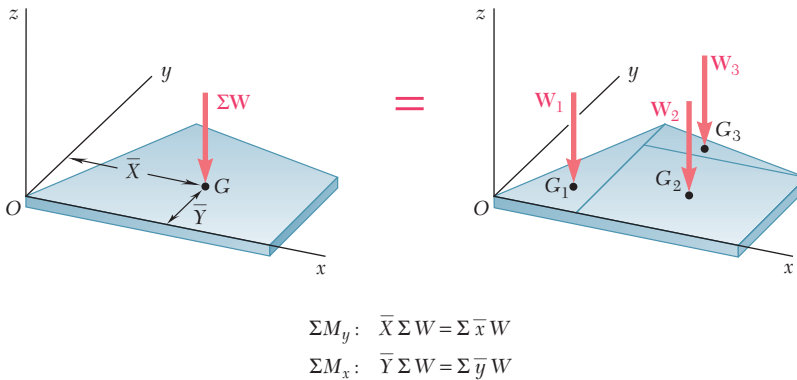


Fig. 5.9. We can determine the location of the center of gravity G of a composite plate from the centers of gravity $G_1, G_2, \dots$ of the component plates.

In more condensed notation, this is

**Center of gravity
of a composite plate**

$$\bar{X} = \frac{\Sigma \bar{x} W}{W} \quad \bar{Y} = \frac{\Sigma \bar{y} W}{W} \quad (5.7)$$

We can use these equations to find the coordinates $\bar{X}$ and $\bar{Y}$ of the center of gravity of the plate from the centers of gravity of its component parts.

If the plate is homogeneous and of uniform thickness, the center of gravity coincides with the centroid C of its area. We can determine the abscissa $\bar{X}$ of the centroid of the area by noting that we can express the first moment Q_y of the composite area with respect to the y axis as (1) the product of $\bar{X}$ and the total area and (2) as the sum of the first moments of the elementary areas with respect to the y axis (Fig. 5.10). We obtain the

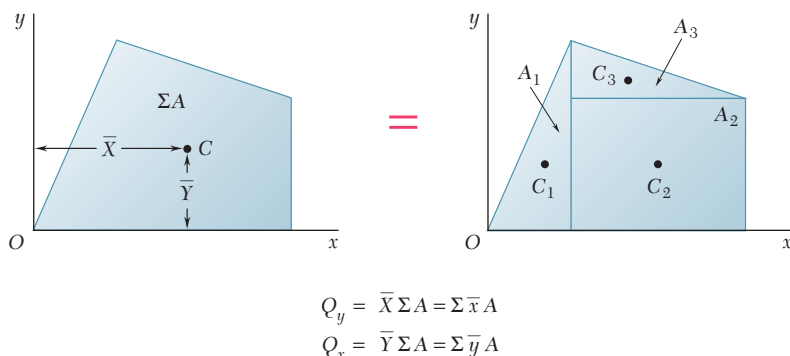
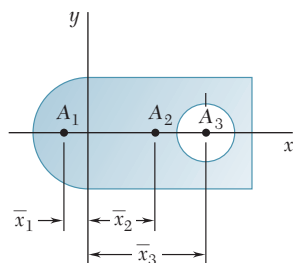
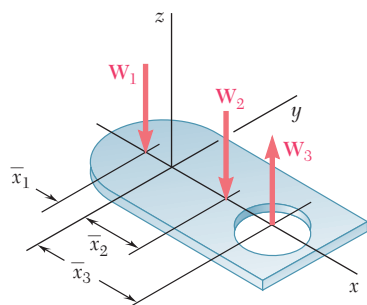


Fig. 5.10 We can find the location of the centroid of a composite area from the centroids of the component areas.



	$\bar{x}$	A	$\bar{x}A$
A_1 Semicircle	-	+	-
A_2 Full rectangle	+	+	+
A_3 Circular hole	+	-	-

Fig. 5.11 When calculating the centroid of a composite area, note that if the centroid of a component area has a negative coordinate distance relative to the origin, or if the area represents a hole, then the first moment is negative.

ordinate $\bar{Y}$ of the centroid in a similar way by considering the first moment Q_x of the composite area. We have

$$Q_y = \bar{X}(A_1 + A_2 + \cdots + A_n) = \bar{x}_1 A_1 + \bar{x}_2 A_2 + \cdots + \bar{x}_n A_n$$

$$Q_x = \bar{Y}(A_1 + A_2 + \cdots + A_n) = \bar{y}_1 A_1 + \bar{y}_2 A_2 + \cdots + \bar{y}_n A_n$$

Again, in shorter form,

Centroid of a composite area

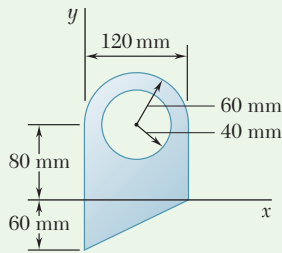
$$Q_y = \bar{X} \Sigma A = \Sigma \bar{x} A \quad Q_x = \bar{Y} \Sigma A = \Sigma \bar{y} A \quad (5.8)$$

These equations yield the first moments of the composite area, or we can use them to obtain the coordinates $\bar{X}$ and $\bar{Y}$ of its centroid.

First moments of areas, like moments of forces, can be positive or negative. Thus, you need to take care to assign the appropriate sign to the moment of each area. For example, an area whose centroid is located to the left of the y axis has a negative first moment with respect to that axis. Also, the area of a hole should be assigned a negative sign (Fig. 5.11).

Similarly, it is possible in many cases to determine the center of gravity of a composite wire or the centroid of a composite line by dividing the wire or line into simpler elements (see Sample Prob. 5.2).

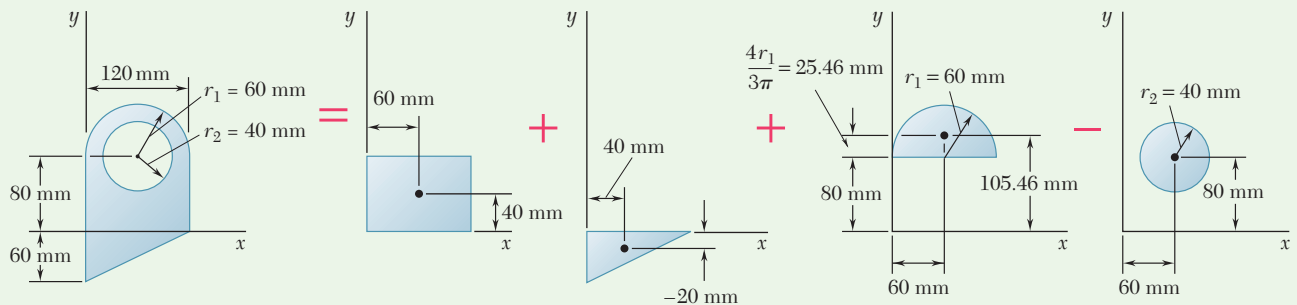
Sample Problem 5.1



For the plane area shown, determine (a) the first moments with respect to the x and y axes; (b) the location of the centroid.

STRATEGY: Break up the given area into simple components, find the centroid of each component, and then find the overall first moments and centroid.

MODELING: As shown in Fig. 1, you obtain the given area by adding a rectangle, a triangle, and a semicircle and then subtracting a circle. Using the coordinate axes shown, find the area and the coordinates of the centroid of each of the component areas. To keep track of the data, enter them in a table. The area of the circle is indicated as negative because it is subtracted from the other areas. The coordinate $\bar{y}$ of the centroid of the triangle is negative for the axes shown. Compute the first moments of the component areas with respect to the coordinate axes and enter them in your table.



Component	A, mm^2	$\bar{x}, \text{mm}$	$\bar{y}, \text{mm}$	$\bar{x}A, \text{mm}^3$	$\bar{y}A, \text{mm}^3$
Rectangle	$(120)(80) = 9.6 \times 10^3$	60	40	$+576 \times 10^3$	$+384 \times 10^3$
Triangle	$\frac{1}{2}(120)(60) = 3.6 \times 10^3$	40	-20	$+144 \times 10^3$	-72×10^3
Semicircle	$\frac{1}{2}\pi(60)^2 = 5.655 \times 10^3$	60	105.46	$+339.3 \times 10^3$	$+596.4 \times 10^3$
Circle	$-\pi(40)^2 = -5.027 \times 10^3$	60	80	-301.6×10^3	-402.2×10^3
	$\Sigma A = 13.828 \times 10^3$			$\Sigma \bar{x}A = +757.7 \times 10^3$	$\Sigma \bar{y}A = +506.2 \times 10^3$

Fig. 1 Given area modeled as the combination of simple geometric shapes.

ANALYSIS:

a. First Moments of the Area. Using Eqs. (5.8), you obtain

$$Q_x = \Sigma \bar{y}A = 506.2 \times 10^3 \text{ mm}^3 \quad Q_x = 506 \times 10^3 \text{ mm}^3 \quad \blacktriangleleft$$

$$Q_y = \Sigma \bar{x}A = 757.7 \times 10^3 \text{ mm}^3 \quad Q_y = 758 \times 10^3 \text{ mm}^3 \quad \blacktriangleleft$$

b. Location of Centroid. Substituting the values given in the table into the equations defining the centroid of a composite area yields (Fig. 2)

$$\bar{X}\Sigma A = \Sigma \bar{x}A: \quad \bar{X}(13.828 \times 10^3 \text{ mm}^2) = 757.7 \times 10^3 \text{ mm}^3 \quad \bar{X} = 54.8 \text{ mm} \quad \blacktriangleleft$$

$$\bar{Y}\Sigma A = \Sigma \bar{y}A: \quad \bar{Y}(13.828 \times 10^3 \text{ mm}^2) = 506.2 \times 10^3 \text{ mm}^3 \quad \bar{Y} = 36.6 \text{ mm} \quad \blacktriangleleft$$

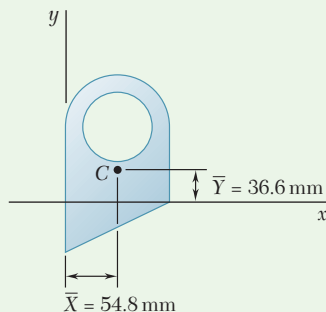
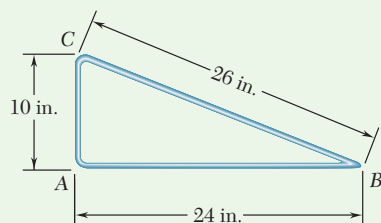


Fig. 2 Centroid of composite area.

REFLECT and THINK: Given that the lower portion of the shape has more area to the left and that the upper portion has a hole, the location of the centroid seems reasonable upon visual inspection.



Sample Problem 5.2

The figure shown is made from a piece of thin, homogeneous wire. Determine the location of its center of gravity.

STRATEGY: Since the figure is formed of homogeneous wire, its center of gravity coincides with the centroid of the corresponding line. Therefore, you can simply determine that centroid.

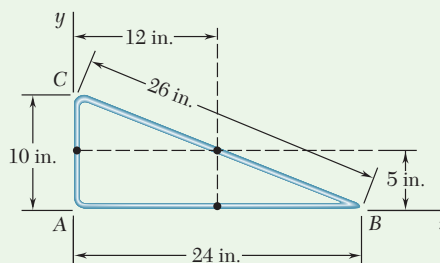


Fig. 1 Location of each line segment's centroid.

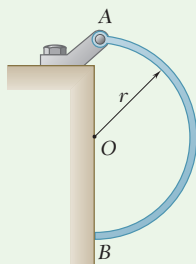
MODELING: Choosing the coordinate axes shown in Fig. 1 with the origin at A, determine the coordinates of the centroid of each line segment and compute the first moments with respect to the coordinate axes. You may find it convenient to list the data in a table.

Segment	L , in.	$\bar{x}$, in.	$\bar{y}$, in.	$\bar{x}L$, in ²	$\bar{y}L$, in ²
AB	24	12	0	288	0
BC	26	12	5	312	130
CA	10	0	5	0	50
$\Sigma L = 60$				$\Sigma \bar{x}L = 600$	$\Sigma \bar{y}L = 180$

ANALYSIS: Substituting the values obtained from the table into the equations defining the centroid of a composite line gives

$$\begin{aligned}\bar{X}\Sigma L &= \Sigma \bar{x}L: & \bar{X}(60 \text{ in.}) &= 600 \text{ in}^2 & \bar{X} &= 10 \text{ in.} \quad \blacktriangleleft \\ \bar{Y}\Sigma L &= \Sigma \bar{y}L: & \bar{Y}(60 \text{ in.}) &= 180 \text{ in}^2 & \bar{Y} &= 3 \text{ in.} \quad \blacktriangleleft\end{aligned}$$

REFLECT and THINK: The centroid is not on the wire itself, but it is within the area enclosed by the wire.



Sample Problem 5.3

A uniform semicircular rod of weight W and radius r is attached to a pin at A and rests against a frictionless surface at B . Determine the reactions at A and B .

STRATEGY: The key to solving the problem is finding where the weight W of the rod acts. Since the rod is a simple geometrical shape, you can look in Fig. 5.8 for the location of the wire's centroid.

MODELING: Draw a free-body diagram of the rod (Fig. 1). The forces acting on the rod are its weight $\mathbf{W}$, which is applied at the center of gravity G (whose position is obtained from Fig. 5.8B); a reaction at A , represented by its components $\mathbf{A}_x$ and $\mathbf{A}_y$; and a horizontal reaction at B .

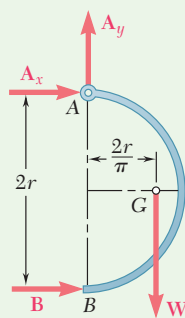


Fig. 1 Free-body diagram of rod.

ANALYSIS:

$$+\circlearrowleft \Sigma M_A = 0: \quad B(2r) - W\left(\frac{2r}{\pi}\right) = 0$$

$$B = +\frac{W}{\pi}$$

$$\mathbf{B} = \frac{W}{\pi} \rightarrow$$

$$+\rightarrow \Sigma F_x = 0: \quad A_x + B = 0$$

$$A_x = -B = -\frac{W}{\pi} \quad \mathbf{A}_x = \frac{W}{\pi} \leftarrow$$

$$+\uparrow \Sigma F_y = 0: \quad A_y - W = 0 \quad \mathbf{A}_y = W \uparrow$$

Adding the two components of the reaction at A (Fig. 2), we have

$$A = \left[W^2 + \left(\frac{W}{\pi} \right)^2 \right]^{1/2}$$

$$\mathbf{A} = W \left(1 + \frac{1}{\pi^2} \right)^{1/2}$$

$$\tan \alpha = \frac{W}{W/\pi} = \pi$$

$$\alpha = \tan^{-1} \pi$$

The answers can also be expressed as

$$\mathbf{A} = 1.049W \nearrow 72.3^\circ \quad \mathbf{B} = 0.318W \rightarrow$$

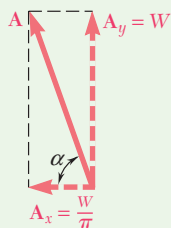


Fig. 2 Reaction at A .

REFLECT and THINK: Once you know the location of the rod's center of gravity, the problem is a straightforward application of the concepts in Chapter 4.

SOLVING PROBLEMS ON YOUR OWN

In this section, we developed the general equations for locating the centers of gravity of two-dimensional bodies and wires [Eqs. (5.2)] and the centroids of plane areas [Eqs. (5.3)] and lines [Eqs. (5.4)]. In the following problems, you will have to locate the centroids of composite areas and lines or determine the first moments of the area for composite plates [Eqs. (5.8)].

1. Locating the centroids of composite areas and lines. Sample Problems 5.1 and 5.2 illustrate the procedure you should follow when solving problems of this type. However, several points are worth emphasizing.

a. The first step in your solution should be to decide how to construct the given area or line from the common shapes of Fig. 5.8. You should recognize that for plane areas it is often possible to construct a particular shape in more than one way. Also, showing the different components (as is done in Sample Prob. 5.1) can help you correctly establish their centroids and areas or lengths. Do not forget that you can subtract areas as well as add them to obtain a desired shape.

b. We strongly recommend that for each problem you construct a table listing the areas or lengths and the respective coordinates of the centroids. Remember, any areas that are “removed” (such as holes) are treated as negative. Also, the sign of negative coordinates must be included. Therefore, you should always carefully note the location of the origin of the coordinate axes.

c. When possible, use symmetry [Sec. 5.1C] to help you determine the location of a centroid.

d. In the formulas for the circular sector and for the arc of a circle in Fig. 5.8, the angle α must always be expressed in radians.

2. Calculating the first moments of an area. The procedures for locating the centroid of an area and for determining the first moments of an area are similar; however, it is not necessary to compute the total area for finding first moments. Also, as noted in Sec. 5.1C, you should recognize that the first moment of an area relative to a centroidal axis is zero.

3. Solving problems involving the center of gravity. The bodies considered in the following problems are homogeneous; thus, their centers of gravity and centroids coincide. In addition, when a body that is suspended from a single pin is in equilibrium, the pin and the body’s center of gravity must lie on the same vertical line.

It may appear that many of the problems in this section have little to do with the study of mechanics. However, being able to locate the centroid of composite shapes will be essential in several topics that you will study later in this course.

Problems

5.1 through 5.9 Locate the centroid of the plane area shown.

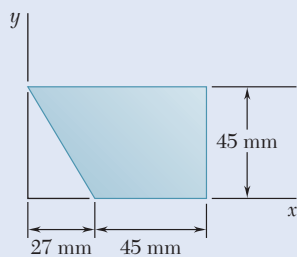


Fig. P5.1

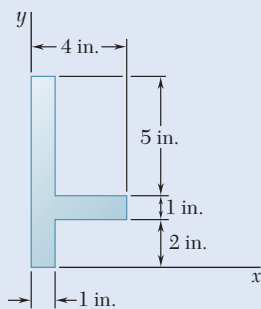


Fig. P5.2

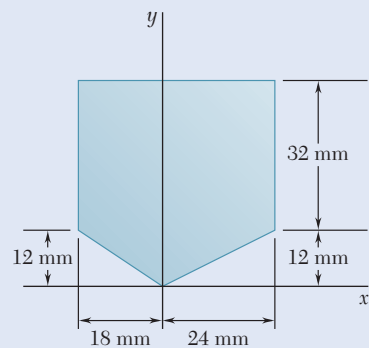


Fig. P5.3

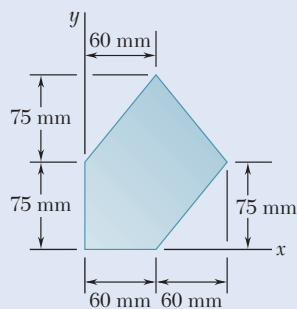


Fig. P5.4

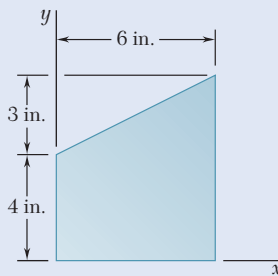


Fig. P5.5

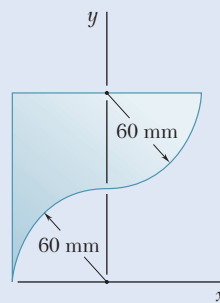


Fig. P5.6

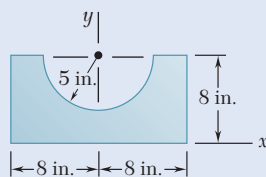


Fig. P5.7

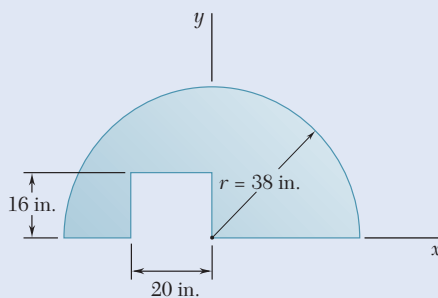


Fig. P5.8

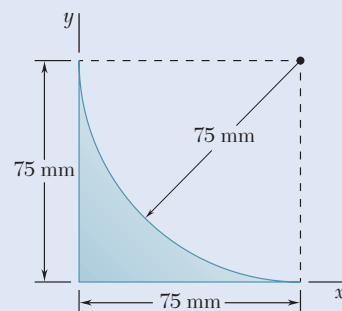


Fig. P5.9

5.10 through 5.15 Locate the centroid of the plane area shown.

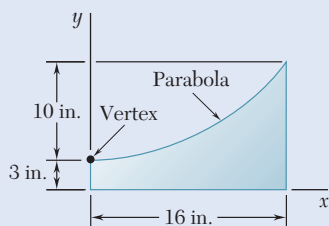


Fig. P5.10

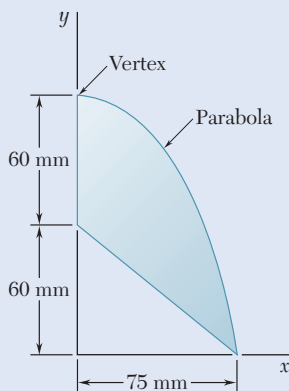


Fig. P5.11

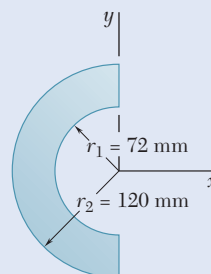


Fig. P5.12

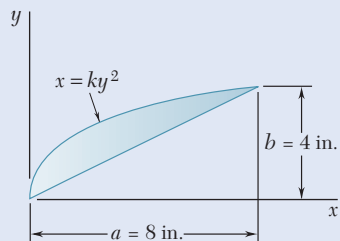


Fig. P5.13

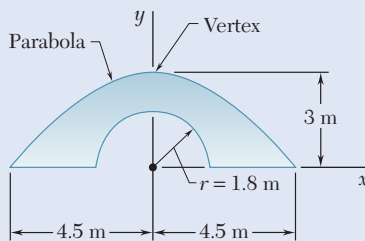


Fig. P5.14

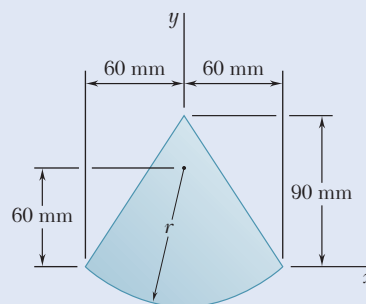


Fig. P5.15

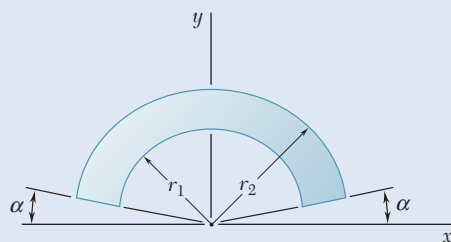


Fig. P5.16 and Fig. P5.17

5.16 Determine the y coordinate of the centroid of the shaded area in terms of r_1 , r_2 , and α .

5.17 Show that as r_1 approaches r_2 , the location of the centroid approaches that for an arc of circle of radius $(r_1 + r_2)/2$.

5.18 Determine the x coordinate of the centroid of the trapezoid shown in terms of h_1 , h_2 , and a .

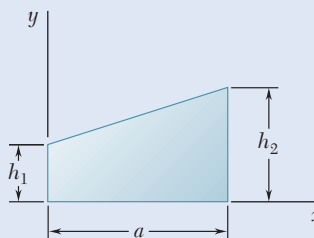


Fig. P5.18

5.19 For the semiannular area of Prob. 5.12, determine the ratio r_1 to r_2 so that the centroid of the area is located at $x = -\frac{1}{2}r_2$ and $y = 0$.

- 5.20** A composite beam is constructed by bolting four plates to four $60 \times 60 \times 12$ -mm angles as shown. The bolts are equally spaced along the beam, and the beam supports a vertical load. As proved in mechanics of materials, the shearing forces exerted on the bolts at A and B are proportional to the first moments with respect to the centroidal x axis of the red shaded areas shown, respectively, in parts a and b of the figure. Knowing that the force exerted on the bolt at A is 280 N, determine the force exerted on the bolt at B .

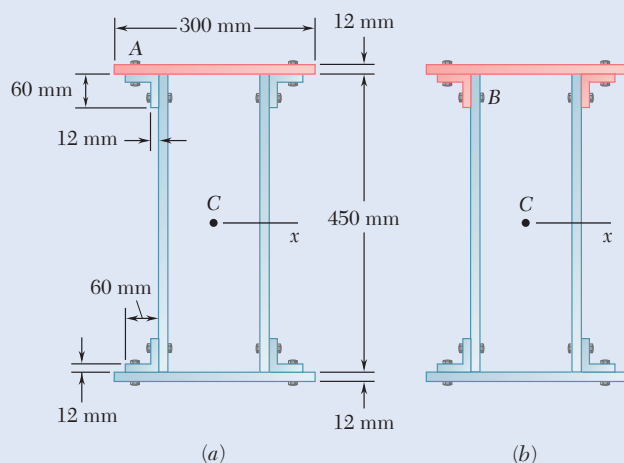


Fig. P5.20

- 5.21 and 5.22** The horizontal x axis is drawn through the centroid C of the area shown, and it divides the area into two component areas A_1 and A_2 . Determine the first moment of each component area with respect to the x axis, and explain the results obtained.

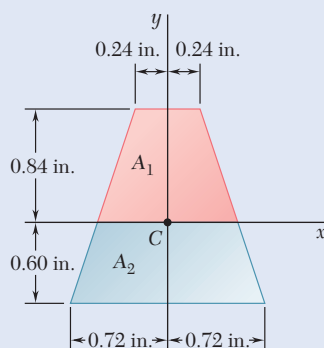


Fig. P5.21

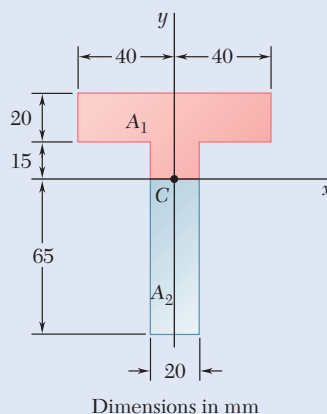


Fig. P5.22

- 5.23** The first moment of the shaded area with respect to the x axis is denoted by Q_x . (a) Express Q_x in terms of b , c , and the distance y from the base of the shaded area to the x axis. (b) For what value of y is Q_x maximum, and what is that maximum value?

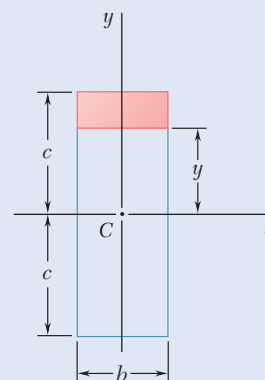


Fig. P5.23

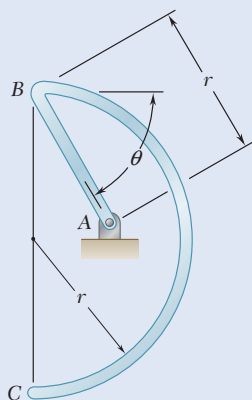


Fig. P5.28

5.24 through 5.27 A thin, homogeneous wire is bent to form the perimeter of the figure indicated. Locate the center of gravity of the wire figure thus formed.

5.24 Fig. P5.1.

5.25 Fig. P5.3.

5.26 Fig. P5.5.

5.27 Fig. P5.8.

5.28 The homogeneous wire ABC is bent into a semicircular arc and a straight section as shown and is attached to a hinge at A . Determine the value of θ for which the wire is in equilibrium for the indicated position.

5.29 The frame for a sign is fabricated from thin, flat steel bar stock of mass per unit length 4.73 kg/m . The frame is supported by a pin at C and by a cable AB . Determine (a) the tension in the cable, (b) the reaction at C .

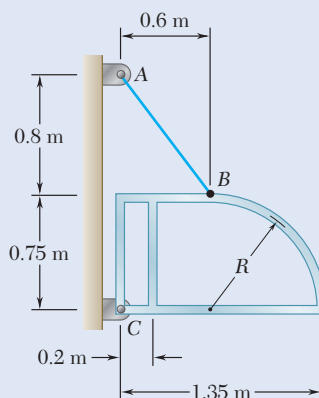


Fig. P5.29

5.30 The homogeneous wire $ABCD$ is bent as shown and is attached to a hinge at C . Determine the length L for which portion BCD of the wire is horizontal.

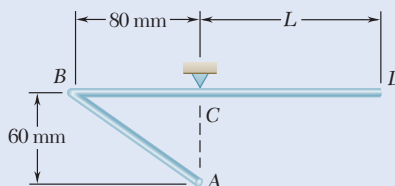


Fig. P5.30 and P5.31

5.31 The homogeneous wire $ABCD$ is bent as shown and is attached to a hinge at C . Determine the length L for which portion AB of the wire is horizontal.

5.32 Determine the distance h for which the centroid of the shaded area is as far above line BB' as possible when (a) $k = 0.10$, (b) $k = 0.80$.

5.33 Knowing that the distance h has been selected to maximize the distance $\bar{y}$ from line BB' to the centroid of the shaded area, show that $\bar{y} = 2h/3$.

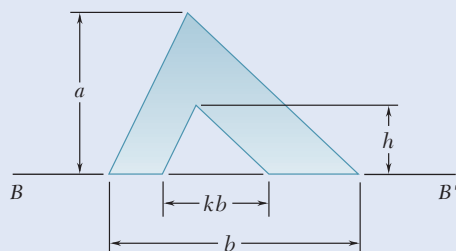


Fig. P5.32 and P5.33

5.2 FURTHER CONSIDERATIONS OF CENTROIDS

The objects we analyzed in Sec. 5.1 were composites of basic geometric shapes like rectangles, triangles, and circles. The same idea of locating a center of gravity or centroid applies for an object with a more complicated shape, but the mathematical techniques for finding the location are a little more difficult.

5.2A Determination of Centroids by Integration

For an area bounded by analytical curves (i.e., curves defined by algebraic equations), we usually determine the centroid by evaluating the integrals in Eqs. (5.3'):

$$\bar{x} = \frac{\int x dA}{A} \quad \bar{y} = \frac{\int y dA}{A} \quad (5.3')$$

If the element of area dA is a small rectangle of sides dx and dy , evaluating each of these integrals requires a *double integration* with respect to x and y . A double integration is also necessary if we use polar coordinates for which dA is a small element with sides dr and $r d\theta$.

In most cases, however, it is possible to determine the coordinates of the centroid of an area by performing a single integration. We can achieve this by choosing dA to be a thin rectangle or strip, or it can be a thin sector or pie-shaped element (Fig. 5.12). The centroid of the thin rectangle is located at its center, and the centroid of the thin sector is located at a distance $(2/3)r$ from its vertex (as it is for a triangle). Then we obtain the coordinates of the centroid of the area under consideration

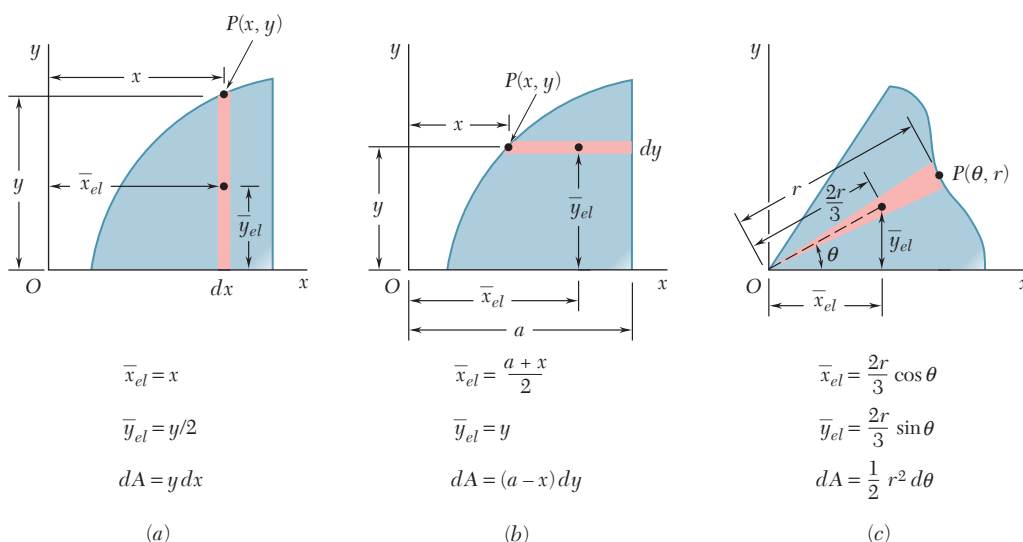


Fig. 5.12 Centroids and areas of differential elements. (a) Vertical rectangular strip; (b) horizontal rectangular strip; (c) triangular sector.

by setting the first moment of the entire area with respect to each of the coordinate axes equal to the sum (or integral) of the corresponding moments of the elements of the area. Denoting the coordinates of the centroid of the element dA by $\bar{x}_{el}$ and $\bar{y}_{el}$, we have

First moments of area

$$\begin{aligned} Q_y = \bar{x}A &= \int \bar{x}_{el} dA \\ Q_x = \bar{y}A &= \int \bar{y}_{el} dA \end{aligned} \quad (5.9)$$

If we do not already know the area A , we can also compute it from these elements.

In order to carry out the integration, we need to express the coordinates $\bar{x}_{el}$ and $\bar{y}_{el}$ of the centroid of the element of area dA in terms of the coordinates of a point located on the curve bounding the area under consideration. Also, we should express the area of the element dA in terms of the coordinates of that point and the appropriate differentials. This has been done in Fig. 5.12 for three common types of elements; the pie-shaped element of part (c) should be used when the equation of the curve bounding the area is given in polar coordinates. You can substitute the appropriate expressions into formulas (5.9), and then use the equation of the bounding curve to express one of the coordinates in terms of the other. This process reduces the double integration to a single integration. Once you have determined the area and evaluated the integrals in Eqs. (5.9), you can solve these equations for the coordinates $\bar{x}$ and $\bar{y}$ of the centroid of the area.

When a line is defined by an algebraic equation, you can determine its centroid by evaluating the integrals in Eqs. (5.4'):

$$\bar{x} = \frac{\int x dL}{L} \quad \bar{y} = \frac{\int y dL}{L} \quad (5.4')$$

You can replace the differential length dL with one of the following expressions, depending upon which coordinate, x , y , or θ , is chosen as the independent variable in the equation used to define the line (these expressions can be derived using the Pythagorean theorem):

$$\begin{aligned} dL &= \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx & dL &= \sqrt{1 + \left(\frac{dx}{dy}\right)^2} dy \\ dL &= \sqrt{r^2 + \left(\frac{dr}{d\theta}\right)^2} d\theta \end{aligned}$$

After you have used the equation of the line to express one of the coordinates in terms of the other, you can perform the integration and solve Eqs. (5.4) for the coordinates $\bar{x}$ and $\bar{y}$ of the centroid of the line.

5.2B Theorems Of Pappus-Guldinus

These two theorems, which were first formulated by the Greek geometer Pappus during the third century C.E. and later restated by the Swiss mathematician Guldinus or Guldin (1577–1643), deal with surfaces and bodies

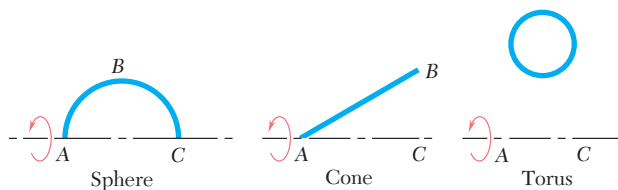


Fig. 5.13. Rotating plane curves about an axis generates surfaces of revolution.

of revolution. A **surface of revolution** is a surface that can be generated by rotating a plane curve about a fixed axis. For example, we can obtain the surface of a sphere by rotating a semicircular arc ABC about the diameter AC (Fig. 5.13). Similarly, rotating a straight line AB about an axis AC produces the surface of a cone, and rotating the circumference of a circle about a nonintersecting axis generates the surface of a torus or ring. A **body of revolution** is a body that can be generated by rotating a plane area about a fixed axis. As shown in Fig. 5.14, we can generate a sphere, a cone, and a torus by rotating the appropriate shape about the indicated axis.

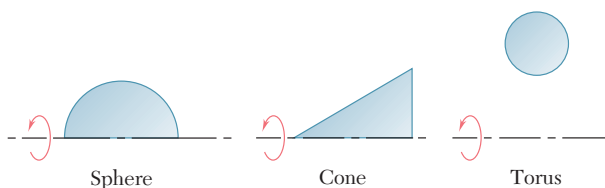


Fig. 5.14. Rotating plane areas about an axis generates volumes of revolution.

Theorem I. *The area of a surface of revolution is equal to the length of the generating curve times the distance traveled by the centroid of the curve while the surface is being generated.*

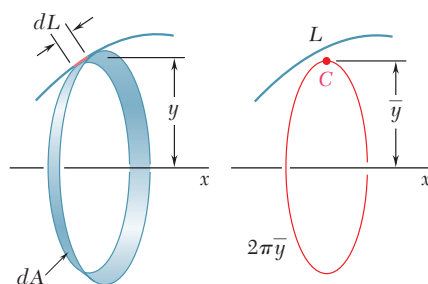


Fig. 5.15 An element of length dL rotated about the x axis generates a circular strip of area dA . The area of the entire surface of revolution equals the length of the line L multiplied by the distance traveled by the centroid C of the line during one revolution.

Proof. Consider an element dL of the line L (Fig. 5.15) that is revolved about the x axis. The circular strip generated by the element dL has an area



Photo 5.3 The storage tanks shown are bodies of revolution. Thus, their surface areas and volumes can be determined using the theorems of Pappus-Guldinus.

dA equal to $2\pi y dL$. Thus, the entire area generated by L is $A = \int 2\pi y dL$. Recall our earlier result that the integral $\int y dL$ is equal to $\bar{y}L$. Therefore, we have

$$A = 2\pi\bar{y}L \quad (5.10)$$

Here $2\pi\bar{y}$ is the distance traveled by the centroid C of L (Fig. 5.15). $\square$

Note that the generating curve must not cross the axis about which it is rotated; if it did, the two sections on either side of the axis would generate areas having opposite signs, and the theorem would not apply.

Theorem II. *The volume of a body of revolution is equal to the generating area times the distance traveled by the centroid of the area while the body is being generated.*

Proof. Consider an element dA of the area A that is revolved about the x axis (Fig. 5.16). The circular ring generated by the element dA has a volume dV equal to $2\pi y dA$. Thus, the entire volume generated by A is $V = \int 2\pi y dA$, and since we showed earlier that the integral $\int y dA$ is equal to $\bar{y}A$, we have

$$V = 2\pi\bar{y}A \quad (5.11)$$

Here $2\pi\bar{y}$ is the distance traveled by the centroid of A . $\square$

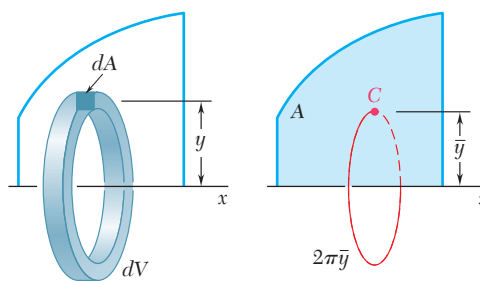


Fig. 5.16 An element of area dA rotated about the x axis generates a circular ring of volume dV . The volume of the entire body of revolution equals the area of the region A multiplied by the distance traveled by the centroid C of the region during one revolution.

Again, note that the theorem does not apply if the axis of rotation intersects the generating area.

The theorems of Pappus-Guldinus offer a simple way to compute the areas of surfaces of revolution and the volumes of bodies of revolution. Conversely, they also can be used to determine the centroid of a plane curve if you know the area of the surface generated by the curve or to determine the centroid of a plane area if you know the volume of the body generated by the area (see Sample Prob. 5.8).

Sample Problem 5.4

Determine the location of the centroid of a parabolic spandrel by direct integration.

STRATEGY: First express the parabolic curve using the parameters a and b . Then choose a differential element of area and express its area in terms of a , b , x , and y . We illustrate the solution first with a vertical element and then a horizontal element.

MODELING:

Determination of the Constant k . Determine the value of k by substituting $x = a$ and $y = b$ into the given equation. We have $b = ka^2$ or $k = b/a^2$. The equation of the curve is thus

$$y = \frac{b}{a^2}x^2 \quad \text{or} \quad x = \frac{a}{b^{1/2}}y^{1/2}$$

ANALYSIS:

Vertical Differential Element. Choosing the differential element shown in Fig. 1, the total area of the region is

$$A = \int dA = \int y \, dx = \int_0^a \frac{b}{a^2} x^2 \, dx = \left[\frac{b}{a^2} \frac{x^3}{3} \right]_0^a = \frac{ab}{3}$$

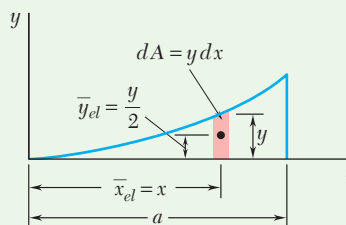


Fig. 1 Vertical differential element used to determine centroid.

The first moment of the differential element with respect to the y axis is $\bar{x}_{el}dA$; hence, the first moment of the entire area with respect to this axis is

$$Q_y = \int \bar{x}_{el} dA = \int xy \, dx = \int_0^a x \left(\frac{b}{a^2} x^2 \right) dx = \left[\frac{b}{a^2} \frac{x^4}{4} \right]_0^a = \frac{a^2 b}{4}$$

Since $Q_y = \bar{x}A$, you have

$$\bar{x}A = \int \bar{x}_{el} dA \quad \bar{x} \frac{ab}{3} = \frac{a^2 b}{4} \quad \bar{x} = \frac{3}{4}a \quad \blacktriangleleft$$

Likewise, the first moment of the differential element with respect to the x axis is $\bar{y}_{el}dA$, so the first moment of the entire area about the x axis is

$$Q_x = \int \bar{y}_{el} dA = \int \frac{y}{2} y \, dx = \int_0^a \frac{1}{2} \left(\frac{b}{a^2} x^2 \right)^2 dx = \left[\frac{b^2}{2a^4} \frac{x^5}{5} \right]_0^a = \frac{ab^2}{10}$$

Since $Q_x = \bar{y}A$, you get

$$\bar{y}A = \int \bar{y}_{el} dA \quad \bar{y} \frac{ab}{3} = \frac{ab^2}{10} \quad \bar{y} = \frac{3}{10}b \quad \blacktriangleleft$$

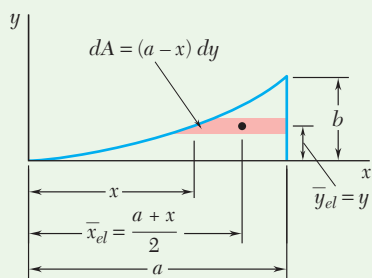


Fig. 2 Horizontal differential element used to determine centroid.

Horizontal Differential Element. You obtain the same results by considering a horizontal element (Fig. 2). The first moments of the area are

$$\begin{aligned} Q_y &= \int \bar{x}_{el} dA = \int \frac{a+x}{2} (a-x) dy = \int_0^b \frac{a^2 - x^2}{2} dy \\ &= \frac{1}{2} \int_0^b \left(a^2 - \frac{a^2}{b} y \right) dy = \frac{a^2 b}{4} \\ Q_x &= \int \bar{y}_{el} dA = \int y(a-x) dy = \int y \left(a - \frac{a}{b^{1/2}} y^{1/2} \right) dy \\ &= \int_0^b \left(ay - \frac{a}{b^{1/2}} y^{3/2} \right) dy = \frac{ab^2}{10} \end{aligned}$$

To determine $\bar{x}$ and $\bar{y}$, again substitute these expressions into the equations defining the centroid of the area.

REFLECT and THINK: You obtain the same results whether you choose a vertical or a horizontal element of area, as you should. You can use both methods as a check against making a mistake in your calculations.

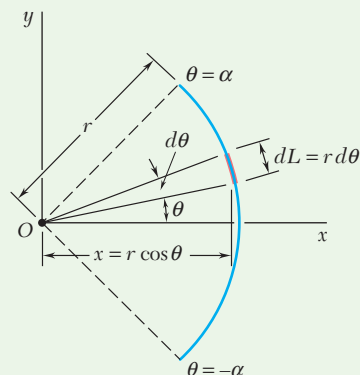
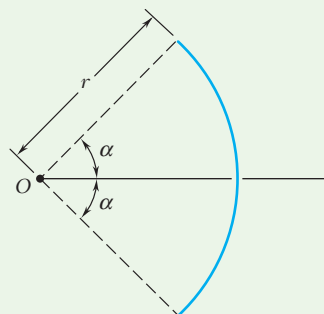


Fig. 1 Differential element used to determine centroid.

Sample Problem 5.5

Determine the location of the centroid of the circular arc shown.

STRATEGY: For a simple figure with circular geometry, you should use polar coordinates.

MODELING: The arc is symmetrical with respect to the x axis, so $\bar{y} = 0$. Choose a differential element, as shown in Fig. 1.

ANALYSIS: Determine the length of the arc by integration.

$$L = \int dL = \int_{-\alpha}^{\alpha} r d\theta = r \int_{-\alpha}^{\alpha} d\theta = 2r\alpha$$

The first moment of the arc with respect to the y axis is

$$\begin{aligned} Q_y &= \int x dL = \int_{-\alpha}^{\alpha} (r \cos \theta) (r d\theta) = r^2 \int_{-\alpha}^{\alpha} \cos \theta d\theta \\ &= r^2 [\sin \theta]_{-\alpha}^{\alpha} = 2r^2 \sin \alpha \end{aligned}$$

Since $Q_y = \bar{x}L$, you obtain

$$\bar{x}(2r\alpha) = 2r^2 \sin \alpha \quad \bar{x} = \frac{r \sin \alpha}{\alpha} \quad \blacktriangleleft$$

REFLECT and THINK: Observe that this result matches that given for this case in Fig. 5.8B.

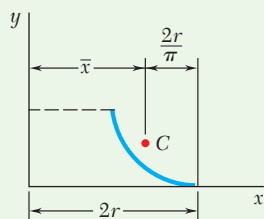
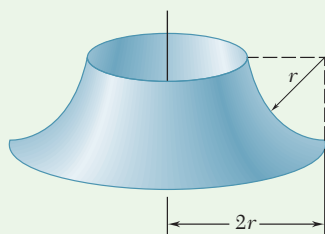


Fig. 1 Centroid location of arc.

Sample Problem 5.6

Determine the area of the surface of revolution shown that is obtained by rotating a quarter-circular arc about a vertical axis.

STRATEGY: According to the first Pappus-Guldinus theorem, the area of the surface of revolution is equal to the product of the length of the arc and the distance traveled by its centroid.

MODELING and ANALYSIS: Referring to Fig. 5.8B and Fig. 1, you have

$$\bar{x} = 2r - \frac{2r}{\pi} = 2r \left(1 - \frac{1}{\pi} \right)$$

$$A = 2\pi\bar{x}L = 2\pi \left[2r \left(1 - \frac{1}{\pi} \right) \right] \left(\frac{\pi r}{2} \right)$$

$$A = 2\pi r^2(\pi - 1) \quad \blacktriangleleft$$

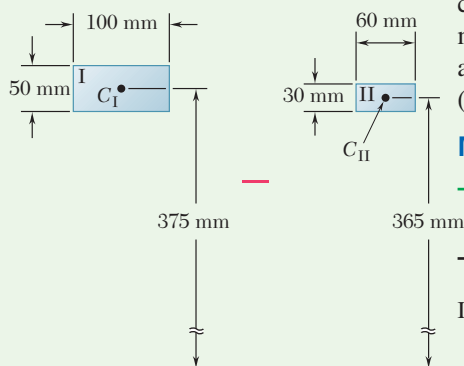
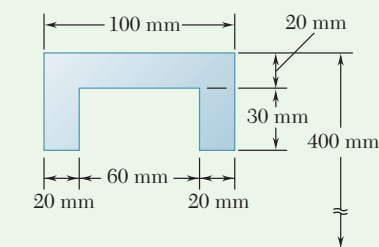


Fig. 1 Modeling the given area by subtracting area II from area I.

Sample Problem 5.7

The outside diameter of a pulley is 0.8 m, and the cross section of its rim is as shown. Knowing that the pulley is made of steel and that the density of steel is $\rho = 7.85 \times 10^3 \text{ kg/m}^3$, determine the mass and weight of the rim.

STRATEGY: You can determine the volume of the rim by applying the second Pappus-Guldinus theorem, which states that the volume equals the product of the given cross-sectional area and the distance traveled by its centroid in one complete revolution. However, you can find the volume more easily by observing that the cross section can be formed from rectangle I with a positive area and from rectangle II with a negative area (Fig. 1).

MODELING: Use a table to keep track of the data, as you did in Sec. 5.1.

	Area, mm ²	$\bar{y}$, mm	Distance Traveled by C, mm	Volume, mm ³
I	+5000	375	$2\pi(375) = 2356$	$(5000)(2356) = 11.78 \times 10^6$
II	-1800	365	$2\pi(365) = 2293$	$(-1800)(2293) = -4.13 \times 10^6$
				Volume of rim = 7.65×10^6

ANALYSIS: Since $1 \text{ mm} = 10^{-3} \text{ m}$, you have $1 \text{ mm}^{-3} = (10^{-3} \text{ m})^3 = 10^{-9} \text{ m}^3$. Thus you obtain $V = 7.65 \times 10^6 \text{ mm}^3 = (7.65 \times 10^6)(10^{-9} \text{ m}^3) = 7.65 \times 10^{-3} \text{ m}^3$.

$$m = \rho V = (7.85 \times 10^3 \text{ kg/m}^3)(7.65 \times 10^{-3} \text{ m}^3) \quad m = 60.0 \text{ kg} \quad \blacktriangleleft$$

$$W = mg = (60.0 \text{ kg})(9.81 \text{ m/s}^2) = 589 \text{ kg} \cdot \text{m/s}^2 \quad W = 589 \text{ N} \quad \blacktriangleleft$$

REFLECT and THINK: When a cross section can be broken down into multiple common shapes, you can apply Theorem II of Pappus–Guldinus in a manner that involves finding the products of the centroid ($\bar{y}$) and area (A), or the first moments of area ($\bar{y}A$), for each shape. Thus, it was not necessary to find the centroid or the area of the overall cross section.

Sample Problem 5.8

Using the theorems of Pappus–Guldinus, determine (a) the centroid of a semicircular area and (b) the centroid of a semicircular arc. Recall that the volume and the surface area of a sphere are $\frac{4}{3}\pi r^3$ and $4\pi r^2$, respectively.

STRATEGY: The volume of a sphere is equal to the product of the area of a semicircle and the distance traveled by the centroid of the semicircle in one revolution about the x axis. Given the volume, you can determine the distance traveled by the centroid and thus the distance of the centroid from the axis. Similarly, the area of a sphere is equal to the product of the length of the generating semicircle and the distance traveled by its centroid in one revolution. You can use this to find the location of the centroid of the arc.

MODELING: Draw diagrams of the semicircular area and the semicircular arc (Fig. 1) and label the important geometries.

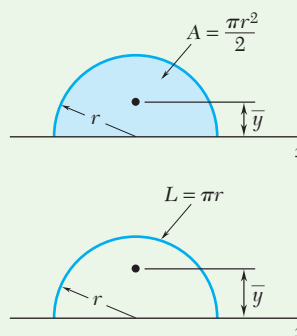


Fig. 1 Semicircular area and semicircular arc.

ANALYSIS: Set up the equalities described in the theorems of Pappus–Guldinus and solve for the location of the centroid.

$$V = 2\pi\bar{y}A \quad \frac{4}{3}\pi r^3 = 2\pi\bar{y}\left(\frac{1}{2}\pi r^2\right) \quad \bar{y} = \frac{4r}{3\pi} \quad \blacktriangleleft$$

$$A = 2\pi\bar{y}L \quad 4\pi r^2 = 2\pi\bar{y}(\pi r) \quad \bar{y} = \frac{2r}{\pi} \quad \blacktriangleleft$$

REFLECT and THINK: Observe that this result matches those given for these cases in Fig. 5.8.

SOLVING PROBLEMS ON YOUR OWN

In the problems for this section, you will use the equations

$$\bar{x} = \frac{\int x dA}{A} \quad \bar{y} = \frac{\int y dA}{A} \quad (5.3')$$

$$\bar{x} = \frac{\int x dL}{L} \quad \bar{y} = \frac{\int y dL}{L} \quad (5.4')$$

to locate the centroids of plane areas and lines, respectively. You will also apply the theorems of Pappus-Guldinus to determine the areas of surfaces of revolution and the volumes of bodies of revolution.

1. Determining the centroids of areas and lines by direct integration.

When solving problems of this type, you should follow the method of solution shown in Sample Probs. 5.4 and 5.5. To compute A or L , determine the first moments of the area or the line, and solve Eqs. (5.3) or (5.4) for the coordinates of the centroid. In addition, you should pay particular attention to the following points.

a. Begin your solution by carefully defining or determining each term in the applicable integral formulas. We strongly encourage you to show on your sketch of the given area or line your choice for dA or dL and the distances to its centroid.

b. As explained in Sec. 5.2A, x and y in Eqs. (5.3) and (5.4) represent the *coordinates of the centroid* of the differential elements dA and dL . It is important to recognize that the coordinates of the centroid of dA are not equal to the coordinates of a point located on the curve bounding the area under consideration. You should carefully study Fig. 5.12 until you fully understand this important point.

c. To possibly simplify or minimize your computations, always examine the shape of the given area or line before defining the differential element that you will use. For example, sometimes it may be preferable to use horizontal rectangular elements instead of vertical ones. Also, it is usually advantageous to use polar coordinates when a line or an area has circular symmetry.

d. Although most of the integrations in this section are straightforward, at times it may be necessary to use more advanced techniques, such as trigonometric substitution or integration by parts. Using a table of integrals is often the fastest method to evaluate difficult integrals.

2. Applying the theorems of Pappus-Guldinus. As shown in Sample Probs. 5.6 through 5.8, these simple, yet very useful theorems allow you to apply your knowledge of centroids to the computation of areas and volumes. Although the theorems refer to the distance traveled by the centroid and to the length of the generating curve or to the generating area, the resulting equations [Eqs. (5.10) and (5.11)] contain the products of these quantities, which are simply the first moments of a line ($\bar{y}L$) and an area ($\bar{y}A$), respectively. Thus, for those problems for which the generating line or area consists of more than one common shape, you need only determine $\bar{y}L$ or $\bar{y}A$; you do not have to calculate the length of the generating curve or the generating area.

Problems

5.34 through 5.36 Determine by direct integration the centroid of the area shown. Express your answer in terms of a and h .

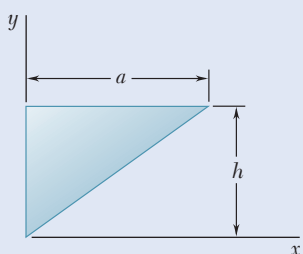


Fig. P5.34

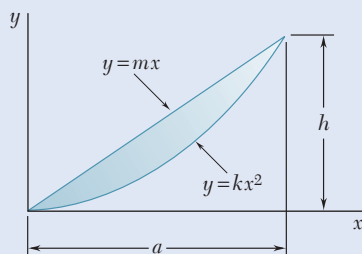


Fig. P5.35

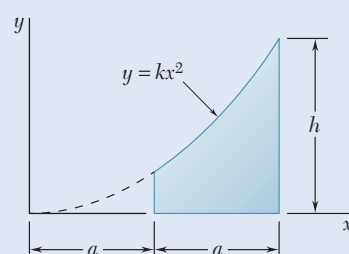


Fig. P5.36

5.37 through 5.39 Determine by direct integration the centroid of the area shown.

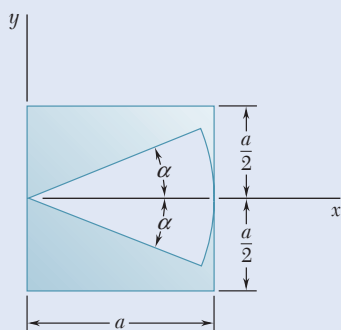


Fig. P5.37

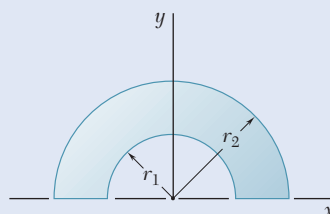


Fig. P5.38

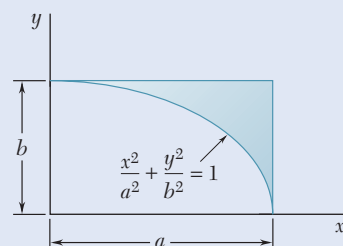


Fig. P5.39

5.40 and 5.41 Determine by direct integration the centroid of the area shown. Express your answer in terms of a and b .

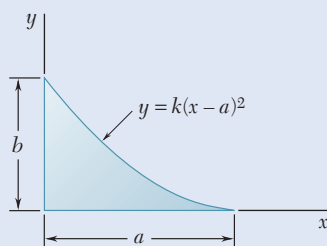


Fig. P5.40

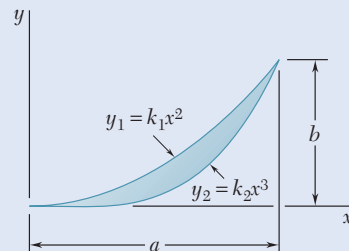


Fig. P5.41

5.42 Determine by direct integration the centroid of the area shown.

5.43 and 5.44 Determine by direct integration the centroid of the area shown. Express your answer in terms of a and b .

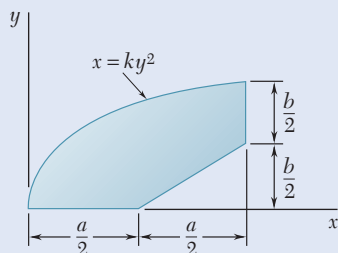


Fig. P5.43

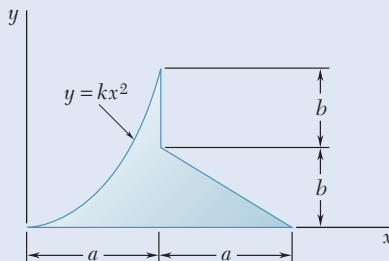


Fig. P5.44

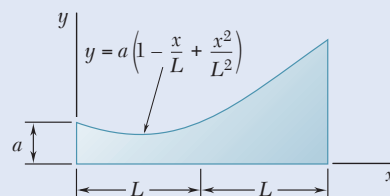


Fig. P5.42

5.45 and 5.46 A homogeneous wire is bent into the shape shown. Determine by direct integration the x coordinate of its centroid.

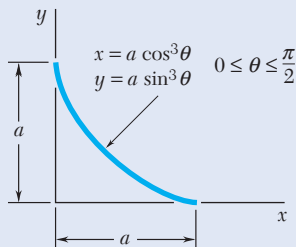


Fig. P5.45

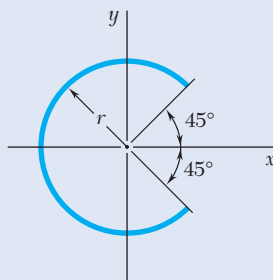


Fig. P5.46

***5.47** A homogeneous wire is bent into the shape shown. Determine by direct integration the x coordinate of its centroid. Express your answer in terms of a .

***5.48 and *5.49** Determine by direct integration the centroid of the area shown.

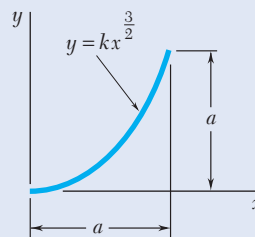


Fig. P5.47

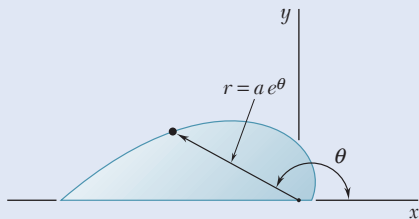


Fig. P5.48

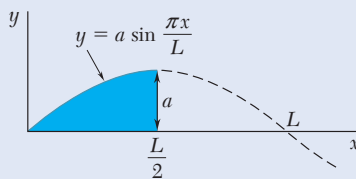


Fig. P5.49

5.50 Determine the centroid of the area shown in terms of a .

5.51 Determine the centroid of the area shown when $a = 4$ in.

5.52 Determine the volume and the surface area of the solid obtained by rotating the area of Prob. 5.1 about (a) the x axis, (b) the line $x = 72$ mm.

5.53 Determine the volume and the surface area of the solid obtained by rotating the area of Prob. 5.2 about (a) the x axis, (b) the y axis.

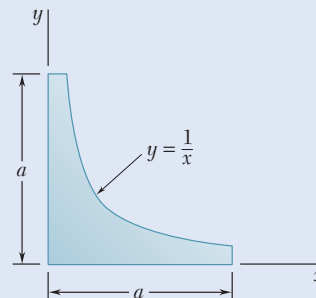


Fig. P5.50 and P5.51

- 5.54** Determine the volume and the surface area of the solid obtained by rotating the area of Prob. 5.6 about (a) the line $x = -60$ mm, (b) the line $y = 120$ mm.
- 5.55** Determine the volume and the surface area of the chain link shown, which is made from a 6-mm-diameter bar, if $R = 10$ mm and $L = 30$ mm.

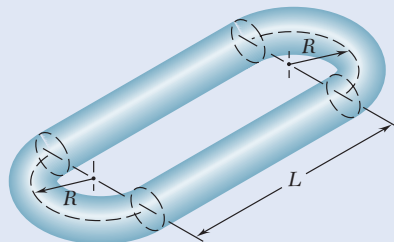


Fig. P5.55

- 5.56** Determine the volume of the solid generated by rotating the parabolic area shown about (a) the x axis, (b) the axis AA' .

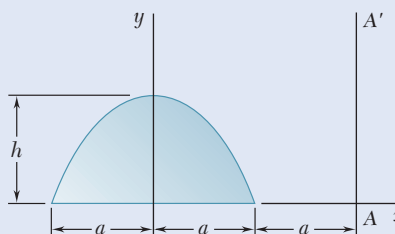


Fig. P5.56

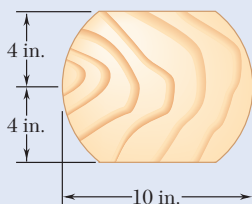


Fig. P5.58

- 5.57** Verify that the expressions for the volumes of the first four shapes in Fig. 5.21 are correct.
- 5.58** Knowing that two equal caps have been removed from a 10-in.-diameter wooden sphere, determine the total surface area of the remaining portion.
- 5.59** Three different drive belt profiles are to be studied. If at any given time each belt makes contact with one-half of the circumference of its pulley, determine the *contact area* between the belt and the pulley for each design.

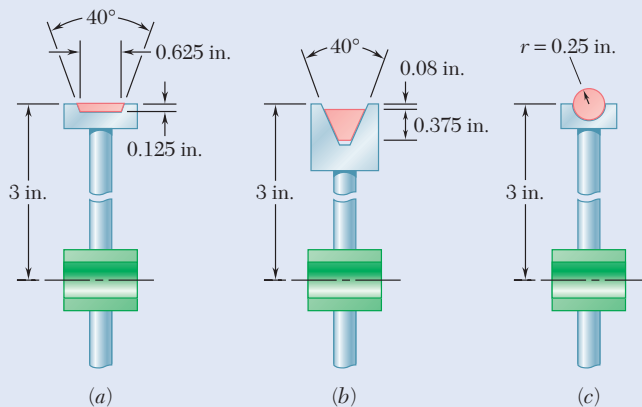


Fig. P5.59

- 5.60** Determine the capacity, in liters, of the punch bowl shown if $R = 250$ mm.

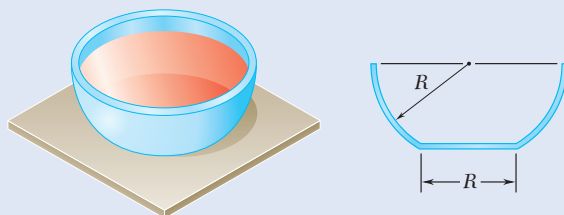


Fig. P5.60

- 5.61** Determine the volume and total surface area of the bushing shown.

- 5.62** Determine the volume and weight of the solid brass knob shown, knowing that the specific weight of brass is 0.306 lb/in³.

- 5.63** Determine the total surface area of the solid brass knob shown.

- 5.64** The aluminum shade for the small high-intensity lamp shown has a uniform thickness of 1 mm. Knowing that the density of aluminum is 2800 kg/m³, determine the mass of the shade.

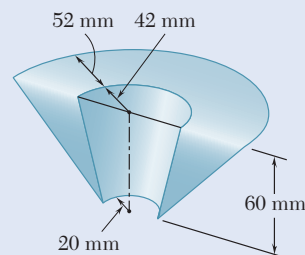


Fig. P5.61

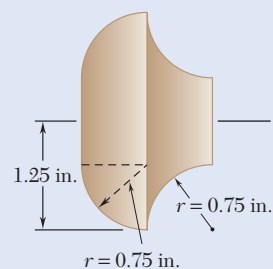


Fig. P5.62 and P5.63

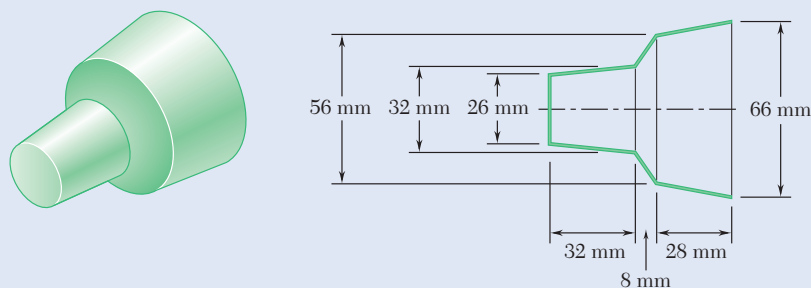


Fig. P5.64

- *5.65** The shade for a wall-mounted light is formed from a thin sheet of translucent plastic. Determine the surface area of the outside of the shade, knowing that it has the parabolic cross section shown.

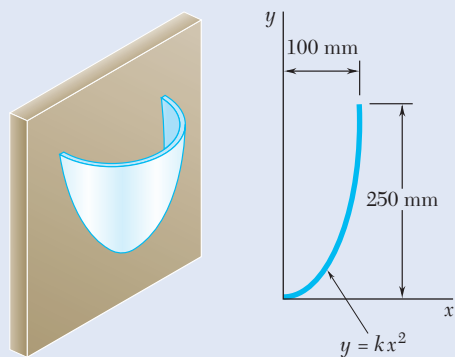


Fig. P5.65

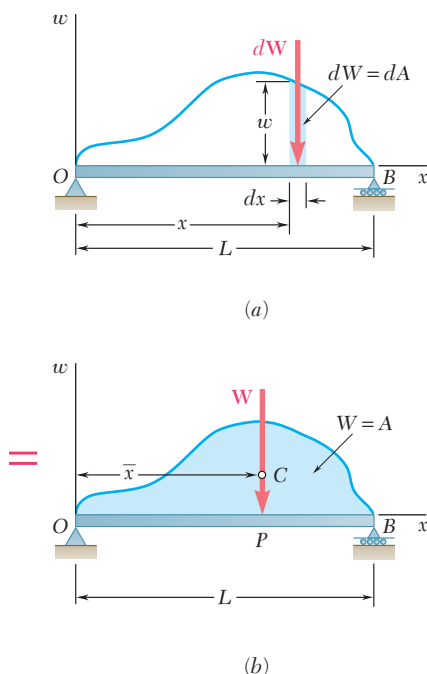


Fig. 5.17 (a) A load curve representing the distribution of load forces along a horizontal beam, with an element of length dx ; (b) the resultant load W has magnitude equal to the area A under the load curve and acts through the centroid of the area.



Photo 5.4 The roof of a building shown must be able to support not only the total weight of the snow but also the nonsymmetric distributed loads resulting from drifting of the snow.

5.3 ADDITIONAL APPLICATIONS OF CENTROIDS

We can use the concept of the center of gravity or the centroid of an area to solve other problems besides those dealing with the weights of flat plates. The same techniques allow us to deal with other kinds of distributed loads on objects, such as the forces on a straight beam (a bridge girder or the main carrying beam of a house floor) or a flat plate under water (the side of a dam or a window in an aquarium tank).

5.3A Distributed Loads on Beams

Consider a beam supporting a **distributed load**; this load may consist of the weight of materials supported directly or indirectly by the beam, or it may be caused by wind or hydrostatic pressure. We can represent the distributed load by plotting the load w supported per unit length (Fig. 5.17); this load is expressed in N/m or in lb/ft. The magnitude of the force exerted on an element of the beam with length dx is $dW = w dx$, and the total load supported by the beam is

$$W = \int_0^L w dx$$

Note that the product $w dx$ is equal in magnitude to the element of area dA shown in Fig. 5.17a. The load W is thus equal in magnitude to the total area A under the load curve, as

$$W = \int dA = A$$

We now want to determine where a *single concentrated load* W , of the same magnitude W as the total distributed load, should be applied on the beam if it is to produce the same reactions at the supports (Fig. 5.17b). However, this concentrated load W , which represents the resultant of the given distributed loading, is equivalent to the loading only when considering the free-body diagram of the entire beam. We obtain the point of application P of the equivalent concentrated load W by setting the moment of W about point O equal to the sum of the moments of the elemental loads dW about O . Thus,

$$(OP)W = \int x dW$$

Then, since $dW = w dx = dA$ and $W = A$, we have

$$(OP)A = \int_0^L x dA \quad (5.12)$$

Since this integral represents the first moment with respect to the w axis of the area under the load curve, we can replace it with the product $\bar{x}A$. We therefore have $OP = \bar{x}$, where $\bar{x}$ is the distance from the w axis to the centroid C of the area A (this is *not* the centroid of the beam).

We can summarize this result:

We can replace a distributed load on a beam by a concentrated load; the magnitude of this single load is equal to the area under the load curve, and its line of action passes through the centroid of that area.

Note, however, that the concentrated load is equivalent to the given loading only so far as external forces are concerned. It can be used to determine reactions, but should not be used to compute internal forces and deflections.

*5.3B Forces on Submerged Surfaces

The approach used for distributed loads on beams works in other applications as well. Here, we use it to determine the resultant of the hydrostatic pressure forces exerted on a *rectangular surface* submerged in a liquid. We can use these methods to determine the resultant of the hydrostatic forces exerted on the surfaces of dams, rectangular gates, and vanes. In Chap. 9, we discuss the resultants of forces on submerged surfaces of variable width.

Consider a rectangular plate with a length of L and width of b , where b is measured perpendicular to the plane of the figure (Fig. 5.18). As for the case of distributed loads on a beam, the load exerted on an element of the plate with a length of dx is $w dx$, where w is the load per unit length and x is the distance along the length. However, this load also can be expressed as $p dA = pb dx$, where p is the gage pressure in the liquid[†] and b is the width of the plate; thus, $w = bp$. Since the gage pressure in a liquid is $p = \gamma h$, where γ is the specific weight of the liquid and h is the vertical distance from the free surface, it follows that

$$w = bp = b\gamma h \quad (5.13)$$

This equation shows that the load per unit length w is proportional to h and, thus, varies linearly with x .

From the results of Sec. 5.3A, the resultant $\mathbf{R}$ of the hydrostatic forces exerted on one side of the plate is equal in magnitude to the trapezoidal area under the load curve, and its line of action passes through the centroid C of that area. The point P of the plate where $\mathbf{R}$ is applied is known as the *center of pressure*.[‡]

Now consider the forces exerted by a liquid on a curved surface of constant width (Fig. 5.19a). Since determining the resultant $\mathbf{R}$ of these forces by direct integration would not be easy, we consider the free body obtained by detaching the volume of liquid ABD bounded by the curved surface AB and by the two plane surfaces AD and DB shown in Fig. 5.19b. The forces acting on the free body ABD are the weight $\mathbf{W}$ of the detached volume of liquid, the resultant $\mathbf{R}_1$ of the forces exerted on AD , the resultant $\mathbf{R}_2$ of the forces exerted on BD , and the resultant $-\mathbf{R}$ of the forces exerted by the curved surface on the liquid. The resultant $-\mathbf{R}$ is both equal and opposite to and has the same line of action as the resultant $\mathbf{R}$ of the forces exerted by the liquid on the curved surface. We can determine the forces $\mathbf{W}$, $\mathbf{R}_1$, and $\mathbf{R}_2$ by standard methods. After their values have been found, we obtain the force $-\mathbf{R}$ by solving the equations of equilibrium for the free body of Fig. 5.19b. The resultant $\mathbf{R}$ of the hydrostatic forces exerted on the curved surface is just the reverse of $-\mathbf{R}$.

[†]The pressure p , which represents a load per unit area, is measured in N/m^2 or in lb/ft^2 . The derived SI unit N/m^2 is called a pascal (Pa).

[‡]The area under the load curve is equal to $w_E L$, where w_E is the load per unit length at the center E of the plate. Then from Eq. (5.13), we have

$$R = w_E L = (bp_E)L = p_E(bL) = p_E A$$

where A denotes the area of the plate. Thus, we can obtain the magnitude of $\mathbf{R}$ by multiplying the area of the plate by the pressure at its center E . Note, however, that the resultant $\mathbf{R}$ should be applied at P , not at E .

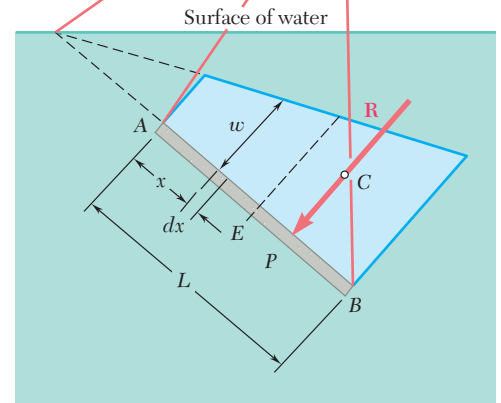
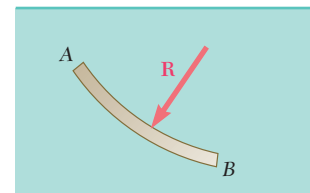
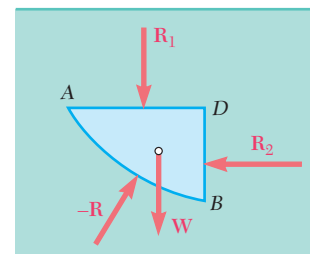


Fig. 5.18 The waterside face of a hydroelectric dam can be modeled as a rectangular plate submerged under water. Shown is a side view of the plate.



(a)



(b)

Fig. 5.19 (a) Force $\mathbf{R}$ exerted by a liquid on a submerged curved surface of constant width; (b) free-body diagram of the volume of liquid ABD .

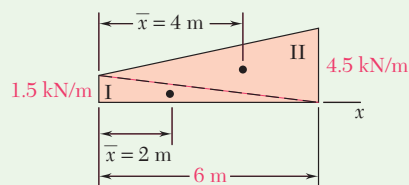
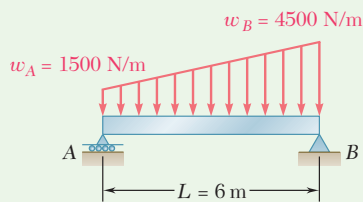


Fig. 1 The load modeled as two triangular areas.

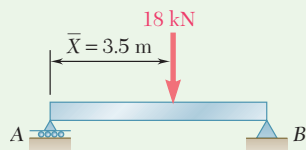


Fig. 2 Equivalent concentrated load.

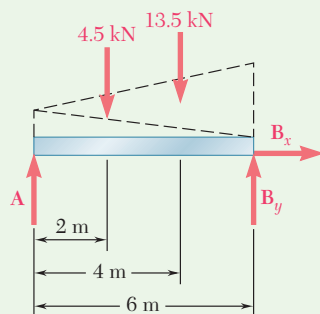


Fig. 3 Free-body diagram of beam.

Sample Problem 5.9

A beam supports a distributed load as shown. (a) Determine the equivalent concentrated load. (b) Determine the reactions at the supports.

STRATEGY: The magnitude of the resultant of the load is equal to the area under the load curve, and the line of action of the resultant passes through the centroid of the same area. Break down the area into pieces for easier calculation, and determine the resultant load. Then, use the calculated forces or their resultant to determine the reactions.

MODELING and ANALYSIS:

a. Equivalent Concentrated Load. Divide the area under the load curve into two triangles (Fig. 1), and construct the table below. To simplify the computations and tabulation, the given loads per unit length have been converted into kN/m.

Component	A, kN	$\bar{x}$, m	$\bar{x}A$, kN·m
Triangle I	4.5	2	9
Triangle II	13.5	4	54
	$\Sigma A = 18.0$		$\Sigma \bar{x}A = 63$

$$\text{Thus, } \bar{X}\Sigma A = \Sigma \bar{x}A: \quad \bar{X}(18 \text{ kN}) = 63 \text{ kN}\cdot\text{m} \quad \bar{X} = 3.5 \text{ m}$$

The equivalent concentrated load (Fig. 2) is

$$W = 18 \text{ kN} \downarrow \quad \blacktriangleleft$$

Its line of action is located at a distance

$$\bar{X} = 3.5 \text{ m to the right of A} \quad \blacktriangleleft$$

b. Reactions. The reaction at A is vertical and is denoted by **A**. Represent the reaction at B by its components **B_x** and **B_y**. Consider the given load to be the sum of two triangular loads (see the free-body diagram, Fig. 3). The resultant of each triangular load is equal to the area of the triangle and acts at its centroid.

Write the following equilibrium equations from the free-body diagram:

$$\rightarrow \Sigma F_x = 0: \quad \mathbf{B}_x = 0 \quad \blacktriangleleft$$

$$+\uparrow \Sigma M_A = 0: \quad -(4.5 \text{ kN})(2 \text{ m}) - (13.5 \text{ kN})(4 \text{ m}) + B_y(6 \text{ m}) = 0$$

$$\mathbf{B}_y = 10.5 \text{ kN} \uparrow \quad \blacktriangleleft$$

$$+\uparrow \Sigma M_B = 0: \quad +(4.5 \text{ kN})(4 \text{ m}) + (13.5 \text{ kN})(2 \text{ m}) - A(6 \text{ m}) = 0$$

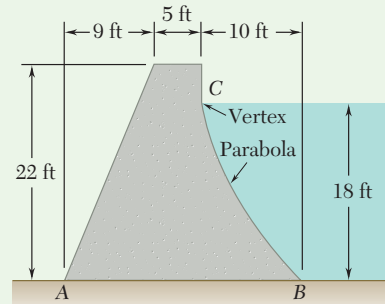
$$\mathbf{A} = 7.5 \text{ kN} \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: You can replace the given distributed load by its resultant, which you found in part a. Then you can determine the reactions from the equilibrium equations $\Sigma F_x = 0$, $\Sigma M_A = 0$, and $\Sigma M_B = 0$. Again the results are

$$\mathbf{B}_x = 0 \quad \mathbf{B}_y = 10.5 \text{ kN} \uparrow \quad \mathbf{A} = 7.5 \text{ kN} \uparrow \quad \blacktriangleleft$$

Sample Problem 5.10

The cross section of a concrete dam is shown. Consider a 1-ft-thick section of the dam, and determine (a) the resultant of the reaction forces exerted by the ground on the base AB of the dam, (b) the resultant of the pressure forces exerted by the water on the face BC of the dam. The specific weights of concrete and water are 150 lb/ft^3 and 62.4 lb/ft^3 , respectively.



STRATEGY: Draw a free-body diagram of the section of the dam, breaking it into parts to simplify the calculations. Model the resultant of the reactions as a force-couple system at A . Use the method described in Sec. 5.3B to find the force exerted by the dam on the water and reverse it to find the force exerted by the water on face BC .

MODELING and ANALYSIS:

a. Ground Reaction. Choose as a free body the 1-ft-thick section $AEFCDB$ of the dam and water (Fig. 1). The reaction forces exerted by the ground on the base AB are represented by an equivalent force-couple system at A . Other forces acting on the free body are the weight of the dam represented by the weights of its components $\mathbf{W}_1$, $\mathbf{W}_2$, and $\mathbf{W}_3$; the weight of the water $\mathbf{W}_4$; and the resultant $\mathbf{P}$ of the pressure forces exerted on section BD by the water to the right of section BD .

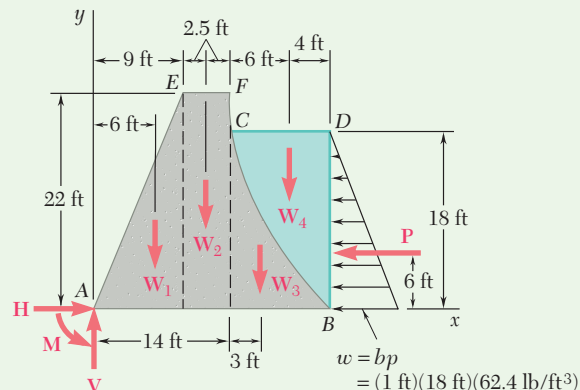


Fig. 1 Free-body diagram of dam and water.

Calculate each of the forces that appear in the free-body diagram, Fig. 1:

$$W_1 = \frac{1}{2}(9 \text{ ft})(22 \text{ ft})(1 \text{ ft})(150 \text{ lb/ft}^3) = 14,850 \text{ lb}$$

$$W_2 = (5 \text{ ft})(22 \text{ ft})(1 \text{ ft})(150 \text{ lb/ft}^3) = 16,500 \text{ lb}$$

$$W_3 = \frac{1}{3}(10 \text{ ft})(18 \text{ ft})(1 \text{ ft})(150 \text{ lb/ft}^3) = 9000 \text{ lb}$$

$$W_4 = \frac{2}{3}(10 \text{ ft})(18 \text{ ft})(1 \text{ ft})(62.4 \text{ lb/ft}^3) = 7488 \text{ lb}$$

$$P = \frac{1}{2}(18 \text{ ft})(1 \text{ ft})(18 \text{ ft})(62.4 \text{ lb/ft}^3) = 10,109 \text{ lb}$$

Equilibrium Equations. Write the equilibrium equations for the section of the dam, and calculate the forces and moment labeled at A in Fig. 1.

$$\rightarrow \Sigma F_x = 0: \quad H - 10,109 \text{ lb} = 0 \quad \mathbf{H = 10,110 \text{ lb} \rightarrow} \quad \blacktriangleleft$$

$$+\uparrow \Sigma F_y = 0: \quad V - 14,850 \text{ lb} - 16,500 \text{ lb} - 9000 \text{ lb} - 7488 \text{ lb} = 0$$

$$\mathbf{V = 47,840 \text{ lb} \uparrow} \quad \blacktriangleleft$$

$$+\curvearrowright \Sigma M_A = 0: \quad -(14,850 \text{ lb})(6 \text{ ft}) - (16,500 \text{ lb})(11.5 \text{ ft})$$

$$- (9000 \text{ lb})(17 \text{ ft}) - (7488 \text{ lb})(20 \text{ ft}) + (10,109 \text{ lb})(6 \text{ ft}) + M = 0$$

$$\mathbf{M = 520,960 \text{ lb}\cdot\text{ft} \curvearrowright} \quad \blacktriangleleft$$

You can replace the force-couple system by a single force acting at a distance d to the right of A, where

$$d = \frac{520,960 \text{ lb}\cdot\text{ft}}{47,840 \text{ lb}} = 10.89 \text{ ft}$$

b. Resultant R of Water Forces. Draw a free-body diagram for the parabolic section of water BCD (Fig. 2). The forces involved are the resultant $-\mathbf{R}$ of the forces exerted by the dam on the water, the weight $\mathbf{W}_4$, and the force $\mathbf{P}$. Since these forces must be concurrent, $-\mathbf{R}$ passes through the point of intersection G of $\mathbf{W}_4$ and $\mathbf{P}$. Draw a force triangle to determine the magnitude and direction of $-\mathbf{R}$. The resultant $\mathbf{R}$ of the forces exerted by the water on the face BC is equal and opposite. Hence,

$$\mathbf{R = 12,580 \text{ lb} \nearrow 36.5^\circ} \quad \blacktriangleleft$$

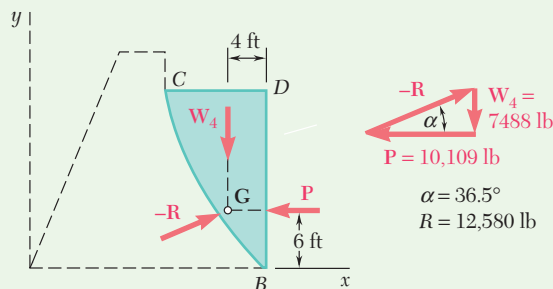


Fig. 2 Free-body diagram of parabolic section of water BCD .

REFLECT and THINK: Note that if you found the distance d to be negative—that is, if the moment reaction at A had been acting in the opposite direction—this would have indicated an instability condition of the dam. In this situation, the effects of the water pressure would overcome the weight of the dam, causing it to tip about A.

SOLVING PROBLEMS ON YOUR OWN

The problems in this section involve two common and very important types of loading: distributed loads on beams and forces on submerged surfaces of constant width. As we discussed in Sec. 5.3 and illustrated in Sample Probs. 5.9 and 5.10, determining the single equivalent force for each of these loadings requires a knowledge of centroids.

1. Analyzing beams subjected to distributed loads. In Sec. 5.3A, we showed that a distributed load on a beam can be replaced by a single equivalent force. The magnitude of this force is equal to the area under the distributed load curve, and its line of action passes through the centroid of that area. Thus, you should begin solving this kind of problem by replacing the various distributed loads on a given beam by their respective single equivalent forces. You can then determine the reactions at the supports of the beam by using the methods of Chap. 4.

When possible, divide complex distributed loads into the common-shape areas shown in Fig. 5.8A (Sample Prob. 5.9). You can replace each of these areas under the loading curve by a single equivalent force. If required, you can further reduce the system of equivalent forces to a single equivalent force. As you study Sample Prob. 5.9, note how we used the analogy between force and area under the loading curve and applied the techniques for locating the centroid of a composite area to analyze a beam subjected to a distributed load.

2. Solving problems involving forces on submerged bodies. Remember the following points and techniques when solving problems of this type.

a. The pressure p at a depth h below the free surface of a liquid is equal to γh or ρgh , where γ and ρ are the specific weight and the density of the liquid, respectively. The load per unit length w acting on a submerged surface of constant width b is then

$$w = bp = b\gamma h = b\rho gh$$

b. The line of action of the resultant force $\mathbf{R}$ acting on a submerged plane surface is perpendicular to the surface.

c. For a vertical or inclined plane rectangular surface with a width of b , you can represent the loading on the surface using a linearly distributed load that is trapezoidal in shape (Fig. 5.18). The magnitude of the resultant $\mathbf{R}$ is given by

$$R = \gamma h_E A$$

where h_E is the vertical distance to the center of the surface and A is the area of the surface.

d. The load curve is triangular (rather than trapezoidal) when the top edge of a plane rectangular surface coincides with the free surface of the liquid, since the pressure of the liquid at the free surface is zero. For this case, it is straightforward to determine the line of action of $\mathbf{R}$, because it passes through the centroid of a *triangular* distributed load.

e. For the general case, rather than analyzing a trapezoid, we suggest that you use the method indicated in part *b* of Sample Prob. 5.9. First divide the trapezoidal distributed load into two triangles, and then compute the magnitude of the resultant of each triangular load. (The magnitude is equal to the area of the triangle times the width of the plate.) Note that the line of action of each resultant force passes through the centroid of the corresponding triangle and that the sum of these forces is equivalent to $\mathbf{R}$. Thus, rather than using $\mathbf{R}$, you can use the two equivalent resultant forces whose points of application are easily calculated. You should use the equation given for R here in paragraph **c** when you need only the magnitude of $\mathbf{R}$.

f. When the submerged surface of a constant width is curved, you can obtain the resultant force acting on the surface by considering the equilibrium of the volume of liquid bounded by the curved surface and by using horizontal and vertical planes (Fig. 5.19). Observe that the force $\mathbf{R}_1$ of Fig. 5.19 is equal to the weight of the liquid lying above the plane AD . The method of solution for problems involving curved surfaces is shown in part *b* of Sample Prob. 5.10.

In subsequent mechanics courses (in particular, mechanics of materials and fluid mechanics), you will have ample opportunity to use the ideas introduced in this section.

Problems

5.66 and 5.67 For the beam and loading shown, determine (a) the magnitude and location of the resultant of the distributed load, (b) the reactions at the beam supports.

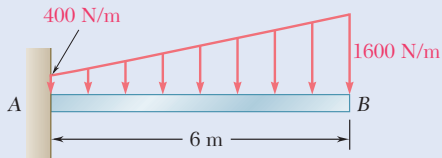


Fig. P5.66

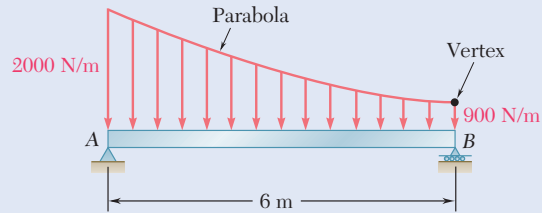


Fig. P5.67

5.68 through 5.73 Determine the reactions at the beam supports for the given loading.

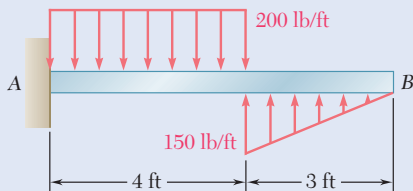


Fig. P5.68

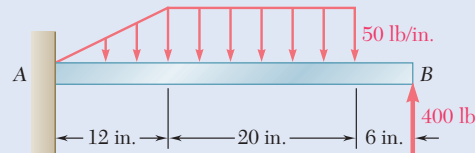


Fig. P5.69

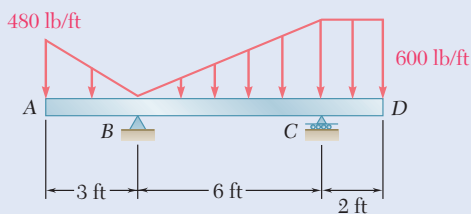


Fig. P5.70

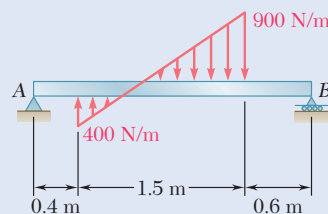


Fig. P5.71

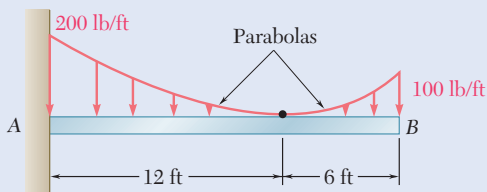


Fig. P5.72

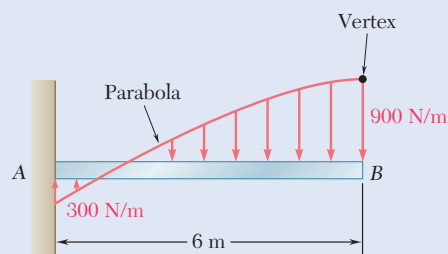


Fig. P5.73

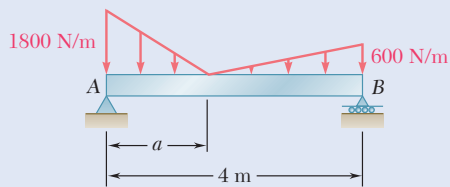


Fig. P5.74 and P5.75

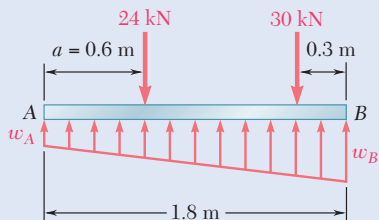


Fig. P5.78

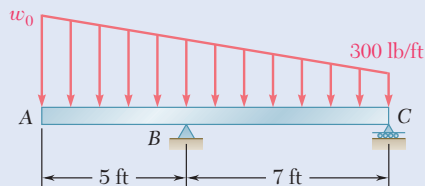


Fig. P5.76 and P5.77

- 5.74** Determine (a) the distance a so that the vertical reactions at supports A and B are equal, (b) the corresponding reactions at the supports.
- 5.75** Determine (a) the distance a so that the reaction at support B is minimum, (b) the corresponding reactions at the supports.
- 5.76** Determine the reactions at the beam supports for the given loading when $w_0 = 400$ lb/ft.
- 5.77** Determine (a) the distributed load w_0 at the end A of the beam ABC for which the reaction at C is zero, (b) the corresponding reaction at B .
- 5.78** The beam AB supports two concentrated loads and rests on soil that exerts a linearly distributed upward load as shown. Determine the values of w_A and w_B corresponding to equilibrium.
- 5.79** For the beam and loading of Prob. 5.78, determine (a) the distance a for which $w_A = 20$ kN/m, (b) the corresponding value of w_B .

In the following problems, use $\gamma = 62.4$ lb/ft³ for the specific weight of fresh water and $\gamma_c = 150$ lb/ft³ for the specific weight of concrete if U.S. customary units are used. With SI units, use $\rho = 10^3$ kg/m³ for the density of fresh water and $\rho_c = 2.40 \times 10^3$ kg/m³ for the density of concrete. (See the footnote on page 234 for how to determine the specific weight of a material given its density.)

- 5.80** The cross section of a concrete dam is as shown. For a 1-ft-wide dam section determine (a) the resultant of the reaction forces exerted by the ground on the base AB of the dam, (b) the point of application of the resultant of part a, (c) the resultant of the pressure forces exerted by the water on the face BC of the dam.

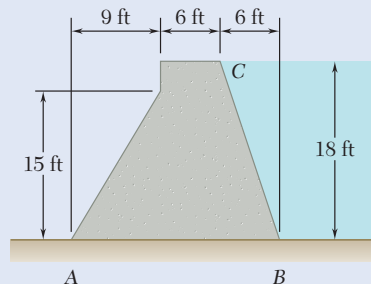


Fig. P5.80

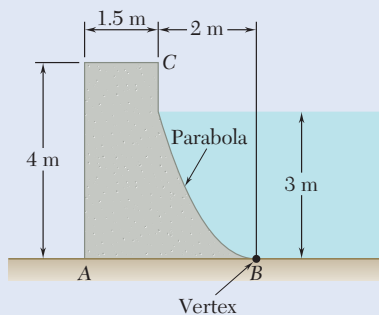


Fig. P5.81

- 5.81** The cross section of a concrete dam is as shown. For a 1-m-wide dam section determine (a) the resultant of the reaction forces exerted by the ground on the base AB of the dam, (b) the point of application of the resultant of part a, (c) the resultant of the pressure forces exerted by the water on the face BC of the dam.

5.82 The dam for a lake is designed to withstand the additional force caused by silt that has settled on the lake bottom. Assuming that silt is equivalent to a liquid of density $\rho_s = 1.76 \times 10^3 \text{ kg/m}^3$ and considering a 1-m-wide section of dam, determine the percentage increase in the force acting on the dam face for a silt accumulation of depth 2 m.

5.83 The base of a dam for a lake is designed to resist up to 120 percent of the horizontal force of the water. After construction, it is found that silt (that is equivalent to a liquid of density $\rho_s = 1.76 \times 10^3 \text{ kg/m}^3$) is settling on the lake bottom at the rate of 12 mm/year. Considering a 1-m-wide section of dam, determine the number of years of service until the dam becomes unsafe.

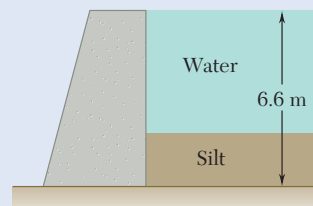


Fig. P5.82 and P5.83

5.84 An automatic valve consists of a 9×9 -in. square plate that is pivoted about a horizontal axis through A located at a distance $h = 3.6$ in. above the lower edge. Determine the depth of water d for which the valve will open.

5.85 An automatic valve consists of a 9×9 -in. square plate that is pivoted about a horizontal axis through A . If the valve is to open when the depth of water is $d = 18$ in., determine the distance h from the bottom of the valve to the pivot A .

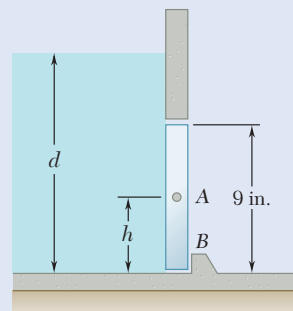


Fig. P5.84 and P5.85

5.86 The 3×4 -m side AB of a tank is hinged at its bottom A and is held in place by a thin rod BC . The maximum tensile force the rod can withstand without breaking is 200 kN, and the design specifications require the force in the rod not to exceed 20 percent of this value. If the tank is slowly filled with water, determine the maximum allowable depth of water d in the tank.

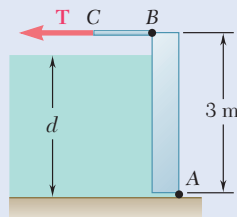


Fig. P5.86 and P5.87

5.87 The 3×4 -m side of an open tank is hinged at its bottom A and is held in place by a thin rod BC . The tank is to be filled with glycerine with a density of 1263 kg/m^3 . Determine the force T in the rod and the reaction at the hinge after the tank is filled to a depth of 2.9 m.

5.88 A 0.5×0.8 -m gate AB is located at the bottom of a tank filled with water. The gate is hinged along its top edge A and rests on a frictionless stop at B . Determine the reactions at A and B when cable BCD is slack.

5.89 A 0.5×0.8 -m gate AB is located at the bottom of a tank filled with water. The gate is hinged along its top edge A and rests on a frictionless stop at B . Determine the minimum tension required in cable BCD to open the gate.

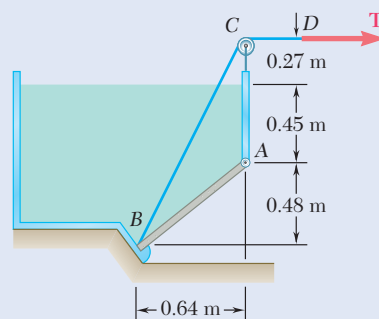


Fig. P5.88 and P5.89

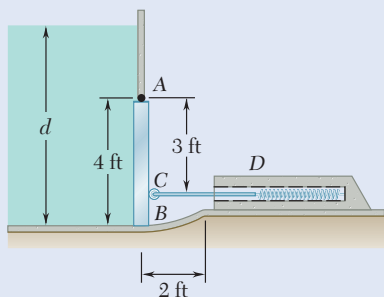


Fig. P5.90

5.90 A 4×2 -ft gate is hinged at A and is held in position by rod CD . End D rests against a spring whose constant is 828 lb/ft. The spring is undeformed when the gate is vertical. Assuming that the force exerted by rod CD on the gate remains horizontal, determine the minimum depth of water d for which the bottom B of the gate will move to the end of the cylindrical portion of the floor.

5.91 Solve Prob. 5.90 if the gate weighs 1000 lb.

5.92 A prismatically shaped gate placed at the end of a freshwater channel is supported by a pin and bracket at A and rests on a frictionless support at B . The pin is located at a distance $h = 0.10$ m below the center of gravity C of the gate. Determine the depth of water d for which the gate will open.

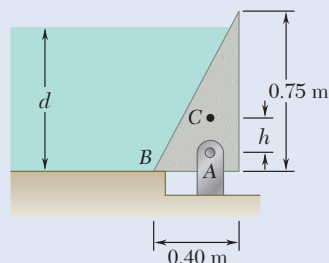


Fig. P5.92 and P5.93

5.93 A prismatically shaped gate placed at the end of a freshwater channel is supported by a pin and bracket at A and rests on a frictionless support at B . The pin is located at a distance h below the center of gravity C of the gate. Determine the distance h if the gate is to open when $d = 0.75$ m.

5.94 A long trough is supported by a continuous hinge along its lower edge and by a series of horizontal cables attached to its upper edge. Determine the tension in each of the cables at a time when the trough is completely full of water.

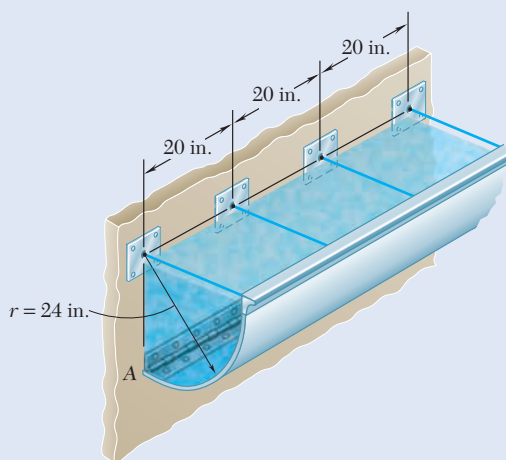


Fig. P5.94

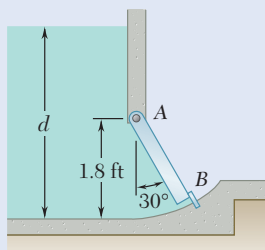


Fig. P5.95

5.95 The square gate AB is held in the position shown by hinges along its top edge A and by a shear pin at B . For a depth of water $d = 3.5$ ft, determine the force exerted on the gate by the shear pin.

5.4 CENTERS OF GRAVITY AND CENTROIDS OF VOLUMES

So far in this chapter, we have dealt with finding centers of gravity and centroids of two-dimensional areas and objects such as flat plates and plane surfaces. However, the same ideas apply to three-dimensional objects as well. The most general situations require the use of multiple integration for analysis, but we can often use symmetry considerations to simplify the calculations. In this section, we show how to do this.

5.4A Three-Dimensional Centers of Gravity and Centroids

For a three-dimensional body, we obtain the center of gravity G by dividing the body into small elements. The weight $\mathbf{W}$ of the body acting at G is equivalent to the system of distributed forces $\Delta\mathbf{W}$ representing the weights of the small elements. Choosing the y axis to be vertical with positive sense upward (Fig. 5.20) and denoting the position vector of G to be $\bar{\mathbf{r}}$, we set $\mathbf{W}$ equal to the sum of the elemental weights $\Delta\mathbf{W}$ and set its moment about O equal to the sum of the moments about O of the elemental weights. Thus,

$$\begin{aligned}\Sigma\mathbf{F}: \quad & -W\mathbf{j} = \Sigma(-\Delta W\mathbf{j}) \\ \Sigma\mathbf{M}_O: \quad & \bar{\mathbf{r}} \times (-W\mathbf{j}) = \Sigma[\mathbf{r} \times (-\Delta W\mathbf{j})]\end{aligned}\tag{5.14}$$

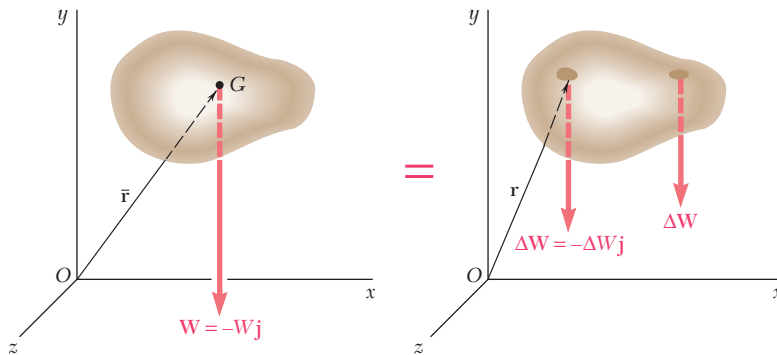


Fig. 5.20 For a three-dimensional body, the weight $\mathbf{W}$ acting through the center of gravity G and its moment about O is equivalent to the system of distributed weights acting on all the elements of the body and the sum of their moments about O .

We can rewrite the last equation in the form

$$\bar{\mathbf{r}}W \times (-\mathbf{j}) = (\Sigma\mathbf{r}\Delta W) \times (-\mathbf{j})\tag{5.15}$$

From these equations, we can see that the weight $\mathbf{W}$ of the body is equivalent to the system of the elemental weights $\Delta\mathbf{W}$ if the following conditions are satisfied:

$$W = \Sigma \Delta W \quad \bar{\mathbf{r}}W = \Sigma\mathbf{r}\Delta W$$



Photo 5.5 To predict the flight characteristics of the modified Boeing 747 when used to transport a space shuttle, engineers had to determine the center of gravity of each craft.

Increasing the number of elements and simultaneously decreasing the size of each element, we obtain in the limit as

**Weight, center of gravity
of a three-dimensional body**

$$W = \int dW \quad \bar{\mathbf{r}} W = \int \mathbf{r} dW \quad (5.16)$$

Note that these relations are independent of the orientation of the body. For example, if the body and the coordinate axes were rotated so that the z axis pointed upward, the unit vector $-\mathbf{j}$ would be replaced by $-\mathbf{k}$ in Eqs. (5.14) and (5.15), but the relations in Eqs. (5.16) would remain unchanged.

Resolving the vectors $\bar{\mathbf{r}}$ and $\mathbf{r}$ into rectangular components, we note that the second of the relations in Eqs. (5.16) is equivalent to the three scalar equations

$$\bar{x} W = \int x dW \quad \bar{y} W = \int y dW \quad \bar{z} W = \int z dW \quad (5.17)$$

or

$$\bar{x} = \frac{\int x dW}{W} \quad \bar{y} = \frac{\int y dW}{W} \quad \bar{z} = \frac{\int z dW}{W} \quad (5.17')$$

If the body is made of a homogeneous material of specific weight γ , we can express the magnitude dW of the weight of an infinitesimal element in terms of the volume dV of the element and express the magnitude W of the total weight in terms of the total volume V . We obtain

$$dW = \gamma dV \quad W = \gamma V$$

Substituting for dW and W in the second of the relations in Eqs. (5.16), we have

$$\bar{\mathbf{r}} V = \int \mathbf{r} dV \quad (5.18)$$

In scalar form, this becomes

**Centroid of a
volume V**

$$\bar{x} V = \int x dV \quad \bar{y} V = \int y dV \quad \bar{z} V = \int z dV \quad (5.19)$$

or

$$\bar{x} = \frac{\int x dV}{V} \quad \bar{y} = \frac{\int y dV}{V} \quad \bar{z} = \frac{\int z dV}{V} \quad (5.19')$$

The center of gravity of a homogeneous body whose coordinates are $\bar{x}, \bar{y}, \bar{z}$ is also known as the **centroid C of the volume V** of the body. If the body is not homogeneous, we cannot use Eqs. (5.19) to determine the center of gravity of the body; however, Eqs. (5.19) still define the centroid of the volume.

The integral $\int x dV$ is known as the **first moment of the volume with respect to the yz plane**. Similarly, the integrals $\int y dV$ and $\int z dV$ define the first moments of the volume with respect to the zx plane and

the xy plane, respectively. You can see from Eqs. (5.19) that if the centroid of a volume is located in a coordinate plane, the first moment of the volume with respect to that plane is zero.

A volume is said to be symmetrical with respect to a given plane if, for every point P of the volume, there exists a point P' of the same volume such that the line PP' is perpendicular to the given plane and is bisected by that plane. We say the plane is a **plane of symmetry** for the given volume. When a volume V possesses a plane of symmetry, the first moment of V with respect to that plane is zero, and the centroid of the volume is located in the plane of symmetry. If a volume possesses two planes of symmetry, the centroid of the volume is located on the line of intersection of the two planes. Finally, if a volume possesses three planes of symmetry that intersect at a well-defined point (i.e., not along a common line), the point of intersection of the three planes coincides with the centroid of the volume. This property enables us to determine immediately the locations of the centroids of spheres, ellipsoids, cubes, rectangular parallelepipeds, etc.

For unsymmetrical volumes or volumes possessing only one or two planes of symmetry, we can determine the location of the centroid by integration (Sec. 5.4C). The centroids of several common volumes are shown in Fig. 5.21. Note that, in general, the centroid of a volume of revolution *does not coincide* with the centroid of its cross section. Thus, the centroid of a hemisphere is different from that of a semicircular area, and the centroid of a cone is different from that of a triangle.

5.4B Composite Bodies

If a body can be divided into several of the common shapes shown in Fig. 5.21, we can determine its center of gravity G by setting the moment about O of its total weight equal to the sum of the moments about O of the weights of the various component parts. Proceeding in this way, we obtain the following equations defining the coordinates $\bar{X}$, $\bar{Y}$, $\bar{Z}$ of the center of gravity G as

Center of gravity of a body with weight W

$$\bar{X}\Sigma W = \Sigma \bar{x}W \quad \bar{Y}\Sigma W = \Sigma \bar{y}W \quad \bar{Z}\Sigma W = \Sigma \bar{z}W \quad (5.20)$$

or

$$\bar{X} = \frac{\Sigma \bar{x}W}{\Sigma W} \quad \bar{Y} = \frac{\Sigma \bar{y}W}{\Sigma W} \quad \bar{Z} = \frac{\Sigma \bar{z}W}{\Sigma W} \quad (5.20')$$

If the body is made of a homogeneous material, its center of gravity coincides with the centroid of its volume, and we obtain

Centroid of a volume V

$$\bar{X}\Sigma V = \Sigma \bar{x}V \quad \bar{Y}\Sigma V = \Sigma \bar{y}V \quad \bar{Z}\Sigma V = \Sigma \bar{z}V \quad (5.21)$$

or

$$\bar{X} = \frac{\Sigma \bar{x}V}{\Sigma V} \quad \bar{Y} = \frac{\Sigma \bar{y}V}{\Sigma V} \quad \bar{Z} = \frac{\Sigma \bar{z}V}{\Sigma V} \quad (5.21')$$

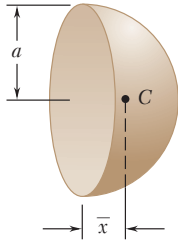
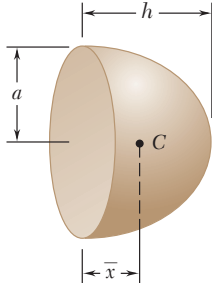
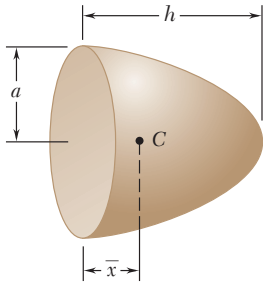
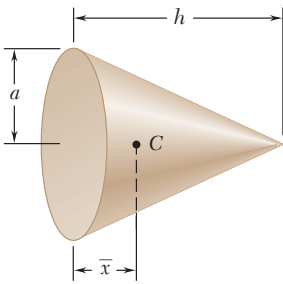
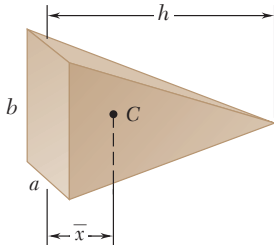
Shape		$\bar{x}$	Volume
Hemisphere		$\frac{3a}{8}$	$\frac{2}{3}\pi a^3$
Semiellipsoid of revolution		$\frac{3h}{8}$	$\frac{2}{3}\pi a^2h$
Paraboloid of revolution		$\frac{h}{3}$	$\frac{1}{2}\pi a^2h$
Cone		$\frac{h}{4}$	$\frac{1}{3}\pi a^2h$
Pyramid		$\frac{h}{4}$	$\frac{1}{3}abh$

Fig. 5.21 Centroids of common shapes and volumes.

5.4C Determination of Centroids of Volumes by Integration

We can determine the centroid of a volume bounded by analytical surfaces by evaluating the integrals given earlier in this section:

$$\bar{x}V = \int x dV \quad \bar{y}V = \int y dV \quad \bar{z}V = \int z dV \quad (5.22)$$

If we choose the element of volume dV to be equal to a small cube with sides dx , dy , and dz , the evaluation of each of these integrals requires a *triple integration*. However, it is possible to determine the coordinates of the centroid of most volumes by *double integration* if we choose dV to be equal to the volume of a thin filament (Fig. 5.22). We then obtain the coordinates of the centroid of the volume by rewriting Eqs. (5.22) as

$$\bar{x}V = \int \bar{x}_{el} dV \quad \bar{y}V = \int \bar{y}_{el} dV \quad \bar{z}V = \int \bar{z}_{el} dV \quad (5.23)$$

Then we substitute the expressions given in Fig. 5.22 for the volume dV and the coordinates $\bar{x}_{el}$, $\bar{y}_{el}$, $\bar{z}_{el}$. By using the equation of the surface to express z in terms of x and y , we reduce the integration to a double integration in x and y .

If the volume under consideration possesses *two planes of symmetry*, its centroid must be located on the line of intersection of the two planes. Choosing the x axis to lie along this line, we have

$$\bar{y} = \bar{z} = 0$$

and the only coordinate to determine is $\bar{x}$. This can be done with a *single integration* by dividing the given volume into thin slabs parallel to the yz plane and expressing dV in terms of x and dx in the equation

$$\bar{x}V = \int \bar{x}_{el} dV \quad (5.24)$$

For a body of revolution, the slabs are circular, and their volume is given in Fig. 5.23.

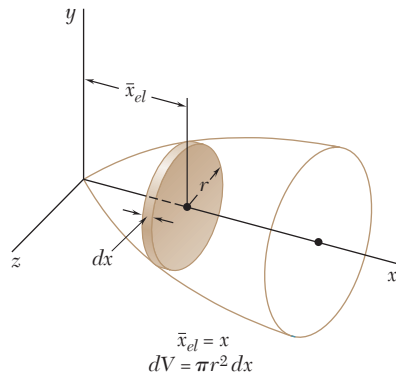


Fig. 5.23 Determining the centroid of a body of revolution.

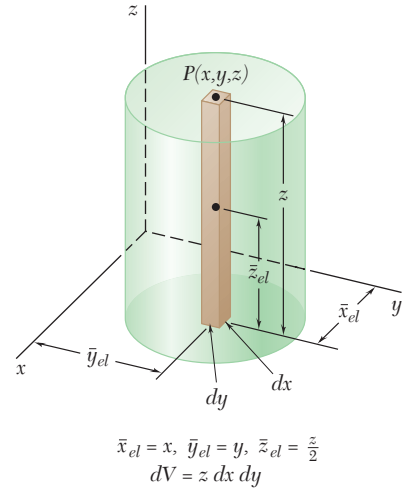
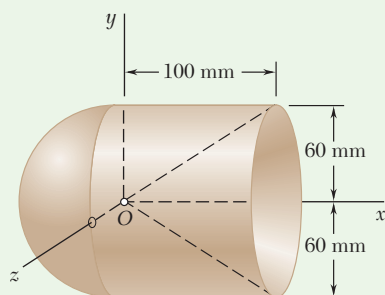


Fig. 5.22 Determining the centroid of a volume by double integration.



Sample Problem 5.11

Determine the location of the center of gravity of the homogeneous body of revolution shown that was obtained by joining a hemisphere and a cylinder and carving out a cone.

STRATEGY: The body is homogeneous, so the center of gravity coincides with the centroid. Since the body was formed from a composite of three simple shapes, you can find the centroid of each shape and combine them using Eq. (5.21).

MODELING: Because of symmetry, the center of gravity lies on the x axis. As shown in Fig. 1, the body is formed by adding a hemisphere to a cylinder and then subtracting a cone. Find the volume and the abscissa of the centroid of each of these components from Fig. 5.21 and enter them in a table (below). Then you can determine the total volume of the body and the first moment of its volume with respect to the yz plane.

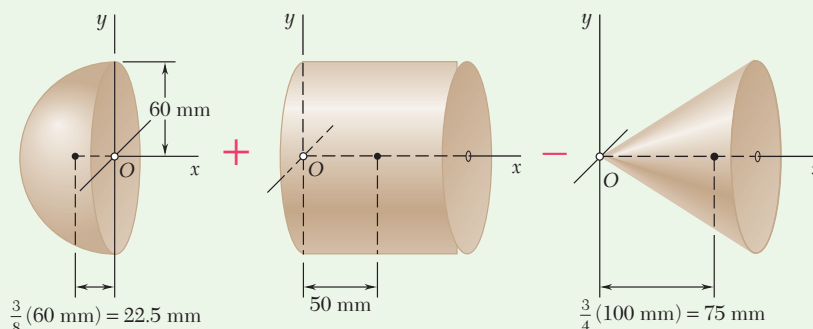


Fig. 1 The given body modeled as the combination of simple geometric shapes.

ANALYSIS: Note that the location of the centroid of the hemisphere is negative because it lies to the left of the origin.

Component	Volume, mm^3	$\bar{x}$, mm	$\bar{x}V$, mm^4
Hemisphere	$\frac{1}{2} \frac{4\pi}{3} (60)^3 = 0.4524 \times 10^6$	-22.5	-10.18×10^6
Cylinder	$\pi(60)^2(100) = 1.1310 \times 10^6$	+50	$+56.55 \times 10^6$
Cone	$-\frac{\pi}{3} (60)^2(100) = -0.3770 \times 10^6$	+75	-28.28×10^6
	$\Sigma V = 1.206 \times 10^6$		$\Sigma \bar{x}V = +18.09 \times 10^6$

Thus,

$$\bar{X} \Sigma V = \Sigma \bar{x}V: \quad \bar{X}(1.206 \times 10^6 \text{ mm}^3) = 18.09 \times 10^6 \text{ mm}^4$$

$$\bar{X} = 15 \text{ mm} \quad \blacktriangleleft$$

REFLECT and THINK: Adding the hemisphere and subtracting the cone have the effect of shifting the centroid of the composite shape to the left of that for the cylinder (50 mm). However, because the first moment of volume

for the cylinder is larger than for the hemisphere and cone combined, you should expect the centroid for the composite to still be in the positive x domain. Thus, as a rough visual check, the result of +15 mm is reasonable.

Sample Problem 5.12

Locate the center of gravity of the steel machine part shown. The diameter of each hole is 1 in.

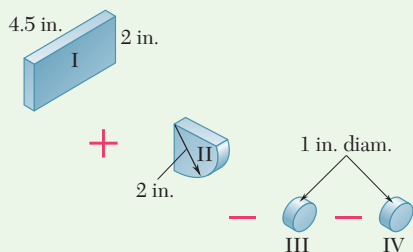
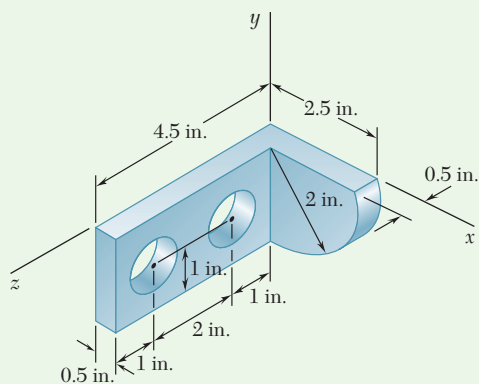


Fig. 1 The given body modeled as the combination of simple geometric shapes.

STRATEGY: This part can be broken down into the sum of two volumes minus two smaller volumes (holes). Find the volume and centroid of each volume and combine them using Eq. (5.21) to find the overall centroid.

MODELING: As shown in Fig. 1, the machine part can be obtained by adding a rectangular parallelepiped (I) to a quarter cylinder (II) and then subtracting two 1-in.-diameter cylinders (III and IV). Determine the volume and the coordinates of the centroid of each component and enter them in a table (below). Using the data in the table, determine the total volume and the moments of the volume with respect to each of the coordinate planes.

ANALYSIS: You can treat each component volume as a planar shape using Fig. 5.8A to find the volumes and centroids, but the right-angle joining of components I and II requires calculations in three dimensions. You may find it helpful to draw more detailed sketches of components with the centroids carefully labeled (Fig. 2).

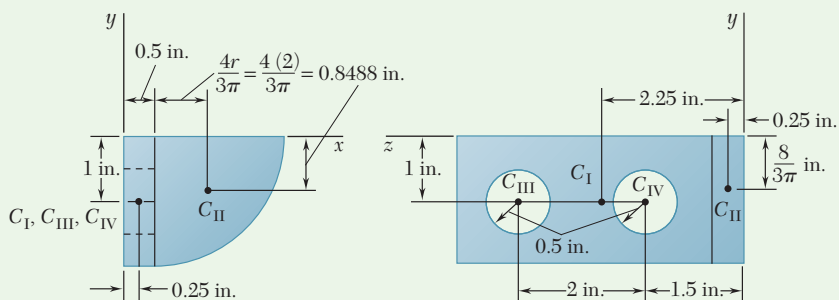


Fig. 2 Centroids of components.

	V, in^3	$\bar{x}, \text{in.}$	$\bar{y}, \text{in.}$	$\bar{z}, \text{in.}$	$\bar{x}V, \text{in}^4$	$\bar{y}V, \text{in}^4$	$\bar{z}V, \text{in}^4$
I	$(4.5)(2)(0.5) = 4.5$	0.25	-1	2.25	1.125	-4.5	10.125
II	$\frac{1}{4}\pi(2)^2(0.5) = 1.571$	1.3488	-0.8488	0.25	2.119	-1.333	0.393
III	$-\pi(0.5)^2(0.5) = -0.3927$	0.25	-1	3.5	-0.098	0.393	-1.374
IV	$-\pi(0.5)^2(0.5) = -0.3927$	0.25	-1	1.5	-0.098	0.393	-0.589
	$\Sigma V = 5.286$				$\Sigma \bar{x}V = 3.048$	$\Sigma \bar{y}V = -5.047$	$\Sigma \bar{z}V = 8.555$

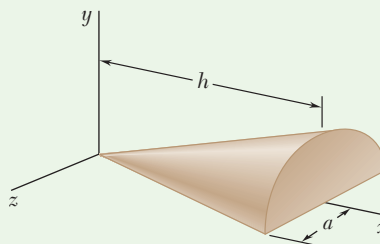
Thus,

$$\begin{aligned}\bar{X}\Sigma V &= \Sigma \bar{x}V: & \bar{X}(5.286 \text{ in}^3) &= 3.048 \text{ in}^4 & \bar{X} &= 0.577 \text{ in.} \quad \blacktriangleleft \\ \bar{Y}\Sigma V &= \Sigma \bar{y}V: & \bar{Y}(5.286 \text{ in}^3) &= -5.047 \text{ in}^4 & \bar{Y} &= -0.955 \text{ in.} \quad \blacktriangleleft \\ \bar{Z}\Sigma V &= \Sigma \bar{z}V: & \bar{Z}(5.286 \text{ in}^3) &= 8.555 \text{ in}^4 & \bar{Z} &= 1.618 \text{ in.} \quad \blacktriangleleft\end{aligned}$$

REFLECT and THINK: By inspection, you should expect $\bar{X}$ and $\bar{Z}$ to be considerably less than $(1/2)(2.5 \text{ in.})$ and $(1/2)(4.5 \text{ in.})$, respectively, and $\bar{Y}$ to be slightly less in magnitude than $(1/2)(2 \text{ in.})$. Thus, as a rough visual check, the results obtained are as expected.

Sample Problem 5.13

Determine the location of the centroid of the half right circular cone shown.



STRATEGY: This is not one of the shapes in Fig. 5.21, so you have to determine the centroid by using integration.

MODELING: Since the xy plane is a plane of symmetry, the centroid lies in this plane, and $\bar{z} = 0$. Choose a slab of thickness dx as a differential element. The volume of this element is

$$dV = \frac{1}{2}\pi r^2 dx$$

Obtain the coordinates $\bar{x}_{el}$ and $\bar{y}_{el}$ of the centroid of the element from Fig. 5.8 (semicircular area):

$$\bar{x}_{el} = x \quad \bar{y}_{el} = \frac{4r}{3\pi}$$

Noting that r is proportional to x , use similar triangles (Fig. 1) to write

$$\frac{r}{x} = \frac{a}{h} \quad r = \frac{a}{h}x$$

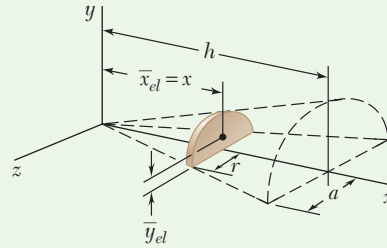


Fig. 1 Geometry of the differential element.

ANALYSIS: The volume of the body is

$$V = \int dV = \int_0^h \frac{1}{2} \pi r^2 dx = \int_0^h \frac{1}{2} \pi \left(\frac{a}{h} x \right)^2 dx = \frac{\pi a^2 h}{6}$$

The moment of the differential element with respect to the yz plane is $\bar{x}_{el} dV$; the total moment of the body with respect to this plane is

$$\int \bar{x}_{el} dV = \int_0^h x \left(\frac{1}{2} \pi r^2 \right) dx = \int_0^h x \left(\frac{1}{2} \pi \right) \left(\frac{a}{h} x \right)^2 dx = \frac{\pi a^2 h^2}{8}$$

Thus,

$$\bar{x}V = \int \bar{x}_{el} dV \quad \bar{x} \frac{\pi a^2 h}{6} = \frac{\pi a^2 h^2}{8} \quad \bar{x} = \frac{3}{4}h \quad \blacktriangleleft$$

Similarly, the moment of the differential element with respect to the zx plane is $\bar{y}_{el} dV$; the total moment is

$$\int \bar{y}_{el} dV = \int_0^h \frac{4r}{3\pi} \left(\frac{1}{2} \pi r^2 \right) dx = \frac{2}{3} \int_0^h \left(\frac{a}{h} x \right)^3 dx = \frac{a^3 h}{6}$$

Thus,

$$\bar{y}V = \int \bar{y}_{el} dV \quad \bar{y} \frac{\pi a^2 h}{6} = \frac{a^3 h}{6} \quad \bar{y} = \frac{a}{\pi} \quad \blacktriangleleft$$

REFLECT and THINK: Since a full right circular cone is a body of revolution, its $\bar{x}$ is unchanged for any portion of the cone bounded by planes intersecting along the x axis. The same centroid location in the x direction was therefore obtained for the half cone that Fig. 5.21 shows for the full cone. Similarly, the same $\bar{x}$ result would be obtained for a quarter cone.

SOLVING PROBLEMS ON YOUR OWN

In the problems for this section, you will be asked to locate the centers of gravity of three-dimensional bodies or the centroids of their volumes. All of the techniques we previously discussed for two-dimensional bodies—using symmetry, dividing the body into common shapes, choosing the most efficient differential element, etc.—also may be applied to the general three-dimensional case.

1. Locating the centers of gravity of composite bodies. In general, you must use Eqs. (5.20):

$$\bar{X}\Sigma W = \Sigma \bar{x}W \quad \bar{Y}\Sigma W = \Sigma \bar{y}W \quad \bar{Z}\Sigma W = \Sigma \bar{z}W \quad (5.20)$$

However, for the case of a *homogeneous body*, the center of gravity of the body coincides with the *centroid of its volume*. Therefore, for this special case, you can also use Eqs. (5.21) to locate the center of gravity of the body:

$$\bar{X}\Sigma V = \Sigma \bar{x}V \quad \bar{Y}\Sigma V = \Sigma \bar{y}V \quad \bar{Z}\Sigma V = \Sigma \bar{z}V \quad (5.21)$$

You should realize that these equations are simply an extension of the equations used for the two-dimensional problems considered earlier in the chapter. As the solutions of Sample Probs. 5.11 and 5.12 illustrate, the methods of solution for two- and three-dimensional problems are identical. Thus, we once again strongly encourage you to construct appropriate diagrams and tables when analyzing composite bodies. Also, as you study Sample Prob. 5.12, observe how we obtained the x and y coordinates of the centroid of the quarter cylinder using the equations for the centroid of a quarter circle.

Two *special cases* of interest occur when the given body consists of either uniform wires or uniform plates made of the same material.

a. For a body made of *several wire elements* of the *same uniform cross section*, the cross-sectional area A of the wire elements factors out of Eqs. (5.21) when V is replaced with the product AL , where L is the length of a given element. Equations (5.21) thus reduce in this case to

$$\bar{X}\Sigma L = \Sigma \bar{x}L \quad \bar{Y}\Sigma L = \Sigma \bar{y}L \quad \bar{Z}\Sigma L = \Sigma \bar{z}L$$

b. For a body made of *several plates* of the *same uniform thickness*, the thickness t of the plates factors out of Eqs. (5.21) when V is replaced with the product tA , where A is the area of a given plate. Equations (5.21) thus reduce in this case to

$$\bar{X}\Sigma A = \Sigma \bar{x}A \quad \bar{Y}\Sigma A = \Sigma \bar{y}A \quad \bar{Z}\Sigma A = \Sigma \bar{z}A$$

2. Locating the centroids of volumes by direct integration. As explained in Sec. 5.4C, you can simplify evaluating the integrals of Eqs. (5.22) by choosing either a thin filament (Fig. 5.22) or a thin slab (Fig. 5.23) for the element of volume dV . Thus, you should begin your solution by identifying, if possible, the dV that produces the single or double integrals that are easiest to compute. For bodies of revolution, this may be a thin slab (as in Sample Prob. 5.13) or a thin cylindrical shell. However, it is important to remember that the relationship that you establish among the variables (like the relationship between r and x in Sample Prob. 5.13) directly affects the complexity of the integrals you have to compute. Finally, we again remind you that $\bar{x}_{el}$, $\bar{y}_{el}$, and $\bar{z}_{el}$ in Eqs. (5.23) are the coordinates of the centroid of dV .

Problems

- 5.96** Consider the composite body shown. Determine (a) the value of $\bar{x}$ when $h = L/2$, (b) the ratio h/L for which $\bar{x} = L$.

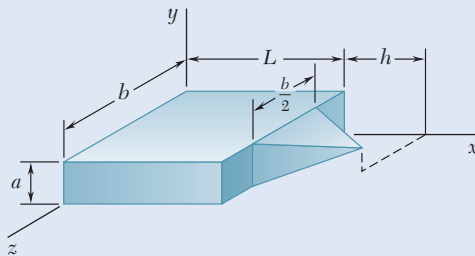


Fig. P5.96

- 5.97** Determine the location of the centroid of the composite body shown when (a) $h = 2b$, (b) $h = 2.5b$.

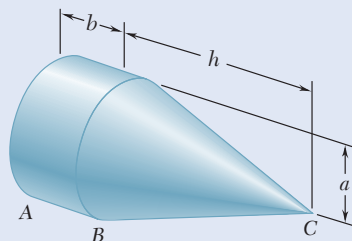


Fig. P5.97

- 5.98** The composite body shown is formed by removing a semiellipsoid of revolution of semimajor axis h and semiminor axis $a/2$ from a hemisphere of radius a . Determine (a) the y coordinate of the centroid when $h = a/2$, (b) the ratio h/a for which $\bar{y} = -0.4a$.

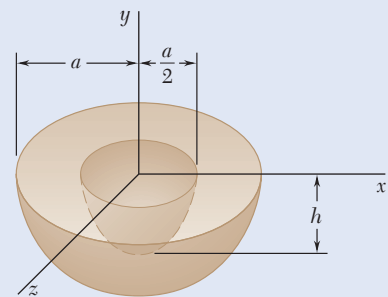


Fig. P5.98

- 5.99** Locate the centroid of the frustum of a right circular cone when $r_1 = 40$ mm, $r_2 = 50$ mm, and $h = 60$ mm.

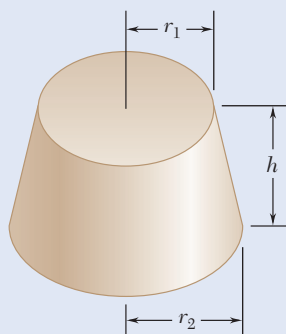


Fig. P5.99

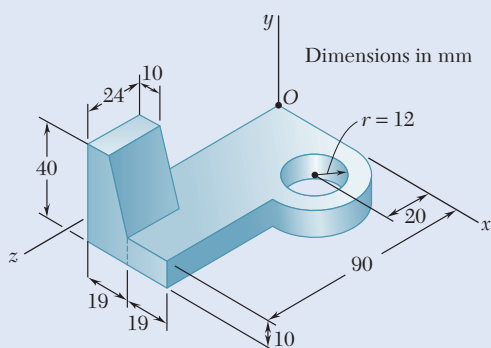


Fig. P5.100 and P5.101

5.100 For the machine element shown, locate the x coordinate of the center of gravity.

5.101 For the machine element shown, locate the z coordinate of the center of gravity.

5.102 For the machine element shown, locate the y coordinate of the center of gravity.

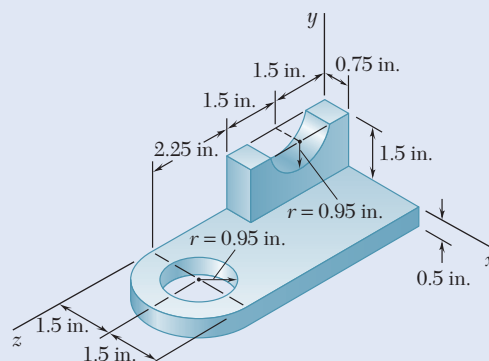


Fig. P5.102 and P5.103

5.103 For the machine element shown, locate the z coordinate of the center of gravity.

5.104 For the machine element shown, locate the y coordinate of the center of gravity.

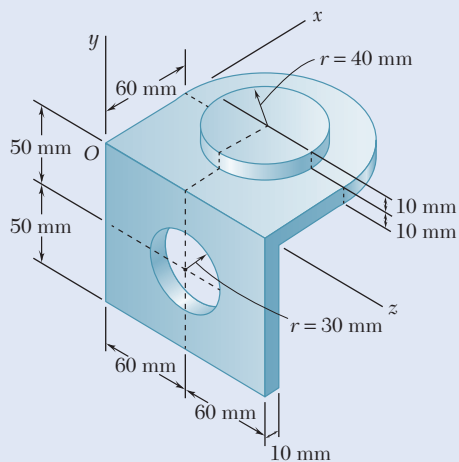


Fig. P5.104 and P5.105

5.105 For the machine element shown, locate the x coordinate of the center of gravity.

5.106 and 5.107 Locate the center of gravity of the sheet-metal form shown.

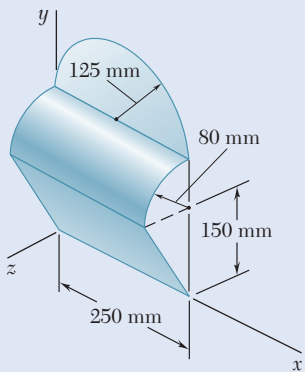


Fig. P5.106

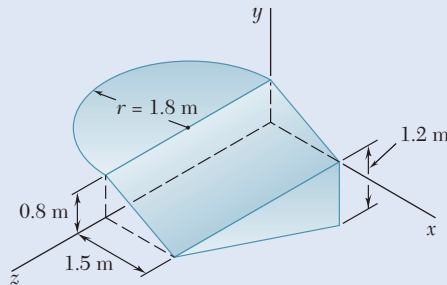


Fig. P5.107

5.108 A corner reflector for tracking by radar has two sides in the shape of a quarter circle with a radius of 15 in. and one side in the shape of a triangle. Locate the center of gravity of the reflector, knowing that it is made of sheet metal with a uniform thickness.

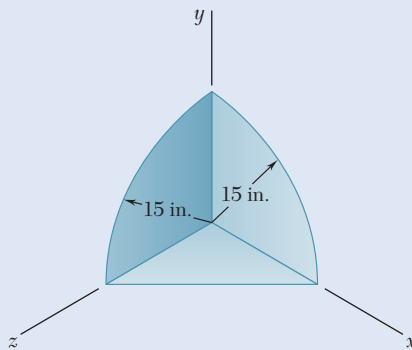


Fig. P5.108

5.109 A wastebasket, designed to fit in the corner of a room, is 16 in. high and has a base in the shape of a quarter circle with a radius of 10 in. Locate the center of gravity of the wastebasket, knowing that it is made of sheet metal with a uniform thickness.

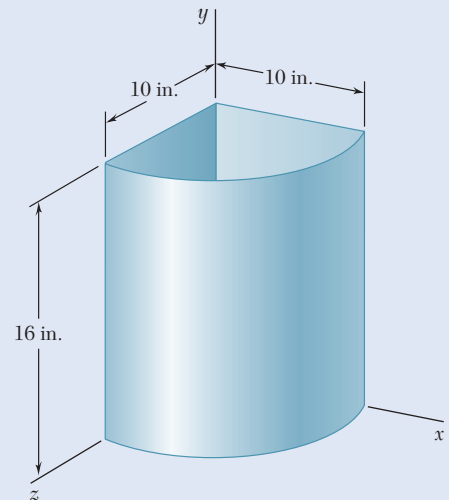


Fig. P5.109

5.110 An elbow for the duct of a ventilating system is made of sheet metal with a uniform thickness. Locate the center of gravity of the elbow.

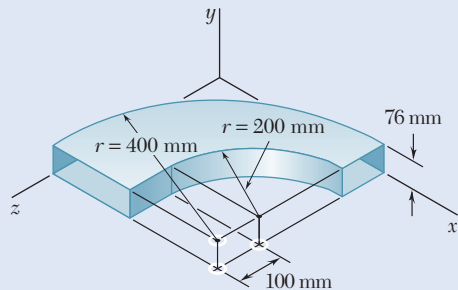


Fig. P5.110

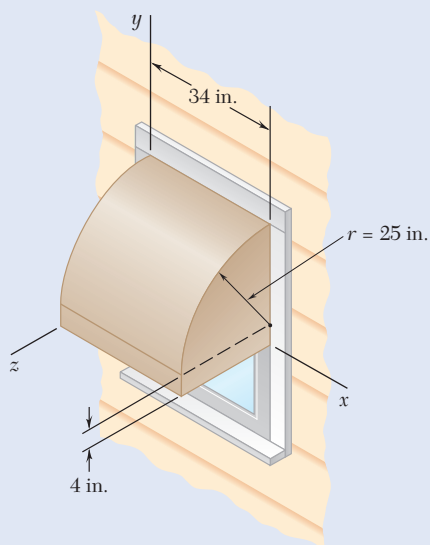


Fig. P5.111

5.111 A window awning is fabricated from sheet metal with a uniform thickness. Locate the center of gravity of the awning.

5.112 A mounting bracket for electronic components is formed from sheet metal with a uniform thickness. Locate the center of gravity of the bracket.

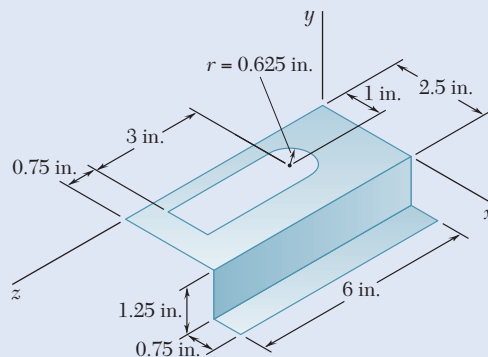


Fig. P5.112

5.113 A thin sheet of plastic with a uniform thickness is bent to form a desk organizer. Locate the center of gravity of the organizer.

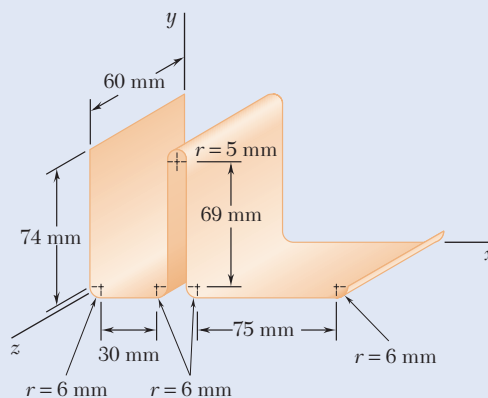


Fig. P5.113

5.114 A thin steel wire with a uniform cross section is bent into the shape shown. Locate its center of gravity.

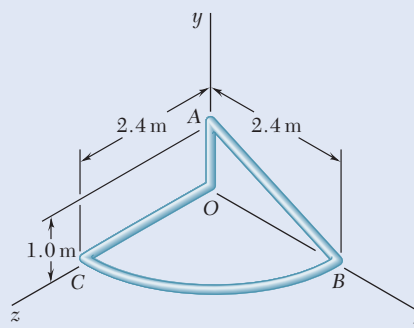


Fig. P5.114

- 5.115** The frame of a greenhouse is constructed from uniform aluminum channels. Locate the center of gravity of the portion of the frame shown.

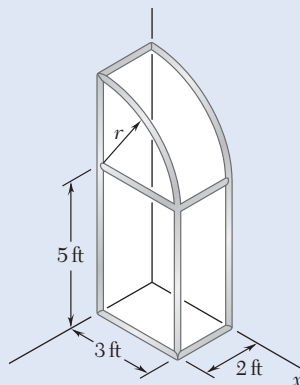


Fig. P5.115

- 5.116 and 5.117** Locate the center of gravity of the figure shown, knowing that it is made of thin brass rods with a uniform diameter.

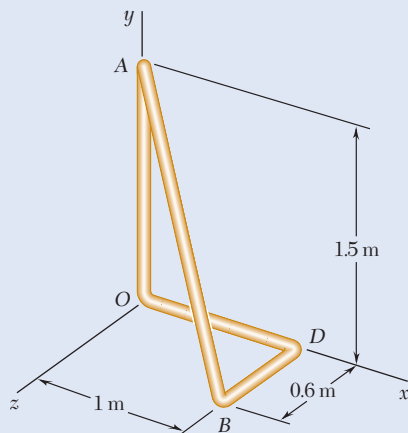


Fig. P5.116

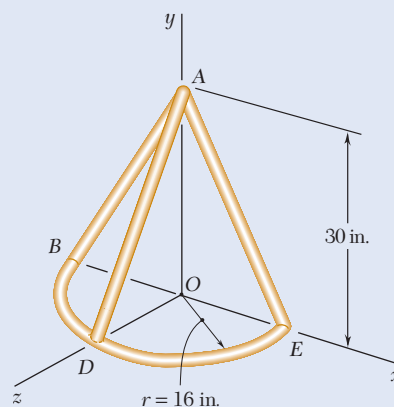


Fig. P5.117

- 5.118** A scratch awl has a plastic handle and a steel blade and shank. Knowing that the density of plastic is 1030 kg/m^3 and of steel is 7860 kg/m^3 , locate the center of gravity of the awl.

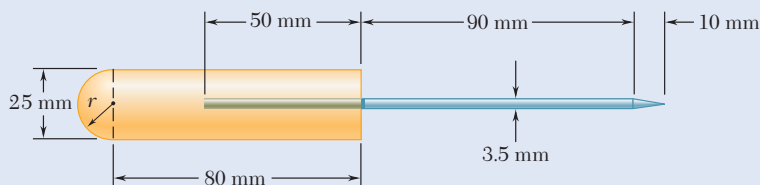


Fig. P5.118

- 5.119** A bronze bushing is mounted inside a steel sleeve. Knowing that the specific weight of bronze is 0.318 lb/in^3 and of steel is 0.284 lb/in^3 , determine the location of the center of gravity of the assembly.

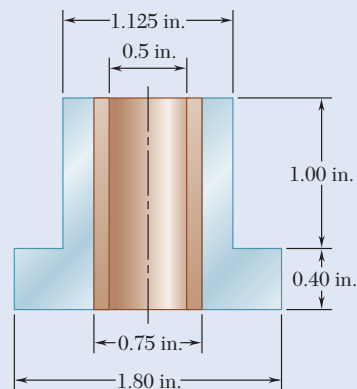


Fig. P5.119

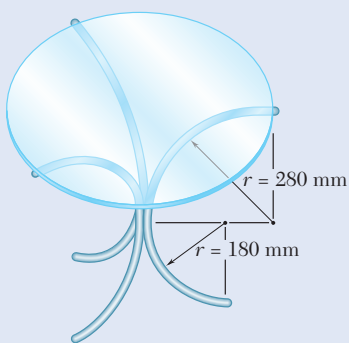


Fig. P5.121

- 5.120** A brass collar with a length of 2.5 in. is mounted on an aluminum rod with a length of 4 in. Locate the center of gravity of the composite body. (Specific weights: brass = 0.306 lb/in³, aluminum = 0.101 lb/in³.)

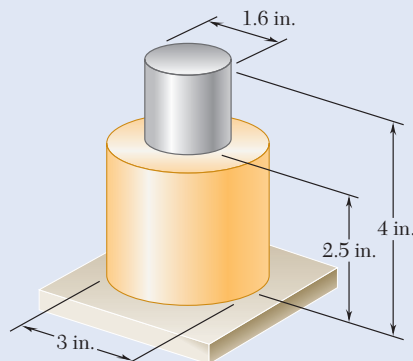


Fig. P5.120

- 5.121** The three legs of a small glass-topped table are equally spaced and are made of steel tubing that has an outside diameter of 24 mm and a cross-sectional area of 150 mm². The diameter and the thickness of the table top are 600 mm and 10 mm, respectively. Knowing that the density of steel is 7860 kg/m³ and of glass is 2190 kg/m³, locate the center of gravity of the table.

- 5.122 through 5.124** Determine by direct integration the values of $\bar{x}$ for the two volumes obtained by passing a vertical cutting plane through the given shape of Fig. 5.21. The cutting plane is parallel to the base of the given shape and divides the shape into two volumes of equal height.

5.122 A hemisphere

5.123 A semiellipsoid of revolution

5.124 A paraboloid of revolution.

- 5.125 and 5.126** Locate the centroid of the volume obtained by rotating the shaded area about the x axis.

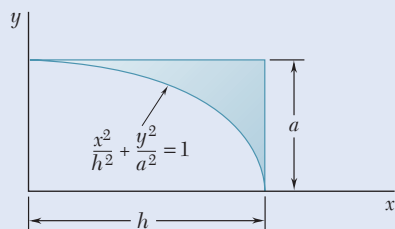


Fig. P5.127

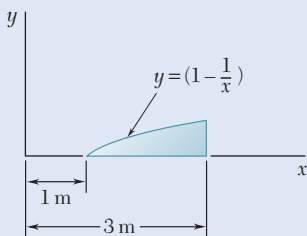


Fig. P5.125

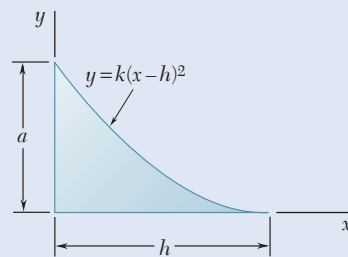


Fig. P5.126

- 5.127** Locate the centroid of the volume obtained by rotating the shaded area about the line $x = h$.

- *5.128** Locate the centroid of the volume generated by revolving the portion of the sine curve shown about the x axis.

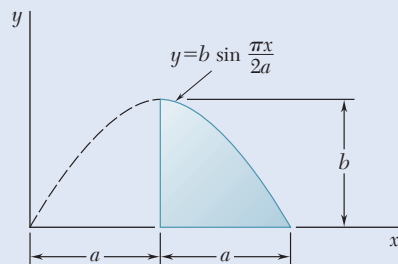


Fig. P5.128 and P5.129

- *5.129** Locate the centroid of the volume generated by revolving the portion of the sine curve shown about the y axis. (*Hint:* Use a thin cylindrical shell of radius r and thickness dr as the element of volume.)

- *5.130** Show that for a regular pyramid of height h and n sides ($n = 3, 4, \dots$) the centroid of the volume of the pyramid is located at a distance $h/4$ above the base.

- 5.131** Determine by direct integration the location of the centroid of one-half of a thin, uniform hemispherical shell of radius R .

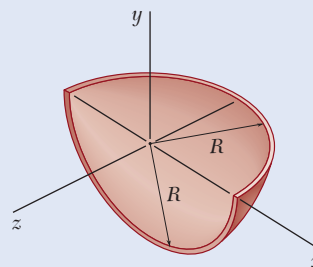


Fig. P5.131

- 5.132** The sides and the base of a punch bowl are of uniform thickness t . If $t \ll R$ and $R = 250$ mm, determine the location of the center of gravity of (a) the bowl, (b) the punch.

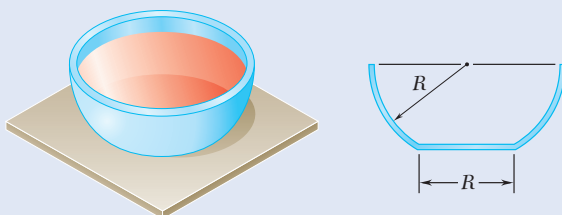


Fig. P5.132

- 5.133** Locate the centroid of the section shown, which was cut from a thin circular pipe by two oblique planes.

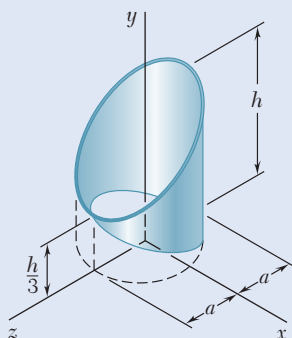


Fig. P5.133

- *5.134** Locate the centroid of the section shown, which was cut from an elliptical cylinder by an oblique plane.

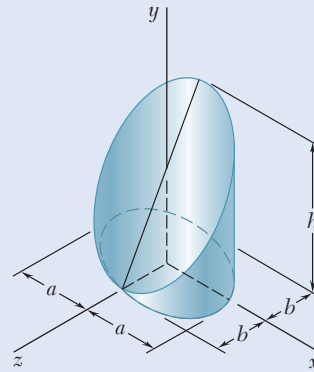


Fig. P5.134

- 5.135** Determine by direct integration the location of the centroid of the volume between the xz plane and the portion shown of the surface $y = 16h(ax - x^2)(bz - z^2)/a^2b^2$.

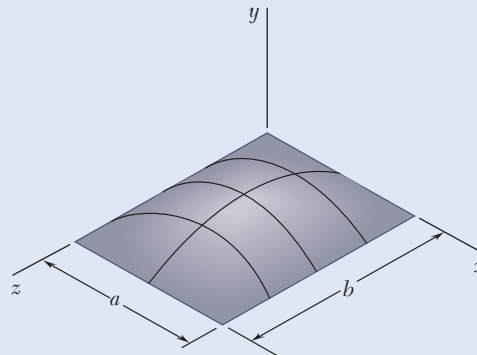


Fig. P5.135

- 5.136** After grading a lot, a builder places four stakes to designate the corners of the slab for a house. To provide a firm, level base for the slab, the builder places a minimum of 3 in. of gravel beneath the slab. Determine the volume of gravel needed and the x coordinate of the centroid of the volume of the gravel. (*Hint:* The bottom surface of the gravel is an oblique plane, which can be represented by the equation $y = a + bx + cz$.)

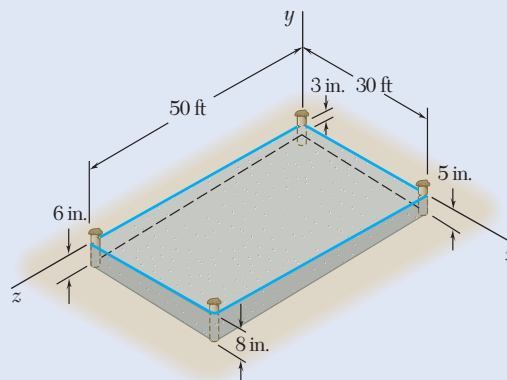


Fig. P5.136

Review and Summary

This chapter was devoted chiefly to determining the **center of gravity** of a rigid body, i.e., to determining the point G where we can apply a single force $\mathbf{W}$ —the *weight* of the body—to represent the effect of Earth's attraction on the body.

Center of Gravity of a Two-Dimensional Body

In the first part of this chapter, we considered *two-dimensional bodies*, such as flat plates and wires contained in the xy plane. By adding force components in the vertical z direction and moments about the horizontal y and x axes [Sec. 5.1A], we derived the relations

$$W = \int dW \quad \bar{x}W = \int x dW \quad \bar{y}W = \int y dW \quad (5.2)$$

These equations define the weight of the body and the coordinates $\bar{x}$ and $\bar{y}$ of its center of gravity.

Centroid of an Area or Line

In the case of a *homogeneous flat* plate of uniform thickness [Sec. 5.1B], the center of gravity G of the plate coincides with the **centroid** C of the area A of the plate. The coordinates are defined by the relations

$$\bar{x}A = \int x dA \quad \bar{y}A = \int y dA \quad (5.3)$$

Similarly, determining the center of gravity of a *homogeneous wire of uniform cross section* contained in a plane reduces to determining the **centroid** C of the line L representing the wire; we have

$$\bar{x}L = \int x dL \quad \bar{y}L = \int y dL \quad (5.4)$$

First Moments

The integrals in Eqs. (5.3) are referred to as the **first moments** of the area A with respect to the y and x axes and are denoted by Q_y and Q_x , respectively [Sec. 5.1C]. We have

$$Q_y = \bar{x}A \quad Q_x = \bar{y}A \quad (5.6)$$

The first moments of a line can be defined in a similar way.

Properties of Symmetry

Determining the centroid C of an area or line is simplified when the area or line possesses certain properties of symmetry. If the area or line is symmetric with respect to an axis, its centroid C lies on that axis; if it is symmetric with respect to two axes, C is located at the intersection of the two axes; if it is symmetric with respect to a center O , C coincides with O .

Center of Gravity of a Composite Body

The areas and the centroids of various common shapes are tabulated in Fig. 5.8. When a flat plate can be divided into several of these shapes, the coordinates $\bar{X}$ and $\bar{Y}$ of its center of gravity G can be determined from the coordinates $\bar{x}_1, \bar{x}_2, \dots$ and $\bar{y}_1, \bar{y}_2, \dots$ of the centers of gravity $G_1, G_2, \dots$ of the various parts [Sec. 5.1D]. Equating moments about the y and x axes, respectively (Fig. 5.24), we have

$$\bar{X}\Sigma W = \Sigma \bar{x}W \quad \bar{Y}\Sigma W = \Sigma \bar{y}W \quad (5.7)$$

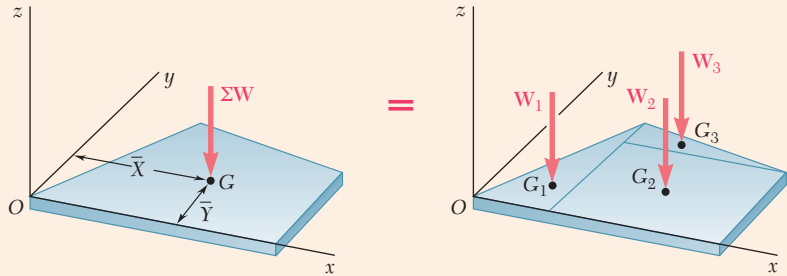


Fig. 5.24

If the plate is homogeneous and of uniform thickness, its center of gravity coincides with the centroid C of the area of the plate, and Eqs. (5.7) reduce to

$$Q_y = \bar{X}\Sigma A = \Sigma \bar{x}A \quad Q_x = \bar{Y}\Sigma A = \Sigma \bar{y}A \quad (5.8)$$

These equations yield the first moments of the composite area, or they can be solved for the coordinates $\bar{X}$ and $\bar{Y}$ of its centroid [Sample Prob. 5.1]. Determining the center of gravity of a composite wire is carried out in a similar fashion [Sample Prob. 5.2].

Determining a Centroid by Integration

When an area is bounded by analytical curves, you can determine the coordinates of its centroid by *integration* [Sec. 5.2A]. This can be done by evaluating either the double integrals in Eqs. (5.3) or a single integral that uses one of the thin rectangular or pie-shaped elements of area shown in Fig. 5.12. Denoting by $\bar{x}_{el}$ and $\bar{y}_{el}$ the coordinates of the centroid of the element dA , we have

$$Q_y = \bar{x}A = \int \bar{x}_{el} dA \quad Q_x = \bar{y}A = \int \bar{y}_{el} dA \quad (5.9)$$

It is advantageous to use the same element of area to compute both of the first moments Q_y and Q_x ; we can also use the same element to determine the area A [Sample Prob. 5.4].

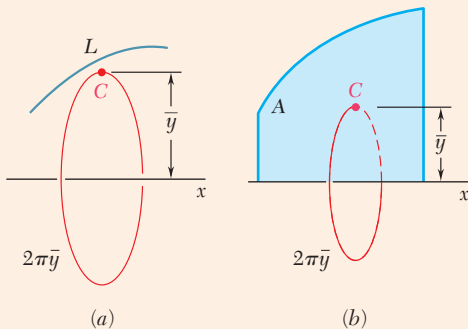


Fig. 5.25

Theorems of Pappus–Guldinus

The **theorems of Pappus–Guldinus** relate the area of a surface of revolution or the volume of a body of revolution to the centroid of the generating curve or area [Sec. 5.2B]. The area A of the surface generated by rotating a curve of length L about a fixed axis (Fig. 5.25a) is

$$A = 2\pi\bar{y}L \quad (5.10)$$

where $\bar{y}$ represents the distance from the centroid C of the curve to the fixed axis. Similarly, the volume V of the body generated by rotating an area A

about a fixed axis (Fig. 5.25b) is

$$V = 2\pi\bar{y}A \quad (5.11)$$

where $\bar{y}$ represents the distance from the centroid C of the area to the fixed axis.

Distributed Loads

The concept of centroid of an area also can be used to solve problems other than those dealing with the weight of flat plates. For example, to determine the reactions at the supports of a beam [Sec. 5.3A], we can replace a **distributed load** w by a concentrated load W equal in magnitude to the area A under the load curve and passing through the centroid C of that area (Fig. 5.26). We can use this same approach to determine the resultant of the hydrostatic forces exerted on a **rectangular plate submerged in a liquid** [Sec. 5.3B].

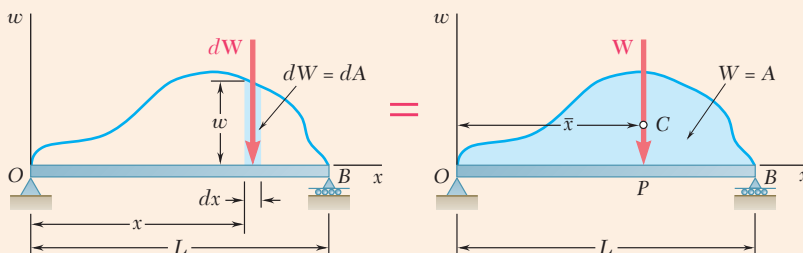


Fig. 5.26

Center of Gravity of a Three-Dimensional Body

The last part of this chapter was devoted to determining the center of gravity G of a three-dimensional body. We defined the coordinates $\bar{x}$, $\bar{y}$, $\bar{z}$ of G by the relations

$$\bar{x}W = \int x dW \quad \bar{y}W = \int y dW \quad \bar{z}W = \int z dW \quad (5.17)$$

Centroid of a Volume

In the case of a homogeneous body, the center of gravity G coincides with the centroid C of the volume V of the body. The coordinates of C are defined by the relations

$$\bar{x}V = \int x dV \quad \bar{y}V = \int y dV \quad \bar{z}V = \int z dV \quad (5.19)$$

If the volume possesses a *plane of symmetry*, its centroid C lies in that plane; if it possesses two planes of symmetry, C is located on the line of intersection of the two planes; if it possesses three planes of symmetry that intersect at only one point, C coincides with that point [Sec. 5.4A].

Center of Gravity of a Composite Body

The volumes and centroids of various common three-dimensional shapes are tabulated in Fig. 5.21. When a body can be divided into several of these shapes, we can determine the coordinates $\bar{X}$, $\bar{Y}$, $\bar{Z}$ of its center of gravity G from the corresponding coordinates of the centers of gravity of its various parts [Sec. 5.4B]. We have

$$\bar{X}\Sigma W = \Sigma \bar{x}W \quad \bar{Y}\Sigma W = \Sigma \bar{y}W \quad \bar{Z}\Sigma W = \Sigma \bar{z}W \quad (5.20)$$

If the body is made of a homogeneous material, its center of gravity coincides with the centroid C of its volume, and we have [Sample Probs. 5.11 and 5.12]

$$\bar{X}\Sigma V = \Sigma \bar{x}V \quad \bar{Y}\Sigma V = \Sigma \bar{y}V \quad \bar{Z}\Sigma V = \Sigma \bar{z}V \quad (5.21)$$

Determining a Centroid by Integration

When a volume is bounded by analytical surfaces, we can find the coordinates of its centroid by *integration* [Sec. 5.4C]. To avoid the computation of triple integrals in Eqs. (5.19), we can use elements of volume in the shape of thin filaments, as shown in Fig. 5.27. Denoting the coordinates of the centroid of the element dV as $\bar{x}_{el}$, $\bar{y}_{el}$, $\bar{z}_{el}$, we rewrite Eqs. (5.19) as

$$\bar{x}V = \int \bar{x}_{el} dV \quad \bar{y}V = \int \bar{y}_{el} dV \quad \bar{z}V = \int \bar{z}_{el} dV \quad (5.23)$$

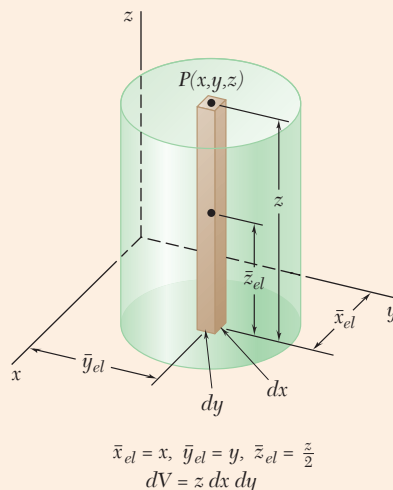


Fig. 5.27

that involve only double integrals. If the volume possesses *two planes of symmetry*, its centroid C is located on their line of intersection. Choosing the x axis to lie along that line and dividing the volume into thin slabs parallel to the yz plane, we can determine C from the relation

$$\bar{x}V = \int \bar{x}_{el} dV \quad (5.24)$$

with a *single integration* [Sample Prob. 5.13]. For a body of revolution, these slabs are circular and their volume is given in Fig. 5.28.

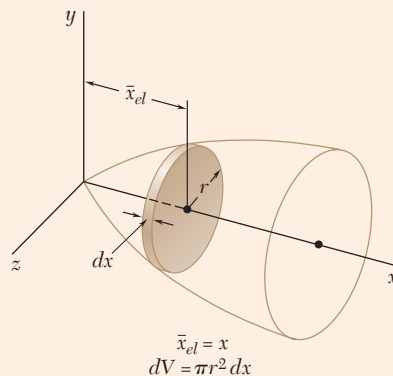


Fig. 5.28

Review Problems

5.137 and 5.138 Locate the centroid of the plane area shown.

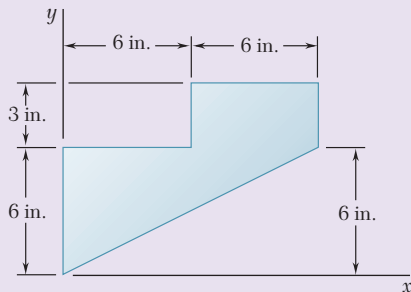


Fig. P5.137

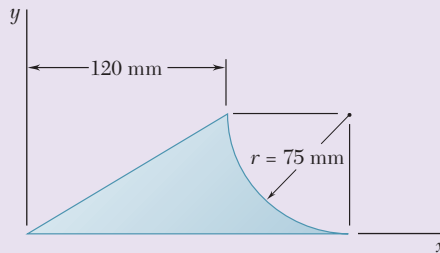


Fig. P5.138

5.139 A uniform circular rod with a weight of 8 lb and radius of 10 in. is attached to a pin at C and to the cable AB . Determine (a) the tension in the cable, (b) the reaction at C .

5.140 Determine by direct integration the centroid of the area shown. Express your answer in terms of a and h .

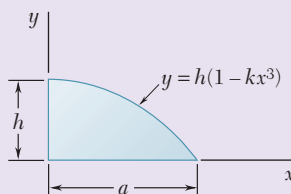


Fig. P5.140

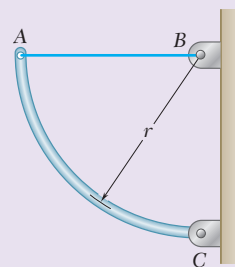


Fig. P5.139

5.141 Determine by direct integration the centroid of the area shown.

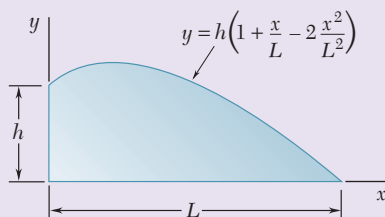


Fig. P5.141

5.142 The escutcheon (a decorative plate placed on a pipe where the pipe exits from a wall) shown is cast from brass. Knowing that the density of brass is 8470 kg/m^3 , determine the mass of the escutcheon.

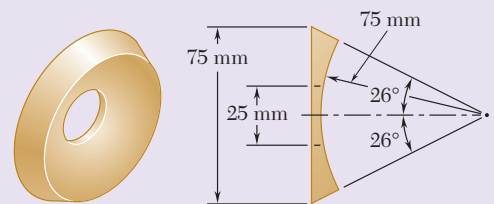


Fig. P5.142

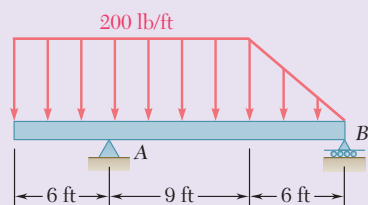


Fig. P5.143

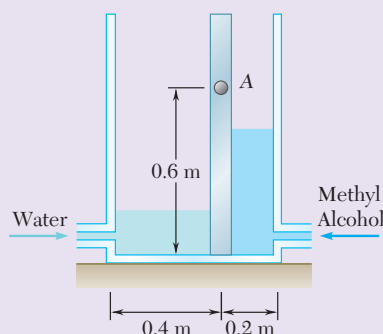


Fig. P5.145

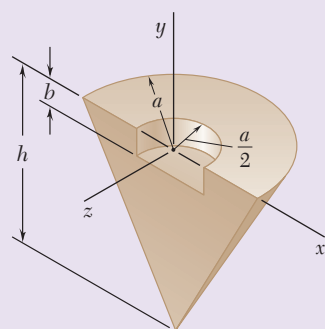


Fig. P5.146

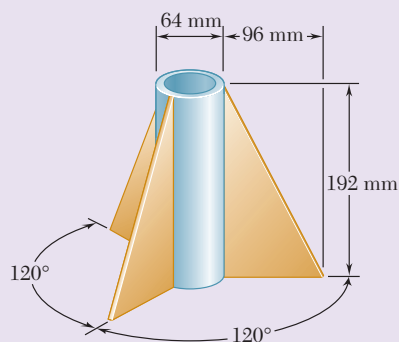


Fig. P5.148

5.143 Determine the reactions at the beam supports for the given loading.

5.144 A beam is subjected to a linearly distributed downward load and rests on two wide supports BC and DE that exert uniformly distributed upward loads as shown. Determine the values of w_{BC} and w_{DE} corresponding to equilibrium when $w_A = 600 \text{ N/m}$.

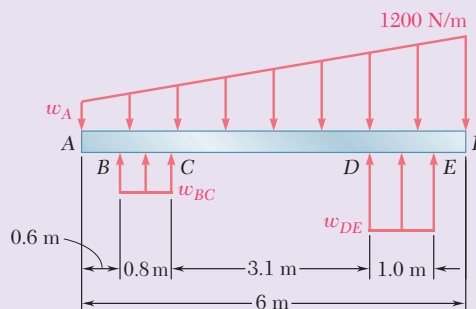


Fig. P5.144

5.145 A tank is divided into two sections by a $1 \times 1\text{-m}$ square gate that is hinged at A . A couple with a magnitude of $490 \text{ N}\cdot\text{m}$ is required for the gate to rotate. If one side of the tank is filled with water at the rate of $0.1 \text{ m}^3/\text{min}$ and the other side is filled simultaneously with methyl alcohol (density $\rho_{ma} = 789 \text{ kg/m}^3$) at the rate of $0.2 \text{ m}^3/\text{min}$, determine at what time and in which direction the gate will rotate.

5.146 Determine the y coordinate of the centroid of the body shown.

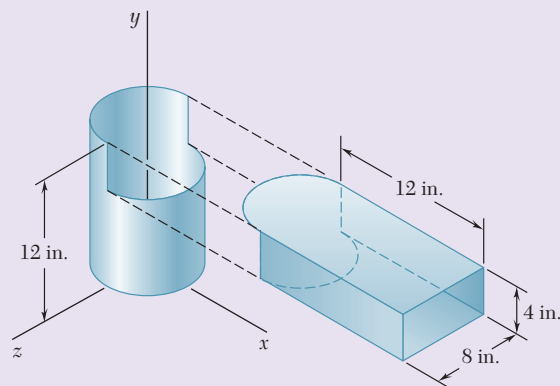


Fig. P5.147

5.147 An 8-in.-diameter cylindrical duct and a $4 \times 8\text{-in.}$ rectangular duct are to be joined as indicated. Knowing that the ducts were fabricated from the same sheet metal, which is of uniform thickness, locate the center of gravity of the assembly.

5.148 Three brass plates are brazed to a steel pipe to form the flagpole base shown. Knowing that the pipe has a wall thickness of 8 mm and that each plate is 6 mm thick, determine the location of the center of gravity of the base. (Densities: brass = 8470 kg/m^3 ; steel = 7860 kg/m^3 .)



6

Analysis of Structures

Trusses, such as this cantilever arch bridge over Deception Pass in Washington State, provide both a practical and an economical solution to many engineering problems.

Introduction

6.1 ANALYSIS OF TRUSSES

6.1A Simple Trusses

6.1B The Method of Joints

*6.1C Joints Under Special Loading Conditions

*6.1D Space Trusses

6.2 OTHER TRUSS ANALYSES

6.2A The Method of Sections

6.2B Trusses Made of Several Simple Trusses

6.3 FRAMES

6.3A Analysis of a Frame

6.3B Frames That Collapse Without Supports

6.4 MACHINES

Objectives

- **Define** an ideal truss, and consider the attributes of simple trusses.
- **Analyze** plane and space trusses by the method of joints.
- **Simplify** certain truss analyses by recognizing special loading and geometry conditions.
- **Analyze** trusses by the method of sections.
- **Consider** the characteristics of compound trusses.
- **Analyze** structures containing multiforce members, such as frames and machines.

Introduction

In the preceding chapters, we studied the equilibrium of a single rigid body, where all forces involved were external to the rigid body. We now consider the equilibrium of structures made of several connected parts. This situation calls for determining not only the external forces acting on the structure, but also the forces that hold together the various parts of the structure. From the point of view of the structure as a whole, these forces are **internal forces**.

Consider, for example, the crane shown in Fig. 6.1a that supports a load W . The crane consists of three beams AD , CF , and BE connected by frictionless pins; it is supported by a pin at A and by a cable DG . The free-body diagram of the crane is drawn in Fig. 6.1b. The external forces shown in the diagram include the weight W , the two components A_x and A_y of the reaction at A , and the force T exerted by the cable at D . The internal forces holding the various parts of the crane together do not appear in the free-body diagram. If, however, we dismember the crane and draw a free-body diagram for each of its component parts, we can see the forces holding the three beams together, since these forces are external forces from the point of view of each component part (Fig. 6.1c).

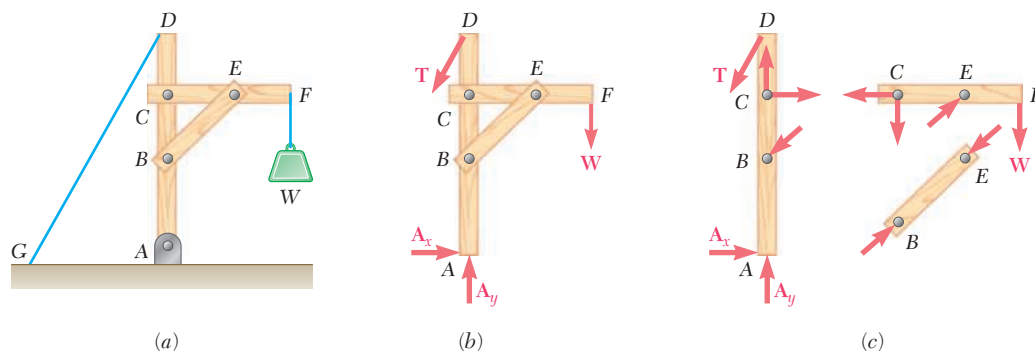


Fig. 6.1 A structure in equilibrium. (a) Diagram of a crane supporting a load; (b) free-body diagram of the crane; (c) free-body diagrams of the components of the crane.

Note that we represent the force exerted at B by member BE on member AD as equal and opposite to the force exerted at the same point by member AD on member BE . Similarly, the force exerted at E by BE on CF is shown equal and opposite to the force exerted by CF on BE , and the components of the force exerted at C by CF on AD are shown equal and opposite to the components of the force exerted by AD on CF . These representations agree with Newton's third law, which states that

The forces of action and reaction between two bodies in contact have the same magnitude, same line of action, and opposite sense.

We pointed out in Chap. 1 that this law, which is based on experimental evidence, is one of the six fundamental principles of elementary mechanics. Its application is essential for solving problems involving connected bodies.

In this chapter, we consider three broad categories of engineering structures:

1. **Trusses**, which are designed to support loads and are usually stationary, fully constrained structures. Trusses consist exclusively of straight members connected at joints located at the ends of each member. Members of a truss, therefore, are **two-force members**, i.e., members acted upon by two equal and opposite forces directed along the member.
2. **Frames**, which are also designed to support loads and are also usually stationary, fully constrained structures. However, like the crane of Fig. 6.1, frames always contain at least one **multi-force member**, i.e., a member acted upon by three or more forces that, in general, are not directed along the member.
3. **Machines**, which are designed to transmit and modify forces and are structures containing moving parts. Machines, like frames, always contain at least one multi-force member.

Two-force member



(a) A truss bridge

Multi-force member



(b) A bicycle frame

Multi-force member



(c) A hydraulic machine arm

Photo 6.1 The structures you see around you to support loads or transmit forces are generally trusses, frames, or machines.

6.1 ANALYSIS OF TRUSSES

The truss is one of the major types of engineering structures. It provides a practical and economical solution to many engineering situations, especially in the design of bridges and buildings. In this section, we describe the basic elements of a truss and study a common method for analyzing the forces acting in a truss.

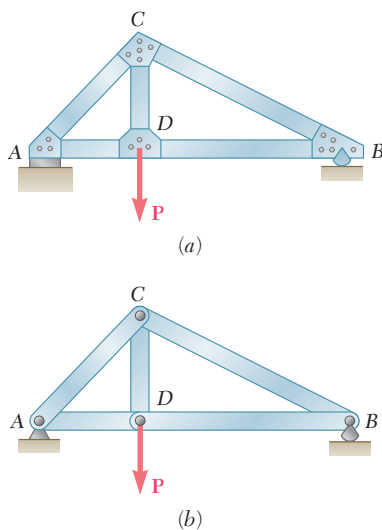


Fig. 6.2 (a) A typical truss consists of straight members connected at joints; (b) we can model a truss as two-force members connected by pins.



Photo 6.2 Shown is a pin-jointed connection on the approach span to the San Francisco–Oakland Bay Bridge.

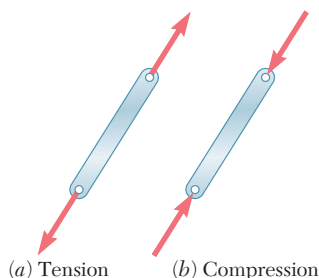


Fig. 6.4 A two-force member of a truss can be in tension or compression.

6.1A Simple Trusses

A truss consists of straight members connected at joints, as shown in Fig. 6.2a. Truss members are connected at their extremities only; no member is continuous through a joint. In Fig. 6.2a, for example, there is no member AB ; instead we have two distinct members AD and DB . Most actual structures are made of several trusses joined together to form a space framework. Each truss is designed to carry those loads that act in its plane and thus may be treated as a two-dimensional structure.

In general, the members of a truss are slender and can support little lateral load; all loads, therefore, must be applied at the various joints and not to the members themselves. When a concentrated load is to be applied between two joints or when the truss must support a distributed load, as in the case of a bridge truss, a floor system must be provided. The floor transmits the load to the joints through the use of stringers and floor beams (Fig. 6.3).

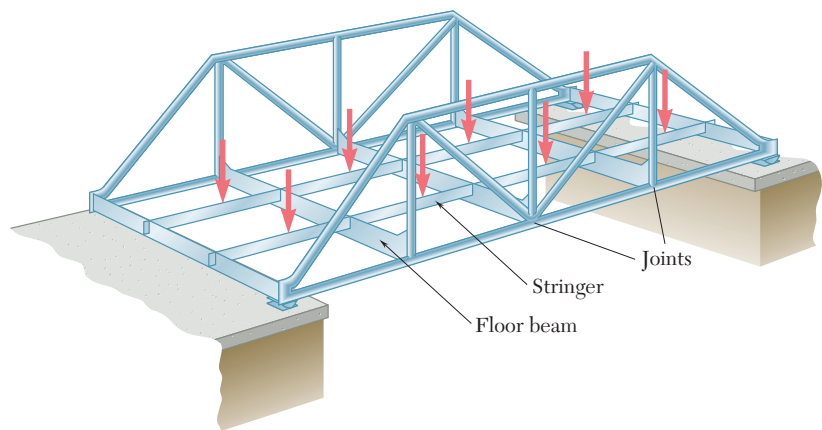


Fig. 6.3 A floor system of a truss uses stringers and floor beams to transmit an applied load to the joints of the truss.

We assume that the weights of the truss members can be applied to the joints, with half of the weight of each member applied to each of the two joints the member connects. Although the members are actually joined together by means of welded, bolted, or riveted connections, it is customary to assume that the members are pinned together; therefore, the forces acting at each end of a member reduce to a single force and no couple. This enables us to model the forces applied to a truss member as a single force at each end of the member. We can then treat each member as a two-force member, and we can consider the entire truss as a group of pins and two-force members (Fig. 6.2b). An individual member can be acted upon as shown in either of the two sketches of Fig. 6.4. In Fig. 6.4a, the forces tend to pull the member apart, and the member is in tension; in Fig. 6.4b, the forces tend to push the member together, and the member is in compression. Some typical trusses are shown in Fig. 6.5.

Consider the truss of Fig. 6.6a, which is made of four members connected by pins at A , B , C , and D . If we apply a load at B , the truss will greatly deform, completely losing its original shape. In contrast, the truss of Fig. 6.6b, which is made of three members connected by pins at A , B , and C , will deform only slightly under a load applied at B . The only possible

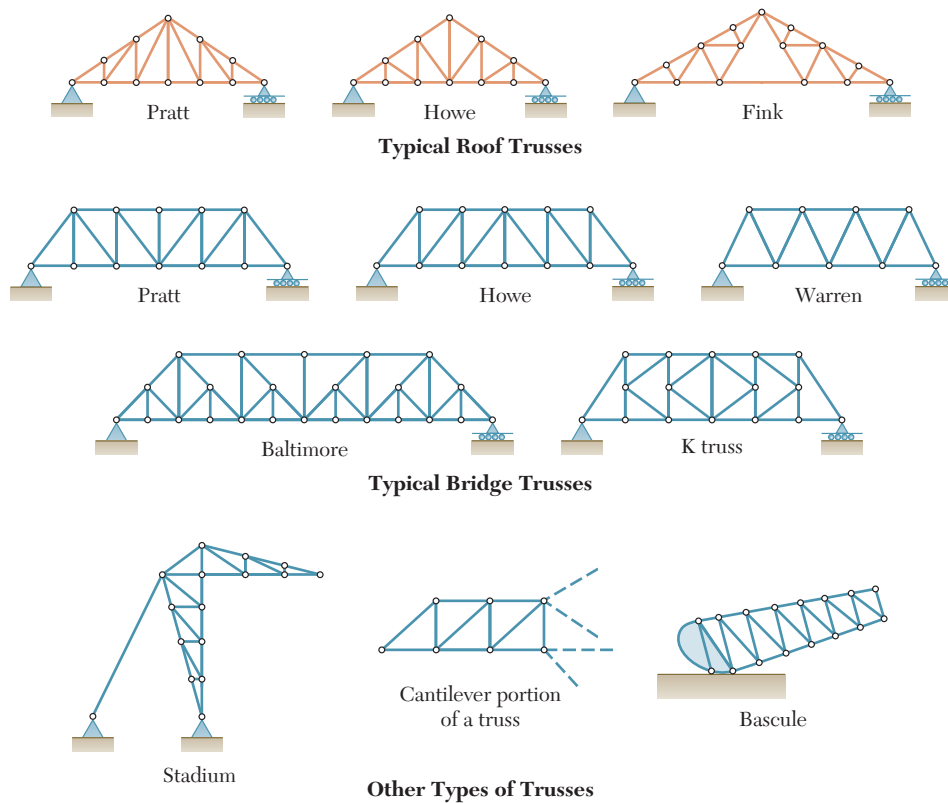


Fig. 6.5 You can often see trusses in the design of a building roof, a bridge, or other other larger structures.

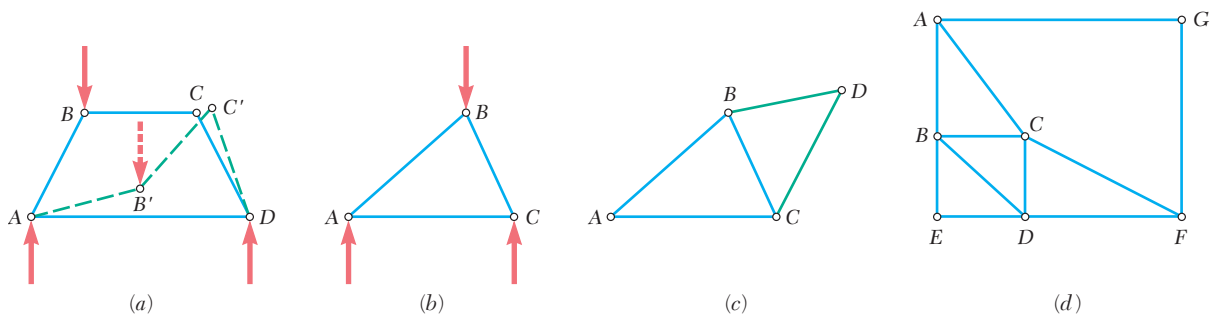


Fig. 6.6 (a) A poorly designed truss that cannot support a load; (b) the most elementary rigid truss consists of a simple triangle; (c) a larger rigid truss built up from the triangle in (b); (d) a rigid truss not made up of triangles alone.

deformation for this truss is one involving small changes in the length of its members. The truss of Fig. 6.6b is said to be a **rigid truss**, the term ‘rigid’ being used here to indicate that the truss *will not collapse*.

As shown in Fig. 6.6c, we can obtain a larger rigid truss by adding two members BD and CD to the basic triangular truss of Fig. 6.6b. We can repeat this procedure as many times as we like, and the resulting truss will be rigid if each time we add two new members they are attached to two existing joints and connected at a new joint. (The three joints must not be in a straight line.) A truss that can be constructed in this manner is called a **simple truss**.

Note that a simple truss is not necessarily made only of triangles. The truss of Fig. 6.6d, for example, is a simple truss that we constructed from triangle ABC by adding successively the joints D , E , F , and G .



Photo 6.3 Two K trusses were used as the main components of the movable bridge shown, which moved above a large stockpile of ore. The bucket below the trusses picked up ore and redeposited it until the ore was thoroughly mixed. The ore was then sent to the mill for processing into steel.

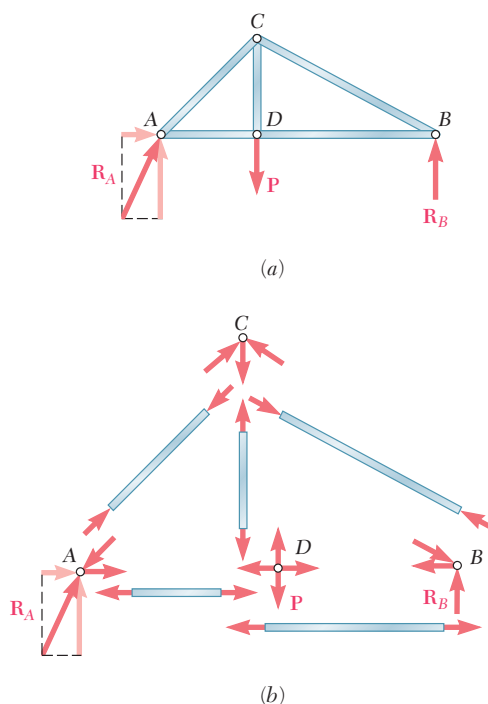


Fig. 6.7 (a) Free-body diagram of the truss as a rigid body; (b) free-body diagrams of the five members and four pins that make up the truss.

On the other hand, rigid trusses are not always simple trusses, even when they appear to be made of triangles. The Fink and Baltimore trusses shown in Fig. 6.5, for instance, are not simple trusses, because they cannot be constructed from a single triangle in the manner just described. All of the other trusses shown in Fig. 6.5 are simple trusses, as you may easily check. (For the K truss, start with one of the central triangles.)

Also note that the basic triangular truss of Fig. 6.6b has three members and three joints. The truss of Fig. 6.6c has two more members and one more joint; i.e., five members and four joints altogether. Observing that every time we add two new members, we increase the number of joints by one, we find that in a simple truss the total number of members is $m = 2n - 3$, where n is the total number of joints.

6.1B The Method of Joints

We have just seen that a truss can be considered as a group of pins and two-force members. Therefore, we can dismember the truss of Fig. 6.2, whose free-body diagram is shown in Fig. 6.7a, and draw a free-body diagram for each pin and each member (Fig. 6.7b). Each member is acted upon by two forces, one at each end; these forces have the same magnitude, same line of action, and opposite sense (Sec. 4.2A). Furthermore, Newton's third law states that the forces of action and reaction between a member and a pin are equal and opposite. Therefore, the forces exerted by a member on the two pins it connects must be directed along that member and be equal and opposite. The common magnitude of the forces exerted by a member on the two pins it connects is commonly referred to as the *force in the member*, even though this quantity is actually a scalar. Since we know the lines of action of all the internal forces in a truss, the analysis of a truss reduces to computing the forces in its various members and determining whether each of its members is in tension or compression.

Since the entire truss is in equilibrium, each pin must be in equilibrium. We can use the fact that a pin is in equilibrium to draw its free-body diagram and write two equilibrium equations (Sec. 2.3A). Thus, if the truss contains n pins, we have $2n$ equations available, which can be solved for $2n$ unknowns. In the case of a simple truss, we have $m = 2n - 3$; that is, $2n = m + 3$, and the number of unknowns that we can determine from the free-body diagrams of the pins is $m + 3$. This means that we can find the forces in all the members, the two components of the reaction $\mathbf{R}_A$, and the reaction $\mathbf{R}_B$ by considering the free-body diagrams of the pins.

We can also use the fact that the entire truss is a rigid body in equilibrium to write three more equations involving the forces shown in the free-body diagram of Fig. 6.7a. Since these equations do not contain any new information, they are not independent of the equations associated with the free-body diagrams of the pins. Nevertheless, we can use them to determine the components of the reactions at the supports. The arrangement of pins and members in a simple truss is such that it is always possible to find a joint involving only two unknown forces. We can determine these forces by using the methods of Sec. 2.3C and then transferring their values to the adjacent joints, treating them as known quantities at these joints. We repeat this procedure until we have determined all unknown forces.

As an example, let's analyze the truss of Fig. 6.7 by considering the equilibrium of each pin successively, starting with a joint at which only

two forces are unknown. In this truss, all pins are subjected to at least three unknown forces. Therefore, we must first determine the reactions at the supports by considering the entire truss as a free body and using the equations of equilibrium of a rigid body. In this way we find that $\mathbf{R}_A$ is vertical, and we determine the magnitudes of $\mathbf{R}_A$ and $\mathbf{R}_B$.

This reduces the number of unknown forces at joint A to two, and we can determine these forces by considering the equilibrium of pin A. The reaction $\mathbf{R}_A$ and the forces $\mathbf{F}_{AC}$ and $\mathbf{F}_{AD}$ exerted on pin A by members AC and AD, respectively, must form a force triangle. First we draw $\mathbf{R}_A$ (Fig. 6.8); noting that $\mathbf{F}_{AC}$ and $\mathbf{F}_{AD}$ are directed along AC and AD, respectively, we complete the triangle and determine the magnitude and sense of $\mathbf{F}_{AC}$ and $\mathbf{F}_{AD}$. The magnitudes F_{AC} and F_{AD} represent the forces in members AC and AD. Since $\mathbf{F}_{AC}$ is directed down and to the left—that is, *toward* joint A—member AC pushes on pin A and is in compression. (From Newton's third law, pin A pushes *on* member AC.) Since $\mathbf{F}_{AD}$ is directed *away* from joint A, member AD pulls on pin A and is in tension. (From Newton's third law, pin A pulls *away* from member AD.)



Photo 6.4 Because roof trusses, such as those shown, require support only at their ends, it is possible to construct buildings with large, unobstructed interiors.

	Free-body diagram	Force polygon
Joint A		
Joint D		
Joint C		
Joint B		

Fig. 6.8 Free-body diagrams and force polygons used to determine the forces on the pins and in the members of the truss in Fig. 6.7.

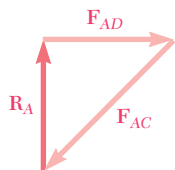


Fig. 6.9 Alternative force polygon for joint A in Fig. 6.8.

We can now proceed to joint D , where only two forces, $\mathbf{F}_{DC}$ and $\mathbf{F}_{DB}$, are still unknown. The other forces are the load $\mathbf{P}$, which is given, and the force $\mathbf{F}_{DA}$ exerted on the pin by member AD . As indicated previously, this force is equal and opposite to the force $\mathbf{F}_{AD}$ exerted by the same member on pin A . We can draw the force polygon corresponding to joint D , as shown in Fig. 6.8, and determine the forces $\mathbf{F}_{DC}$ and $\mathbf{F}_{DB}$ from that polygon. However, when more than three forces are involved, it is usually more convenient to solve the equations of equilibrium $\Sigma F_x = 0$ and $\Sigma F_y = 0$ for the two unknown forces. Since both of these forces are directed away from joint D , members DC and DB pull on the pin and are in tension.

Next, we consider joint C ; its free-body diagram is shown in Fig. 6.8. Both $\mathbf{F}_{CD}$ and $\mathbf{F}_{CA}$ are known from the analysis of the preceding joints, so only $\mathbf{F}_{CB}$ is unknown. Since the equilibrium of each pin provides sufficient information to determine two unknowns, we can check our analysis at this joint. We draw the force triangle and determine the magnitude and sense of $\mathbf{F}_{CB}$. Since $\mathbf{F}_{CB}$ is directed toward joint C , member CB pushes on pin C and is in compression. The check is obtained by verifying that the force $\mathbf{F}_{CB}$ and member CB are parallel.

Finally, at joint B , we know all of the forces. Since the corresponding pin is in equilibrium, the force triangle must close, giving us an additional check of the analysis.

Note that the force polygons shown in Fig. 6.8 are not unique; we could replace each of them by an alternative configuration. For example, the force triangle corresponding to joint A could be drawn as shown in Fig. 6.9. We obtained the triangle actually shown in Fig. 6.8 by drawing the three forces $\mathbf{R}_A$, $\mathbf{F}_{AC}$, and $\mathbf{F}_{AD}$ in tip-to-tail fashion in the order in which we cross their lines of action when moving clockwise around joint A .

*6.1C Joints Under Special Loading Conditions

Some geometric arrangements of members in a truss are particularly simple to analyze by observation. For example, Fig. 6.10a shows a joint connecting four members lying along two intersecting straight lines. The free-body diagram of Fig. 6.10b shows that pin A is subjected to two pairs of directly opposite forces. The corresponding force polygon, therefore, must be a parallelogram (Fig. 6.10c), and **the forces in opposite members must be equal**.

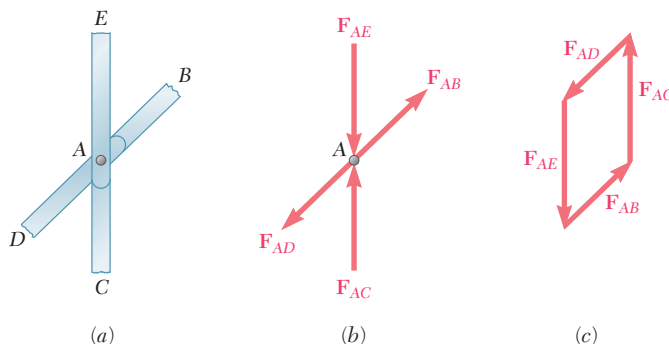


Fig. 6.10 (a) A joint A connecting four members of a truss in two straight lines; (b) free-body diagram of pin A ; (c) force polygon (parallelogram) for pin A . Forces in opposite members are equal.

Consider next Fig. 6.11*a*, in which a joint connects three members and supports a load $\mathbf{P}$. Two members lie along the same line, and load $\mathbf{P}$ acts along the third member. The free-body diagram of pin A and the corresponding force polygon are the same as in Fig. 6.10*b* and *c*, with $\mathbf{F}_{AE}$ replaced by load $\mathbf{P}$. Thus, **the forces in the two opposite members must be equal, and the force in the other member must equal P** . Figure 6.11*b* shows a particular case of special interest. Since, in this case, no external load is applied to the joint, we have $P = 0$, and the force in member AC is zero. Member AC is said to be a **zero-force member**.

Now consider a joint connecting two members only. From Sec. 2.3A, we know that a particle acted upon by two forces is in equilibrium if the two forces have the same magnitude, same line of action, and opposite sense. In the case of the joint of Fig. 6.12*a*, which connects two members AB and AD lying along the same line, the forces in the two members must be equal for pin A to be in equilibrium. In the case of the joint of Fig. 6.12*b*, pin A cannot be in equilibrium unless the forces in both members are zero. Members connected as shown in Fig. 6.12*b*, therefore, must be **zero-force members**.

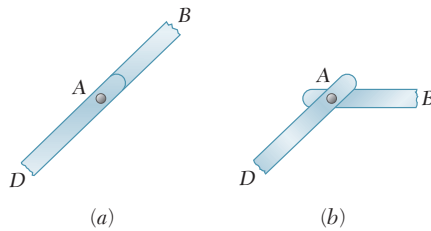


Fig. 6.12 (a) A joint in a truss connecting two members in a straight line. Forces in the members are equal. (b) If the two members are not in a straight line, they must be zero-force members.

Spotting joints that are under the special loading conditions just described will expedite the analysis of a truss. Consider, for example, a Howe truss loaded as shown in Fig. 6.13. We can recognize all of the members represented by green lines as zero-force members. Joint C connects three members, two of which lie in the same line, and is not subjected to any external load; member BC is thus a zero-force member. Applying the same reasoning to joint K , we find that member JK is also a zero-force member. But joint J is now in the same situation as joints C and K , so member IJ also must be a zero-force member. Examining joints C , J , and K also shows that the forces in members AC and CE are equal, that the forces in members HJ and JL are equal, and that the forces in members IK and KL are equal. Turning our attention to joint I , where the 20-kN load and member HI are collinear, we note that the force in member HI is 20 kN (tension) and that the forces in members GI and IK are equal. Hence, the forces in members GI , IK , and KL are equal.

Note that the conditions described here do not apply to joints B and D in Fig. 6.13, so it is wrong to assume that the force in member DE is 25 kN or that the forces in members AB and BD are equal. To determine the forces in these members and in all remaining members, you need to

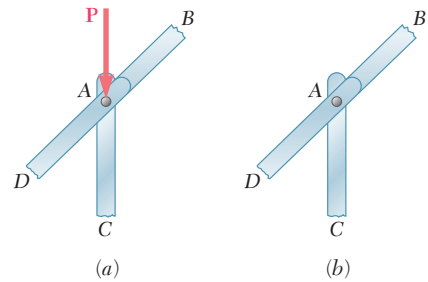


Fig. 6.11 (a) Joint A in a truss connects three members, two in a straight line and the third along the line of a load. Force in the third member equals the load. (b) If the load is zero, the third member is a zero-force member.

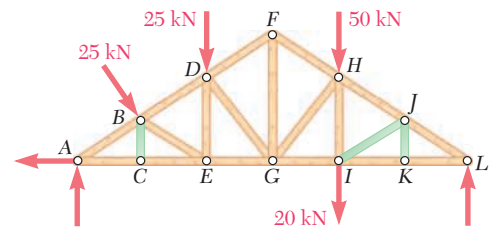


Fig. 6.13 An example of loading on a Howe truss; identifying special loading conditions.

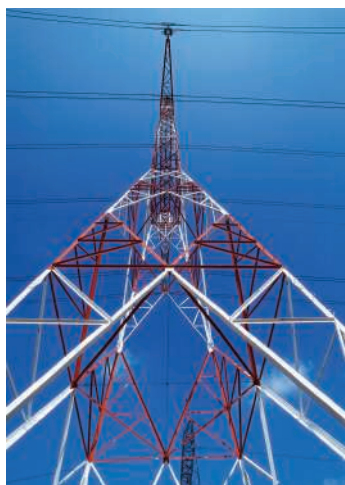


Photo 6.5 Three-dimensional or space trusses are used for broadcast and power transmission line towers, roof framing, and spacecraft applications, such as components of the *International Space Station*.

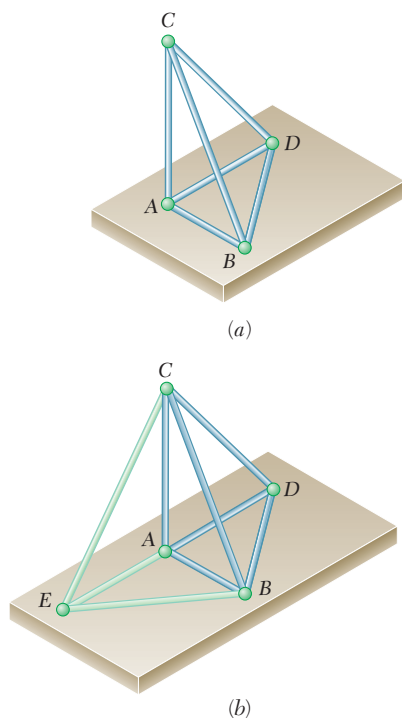


Fig. 6.14 (a) The most elementary space truss consists of six members joined at their ends to form a tetrahedron. (b) We can add three members at a time to three joints of an existing space truss, connecting the new members at a new joint, to build a larger simple space truss.

carry out the analysis of joints A , B , D , E , F , G , H , and L in the usual manner. Thus, until you have become thoroughly familiar with the conditions under which you can apply the rules described in this section, you should draw the free-body diagrams of all pins and write the corresponding equilibrium equations (or draw the corresponding force polygons) whether or not the joints being considered are under one of these special loading conditions.

A final remark concerning zero-force members: These members are not useless. For example, although the zero-force members of Fig. 6.13 do not carry any loads under the loading conditions shown, the same members would probably carry loads if the loading conditions were changed. Besides, even in the case considered, these members are needed to support the weight of the truss and to maintain the truss in the desired shape.

*6.1D Space Trusses

When several straight members of a truss are joined together at their extremities to form a three-dimensional configuration, the resulting structure is called a **space truss**. Recall from Sec. 6.1A that the most elementary two-dimensional rigid truss consists of three members joined at their extremities to form the sides of a triangle. By adding two members at a time to this basic configuration and connecting them at a new joint, we could obtain a larger rigid structure that we defined as a simple truss. Similarly, the most elementary rigid space truss consists of six members joined at their extremities to form the edges of a tetrahedron $ABCD$ (Fig. 6.14a). By adding three members at a time to this basic configuration, such as AE , BE , and CE (Fig. 6.14b), attaching them to three existing joints, and connecting them at a new joint, we can obtain a larger rigid structure that we define as a **simple space truss**. (The four joints must not lie in a plane.) Note that the basic tetrahedron has six members and four joints, and every time we add three members, the number of joints increases by one. Therefore, we conclude that in a simple space truss the total number of members is $m = 3n - 6$, where n is the total number of joints.

If a space truss is to be completely constrained and if the reactions at its supports are to be statically determinate, the supports should consist of a combination of balls, rollers, and balls and sockets, providing six unknown reactions (see Sec. 4.3B). We can determine these unknown reactions by solving the six equations expressing that the three-dimensional truss is in equilibrium.

Although the members of a space truss are actually joined together by means of bolted or welded connections, we assume for analysis purposes that each joint consists of a ball-and-socket connection. Thus, no couple is applied to the members of the truss, and we can treat each member as a two-force member. The conditions of equilibrium for each joint are expressed by the three equations $\Sigma F_x = 0$, $\Sigma F_y = 0$, and $\Sigma F_z = 0$. Thus, in the case of a simple space truss containing n joints, writing the conditions of equilibrium for each joint yields $3n$ equations. Since $m = 3n - 6$, these equations suffice to determine all unknown forces (forces in m members and six reactions at the supports). However, to avoid the necessity of solving simultaneous equations, you should take care to select joints in such an order that no selected joint involves more than three unknown forces.

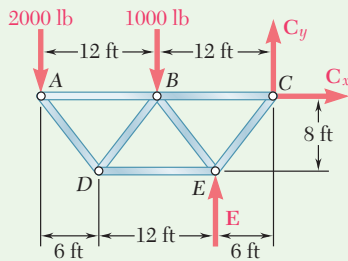
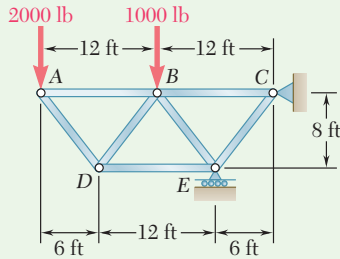


Fig. 1 Free-body diagram of entire truss.

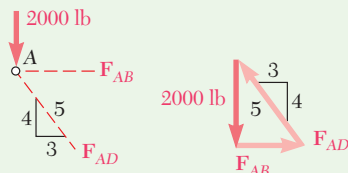


Fig. 2 Free-body diagram of joint A.

Sample Problem 6.1

Using the method of joints, determine the force in each member of the truss shown.

STRATEGY: To use the method of joints, you start with an analysis of the free-body diagram of the entire truss. Then look for a joint connecting only two members as a starting point for the calculations. In this example, we start at joint A and proceed through joints D, B, E, and C, but you could also start at joint C and proceed through joints E, B, D, and A.

MODELING and ANALYSIS: You can combine these steps for each joint of the truss in turn. Draw a free-body diagram; draw a force polygon or write the equilibrium equations; and solve for the unknown forces.

Entire Truss. Draw a free-body diagram of the entire truss (Fig. 1); external forces acting on this free body are the applied loads and the reactions at C and E. Write the equilibrium equations, taking moments about C.

$$\begin{aligned}
 +\curvearrowright \Sigma M_C &= 0: & (2000 \text{ lb})(24 \text{ ft}) + (1000 \text{ lb})(12 \text{ ft}) - E(6 \text{ ft}) &= 0 \\
 & & E &= +10,000 \text{ lb} & E &= 10,000 \text{ lb} \uparrow \\
 +\rightarrow \Sigma F_x &= 0: & C_x &= 0 \\
 +\uparrow \Sigma F_y &= 0: & -2000 \text{ lb} - 1000 \text{ lb} + 10,000 \text{ lb} + C_y &= 0 \\
 & & C_y &= -7000 \text{ lb} & C_y &= 7000 \text{ lb} \downarrow
 \end{aligned}$$

Joint A. This joint is subject to only two unknown forces: the forces exerted by AB and those by AD. Use a force triangle to determine F_{AB} and F_{AD} (Fig. 2). Note that member AB pulls on the joint so AB is in tension, and member AD pushes on the joint so AD is in compression. Obtain the magnitudes of the two forces from the proportion

$$\frac{2000 \text{ lb}}{4} = \frac{F_{AB}}{3} = \frac{F_{AD}}{5}$$

$$\begin{aligned}
 F_{AB} &= 1500 \text{ lb } T & \blacktriangleleft \\
 F_{AD} &= 2500 \text{ lb } C & \blacktriangleleft
 \end{aligned}$$

Joint D. Since you have already determined the force exerted by member AD, only two unknown forces are now involved at this joint. Again, use a force triangle to determine the unknown forces in members DB and DE (Fig. 3).

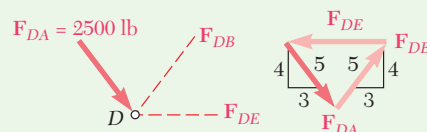


Fig. 3 Free-body diagram of joint D.

(continued)

$$F_{DB} = F_{DA} \quad F_{DB} = 2500 \text{ lb } T \quad \blacktriangleleft$$

$$F_{DE} = 2\left(\frac{3}{5}\right)F_{DA} \quad F_{DE} = 3000 \text{ lb } C \quad \blacktriangleleft$$

Joint B. Since more than three forces act at this joint (Fig. 4), determine the two unknown forces $\mathbf{F}_{BC}$ and $\mathbf{F}_{BE}$ by solving the equilibrium equations $\Sigma F_x = 0$ and $\Sigma F_y = 0$. Suppose you arbitrarily assume that both unknown forces act away from the joint, i.e., that the members are in tension. The positive value obtained for F_{BC} indicates that this assumption is correct; member BC is in tension. The negative value of F_{BE} indicates that the second assumption is wrong; member BE is in compression.

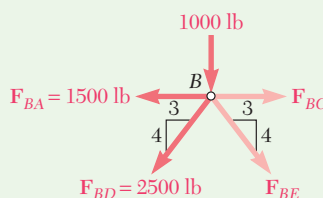


Fig. 4 Free-body diagram of joint B.

$$+\uparrow \Sigma F_y = 0: \quad -1000 - \frac{4}{5}(2500) - \frac{4}{5}F_{BE} = 0$$

$$F_{BE} = -3750 \text{ lb} \quad F_{BE} = 3750 \text{ lb } C \quad \blacktriangleleft$$

$$+\rightarrow \Sigma F_x = 0: \quad F_{BC} - 1500 - \frac{3}{5}(2500) - \frac{3}{5}(3750) = 0$$

$$F_{BC} = +5250 \text{ lb} \quad F_{BC} = 5250 \text{ lb } T \quad \blacktriangleleft$$

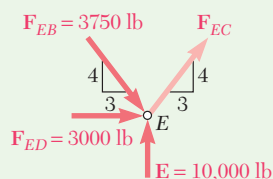


Fig. 5 Free-body diagram of joint E.

Joint E. Assume the unknown force $\mathbf{F}_{EC}$ acts away from the joint (Fig. 5). Summing x components, you obtain

$$+\rightarrow \Sigma F_x = 0: \quad \frac{3}{5}F_{EC} + 3000 + \frac{3}{5}(3750) = 0$$

$$F_{EC} = -8750 \text{ lb} \quad F_{EC} = 8750 \text{ lb } C \quad \blacktriangleleft$$

Summing y components, you obtain a check of your computations:

$$+\uparrow \Sigma F_y = 10,000 - \frac{4}{5}(3750) - \frac{4}{5}(8750)$$

$$= 10,000 - 3000 - 7000 = 0 \quad (\text{checks})$$

REFLECT and THINK: Using the computed values of $\mathbf{F}_{CB}$ and $\mathbf{F}_{CE}$, you can determine the reactions $\mathbf{C}_x$ and $\mathbf{C}_y$ by considering the equilibrium of Joint C (Fig. 6). Since these reactions have already been determined from the equilibrium of the entire truss, this provides two checks of your computations. You can also simply use the computed values of all forces acting on the joint (forces in members and reactions) and check that the joint is in equilibrium:

$$+\rightarrow \Sigma F_x = -5250 + \frac{3}{5}(8750) = -5250 + 5250 = 0 \quad (\text{checks})$$

$$+\uparrow \Sigma F_y = -7000 + \frac{4}{5}(8750) = -7000 + 7000 = 0 \quad (\text{checks})$$

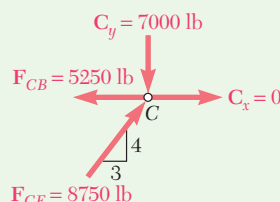


Fig. 6 Free-body diagram of joint C.

SOLVING PROBLEMS ON YOUR OWN

In this section, you learned to use the **method of joints** to determine the forces in the members of a **simple truss**; that is, a truss that can be constructed from a basic triangular truss by adding to it two new members at a time and connecting them at a new joint.

The method consists of the following steps:

1. Draw a free-body diagram of the entire truss, and use this diagram to determine the reactions at the supports.

2. Locate a joint connecting only two members, and draw the free-body diagram of its pin. Use this free-body diagram to determine the unknown force in each of the two members. If only three forces are involved (the two unknown forces and a known one), you will probably find it more convenient to draw and solve the corresponding force triangle. If more than three forces are involved, you should write and solve the equilibrium equations for the pin, $\Sigma F_x = 0$ and $\Sigma F_y = 0$, assuming that the members are in tension. A positive answer means that the member is in tension, a negative answer means that the member is in compression. Once you have found the forces, enter their values on a sketch of the truss with *T* for tension and *C* for compression.

3. Next, locate a joint where the forces in only two of the connected members are still unknown. Draw the free-body diagram of the pin and use it as indicated in Step 2 to determine the two unknown forces.

4. Repeat this procedure until you have found the forces in all the members of the truss. Since you previously used the three equilibrium equations associated with the free-body diagram of the entire truss to determine the reactions at the supports, you will end up with three extra equations. These equations can be used to check your computations.

5. Note that the choice of the first joint is not unique. Once you have determined the reactions at the supports of the truss, you can choose either of two joints as a starting point for your analysis. In Sample Prob. 6.1, we started at joint *A* and proceeded through joints *D*, *B*, *E*, and *C*, but we could also have started at joint *C* and proceeded through joints *E*, *B*, *D*, and *A*. On the other hand, having selected a first joint, you may in some cases reach a point in your analysis beyond which you cannot proceed. You must then start again from another joint to complete your solution.

Keep in mind that the analysis of a simple truss always can be carried out by the method of joints. Also remember that it is helpful to outline your solution *before* starting any computations.

PROBLEMS

6.1 through 6.8 Using the method of joints, determine the force in each member of the truss shown. State whether each member is in tension or compression.

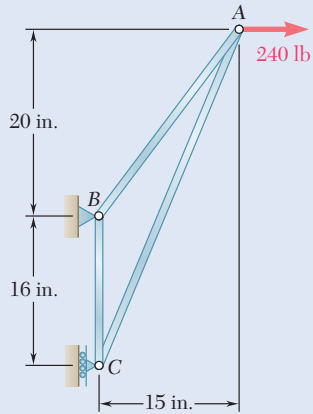


Fig. P6.1

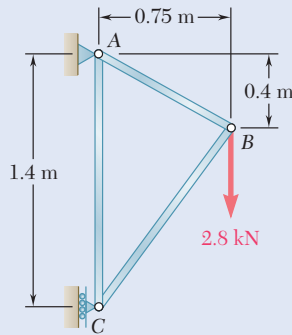


Fig. P6.2

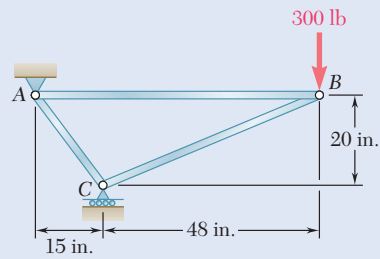


Fig. P6.3

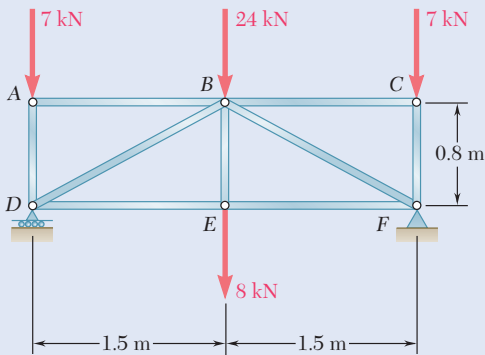


Fig. P6.4

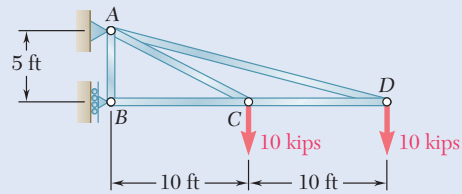


Fig. P6.5

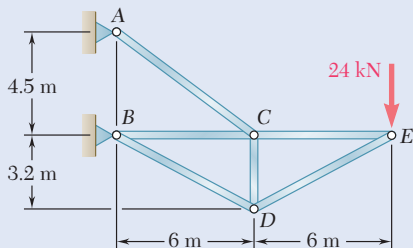


Fig. P6.6

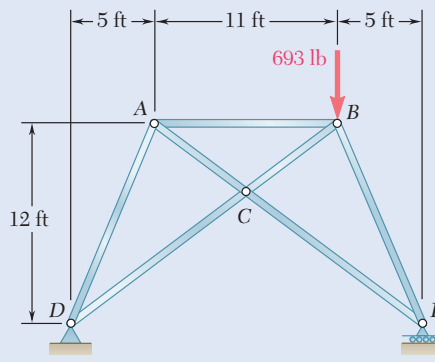


Fig. P6.7

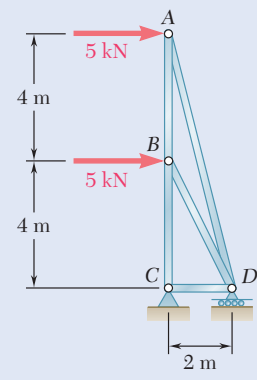


Fig. P6.8

6.9 and 6.10 Determine the force in each member of the truss shown. State whether each member is in tension or compression.

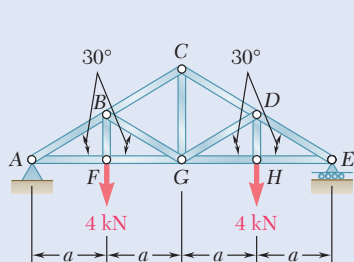


Fig. P6.9

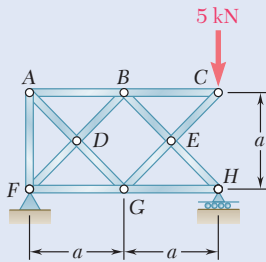


Fig. P6.10

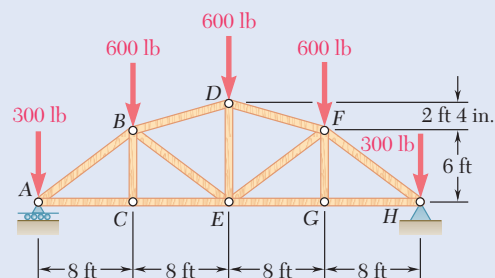


Fig. P6.11

6.11 Determine the force in each member of the Gambrel roof truss shown. State whether each member is in tension or compression.

6.12 Determine the force in each member of the Howe roof truss shown. State whether each member is in tension or compression.

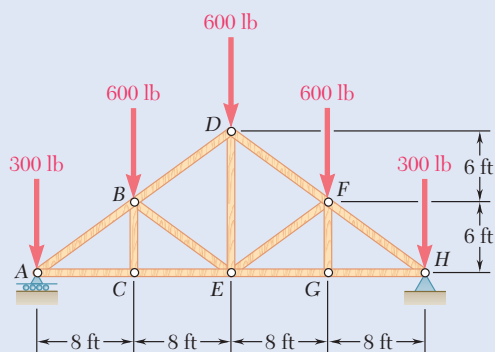


Fig. P6.12

6.13 Determine the force in each member of the roof truss shown. State whether each member is in tension or compression.

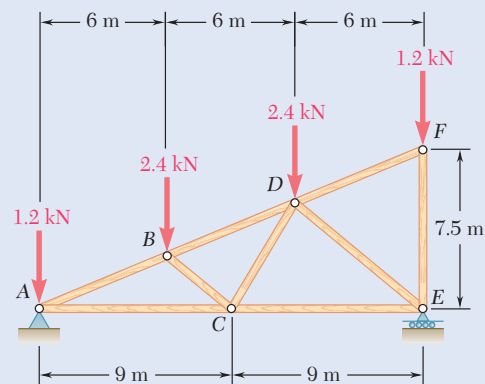


Fig. P6.13

6.14 Determine the force in each member of the Fink roof truss shown. State whether each member is in tension or compression.

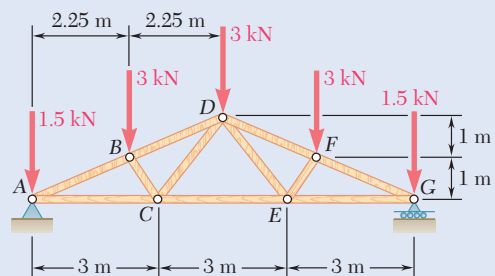


Fig. P6.14

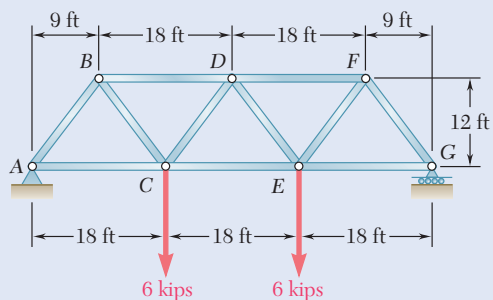


Fig. P6.15

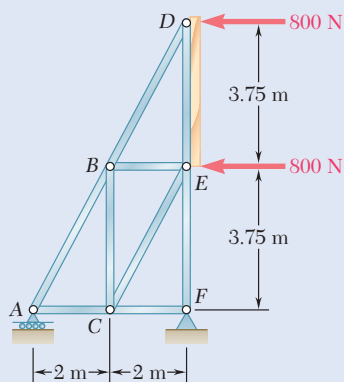


Fig. P6.18

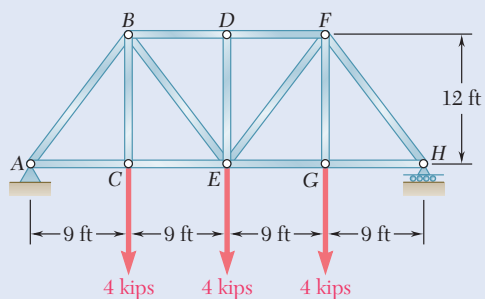


Fig. P6.19

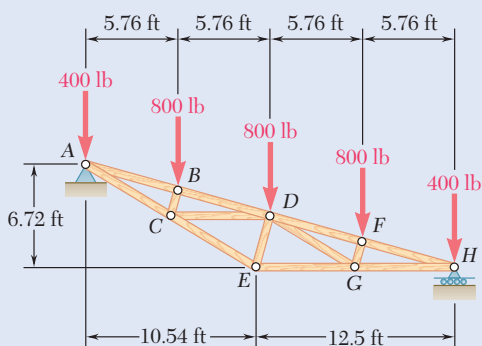


Fig. P6.22 and P6.23

6.15 Determine the force in each member of the Warren bridge truss shown. State whether each member is in tension or compression.

6.16 Solve Prob. 6.15 assuming that the load applied at E has been removed.

6.17 Determine the force in each member of the Pratt roof truss shown. State whether each member is in tension or compression.

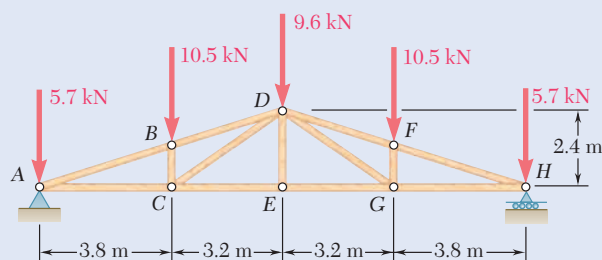


Fig. P6.17

6.18 The truss shown is one of several supporting an advertising panel. Determine the force in each member of the truss for a wind load equivalent to the two forces shown. State whether each member is in tension or compression.

6.19 Determine the force in each member of the Pratt bridge truss shown. State whether each member is in tension or compression.

6.20 Solve Prob. 6.19 assuming that the load applied at G has been removed.

6.21 Determine the force in each of the members located to the left of FG for the scissors roof truss shown. State whether each member is in tension or compression.

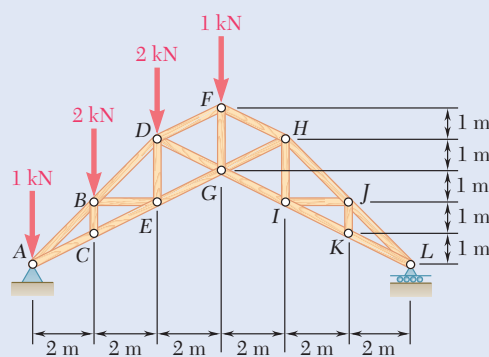


Fig. P6.21

6.22 Determine the force in member DE and in each of the members located to the left of DE for the inverted Howe roof truss shown. State whether each member is in tension or compression.

6.23 Determine the force in each of the members located to the right of DE for the inverted Howe roof truss shown. State whether each member is in tension or compression.

- 6.24** The portion of truss shown represents the upper part of a power transmission line tower. For the given loading, determine the force in each of the members located above HJ . State whether each member is in tension or compression.

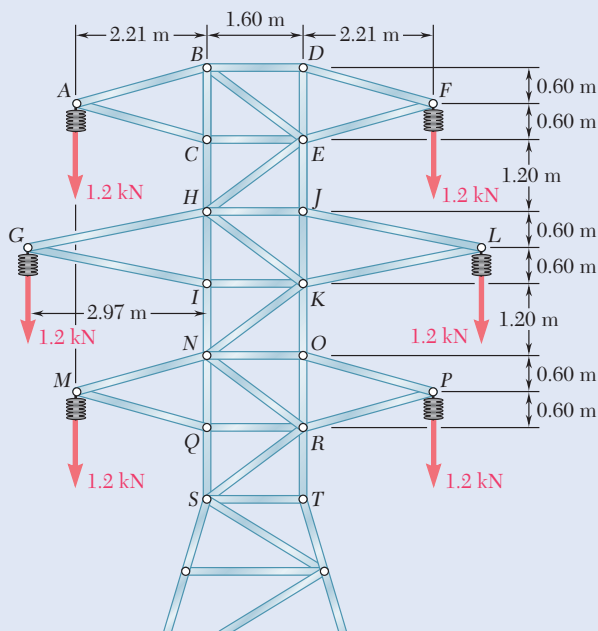


Fig. P6.24

- 6.25** For the tower and loading of Prob. 6.24 and knowing that $F_{CH} = F_{EI} = 1.2 \text{ kN C}$ and $F_{EH} = 0$, determine the force in member HJ and in each of the members located between HJ and NO . State whether each member is in tension or compression.
- 6.26** Solve Prob. 6.24 assuming that the cables hanging from the right side of the tower have fallen to the ground.
- 6.27 and 6.28** Determine the force in each member of the truss shown. State whether each member is in tension or compression.

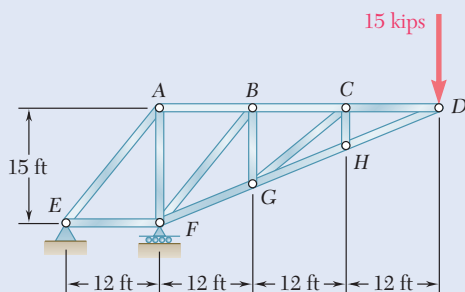


Fig. P6.27

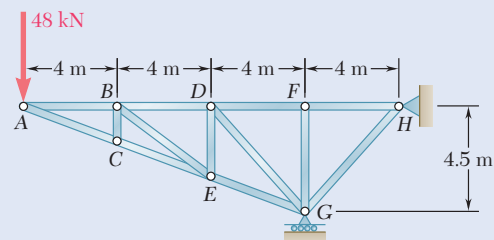
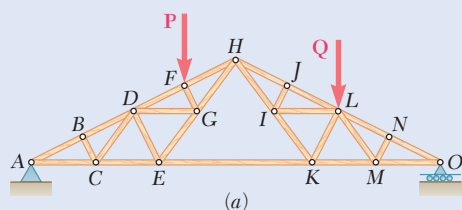
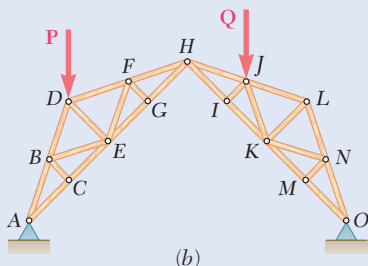


Fig. P6.28



(a)



(b)

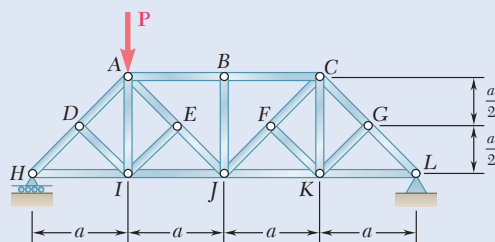
Fig. P6.31

6.29 Determine whether the trusses of Probs. 6.31a, 6.32a, and 6.33a are simple trusses.

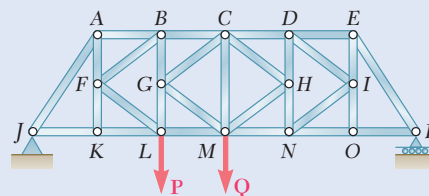
6.30 Determine whether the trusses of Probs. 6.31b, 6.32b, and 6.33b are simple trusses.

6.31 For the given loading, determine the zero-force members in each of the two trusses shown.

6.32 For the given loading, determine the zero-force members in each of the two trusses shown.



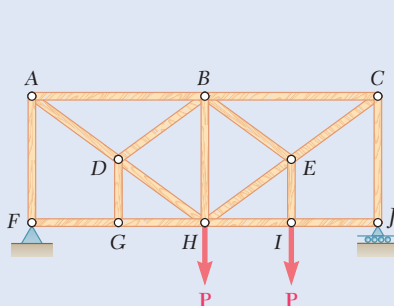
(a)



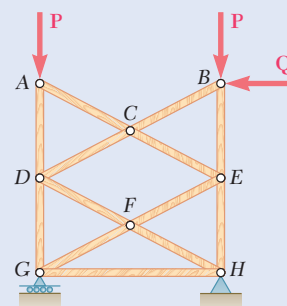
(b)

Fig. P6.32

6.33 For the given loading, determine the zero-force members in each of the two trusses shown.



(a)



(b)

Fig. P6.33

6.34 Determine the zero-force members in the truss of (a) Prob. 6.21, (b) Prob. 6.27.

***6.35** The truss shown consists of six members and is supported by a short link at A , two short links at B , and a ball-and-socket at D . Determine the force in each of the members for the given loading.

***6.36** The truss shown consists of six members and is supported by a ball-and-socket at B , a short link at C , and two short links at D . Determine the force in each of the members for $\mathbf{P} = (-2184 \text{ N})\mathbf{j}$ and $\mathbf{Q} = 0$.

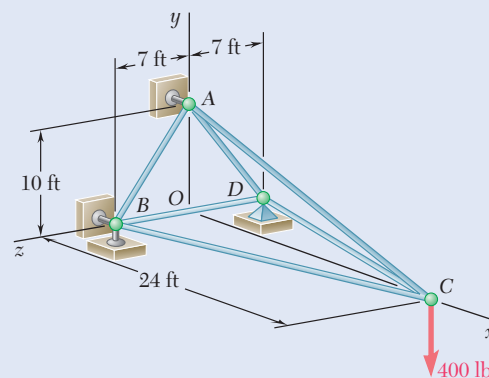


Fig. P6.35

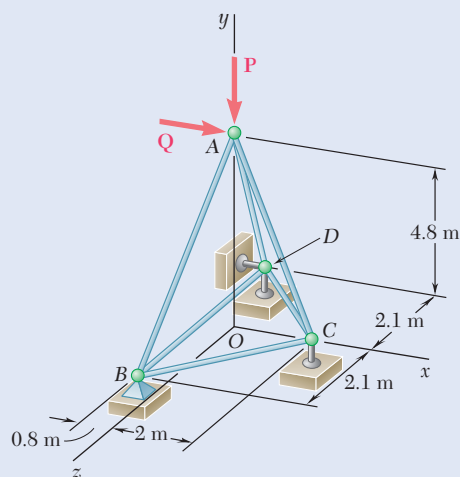


Fig. P6.36 and P6.37

***6.37** The truss shown consists of six members and is supported by a ball-and-socket at B , a short link at C , and two short links at D . Determine the force in each of the members for $\mathbf{P} = 0$ and $\mathbf{Q} = (2968 \text{ N})\mathbf{i}$.

***6.38** The truss shown consists of nine members and is supported by a ball-and-socket at A , two short links at B , and a short link at C . Determine the force in each of the members for the given loading.

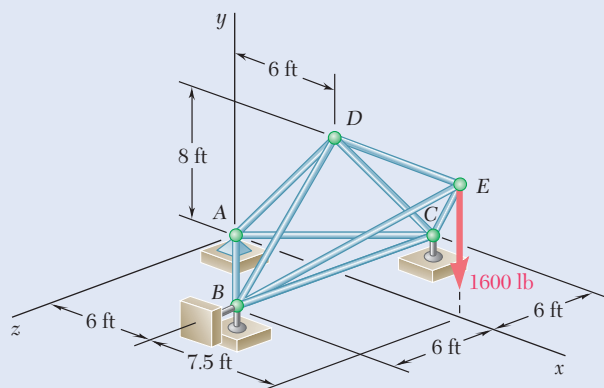


Fig. P6.38

- *6.39** The truss shown consists of nine members and is supported by a ball-and-socket at B , a short link at C , and two short links at D . (a) Check that this truss is a simple truss, that it is completely constrained, and that the reactions at its supports are statically determinate. (b) Determine the force in each member for $\mathbf{P} = (-1200 \text{ N})\mathbf{j}$ and $\mathbf{Q} = 0$.

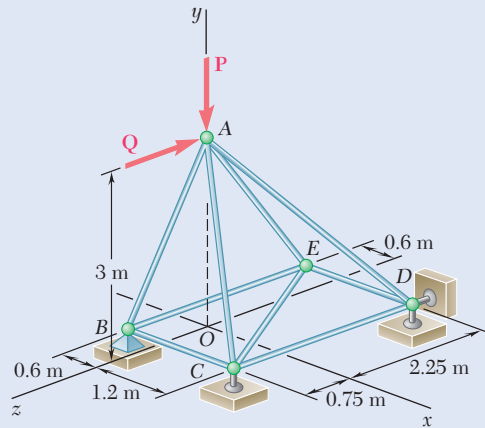


Fig. P6.39

- *6.40** Solve Prob. 6.39 for $\mathbf{P} = 0$ and $\mathbf{Q} = (-900 \text{ N})\mathbf{k}$.
- *6.41** The truss shown consists of 18 members and is supported by a ball-and-socket at A , two short links at B , and one short link at G . (a) Check that this truss is a simple truss, that it is completely constrained, and that the reactions at its supports are statically determinate. (b) For the given loading, determine the force in each of the six members joined at E .

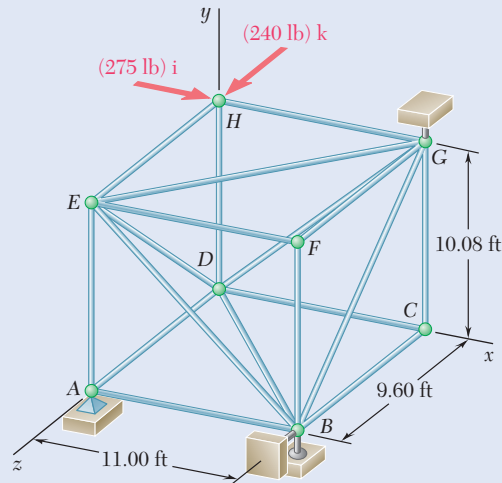


Fig. P6.41 and P6.42

- *6.42** The truss shown consists of 18 members and is supported by a ball-and-socket at A , two short links at B , and one short link at G . (a) Check that this truss is a simple truss, that it is completely constrained, and that the reactions at its supports are statically determinate. (b) For the given loading, determine the force in each of the six members joined at G .

6.2 OTHER TRUSS ANALYSES

The method of joints is most effective when we want to determine the forces in all the members of a truss. If, however, we need to determine the force in only one member or in a very few members, the method of sections is more efficient.

6.2A The Method of Sections

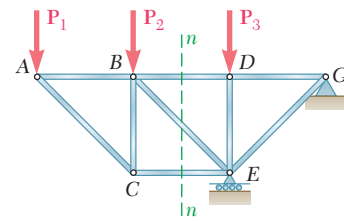
Assume, for example, that we want to determine the force in member BD of the truss shown in Fig. 6.15a. To do this, we must determine the force with which member BD acts on either joint B or joint D . If we were to use the method of joints, we would choose either joint B or joint D as a free body. However, we can also choose a larger portion of the truss that is composed of several joints and members, provided that the force we want to find is one of the external forces acting on that portion. If, in addition, we choose the portion of the truss as a free body where a total of only three unknown forces act upon it, we can obtain the desired force by solving the equations of equilibrium for this portion of the truss. In practice, we isolate a portion of the truss by *passing a section* through three members of the truss, one of which is the desired member. That is, we draw a line that divides the truss into two completely separate parts but does not intersect more than three members. We can then use as a free body either of the two portions of the truss obtained after the intersected members have been removed.[†]

In Fig. 6.15a, we have passed the section nn through members BD , BE , and CE , and we have chosen the portion ABC of the truss as the free body (Fig. 6.15b). The forces acting on this free body are the loads $\mathbf{P}_1$ and $\mathbf{P}_2$ at points A and B and the three unknown forces $\mathbf{F}_{BD}$, $\mathbf{F}_{BE}$, and $\mathbf{F}_{CE}$. Since we do not know whether the members removed are in tension or compression, we have arbitrarily drawn the three forces away from the free body as if the members are in tension.

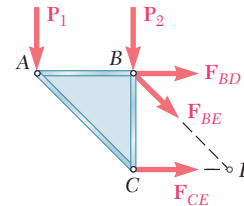
We use the fact that the rigid body ABC is in equilibrium to write three equations that we can solve for the three unknown forces. If we want to determine only force $\mathbf{F}_{BD}$, say, we need write only one equation, provided that the equation does not contain the other unknowns. Thus, the equation $\Sigma M_E = 0$ yields the value of the magnitude F_{BD} (Fig. 6.15b). A positive sign in the answer will indicate that our original assumption regarding the sense of $\mathbf{F}_{BD}$ was correct and that member BD is in tension; a negative sign will indicate that our assumption was incorrect and that BD is in compression.

On the other hand, if we want to determine only force $\mathbf{F}_{CE}$, we need to write an equation that does not involve $\mathbf{F}_{BD}$ or $\mathbf{F}_{BE}$; the appropriate equation is $\Sigma M_B = 0$. Again, a positive sign for the magnitude F_{CE} of the desired force indicates a correct assumption, that is, tension; and a negative sign indicates an incorrect assumption, that is, compression.

If we want to determine only force $\mathbf{F}_{BE}$, the appropriate equation is $\Sigma F_y = 0$. Whether the member is in tension or compression is again determined from the sign of the answer.



(a)



(b)

Fig. 6.15 (a) We can pass a section nn through the truss, dividing the three members BD , BE , and CE . (b) Free-body diagram of portion ABC of the truss. We assume that members BD , BE , and CE are in tension.

[†]In the analysis of some trusses, we can pass sections through more than three members, provided we can write equilibrium equations involving only one unknown that we can use to determine the forces in one, or possibly two, of the intersected members. See Probs. 6.61 through 6.64.

If we determine the force in only one member, no independent check of the computation is available. However, if we calculate all of the unknown forces acting on the free body, we can check the computations by writing an additional equation. For instance, if we determine $\mathbf{F}_{BD}$, $\mathbf{F}_{BE}$, and $\mathbf{F}_{CE}$ as indicated previously, we can check the work by verifying that $\Sigma F_x = 0$.

6.2B Trusses Made of Several Simple Trusses

Consider two simple trusses ABC and DEF . If we connect them by three bars BD , BE , and CE as shown in Fig. 6.16a, together they form a rigid truss $ABDF$. We can also combine trusses ABC and DEF into a single rigid truss by joining joints B and D at a single joint B and connecting joints C and E by a bar CE (Fig. 6.16b). This is known as a *Fink truss*. The trusses of Fig. 6.16a and b are *not* simple trusses; you cannot construct them from a triangular truss by adding successive pairs of members as described in Sec. 6.1A. They are rigid trusses, however, as you can check by comparing the systems of connections used to hold the simple trusses ABC and DEF together (three bars in Fig. 6.16a, one pin and one bar in Fig. 6.16b) with the systems of supports discussed in Sec. 4.1. Trusses made of several simple trusses rigidly connected are known as **compound trusses**.

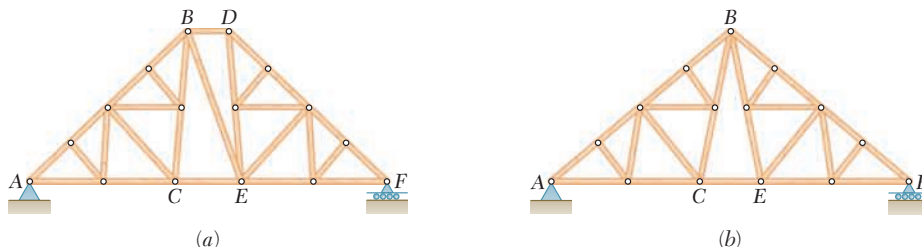


Fig. 6.16 Compound trusses. (a) Two simple trusses ABC and DEF connected by three bars. (b) Two simple trusses ABC and DEF connected by one joint and one bar (a Fink truss).

In a compound truss, the number of members m and the number of joints n are still related by the formula $m = 2n - 3$. You can verify this by observing that if a compound truss is supported by a frictionless pin and a roller (involving three unknown reactions), the total number of unknowns is $m + 3$, and this number must be equal to the number $2n$ of equations obtained by expressing that the n pins are in equilibrium. It follows that $m = 2n - 3$.

Compound trusses supported by a pin and a roller or by an equivalent system of supports are *statically determinate*, *rigid*, and *completely constrained*. This means that we can determine all of the unknown reactions and the forces in all of the members by using the methods of statics, and the truss will neither collapse nor move. However, the only way to determine all of the forces in the members using the method of joints requires solving a large number of simultaneous equations. In the case of the compound truss of Fig. 6.16a, for example, it is more efficient to pass

a section through members BD , BE , and CE to determine the forces in these members.

Suppose, now, that the simple trusses ABC and DEF are connected by *four* bars; BD , BE , CD , and CE (Fig. 6.17). The number of members m is now larger than $2n - 3$. This truss is said to be **overrigid**, and one of the four members BD , BE , CD , or CE is **redundant**. If the truss is supported by a pin at A and a roller at F , the total number of unknowns is $m + 3$. Since $m > 2n - 3$, the number $m + 3$ of unknowns is now larger than the number $2n$ of available independent equations; the truss is *statically indeterminate*.

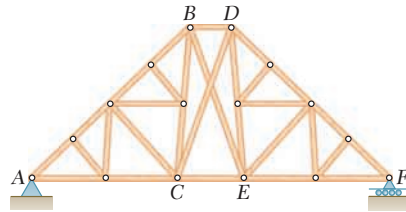


Fig. 6.17 A statically indeterminate, overrigid compound truss, due to a redundant member.

Finally, let us assume that the two simple trusses ABC and DEF are joined by a single pin, as shown in Fig. 6.18a. The number of members, m , is now smaller than $2n - 3$. If the truss is supported by a pin at A and a roller at F , the total number of unknowns is $m + 3$. Since $m < 2n - 3$, the number $m + 3$ of unknowns is now smaller than the number $2n$ of equilibrium equations that need to be satisfied. This truss is **nonrigid** and will collapse under its own weight. However, if two pins are used to support it, the truss becomes *rigid* and will not collapse (Fig. 6.18b). Note that the total number of unknowns is now $m + 4$ and is equal to the number $2n$ of equations.

More generally, if the reactions at the supports involve r unknowns, the condition for a compound truss to be statically determinate, rigid, and completely constrained is $m + r = 2n$. However, although this condition is necessary, it is not sufficient for the equilibrium of a structure that ceases to be rigid when detached from its supports (see Sec. 6.3B).

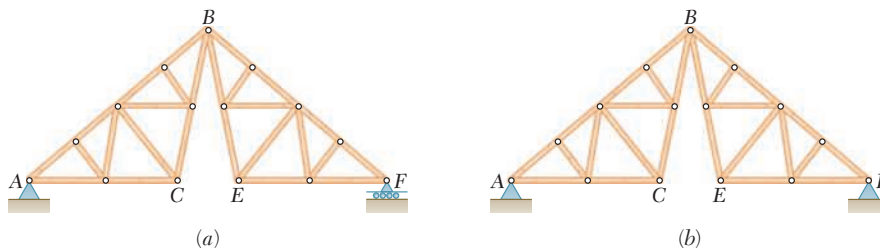
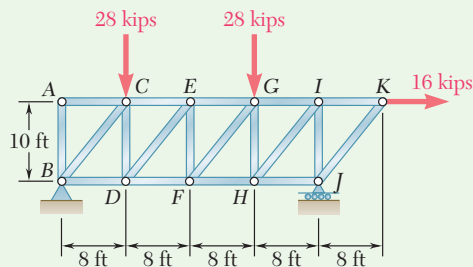


Fig. 6.18 Two simple trusses joined by a pin. (a) Supported by a pin and a roller, the truss will collapse under its own weight. (b) Supported by two pins, the truss becomes rigid and does not collapse.

Sample Problem 6.2

Determine the forces in members EF and GI of the truss shown.



STRATEGY: You are asked to determine the forces in only two of the members in this truss, so the method of sections is more appropriate than the method of joints. You can use a free-body diagram of the entire truss to help determine the reactions, and then pass sections through the truss to isolate parts of it for calculating the desired forces.

MODELING and ANALYSIS: You can go through the steps that follow for the determination of the support reactions, and then for the analysis of portions of the truss.

Free-Body: Entire Truss. Draw a free-body diagram of the entire truss. External forces acting on this free body consist of the applied loads and the reactions at B and J (Fig. 1). Write and solve the following equilibrium equations.

$$+\circlearrowleft \sum M_B = 0:$$

$$-(28 \text{ kips})(8 \text{ ft}) - (28 \text{ kips})(24 \text{ ft}) - (16 \text{ kips})(10 \text{ ft}) + J(32 \text{ ft}) = 0$$

$$J = +33 \text{ kips} \quad \mathbf{J} = 33 \text{ kips} \uparrow$$

$$\rightarrow \sum F_x = 0: \quad B_x + 16 \text{ kips} = 0$$

$$B_x = -16 \text{ kips} \quad \mathbf{B}_x = 16 \text{ kips} \leftarrow$$

$$+\circlearrowleft \sum M_J = 0:$$

$$(28 \text{ kips})(24 \text{ ft}) + (28 \text{ kips})(8 \text{ ft}) - (16 \text{ kips})(10 \text{ ft}) - B_y(32 \text{ ft}) = 0$$

$$B_y = +23 \text{ kips} \quad \mathbf{B}_y = 23 \text{ kips} \uparrow$$

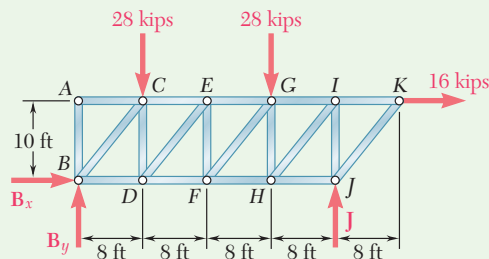


Fig. 1 Free-body diagram of entire truss.

Force in Member EF . Pass section mn through the truss diagonally so that it intersects member EF and only two additional members (Fig. 2).

Remove the intersected members and choose the left-hand portion of the truss as a free body (Fig. 3). Three unknowns are involved; to eliminate the two horizontal forces, we write

$$+\uparrow \Sigma F_y = 0: \quad +23 \text{ kips} - 28 \text{ kips} - F_{EF} = 0$$

$$F_{EF} = -5 \text{ kips}$$

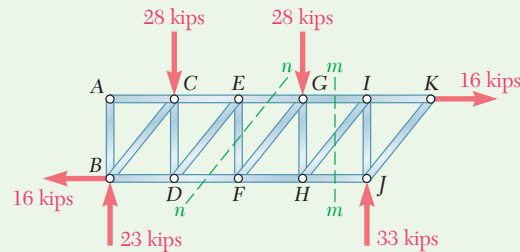


Fig. 2 Sections nn and mm that will be used to analyze members EF and GI .

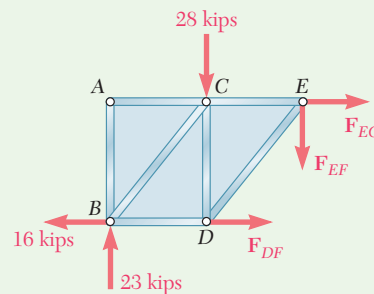


Fig. 3 Free-body diagram to analyze member EF .

The sense of F_{EF} was chosen assuming member EF to be in tension; the negative sign indicates that the member is in compression.

$$F_{EF} = 5 \text{ kips } C \quad \blacktriangleleft$$

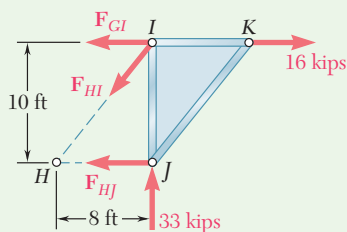


Fig. 4 Free-body diagram to analyze member GI .

Force in Member GI . Pass section mm through the truss vertically so that it intersects member GI and only two additional members (Fig. 2). Remove the intersected members and choose the right-hand portion of the truss as a free body (Fig. 4). Again, three unknown forces are involved; to eliminate the two forces passing through point H , sum the moments about that point.

$$+\curvearrowright \Sigma M_H = 0: \quad (33 \text{ kips})(8 \text{ ft}) - (16 \text{ kips})(10 \text{ ft}) + F_{GI}(10 \text{ ft}) = 0$$

$$F_{GI} = -10.4 \text{ kips} \quad F_{GI} = 10.4 \text{ kips } C \quad \blacktriangleleft$$

REFLECT and THINK: Note that a section passed through a truss does not have to be vertical or horizontal; it can be diagonal as well. Choose the orientation that cuts through no more than three members of unknown force and also gives you the simplest part of the truss for which you can write equilibrium equations and determine the unknowns.

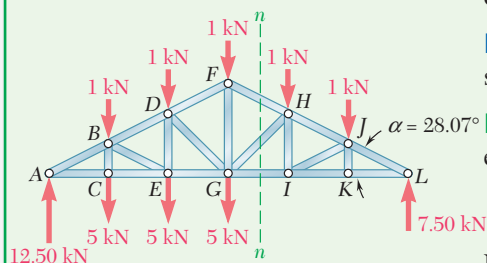
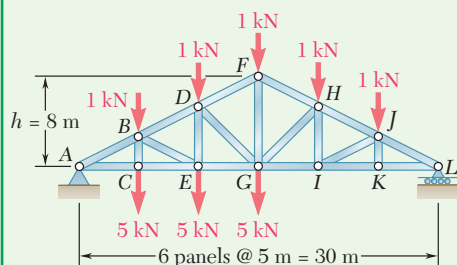


Fig. 1 Free-body diagram of entire truss.

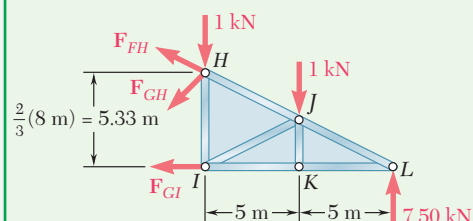


Fig. 2 Free-body diagram to analyze member GI .

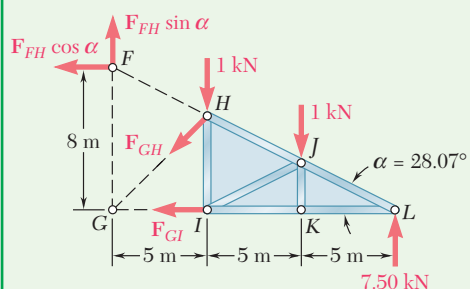


Fig. 3 Simplifying the analysis of member FH by first sliding its force to point F .

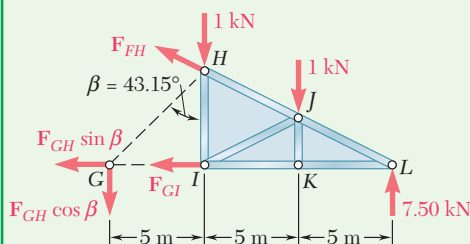


Fig. 4 Simplifying the analysis of member GH by first sliding its force to point G .

Sample Problem 6.3

Determine the forces in members FH , GH , and GI of the roof truss shown.

STRATEGY: You are asked to determine the forces in only three members of the truss, so use the method of sections. Determine the reactions by treating the entire truss as a free body and then isolate part of it for analysis. In this case, you can use the same smaller part of the truss to determine all three desired forces.

MODELING and ANALYSIS: Your reasoning and computation should go something like the sequence given here.

Free Body: Entire Truss. From the free-body diagram of the entire truss (Fig. 1), find the reactions at A and L :

$$A = 12.50 \text{ kN} \uparrow \quad L = 7.50 \text{ kN} \uparrow$$

Note that

$$\tan \alpha = \frac{FG}{GL} = \frac{8 \text{ m}}{15 \text{ m}} = 0.5333 \quad \alpha = 28.07^\circ$$

Force in Member GI . Pass section nn vertically through the truss (Fig. 1). Using the portion HLI of the truss as a free body (Fig. 2), obtain the value of F_{GI} :

$$+\circlearrowleft \Sigma M_H = 0: \quad (7.50 \text{ kN})(10 \text{ m}) - (1 \text{ kN})(5 \text{ m}) - F_{GI}(5.33 \text{ m}) = 0$$

$$F_{GI} = +13.13 \text{ kN} \quad F_{GI} = 13.13 \text{ kN } T \quad \blacktriangleleft$$

Force in Member FH . Determine the value of F_{FH} from the equation $\Sigma M_G = 0$. To do this, move F_{FH} along its line of action until it acts at point F , where you can resolve it into its x and y components (Fig. 3). The moment of F_{FH} with respect to point G is now $(F_{FH} \cos \alpha)(8 \text{ m})$.

$$+\circlearrowleft \Sigma M_G = 0:$$

$$(7.50 \text{ kN})(15 \text{ m}) - (1 \text{ kN})(10 \text{ m}) - (1 \text{ kN})(5 \text{ m}) + (F_{FH} \cos \alpha)(8 \text{ m}) = 0$$

$$F_{FH} = -13.81 \text{ kN} \quad F_{FH} = 13.81 \text{ kN } C \quad \blacktriangleleft$$

Force in Member GH . First note that

$$\tan \beta = \frac{GI}{HI} = \frac{5 \text{ m}}{\frac{2}{3}(8 \text{ m})} = 0.9375 \quad \beta = 43.15^\circ$$

Then determine the value of F_{GH} by resolving the force F_{GH} into x and y components at point G (Fig. 4) and solving the equation $\Sigma M_L = 0$.

$$+\circlearrowleft \Sigma M_L = 0: \quad (1 \text{ kN})(10 \text{ m}) + (1 \text{ kN})(5 \text{ m}) + (F_{GH} \cos \beta)(15 \text{ m}) = 0$$

$$F_{GH} = -1.371 \text{ kN} \quad F_{GH} = 1.371 \text{ kN } C \quad \blacktriangleleft$$

REFLECT and THINK: Sometimes you should resolve a force into components to include it in the equilibrium equations. By first sliding this force along its line of action to a more strategic point, you might eliminate one of its components from a moment equilibrium equation.

SOLVING PROBLEMS ON YOUR OWN

The **method of joints** that you studied in Sec. 6.1 is usually the best method to use when you need to find the forces *in all of the members* of a simple truss. However, the **method of sections**, which was covered in this section, is more efficient when you need to find the force *in only one member* or the forces *in a very few members* of a simple truss. The method of sections also must be used when the truss *is not a simple truss*.

A. To determine the force in a given truss member by the method of sections, follow these steps:

1. Draw a free-body diagram of the entire truss, and use this diagram to determine the reactions at the supports.

2. Pass a section through three members of the truss, one of which is the member whose force you want to find. After you cut through these members, you will have two separate portions of truss.

3. Select one of these two portions of truss and draw its free-body diagram. This diagram should include the external forces applied to the selected portion as well as the forces exerted on it by the intersected members that were removed.

4. You can now write three equilibrium equations that can be solved for the forces in the three intersected members.

5. An alternative approach is to write a single equation that can be solved for the force in the desired member. To do so, first observe whether the forces exerted by the other two members on the free body are parallel or whether their lines of action intersect.

a. If these forces are parallel, you can eliminate them by writing an equilibrium equation involving *components in a direction perpendicular* to these two forces.

b. If their lines of action intersect at a point H , you can eliminate them by writing an equilibrium equation involving *moments about H* .

6. Keep in mind that the section you use must intersect three members only. The reason is that the equilibrium equations in Step 4 can be solved for only three unknowns. However, you can pass a section through more than three members to find the force in one of those members if you can write an equilibrium equation containing only that force as an unknown. Such special situations are found in Probs. 6.61 through 6.64.

B. About completely constrained and determinate trusses:

1. Any simple truss that is simply supported is a completely constrained and determinate truss.

2. To determine whether any other truss is or is not completely constrained and determinate, count the number m of its members, the number n of its joints, and the number r of the reaction components at its supports. Compare the sum $m + r$ representing the number of unknowns and the product $2n$ representing the number of available independent equilibrium equations.

a. If $m + r < 2n$, there are fewer unknowns than equations. Thus, some of the equations cannot be satisfied, and the truss is only *partially constrained*.

b. If $m + r > 2n$, there are more unknowns than equations. Thus, some of the unknowns cannot be determined, and the truss is *indeterminate*.

c. If $m + r = 2n$, there are as many unknowns as there are equations. This, however, does not mean that all of the unknowns can be determined and that all of the equations can be satisfied. To find out whether the truss is *completely* or *improperly constrained*, try to determine the reactions at its supports and the forces in its members. If you can find all of them, the truss is *completely constrained and determinate*.

Problems

- 6.43** A Mansard roof truss is loaded as shown. Determine the force in members DF , DG , and EG .

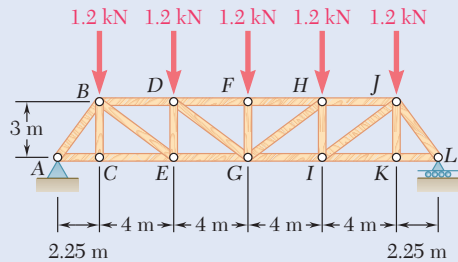


Fig. P6.43 and P6.44

- 6.44** A Mansard roof truss is loaded as shown. Determine the force in members GI , HI , and HJ .

- 6.45** Determine the force in members BD and CD of the truss shown.

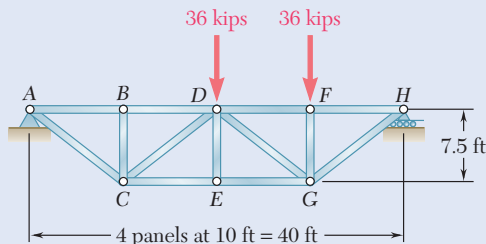


Fig. P6.45 and P6.46

- 6.46** Determine the force in members DF and DG of the truss shown.

- 6.47** Determine the force in members CD and DF of the truss shown.

- 6.48** Determine the force in members FG and FH of the truss shown.

- 6.49** Determine the force in members CD and DF of the truss shown.

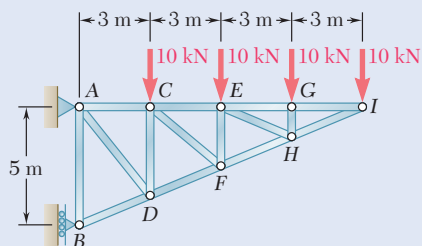


Fig. P6.49 and P6.50

- 6.50** Determine the force in members CE and EF of the truss shown.

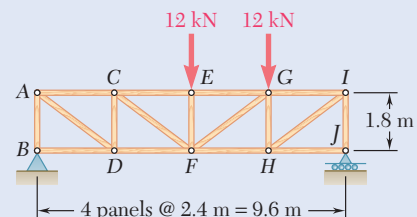


Fig. P6.47 and P6.48

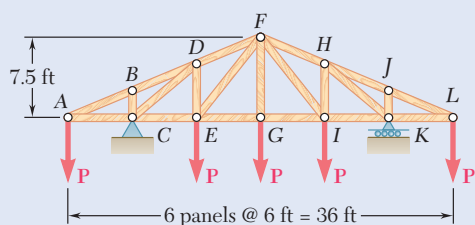


Fig. P6.51 and P6.52

6.51 Determine the force in members DE and DF of the truss shown when $P = 20$ kips.

6.52 Determine the force in members EG and EF of the truss shown when $P = 20$ kips.

6.53 Determine the force in members DF and DE of the truss shown.

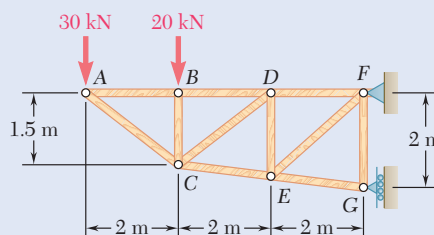


Fig. P6.53 and P6.54

6.54 Determine the force in members CD and CE of the truss shown.

6.55 A monosloped roof truss is loaded as shown. Determine the force in members CE , DE , and DF .

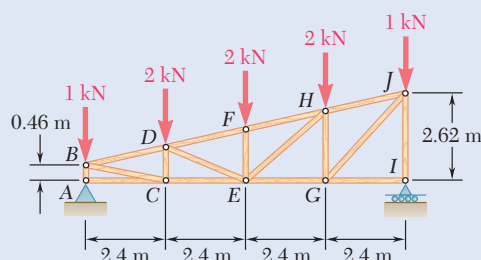


Fig. P6.55 and P6.56

6.56 A monosloped roof truss is loaded as shown. Determine the force in members EG , GH , and HJ .

6.57 A Howe scissors roof truss is loaded as shown. Determine the force in members DF , DG , and EG .

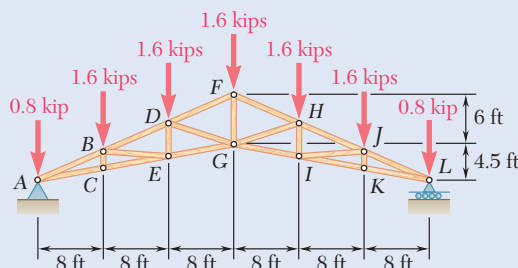


Fig. P6.57 and P6.58

6.58 A Howe scissors roof truss is loaded as shown. Determine the force in members GI , HI , and HJ .

6.59 Determine the force in members AD , CD , and CE of the truss shown.

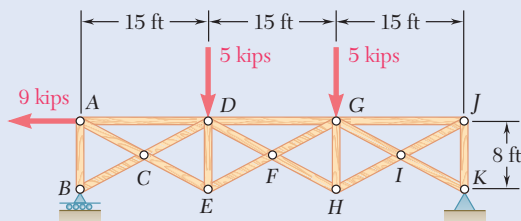


Fig. P6.59 and P6.60

6.60 Determine the force in members DG , FG , and FH of the truss shown.

6.61 Determine the force in members DG and FI of the truss shown. (Hint: Use section aa .)

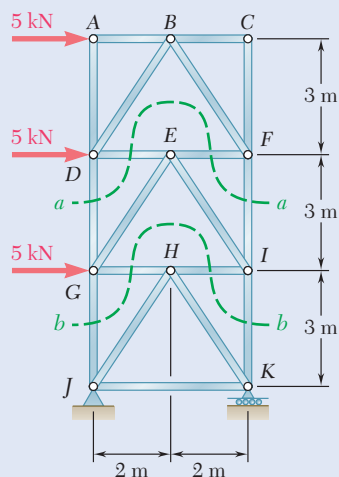


Fig. P6.61 and P6.62

6.62 Determine the force in members GJ and IK of the truss shown. (Hint: Use section bb .)

6.63 Determine the force in members EH and GI of the truss shown. (Hint: Use section aa .)

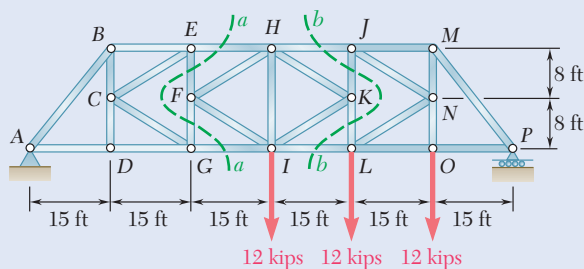


Fig. P6.63 and P6.64

6.64 Determine the force in members HJ and IL of the truss shown. (Hint: Use section bb .)

6.65 and 6.66 The diagonal members in the center panels of the power transmission line tower shown are very slender and can act only in tension; such members are known as *counters*. For the given loading, determine (a) which of the two counters listed below is acting, (b) the force in that counter.

6.65 Counters *CJ* and *HE*.

6.66 Counters *IO* and *KN*.

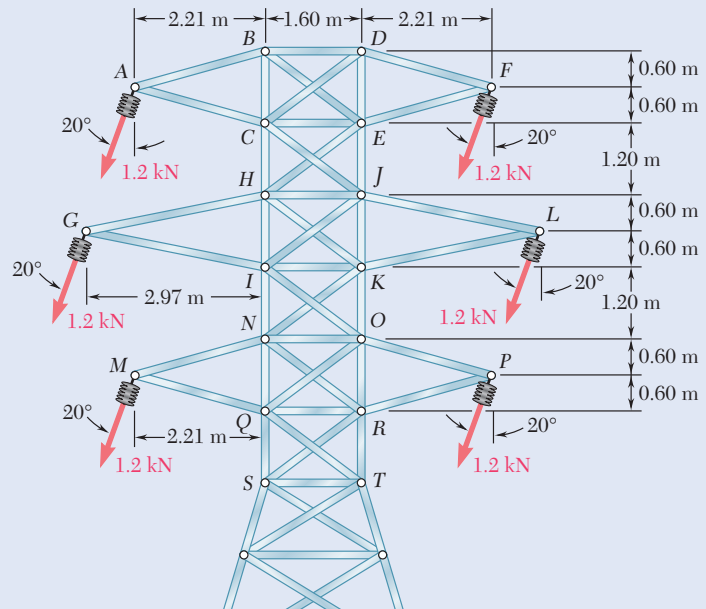


Fig. P6.65 and P6.66

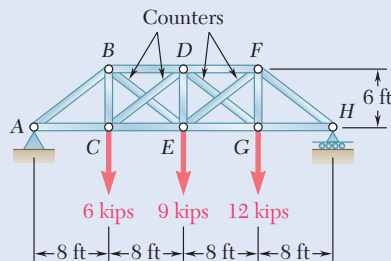


Fig. P6.67

6.67 The diagonal members in the center panels of the truss shown are very slender and can act only in tension; such members are known as *counters*. Determine the force in member *DE* and in the counters that are acting under the given loading.

6.68 Solve Prob. 6.67 assuming that the 9-kip load has been removed.

6.69 Classify each of the structures shown as completely, partially, or improperly constrained; if completely constrained, further classify as determinate or indeterminate. (All members can act both in tension and in compression.)

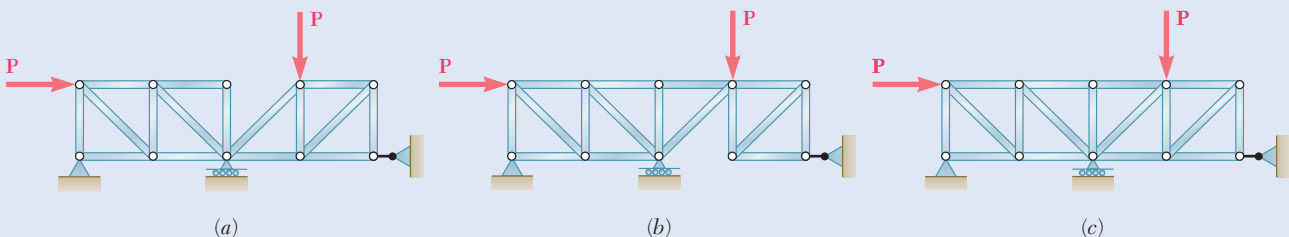


Fig. P6.69

6.70 through 6.74 Classify each of the structures shown as completely, partially, or improperly constrained; if completely constrained, further classify as determinate or indeterminate. (All members can act both in tension and in compression.)

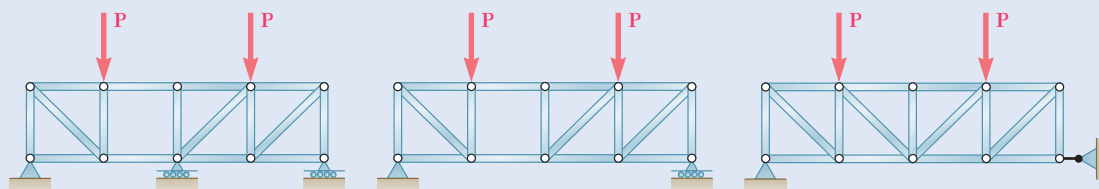


Fig. P6.70

(a)

(b)

(c)

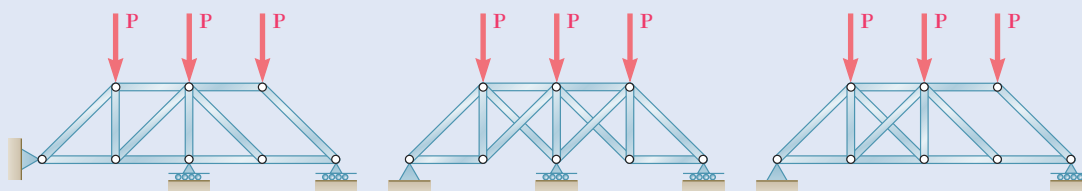


Fig. P6.71

(a)

(b)

(c)

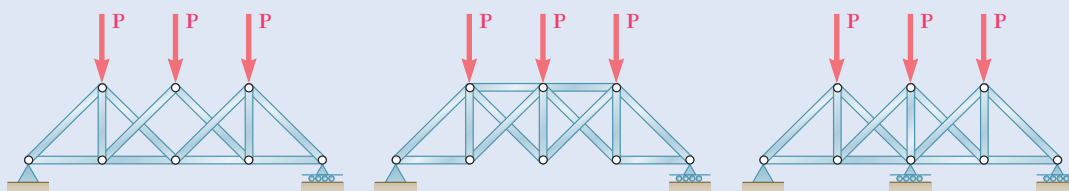


Fig. P6.72

(a)

(b)

(c)

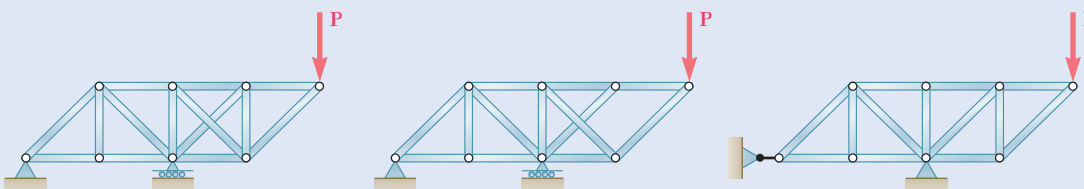


Fig. P6.73

(a)

(b)

(c)

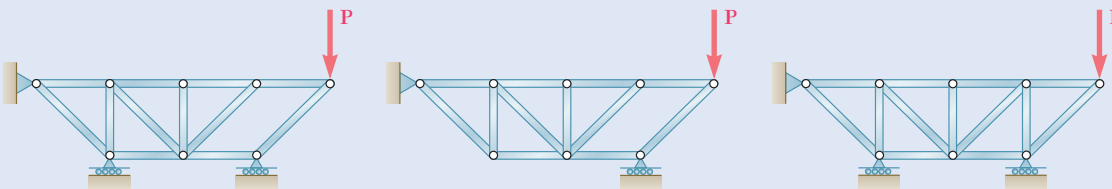


Fig. P6.74

(a)

(b)

(c)

6.3 FRAMES

When we study trusses, we are looking at structures consisting entirely of pins and straight two-force members. The forces acting on the two-force members are directed along the members themselves. We now consider structures in which at least one of the members is a *multi-force* member, i.e., a member acted upon by three or more forces. These forces are generally not directed along the members on which they act; their directions are unknown; therefore, we need to represent them by two unknown components.

Frames and machines are structures containing multi-force members. **Frames** are designed to support loads and are usually stationary, fully constrained structures. **Machines** are designed to transmit and modify forces; they may or may not be stationary and always contain moving parts.



Photo 6.6 Frames and machines contain multi-force members. Frames are fully constrained structures, whereas machines like this prosthetic hand are movable and designed to transmit or modify forces.

6.3A Analysis of a Frame

As the first example of analysis of a frame, we consider again the crane described in Sec. 6.1 that carries a given load W (Fig. 6.19a). The free-body diagram of the entire frame is shown in Fig. 6.19b. We can use this diagram to determine the external forces acting on the frame. Summing moments about A , we first determine the force T exerted by the cable; summing x and y components, we then determine the components A_x and A_y of the reaction at the pin A .

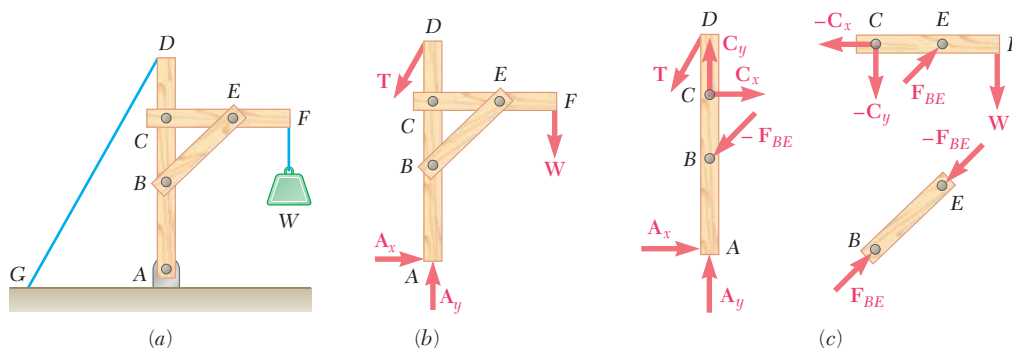


Fig. 6.19 A frame in equilibrium. (a) Diagram of a crane supporting a load; (b) free-body diagram of the crane; (c) free-body diagrams of the components of the crane.

In order to determine the internal forces holding the various parts of a frame together, we must dismember it and draw a free-body diagram for each of its component parts (Fig. 6.19c). First, we examine the two-force members. In this frame, member BE is the only two-force member. The forces acting at each end of this member must have the same magnitude, same line of action, and opposite sense (Sec. 4.2A). They are therefore directed along BE and are denoted, respectively, by $\mathbf{F}_{BE}$ and $-\mathbf{F}_{BE}$. We arbitrarily assume their sense as shown in Fig. 6.19c; the sign obtained for the common magnitude F_{BE} of the two forces will confirm or deny this assumption.

Next, we consider the multi-force members, i.e., the members that are acted upon by three or more forces. According to Newton's third law, the force exerted at B by member BE on member AD must be equal and opposite to the force $\mathbf{F}_{BE}$ exerted by AD on BE . Similarly, the force exerted at E by member BE on member CF must be equal and opposite to the force $-\mathbf{F}_{BE}$ exerted by CF on BE . Thus, the forces that the two-force member BE exerts on AD and CF are, respectively, equal to $-\mathbf{F}_{BE}$ and $\mathbf{F}_{BE}$; they have the same magnitude F_{BE} , opposite sense, and should be directed as shown in Fig. 6.19c.

Joint C connects two multi-force members. Since neither the direction nor the magnitude of the forces acting at C are known, we represent these forces by their x and y components. The components $\mathbf{C}_x$ and $\mathbf{C}_y$ of the force acting on member AD are arbitrarily directed to the right and upward. Since, according to Newton's third law, the forces exerted by member CF on AD and by member AD on CF are equal and opposite, the components of the force acting on member CF must be directed to the left and downward; we denote them, respectively, by $-\mathbf{C}_x$ and $-\mathbf{C}_y$. Whether the force $\mathbf{C}_x$ is actually directed to the right and the force $-\mathbf{C}_x$ is actually directed to the left will be determined later from the sign of their common magnitude C_x with a plus sign indicating that the assumption was correct and a minus sign that it was wrong. We complete the free-body diagrams of the multi-force members by showing the external forces acting at A , D , and F .[†]

We can now determine the internal forces by considering the free-body diagram of either of the two multi-force members. Choosing the free-body diagram of CF , for example, we write the equations $\Sigma M_C = 0$, $\Sigma M_E = 0$, and $\Sigma F_x = 0$, which yield the values of the magnitudes F_{BE} , C_y , and C_x , respectively. We can check these values by verifying that member AD is also in equilibrium.

Note that we assume the pins in Fig. 6.19 form an integral part of one of the two members they connected, so it is not necessary to show their free-body diagrams. We can always use this assumption to simplify the analysis of frames and machines. However, when a pin connects three

[†]It is not strictly necessary to use a minus sign to distinguish the force exerted by one member on another from the equal and opposite force exerted by the second member on the first, since the two forces belong to different free-body diagrams and thus are not easily confused. In the Sample Problems, we use the same symbol to represent equal and opposite forces that are applied to different free bodies. Note that, under these conditions, the sign obtained for a given force component does not directly relate the sense of that component to the sense of the corresponding coordinate axis. Rather, a positive sign indicates that *the sense assumed for that component in the free-body diagram* is correct, and a negative sign indicates that it is wrong.

or more members, connects a support and two or more members, or when a load is applied to a pin, we must make a clear decision in choosing the member to which we assume the pin belongs. (If multi-force members are involved, the pin should be attached to one of these members.) We then need to identify clearly the various forces exerted on the pin. This is illustrated in Sample Prob. 6.6.

6.3B Frames That Collapse Without Supports

The crane we just analyzed was constructed so that it could keep the same shape without the help of its supports; we therefore considered it to be a rigid body. Many frames, however, will collapse if detached from their supports; such frames cannot be considered rigid bodies. Consider, for example, the frame shown in Fig. 6.20a that consists of two members AC and CB carrying loads P and Q at their midpoints. The members are supported by pins at A and B and are connected by a pin at C . If we detach this frame from its supports, it will not maintain its shape. Therefore, we should consider it to be made of *two distinct rigid parts* AC and CB .

The equations $\Sigma F_x = 0$, $\Sigma F_y = 0$, and $\Sigma M = 0$ (about any given point) express the conditions for the *equilibrium of a rigid body* (Chap. 4); we should use them, therefore, in connection with the free-body diagrams of members AC and CB (Fig. 6.20b). Since these members are multi-force members and since pins are used at the supports and at the connection, we represent each of the reactions at A and B and the forces at C by two components. In accordance with Newton's third law, we represent the components of the force exerted by CB on AC and the components of the force exerted by AC on CB by vectors of the same magnitude and opposite sense. Thus, if the first pair of components consists of C_x and C_y , the second pair is represented by $-C_x$ and $-C_y$.

Note that four unknown force components act on free body AC , whereas we need only three independent equations to express that the body is in equilibrium. Similarly, four unknowns, but only three equations, are associated with CB . However, only six different unknowns are involved in the analysis of the two members, and altogether, six equations

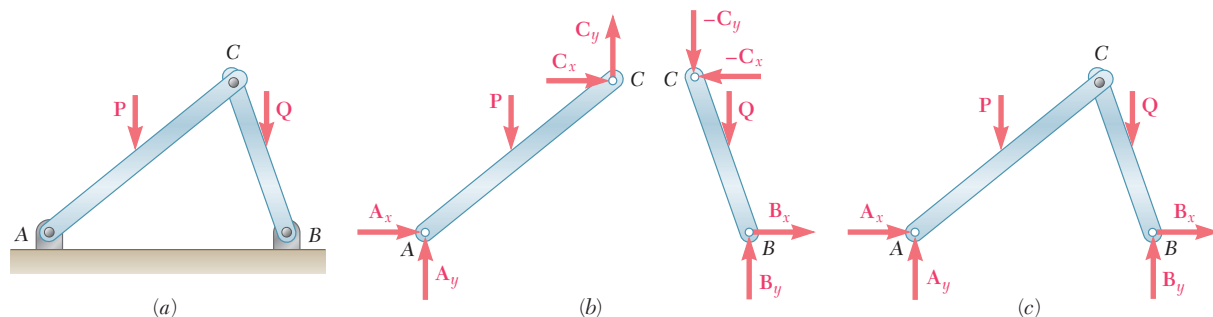


Fig. 6.20 (a) A frame of two members supported by two pins and joined together by a third pin. Without the supports, the frame would collapse and is therefore not a rigid body. (b) Free-body diagrams of the two members. (c) Free-body diagram of the whole frame.

are available to express that the members are in equilibrium. Setting $\Sigma M_A = 0$ for free body AC and $\Sigma M_B = 0$ for CB , we obtain two simultaneous equations that we can solve for the common magnitude C_x of the components C_x and $-C_x$ and for the common magnitude C_y of the components C_y and $-C_y$. We then have $\Sigma F_x = 0$ and $\Sigma F_y = 0$ for each of the two free bodies, successively obtaining the magnitudes A_x , A_y , B_x , and B_y .

Observe that, since the equations of equilibrium $\Sigma F_x = 0$, $\Sigma F_y = 0$, and $\Sigma M = 0$ (about any given point) are satisfied by the forces acting on free body AC and since they are also satisfied by the forces acting on free body CB , they must be satisfied when the forces acting on the two free bodies are considered simultaneously. Since the internal forces at C cancel each other, we find that the equations of equilibrium must be satisfied by the external forces shown on the free-body diagram of the frame ACB itself (Fig. 6.20c), even though the frame is not a rigid body. We can use these equations to determine some of the components of the reactions at A and B . We will find, however, that **the reactions cannot be completely determined from the free-body diagram of the whole frame**. It is thus necessary to dismember the frame and consider the free-body diagrams of its component parts (Fig. 6.20b), even when we are interested in determining external reactions only. The reason is that the equilibrium equations obtained for free body ACB are *necessary conditions* for the equilibrium of a nonrigid structure, *but these are not sufficient conditions*.

The method of solution outlined here involved simultaneous equations. We now present a more efficient method that utilizes the free body ACB as well as the free bodies AC and CB . Writing $\Sigma M_A = 0$ and $\Sigma M_B = 0$ for free body ACB , we obtain B_y and A_y . From $\Sigma M_C = 0$, $\Sigma F_x = 0$, and $\Sigma F_y = 0$ for free body AC , we successively obtain A_x , C_x , and C_y . Finally, setting $\Sigma F_x = 0$ for ACB gives us B_x .

We noted previously that the analysis of the frame in Fig. 6.20 involves six unknown force components and six independent equilibrium equations. (The equilibrium equations for the whole frame were obtained from the original six equations and, therefore, are not independent.) Moreover, we checked that all unknowns could be actually determined and that all equations could be satisfied. This frame is **statically determinate and rigid**. (We use the word “rigid” here to indicate that the frame maintains its shape as long as it remains attached to its supports.) In general, to determine whether a structure is statically determinate and rigid, you should draw a free-body diagram for each of its component parts and count the reactions and internal forces involved. You should then determine the number of independent equilibrium equations (excluding equations expressing the equilibrium of the whole structure or of groups of component parts already analyzed). If you have more unknowns than equations, the structure is *statically indeterminate*. If you have fewer unknowns than equations, the structure is *nonrigid*. If you have as many unknowns as equations *and if all unknowns can be determined and all equations satisfied* under general loading conditions, the structure is statically determinate and rigid. If, however, due to an improper arrangement of members and supports, all unknowns cannot be determined and all equations cannot be satisfied, the structure is **statically indeterminate and nonrigid**.

Sample Problem 6.4

In the frame shown, members ACE and BCD are connected by a pin at C and by the link DE . For the loading shown, determine the force in link DE and the components of the force exerted at C on member BCD .

STRATEGY: Follow the general procedure discussed in this section. First treat the entire frame as a free body, which will enable you to find the reactions at A and B . Then dismember the frame and treat each member as a free body, which will give you the equations needed to find the force at C .

MODELING and ANALYSIS: Since the external reactions involve only three unknowns, compute the reactions by considering the free-body diagram of the entire frame (Fig. 1).

$$\begin{aligned}
 +\uparrow \Sigma F_y = 0: & \quad A_y - 480 \text{ N} = 0 & A_y = +480 \text{ N} & \quad A_y = 480 \text{ N} \uparrow \\
 +\uparrow \Sigma M_A = 0: & \quad -(480 \text{ N})(100 \text{ mm}) + B(160 \text{ mm}) = 0 & B = +300 \text{ N} & \quad B = 300 \text{ N} \rightarrow \\
 \rightarrow \Sigma F_x = 0: & \quad B + A_x = 0 & 300 \text{ N} + A_x = 0 & A_x = -300 \text{ N} & A_x = 300 \text{ N} \leftarrow
 \end{aligned}$$

Now dismember the frame (Figs. 2 and 3). Since only two members are connected at C , the components of the unknown forces acting on ACE and BCD are, respectively, equal and opposite. Assume that link DE is in tension (Fig. 3) and exerts equal and opposite forces at D and E , directed as shown.

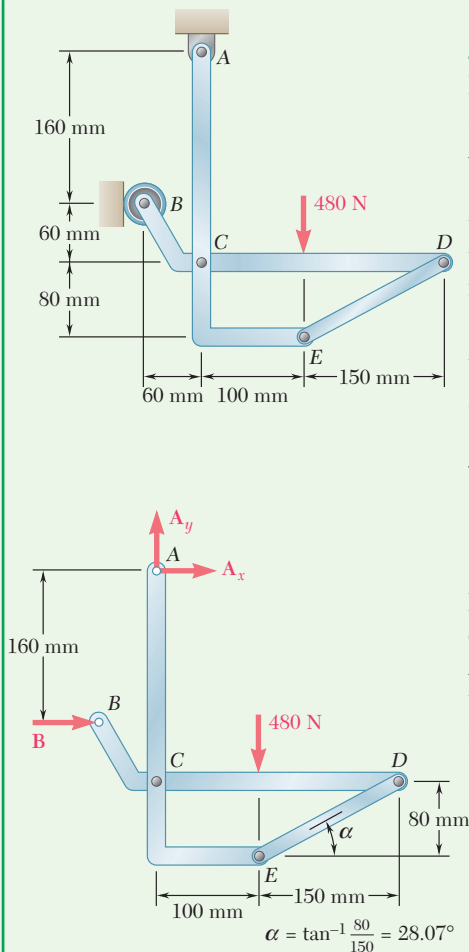


Fig. 1 Free-body diagram of entire frame.

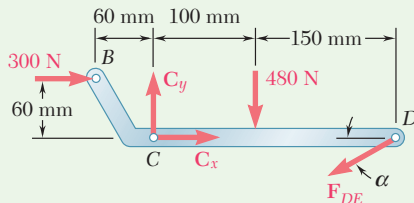


Fig. 2 Free-body diagram of member BCD .

Free Body: Member BCD . Using the free body BCD (Fig. 2), you can write and solve three equilibrium equations:

$$\begin{aligned}
 +\uparrow \Sigma M_C = 0: & \quad (F_{DE} \sin \alpha)(250 \text{ mm}) + (300 \text{ N})(80 \text{ mm}) + (480 \text{ N})(100 \text{ mm}) = 0 \\
 & \quad F_{DE} = -561 \text{ N} & F_{DE} = 561 \text{ N} \text{ C} \quad \blacktriangleleft \\
 \rightarrow \Sigma F_x = 0: & \quad C_x - F_{DE} \cos \alpha + 300 \text{ N} = 0 \\
 & \quad C_x - (-561 \text{ N}) \cos 28.07^\circ + 300 \text{ N} = 0 & C_x = -795 \text{ N} \\
 +\uparrow \Sigma F_y = 0: & \quad C_y - F_{DE} \sin \alpha - 480 \text{ N} = 0 \\
 & \quad C_y - (-561 \text{ N}) \sin 28.07^\circ - 480 \text{ N} = 0 & C_y = +216 \text{ N}
 \end{aligned}$$

From the signs obtained for C_x and C_y , the force components C_x and C_y exerted on member BCD are directed to the left and up, respectively. Thus, you have

$$C_x = 795 \text{ N} \leftarrow, C_y = 216 \text{ N} \uparrow \quad \blacktriangleleft$$

REFLECT and THINK: Check the computations by considering the free body ACE (Fig. 3). For example,

$$\begin{aligned}
 +\uparrow \Sigma M_A = & (F_{DE} \cos \alpha)(300 \text{ mm}) + (F_{DE} \sin \alpha)(100 \text{ mm}) - C_x(220 \text{ mm}) \\
 = & (-561 \cos \alpha)(300) + (-561 \sin \alpha)(100) - (-795)(220) = 0
 \end{aligned}$$

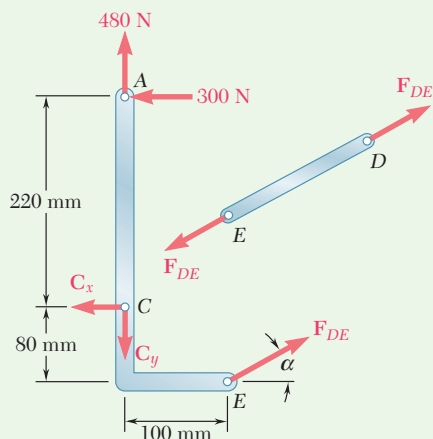


Fig. 3 Free-body diagrams of member ACE and DE .

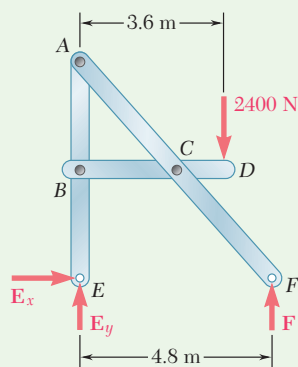
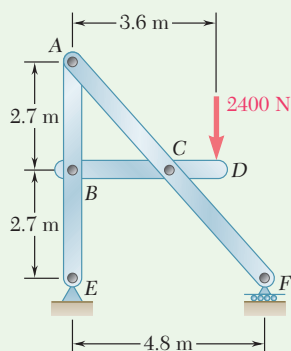


Fig. 1 Free-body diagram of entire frame.

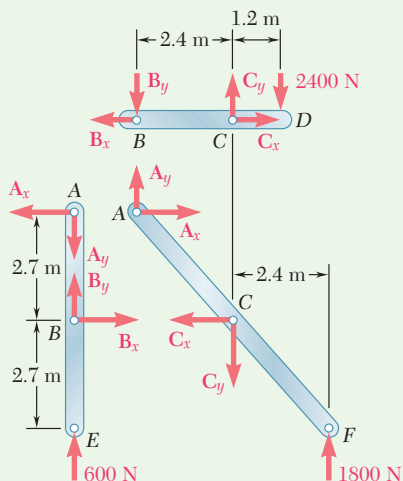


Fig. 2 Free-body diagrams of individual members.

Sample Problem 6.5

Determine the components of the forces acting on each member of the frame shown.

STRATEGY: The approach to this analysis is to consider the entire frame as a free body to determine the reactions, and then consider separate members. However, in this case, you will not be able to determine forces on one member without analyzing a second member at the same time.

MODELING and ANALYSIS: The external reactions involve only three unknowns, so compute the reactions by considering the free-body diagram of the entire frame (Fig. 1).

$$+\uparrow \Sigma M_E = 0: \quad -(2400 \text{ N})(3.6 \text{ m}) + F(4.8 \text{ m}) = 0$$

$$F = +1800 \text{ N}$$

$$F = 1800 \text{ N} \uparrow \quad \blacktriangleleft$$

$$+\uparrow \Sigma F_y = 0: \quad -2400 \text{ N} + 1800 \text{ N} + E_y = 0$$

$$E_y = +600 \text{ N}$$

$$E_y = 600 \text{ N} \uparrow \quad \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0:$$

$$E_x = 0 \quad \blacktriangleleft$$

Now dismember the frame. Since only two members are connected at each joint, force components are equal and opposite on each member at each joint (Fig. 2).

Free Body: Member BCD.

$$+\uparrow \Sigma M_B = 0: \quad -(2400 \text{ N})(3.6 \text{ m}) + C_y(2.4 \text{ m}) = 0 \quad C_y = +3600 \text{ N} \quad \blacktriangleleft$$

$$+\uparrow \Sigma M_C = 0: \quad -(2400 \text{ N})(1.2 \text{ m}) + B_y(2.4 \text{ m}) = 0 \quad B_y = +1200 \text{ N} \quad \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0: \quad -B_x + C_x = 0$$

Neither B_x nor C_x can be obtained by considering only member BCD ; you need to look at member ABE . The positive values obtained for B_y and C_y indicate that the force components B_y and C_y are directed as assumed.

Free Body: Member ABE.

$$+\uparrow \Sigma M_A = 0: \quad B_x(2.7 \text{ m}) = 0 \quad B_x = 0 \quad \blacktriangleleft$$

$$\rightarrow \Sigma F_x = 0: \quad +B_x - A_x = 0 \quad A_x = 0 \quad \blacktriangleleft$$

$$+\uparrow \Sigma F_y = 0: \quad -A_y + B_y + 600 \text{ N} = 0$$

$$-A_y + 1200 \text{ N} + 600 \text{ N} = 0 \quad A_y = +1800 \text{ N} \quad \blacktriangleleft$$

Free Body: Member BCD. Returning now to member BCD , you have

$$\rightarrow \Sigma F_x = 0: \quad -B_x + C_x = 0 \quad 0 + C_x = 0 \quad C_x = 0 \quad \blacktriangleleft$$

REFLECT and THINK: All unknown components have now been found. To check the results, you can verify that member ACF is in equilibrium.

$$\begin{aligned} +\uparrow \Sigma M_C &= (1800 \text{ N})(2.4 \text{ m}) - A_y(2.4 \text{ m}) - A_x(2.7 \text{ m}) \\ &= (1800 \text{ N})(2.4 \text{ m}) - (1800 \text{ N})(2.4 \text{ m}) - 0 = 0 \quad (\text{checks}) \end{aligned}$$

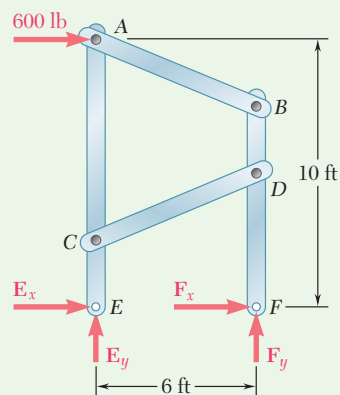
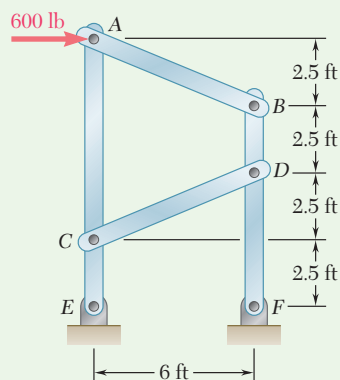


Fig. 1 Free-body diagram of entire frame.

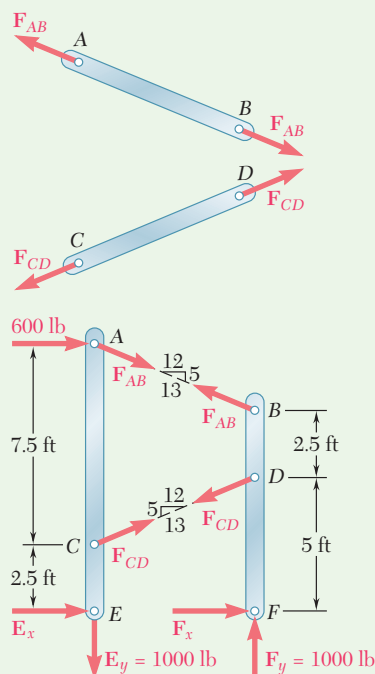


Fig. 2 Free-body diagrams of individual members.

Sample Problem 6.6

A 600-lb horizontal force is applied to pin A of the frame shown. Determine the forces acting on the two vertical members of the frame.

STRATEGY: Begin as usual with a free-body diagram of the entire frame, but this time you will not be able to determine all of the reactions. You will have to analyze a separate member and then return to the entire frame analysis in order to determine the remaining reaction forces.

MODELING and ANALYSIS: Choosing the entire frame as a free body (Fig. 1), you can write equilibrium equations to determine the two force components E_y and F_y . However, these equations are not sufficient to determine E_x and F_x .

$$+\uparrow \Sigma M_E = 0: \quad -(600 \text{ lb})(10 \text{ ft}) + F_y(6 \text{ ft}) = 0$$

$$F_y = +1000 \text{ lb}$$

$$F_y = 1000 \text{ lb} \uparrow \quad \blacktriangleleft$$

$$+\uparrow \Sigma F_y = 0: \quad E_y + F_y = 0$$

$$E_y = -1000 \text{ lb}$$

$$E_y = 1000 \text{ lb} \downarrow \quad \blacktriangleleft$$

To proceed with the solution, now consider the free-body diagrams of the various members (Fig. 2). In dismembering the frame, assume that pin A is attached to the multi-force member ACE so that the 600-lb force is applied to that member. Note that AB and CD are two-force members.

Free Body: Member ACE

$$+\uparrow \Sigma F_y = 0: \quad -\frac{5}{13}F_{AB} + \frac{5}{13}F_{CD} - 1000 \text{ lb} = 0$$

$$+\uparrow \Sigma M_E = 0: \quad -(600 \text{ lb})(10 \text{ ft}) - \left(\frac{12}{13}F_{AB}\right)(10 \text{ ft}) - \left(\frac{12}{13}F_{CD}\right)(2.5 \text{ ft}) = 0$$

Solving these equations simultaneously gives you

$$F_{AB} = -1040 \text{ lb} \quad F_{CD} = +1560 \text{ lb} \quad \blacktriangleleft$$

The signs indicate that the sense assumed for F_{CD} was correct and the sense for F_{AB} was incorrect. Now summing x components, you have

$$+\rightarrow \Sigma F_x = 0: \quad 600 \text{ lb} + \frac{12}{13}(-1040 \text{ lb}) + \frac{12}{13}(+1560 \text{ lb}) + E_x = 0$$

$$E_x = -1080 \text{ lb}$$

$$E_x = 1080 \text{ lb} \leftarrow \quad \blacktriangleleft$$

Free Body: Entire Frame. Now that E_x is determined, you can return to the free-body diagram of the entire frame.

$$+\rightarrow \Sigma F_x = 0: \quad 600 \text{ lb} - 1080 \text{ lb} + F_x = 0$$

$$F_x = +480 \text{ lb}$$

$$F_x = 480 \text{ lb} \rightarrow \quad \blacktriangleleft$$

REFLECT and THINK: Check your computations by verifying that the equation $\Sigma M_B = 0$ is satisfied by the forces acting on member BDF.

$$+\uparrow \Sigma M_B = -\left(\frac{12}{13}F_{CD}\right)(2.5 \text{ ft}) + (F_x)(7.5 \text{ ft})$$

$$= -\frac{12}{13}(1560 \text{ lb})(2.5 \text{ ft}) + (480 \text{ lb})(7.5 \text{ ft})$$

$$= -3600 \text{ lb}\cdot\text{ft} + 3600 \text{ lb}\cdot\text{ft} = 0 \quad (\text{checks})$$

SOLVING PROBLEMS ON YOUR OWN

In this section, we analyzed **frames containing one or more multi-force members**. In the problems that follow, you will be asked to determine the external reactions exerted on the frame and the internal forces that hold together the members of the frame.

In solving problems involving frames containing one or more multi-force members, follow these steps.

1. Draw a free-body diagram of the entire frame. To the greatest extent possible, use this free-body diagram to calculate the reactions at the supports. (In Sample Prob. 6.6 only two of the four reaction components could be found from the free body of the entire frame.)

2. Dismember the frame, and draw a free-body diagram of each member.

3. First consider the two-force members. Equal and opposite forces apply to each two-force member at the points where it is connected to another member. If the two-force member is straight, these forces are directed along the axis of the member. If you cannot tell at this point whether the member is in tension or compression, *assume* that the member is in tension and *direct both of the forces away from the member*. Since these forces have the same unknown magnitude, give them both the *same name* and, to avoid any confusion later, *do not use a plus sign or a minus sign*.

4. Next consider the multi-force members. For each of these members, show all of the forces acting on the member, including *applied loads, reactions, and internal forces at connections*. Clearly indicate the magnitude and direction of any reaction or reaction component found earlier from the free-body diagram of the entire frame.

a. Where a multi-force member is connected to a two-force member, apply a force to the multi-force member that is *equal and opposite* to the force drawn on the free-body diagram of the two-force member, *giving it the same name*.

b. Where a multi-force member is connected to another multi-force member, use *horizontal and vertical components* to represent the internal forces at that point, since the directions and magnitudes of these forces are unknown. The direction you choose for each of the two force components exerted on the first multi-force member is arbitrary, but *you must apply equal and opposite force components of the same name* to the other multi-force member. Again, *do not use a plus sign or a minus sign*.

(continued)

5. Now determine the internal forces as well as any reactions that you have not already found.

a. The free-body diagram of each multi-force member can provide you with *three equilibrium equations*.

b. To simplify your solution, seek a way to write an equation involving a single unknown. If you can locate *a point where all but one of the unknown force components intersect*, you can obtain an equation in a single unknown by summing moments about that point. *If all unknown forces except one are parallel*, you can obtain an equation in a single unknown by summing force components in a direction perpendicular to the parallel forces.

c. Since you arbitrarily chose the direction of each of the unknown forces, you cannot determine whether your guess was correct until the solution is complete. To do that, consider the *sign* of the value found for each of the unknowns: a *positive* sign means that the direction you selected was *correct*; a *negative* sign means that the direction is *opposite* to the direction you assumed.

6. To be more effective and efficient as you proceed through your solution, observe the following rules.

a. If you can find an equation involving only one unknown, write that equation and *solve it for that unknown*. Immediately replace that unknown wherever it appears on other free-body diagrams by the value you have found. Repeat this process by seeking equilibrium equations involving only one unknown until you have found all of the internal forces and unknown reactions.

b. If you cannot find an equation involving only one unknown, you may have to *solve a pair of simultaneous equations*. Before doing so, check that you have included the values of all of the reactions you obtained from the free-body diagram of the entire frame.

c. The total number of equations of equilibrium for the entire frame and for the individual members *will be larger than the number of unknown forces and reactions*. After you have found all of the reactions and all of the internal forces, you can use the remaining equations to check the accuracy of your computations.

Problems

FREE-BODY PRACTICE PROBLEMS

- 6.F1** For the frame and loading shown, draw the free-body diagram(s) needed to determine the force in member BD and the components of the reaction at C .
- 6.F2** For the frame and loading shown, draw the free-body diagram(s) needed to determine the components of all forces acting on member ABC .

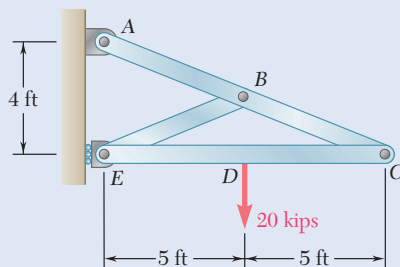


Fig. P6.F2

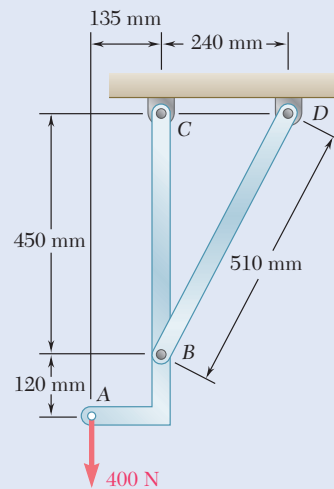


Fig. P6.F1

- 6.F3** Draw the free-body diagram(s) needed to determine all the forces exerted on member AI if the frame is loaded by a clockwise couple of magnitude 1200 lb-in. applied at point D .
- 6.F4** Knowing that the pulley has a radius of 0.5 m, draw the free-body diagram(s) needed to determine the components of the reactions at A and E .

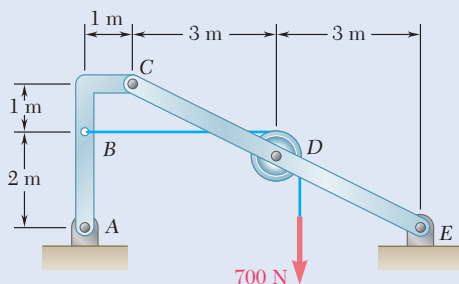


Fig. P6.F4

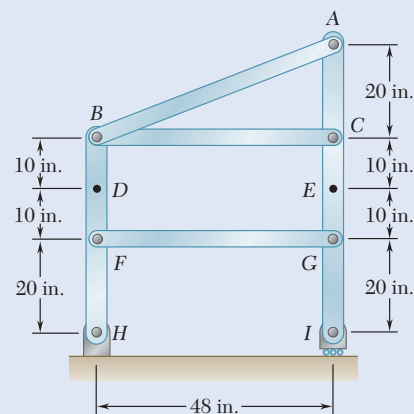


Fig. P6.F3

END-OF-SECTION PROBLEMS

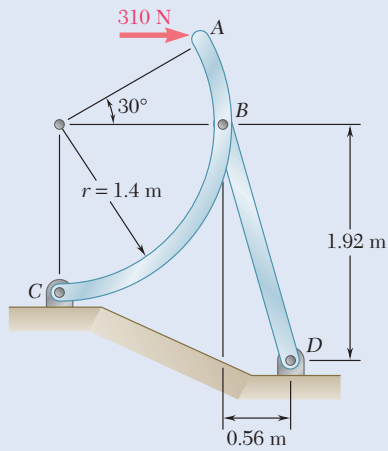


Fig. P6.75

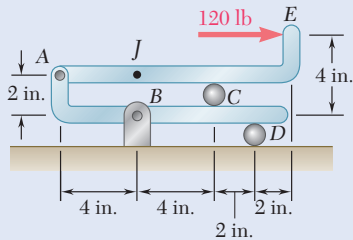


Fig. P6.78

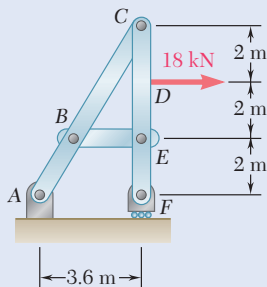


Fig. P6.79

6.75 and 6.76 Determine the force in member BD and the components of the reaction at C .

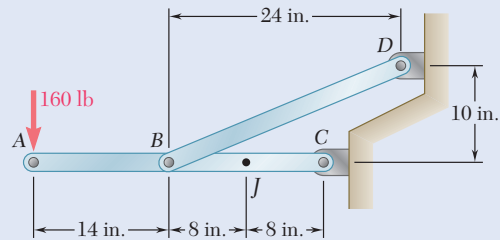


Fig. P6.76

6.77 For the frame and loading shown, determine the force acting on member ABC (a) at B , (b) at C .

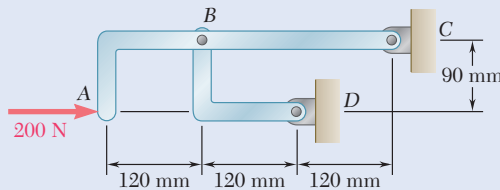


Fig. P6.77

6.78 Determine the components of all forces acting on member $ABCD$ of the assembly shown.

6.79 For the frame and loading shown, determine the components of all forces acting on member ABC .

6.80 Solve Prob. 6.79 assuming that the 18-kN load is replaced by a clockwise couple with a magnitude of 72 kN·m applied to member $CDEF$ at point D .

6.81 Determine the components of all forces acting on member $ABCD$ when $\theta = 0$.

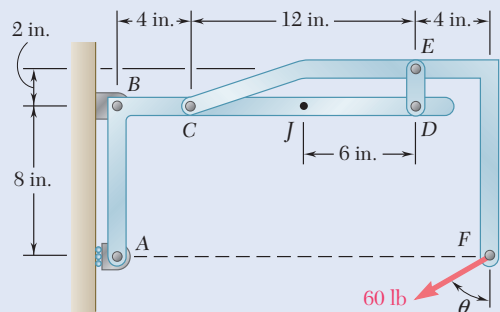


Fig. P6.81 and P6.82

6.82 Determine the components of all forces acting on member $ABCD$ when $\theta = 90^\circ$.

- 6.83** Determine the components of the reactions at A and E , (a) if the 800-N load is applied as shown, (b) if the 800-N load is moved along its line of action and is applied at point D .

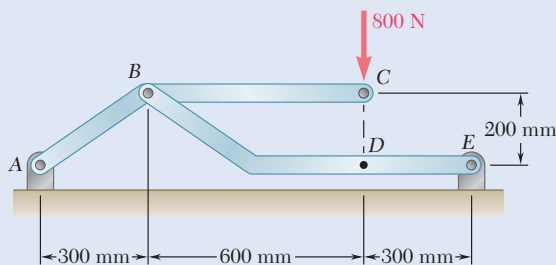


Fig. P6.83

- 6.84** Determine the components of the reactions at D and E if the frame is loaded by a clockwise couple of magnitude 150 N·m applied (a) at A , (b) at B .

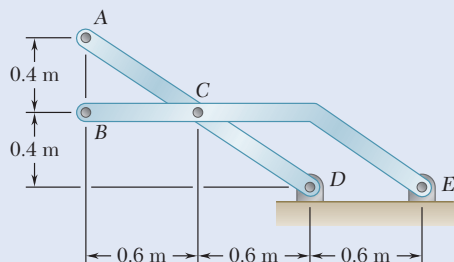


Fig. P6.84

- 6.85** Determine the components of the reactions at A and E if a 750-N force directed vertically downward is applied (a) at B , (b) at D .
- 6.86** Determine the components of the reactions at A and E if the frame is loaded by a clockwise couple with a magnitude of 36 N·m applied (a) at B , (b) at D .
- 6.87** Determine the components of the reactions at A and B , (a) if the 100-lb load is applied as shown, (b) if the 100-lb load is moved along its line of action and is applied at point F .

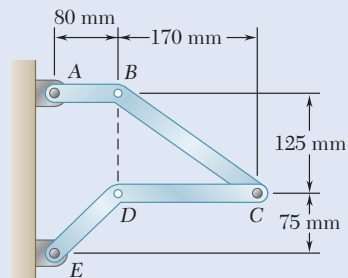


Fig. P6.85 and P6.86

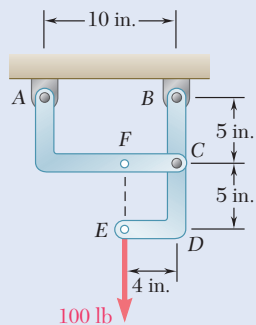


Fig. P6.87

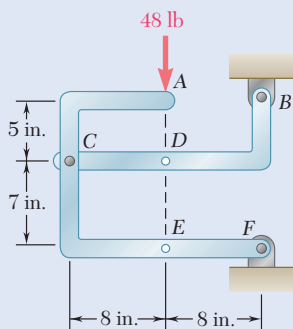


Fig. P6.88 and P6.89

6.88 The 48-lb load can be moved along the line of action shown and applied at A , D , or E . Determine the components of the reactions at B and F if the 48-lb load is applied (a) at A , (b) at D , (c) at E .

6.89 The 48-lb load is removed and a 288-lb·in. clockwise couple is applied successively at A , D , and E . Determine the components of the reactions at B and F if the couple is applied (a) at A , (b) at D , (c) at E .

6.90 (a) Show that, when a frame supports a pulley at A , an equivalent loading of the frame and of each of its component parts can be obtained by removing the pulley and applying at A two forces equal and parallel to the forces that the cable exerted on the pulley. (b) Show that, if one end of the cable is attached to the frame at a point B , a force of magnitude equal to the tension in the cable should also be applied at B .

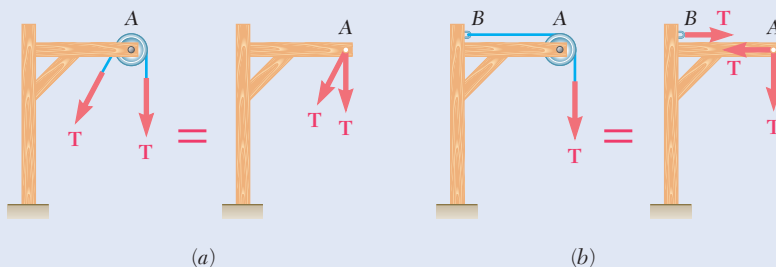


Fig. P6.90

6.91 Knowing that each pulley has a radius of 250 mm, determine the components of the reactions at D and E .

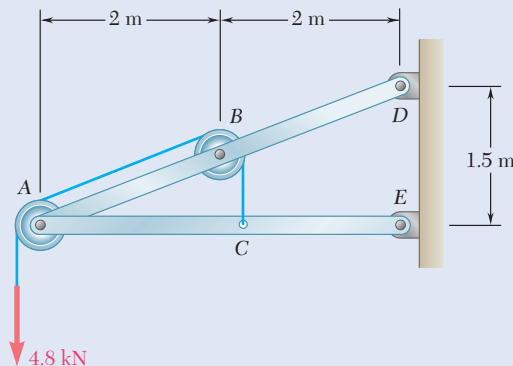


Fig. P6.91

6.92 Knowing that the pulley has a radius of 75 mm, determine the components of the reactions at A and B .

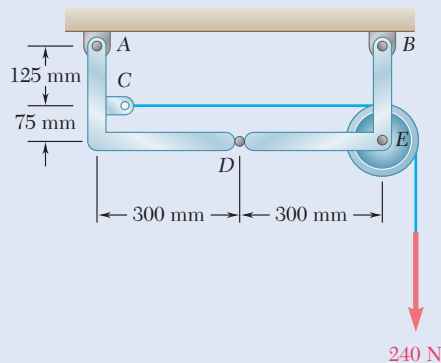


Fig. P6.92

6.93 A 3-ft-diameter pipe is supported every 16 ft by a small frame like that shown. Knowing that the combined weight of the pipe and its contents is 500 lb/ft and assuming frictionless surfaces, determine the components (a) of the reaction at E , (b) of the force exerted at C on member CDE .

6.94 Solve Prob. 6.93 for a frame where $h = 6$ ft.

6.95 A trailer weighing 2400 lb is attached to a 2900-lb pickup truck by a ball-and-socket truck hitch at D . Determine (a) the reactions at each of the six wheels when the truck and trailer are at rest, (b) the additional load on each of the truck wheels due to the trailer.

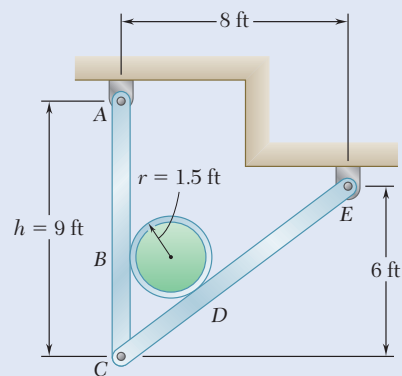


Fig. P6.93

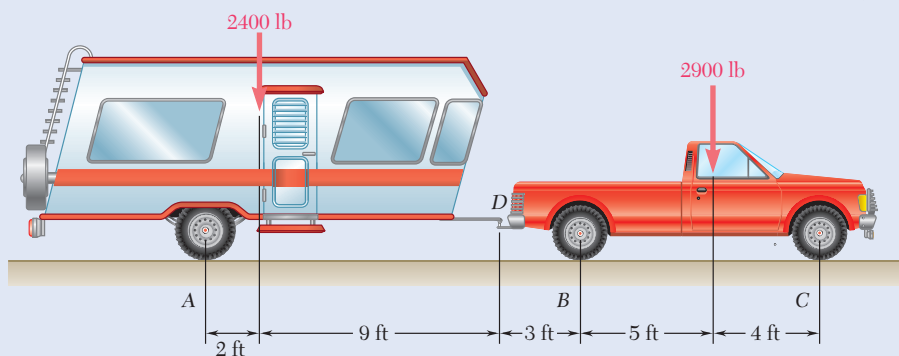


Fig. P6.95

6.96 In order to obtain a better weight distribution over the four wheels of the pickup truck of Prob. 6.95, a compensating hitch of the type shown is used to attach the trailer to the truck. The hitch consists of two bar springs (only one is shown in the figure) that fit into bearings inside a support rigidly attached to the truck. The springs are also connected by chains to the trailer frame, and specially designed hooks make it possible to place both chains in tension. (a) Determine the tension T required in each of the two chains if the additional load due to the trailer is to be evenly distributed over the four wheels of the truck. (b) What are the resulting reactions at each of the six wheels of the trailer-truck combination?

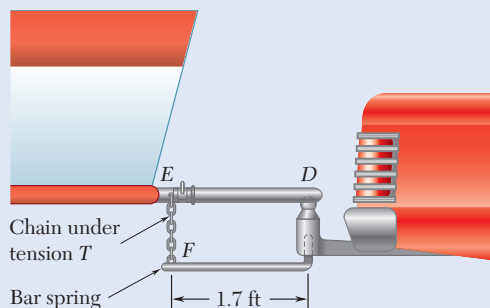


Fig. P6.96

- 6.97** The cab and motor units of the front-end loader shown are connected by a vertical pin located 2 m behind the cab wheels. The distance from C to D is 1 m. The center of gravity of the 300-kN motor unit is located at G_m , while the centers of gravity of the 100-kN cab and 75-kN load are located, respectively, at G_c and G_l . Knowing that the machine is at rest with its brakes released, determine (a) the reactions at each of the four wheels, (b) the forces exerted on the motor unit at C and D .

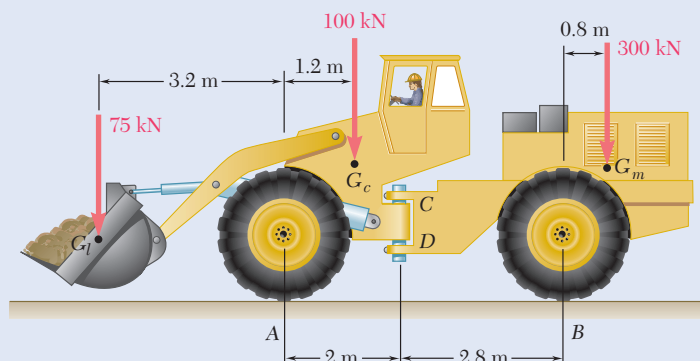


Fig. P6.97

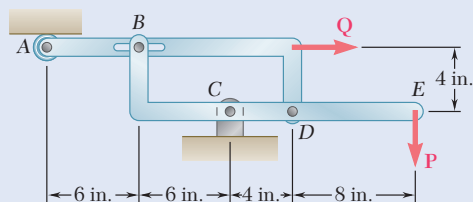


Fig. P6.99 and P6.100

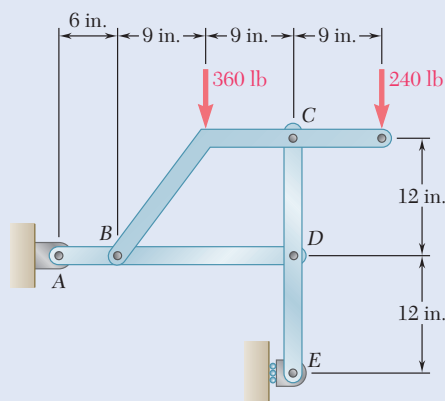


Fig. P6.103

- 6.98** Solve Prob. 6.97 assuming that the 75-kN load has been removed.

- 6.99** Knowing that $P = 90$ lb and $Q = 60$ lb, determine the components of all forces acting on member $BCDE$ of the assembly shown.

- 6.100** Knowing that $P = 60$ lb and $Q = 90$ lb, determine the components of all forces acting on member $BCDE$ of the assembly shown.

- 6.101 and 6.102** For the frame and loading shown, determine the components of all forces acting on member ABE .

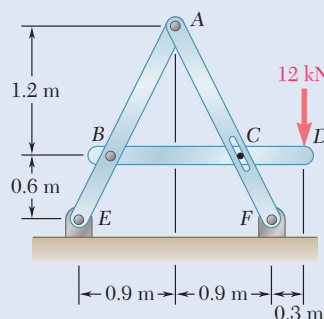


Fig. P6.101

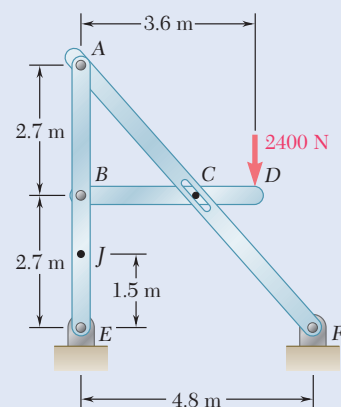


Fig. P6.102

- 6.103** For the frame and loading shown, determine the components of all forces acting on member ABD .

- 6.104** Solve Prob. 6.103 assuming that the 360-lb load has been removed.

6.105 For the frame and loading shown, determine the components of the forces acting on member *DABC* at *B* and *D*.

6.106 Solve Prob. 6.105 assuming that the 6-kN load has been removed.

6.107 The axis of the three-hinge arch *ABC* is a parabola with vertex at *B*. Knowing that $P = 112$ kN and $Q = 140$ kN, determine (a) the components of the reaction at *A*, (b) the components of the force exerted at *B* on segment *AB*.

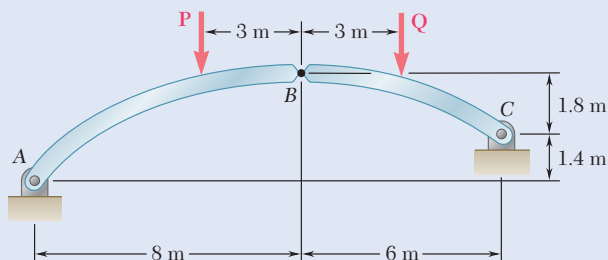


Fig. P6.107 and P6.108

6.108 The axis of the three-hinge arch *ABC* is a parabola with vertex at *B*. Knowing that $P = 140$ kN and $Q = 112$ kN, determine (a) the components of the reaction at *A*, (b) the components of the force exerted at *B* on segment *AB*.

6.109 and 6.110 Neglecting the effect of friction at the horizontal and vertical surfaces, determine the forces exerted at *B* and *C* on member *BCE*.

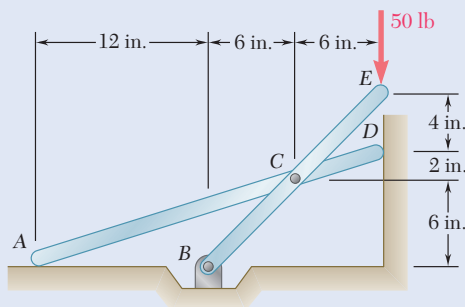


Fig. P6.109

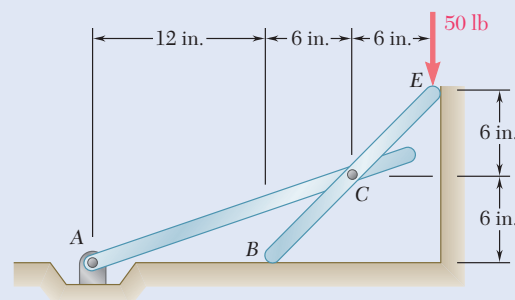


Fig. P6.110

6.111, 6.112, and 6.113 Members *ABC* and *CDE* are pin-connected at *C* and supported by four links. For the loading shown, determine the force in each link.

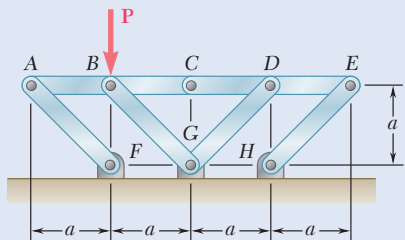


Fig. P6.111

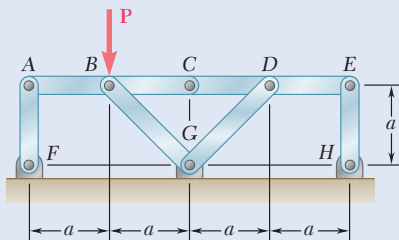


Fig. P6.112

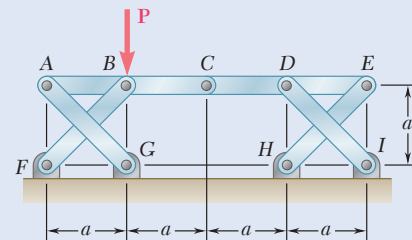


Fig. P6.113

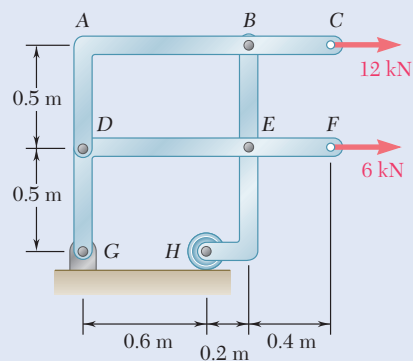


Fig. P6.105

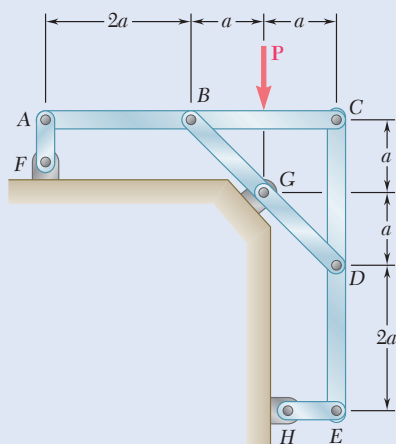


Fig. P6.114

6.114 Members ABC and CDE are pin-connected at C and supported by the four links AF , BG , DG , and EH . For the loading shown, determine the force in each link.

6.115 Solve Prob. 6.112 assuming that the force $\mathbf{P}$ is replaced by a clockwise couple of moment $\mathbf{M}_0$ applied to member CDE at D .

6.116 Solve Prob. 6.114 assuming that the force $\mathbf{P}$ is replaced by a clockwise couple of moment $\mathbf{M}_0$ applied at the same point.

6.117 Four beams, each with a length of $2a$, are nailed together at their midpoints to form the support system shown. Assuming that only vertical forces are exerted at the connections, determine the vertical reactions at A , D , E , and H .

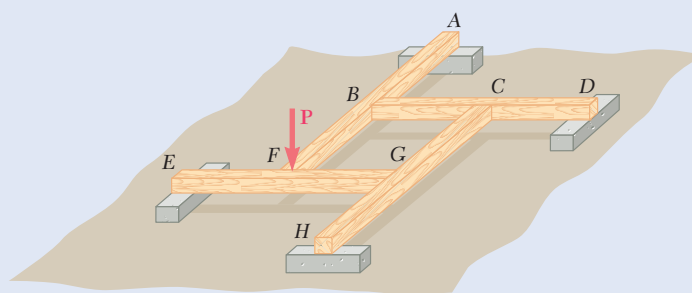


Fig. P6.117

6.118 Four beams, each with a length of $3a$, are held together by single nails at A , B , C , and D . Each beam is attached to a support located at a distance a from an end of the beam as shown. Assuming that only vertical forces are exerted at the connections, determine the vertical reactions at E , F , G , and H .

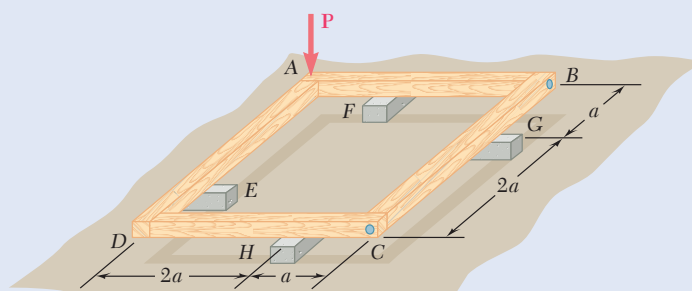


Fig. P6.118

6.119 through 6.121 Each of the frames shown consists of two L-shaped members connected by two rigid links. For each frame, determine the reactions at the supports and indicate whether the frame is rigid.

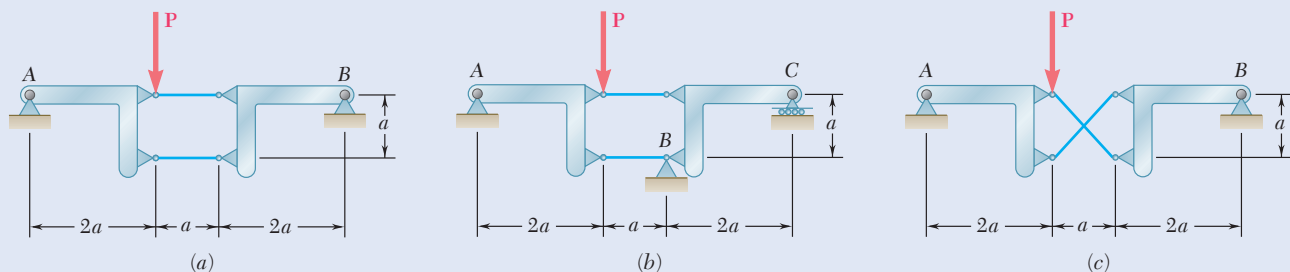


Fig. P6.119

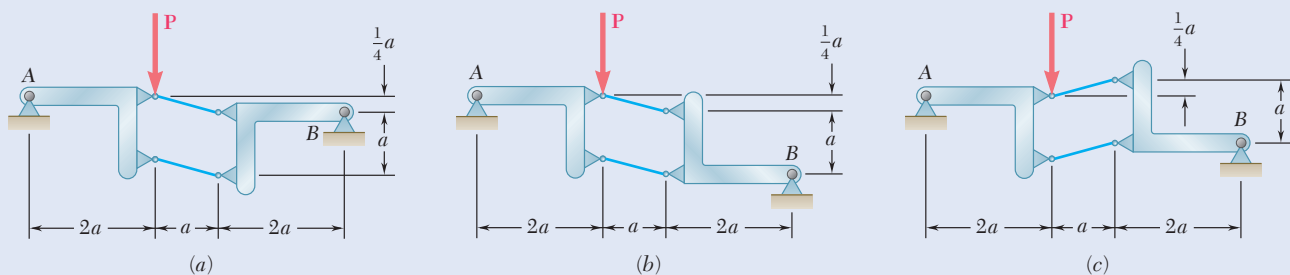


Fig. P6.120

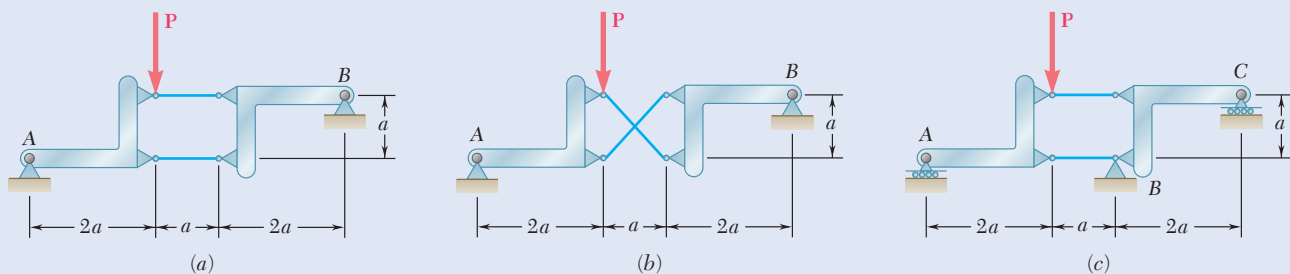


Fig. P6.121



Photo 6.7 This lamp can be placed in many different positions. To determine the forces in the springs and the internal forces at the joints, we need to consider the components of the lamp as free bodies.

6.4 MACHINES

Machines are structures designed to transmit and modify forces. Whether they are simple tools or include complicated mechanisms, their main purpose is to transform **input forces** into **output forces**. Consider, for example, a pair of cutting pliers used to cut a wire (Fig. 6.21a). If we apply two equal and opposite forces $\mathbf{P}$ and $-\mathbf{P}$ on the handles, the pliers will exert two equal and opposite forces $\mathbf{Q}$ and $-\mathbf{Q}$ on the wire (Fig. 6.21b).

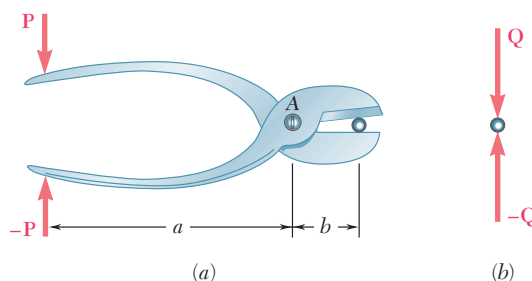


Fig. 6.21 (a) Input forces on the handles of a pair of cutting pliers; (b) output forces cut a wire.

To determine the magnitude Q of the output forces when we know the magnitude P of the input forces (or, conversely, to determine P when Q is known), we draw a free-body diagram of the pliers *alone* (i.e., without the wire), showing the input forces $\mathbf{P}$ and $-\mathbf{P}$ and the *reactions* $-\mathbf{Q}$ and $\mathbf{Q}$ that the wire exerts on the pliers (Fig. 6.22). However, since a pair of pliers forms a nonrigid structure, we must treat one of the component parts as a free body in order to determine the unknown forces. Consider Fig. 6.23a, for example. Taking moments about A , we obtain the relation $Pa = Qb$, which defines the magnitude Q in terms of P (or P in terms of Q). We can use the same free-body diagram to determine the components of the internal force at A ; we find $A_x = 0$ and $A_y = P + Q$.

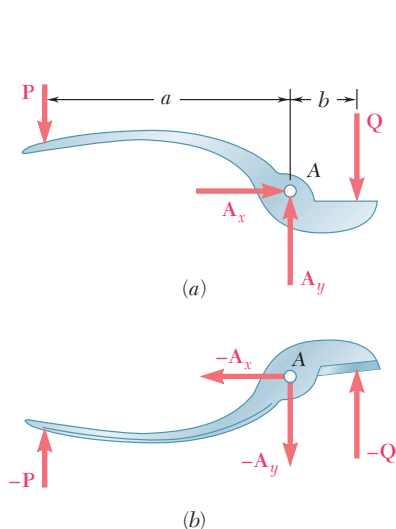


Fig. 6.23 Free-body diagrams of the members of the pliers, showing components of the internal forces at joint A .

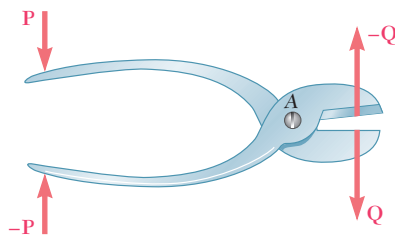
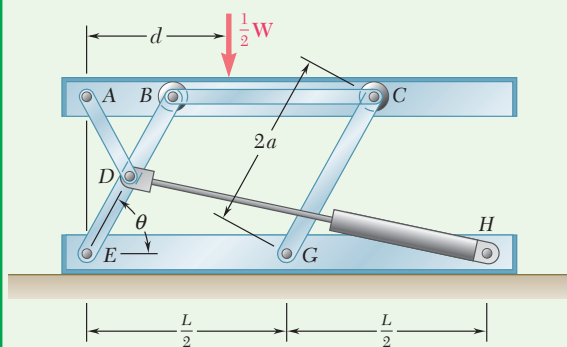


Fig. 6.22 To show a free-body diagram of the pliers in equilibrium, we include the input forces and the reactions to the output forces.

In the case of more complicated machines, it is generally necessary to use several free-body diagrams and, possibly, to solve simultaneous equations involving various internal forces. You should choose the free bodies to include the input forces and the reactions to the output forces, and the total number of unknown force components involved should not exceed the number of available independent equations. It is advisable, before attempting to solve a problem, to determine whether the structure considered is determinate. There is no point, however, in discussing the rigidity of a machine, since a machine includes moving parts and thus *must* be nonrigid.



Sample Problem 6.7

A hydraulic-lift table is used to raise a 1000-kg crate. The table consists of a platform and two identical linkages on which hydraulic cylinders exert equal forces. (Only one linkage and one cylinder are shown.) Members EDB and CG are each of length $2a$, and member AD is pinned to the midpoint of EDB . If the crate is placed on the table so that half of its weight is supported by the system shown, determine the force exerted by each cylinder in raising the crate for $\theta = 60^\circ$, $a = 0.70$ m, and $L = 3.20$ m. Show that the result is independent of the distance d .

STRATEGY: The free-body diagram of the entire frame will involve more than three unknowns, so it alone can not be used to solve this problem. Instead, draw free-body diagrams of each component of the machine and work from them.

MODELING: The machine consists of the platform and the linkage. Its free-body diagram (Fig. 1) includes an input force F_{DH} exerted by the cylinder; the weight $W/2$, which is equal and opposite to the output force; and reactions at E and G , which are assumed to be directed as shown. Dismember the mechanism and draw a free-body diagram for each of its component parts (Fig. 2). Note that AD , BC , and CG are two-force members. Member CG has already been assumed to be in compression; now assume that AD and BC are in tension and direct the forces exerted on them as shown. Use equal and opposite vectors to represent the forces exerted by the two-force members on the platform, on member BDE , and on roller C .

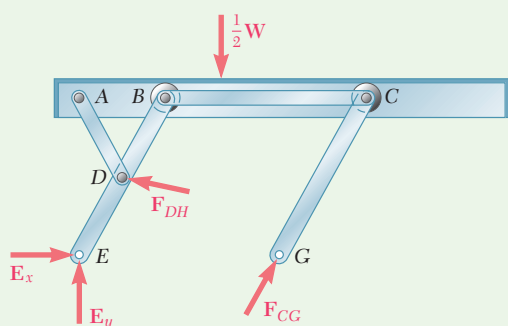


Fig. 1 Free-body diagram of machine.

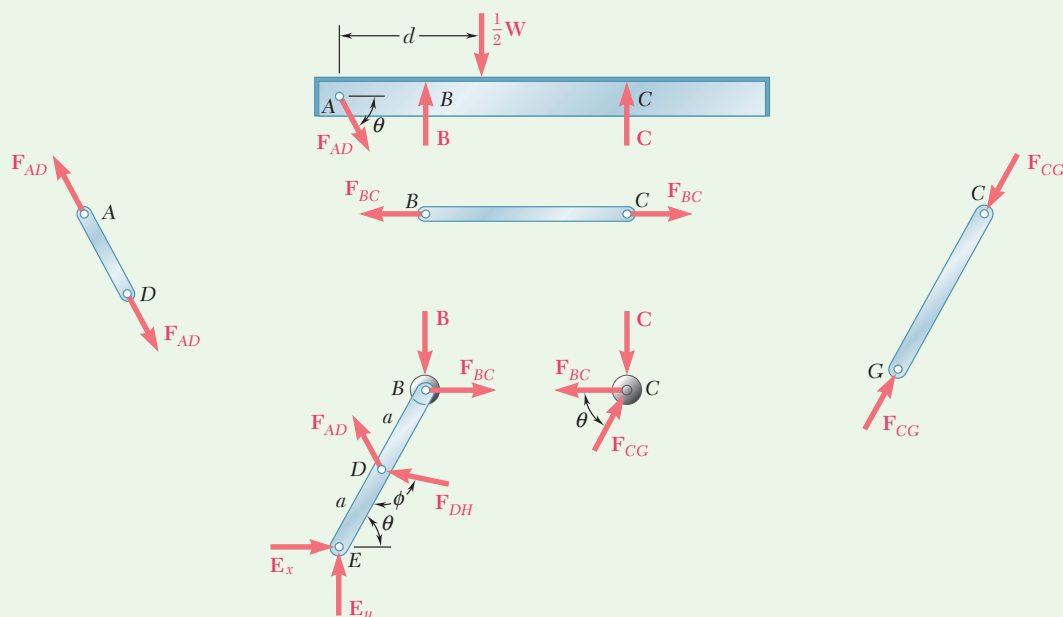


Fig. 2 Free-body diagram of each component part.

(continued)

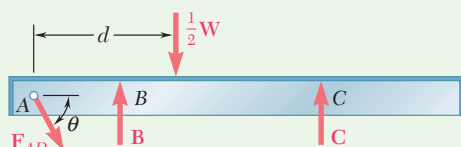


Fig. 3 Free-body diagram of platform ABC.

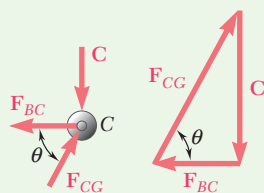


Fig. 4 Free-body diagram of roller C and its force triangle.

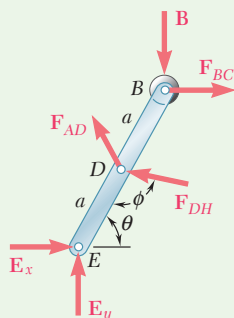


Fig. 5 Free-body diagram of member BDE.

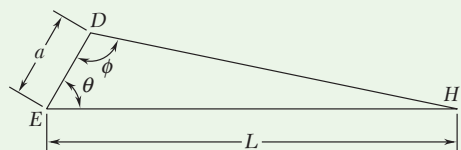


Fig. 6 Geometry of triangle EDH.

ANALYSIS:**Free Body: Platform ABC (Fig. 3).**

$$\begin{aligned} +\rightarrow \Sigma F_x = 0: & \quad F_{AD} \cos \theta = 0 & \quad F_{AD} = 0 \\ +\uparrow \Sigma F_y = 0: & \quad B + C - \frac{1}{2}W = 0 & \quad B + C = \frac{1}{2}W \end{aligned} \quad (1)$$

Free-Body Roller C (Fig. 4). Draw a force triangle and obtain $F_{BC} = C \cot \theta$.

Free Body: Member BDE (Fig. 5). Recalling that $F_{AD} = 0$, you have

$$\begin{aligned} +\curvearrowright \Sigma M_E = 0: & \quad F_{DH} \cos(\phi - 90^\circ)a - B(2a \cos \theta) - F_{BC}(2a \sin \theta) = 0 \\ & \quad F_{DH}a \sin \phi - B(2a \cos \theta) - (C \cot \theta)(2a \sin \theta) = 0 \\ & \quad F_{DH} \sin \phi - 2(B + C) \cos \theta = 0 \end{aligned}$$

From Eq. (1), you obtain

$$F_{DH} = W \frac{\cos \theta}{\sin \phi} \quad (2)$$

Note that *the result obtained is independent of d*. ◀

Applying first the law of sines to triangle EDH (Fig. 6), you have

$$\frac{\sin \phi}{EH} = \frac{\sin \theta}{DH} \quad \sin \phi = \frac{EH}{DH} \sin \theta \quad (3)$$

Now using the law of cosines, you get

$$\begin{aligned} (DH)^2 &= a^2 + L^2 - 2aL \cos \theta \\ &= (0.70)^2 + (3.20)^2 - 2(0.70)(3.20) \cos 60^\circ \\ (DH)^2 &= 8.49 \quad DH = 2.91 \text{ m} \end{aligned}$$

Also note that

$$W = mg = (1000 \text{ kg})(9.81 \text{ m/s}^2) = 9810 \text{ N} = 9.81 \text{ kN}$$

Substituting for $\sin \phi$ from Eq. (3) into Eq. (2) and using the numerical data, your result is

$$F_{DH} = W \frac{DH}{EH} \cot \theta = (9.81 \text{ kN}) \frac{2.91 \text{ m}}{3.20 \text{ m}} \cot 60^\circ$$

$F_{DH} = 5.15 \text{ kN}$ ◀

REFLECT and THINK: Note that link AD ends up having zero force in this situation. However, this member still serves an important function, as it is necessary to enable the machine to support any horizontal load that might be exerted on the platform.

SOLVING PROBLEMS ON YOUR OWN

This section dealt with the analysis of *machines*. Since machines are designed to transmit or modify forces, they always contain moving parts. However, the machines considered here are always at rest, and you will be working with the set of *forces required to maintain the equilibrium of the machine*.

Known forces that act on a machine are called *input forces*. A *machine transforms the input forces into output forces*, such as the cutting forces applied by the pliers of Fig. 6.21. You will determine the output forces by finding the equal and opposite forces that should be applied to the machine to maintain its equilibrium.

In Sec. 6.3, you analyzed frames; you will use almost the same procedure to analyze machines by following these steps.

1. Draw a free-body diagram of the whole machine, and use it to determine as many as possible of the unknown forces exerted on the machine.

2. Dismember the machine and draw a free-body diagram of each member.

3. First consider the two-force members. Apply equal and opposite forces to each two-force member at the points where it is connected to another member. If you cannot tell at this point whether the member is in tension or in compression, *assume* that the member is in tension and *direct both of the forces away from the member*. Since these forces have the same unknown magnitude, *give them both the same name*.

4. Next consider the multi-force members. For each of these members, show all of the forces acting on it, including applied loads and forces, reactions, and internal forces at connections.

a. Where a multi-force member is connected to a two-force member, apply to the multi-force member a force that is *equal and opposite* to the force drawn on the free-body diagram of the two-force member, *giving it the same name*.

b. Where a multi-force member is connected to another multi-force member, use *horizontal and vertical components* to represent the internal forces at that point. The directions you choose for each of the two force components exerted on the first multi-force member are arbitrary, but *you must apply equal and opposite force components of the same name* to the other multi-force member.

5. Write equilibrium equations after you have completed the various free-body diagrams.

a. To simplify your solution, you should, whenever possible, write and solve equilibrium equations involving single unknowns.

b. Since you arbitrarily chose the direction of each of the unknown forces, you must determine at the end of the solution whether your guess was correct. To that effect, *consider the sign* of the value found for each of the unknowns. A *positive* sign indicates that your guess was correct, and a *negative* sign indicates that it was not.

6. Finally, check your solution by substituting the results obtained into an equilibrium equation that you have not previously used.

Problems

FREE-BODY PRACTICE PROBLEMS

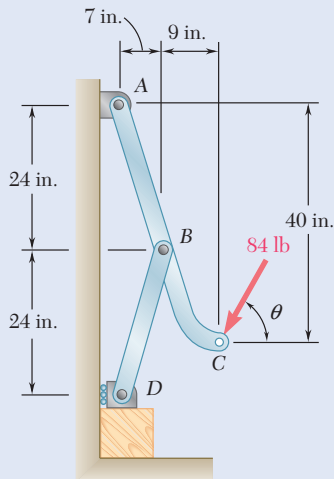


Fig. P6.F5

6.F5 An 84-lb force is applied to the toggle vise at C . Knowing that $\theta = 90^\circ$, draw the free-body diagram(s) needed to determine the vertical force exerted on the block at D .

6.F6 For the system and loading shown, draw the free-body diagram(s) needed to determine the force $\mathbf{P}$ required for equilibrium.

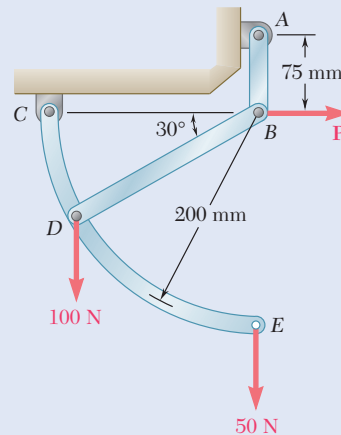


Fig. P6.F6

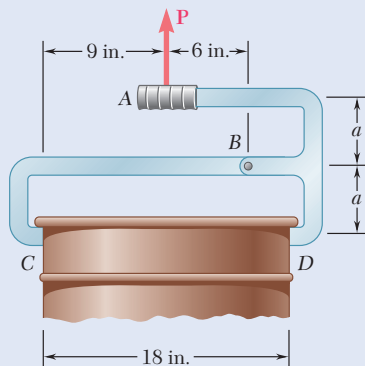


Fig. P6.F7

6.F7 A small barrel weighing 60 lb is lifted by a pair of tongs as shown. Knowing that $a = 5$ in., draw the free-body diagram(s) needed to determine the forces exerted at B and D on tong ABD .

6.F8 The position of member ABC is controlled by the hydraulic cylinder CD . Knowing that $\theta = 30^\circ$, draw the free-body diagram(s) needed to determine the force exerted by the hydraulic cylinder on pin C , and the reaction at B .

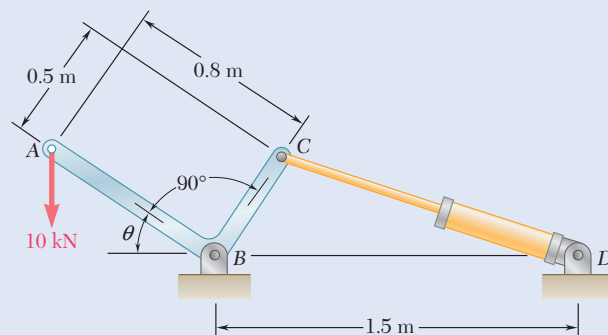


Fig. P6.F8

END-OF-SECTION PROBLEMS

- 6.122** The shear shown is used to cut and trim electronic-circuit-board laminates. For the position shown, determine (a) the vertical component of the force exerted on the shearing blade at D , (b) the reaction at C .
- 6.123** A 100-lb force directed vertically downward is applied to the toggle vise at C . Knowing that link BD is 6 in. long and that $a = 4$ in., determine the horizontal force exerted on block E .

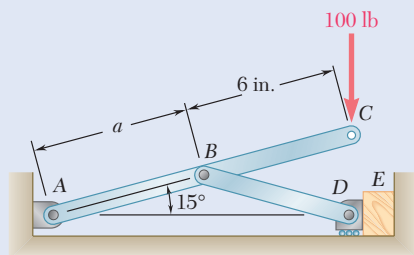


Fig. P6.123 and P6.124

- 6.124** A 100-lb force directed vertically downward is applied to the toggle vise at C . Knowing that link BD is 6 in. long and that $a = 8$ in., determine the horizontal force exerted on block E .
- 6.125** The control rod CE passes through a horizontal hole in the body of the toggle system shown. Knowing that link BD is 250 mm long, determine the force Q required to hold the system in equilibrium when $\beta = 20^\circ$.

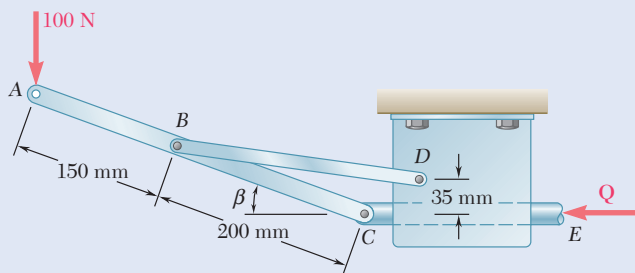


Fig. P6.125

- 6.126** Solve Prob. 6.125 when (a) $\beta = 0$, (b) $\beta = 6^\circ$.
- 6.127** The press shown is used to emboss a small seal at E . Knowing that $P = 250$ N, determine (a) the vertical component of the force exerted on the seal, (b) the reaction at A .
- 6.128** The press shown is used to emboss a small seal at E . Knowing that the vertical component of the force exerted on the seal must be 900 N, determine (a) the required vertical force P , (b) the corresponding reaction at A .

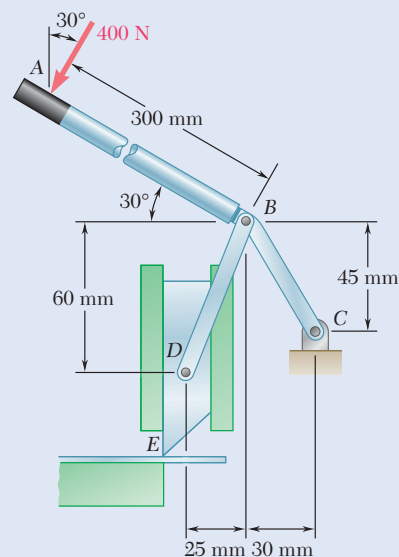


Fig. P6.122

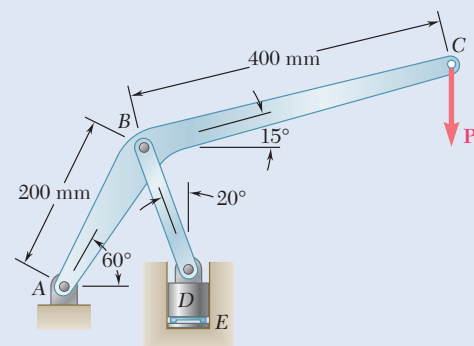


Fig. P6.127 and P6.128

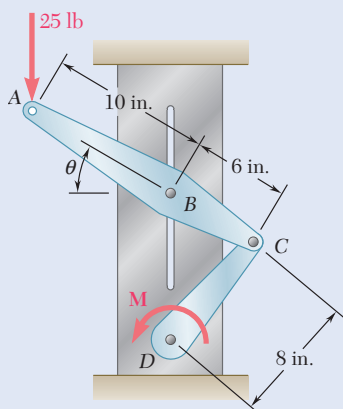


Fig. P6.129 and P6.130

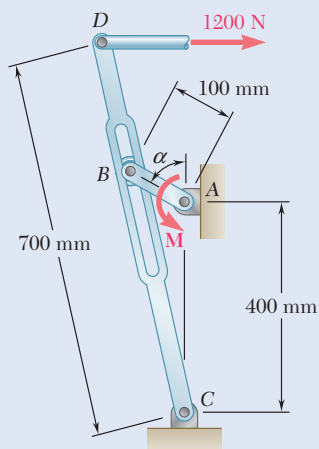


Fig. P6.133

6.129 The pin at B is attached to member ABC and can slide freely along the slot cut in the fixed plate. Neglecting the effect of friction, determine the couple M required to hold the system in equilibrium when $\theta = 30^\circ$.

6.130 The pin at B is attached to member ABC and can slide freely along the slot cut in the fixed plate. Neglecting the effect of friction, determine the couple M required to hold the system in equilibrium when $\theta = 60^\circ$.

6.131 Arm ABC is connected by pins to a collar at B and to crank CD at C . Neglecting the effect of friction, determine the couple M required to hold the system in equilibrium when $\theta = 0$.

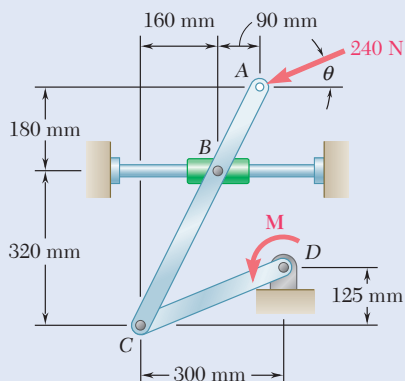


Fig. P6.131 and P6.132

6.132 Arm ABC is connected by pins to a collar at B and to crank CD at C . Neglecting the effect of friction, determine the couple M required to hold the system in equilibrium when $\theta = 90^\circ$.

6.133 The Whitworth mechanism shown is used to produce a quick-return motion of point D . The block at B is pinned to the crank AB and is free to slide in a slot cut in member CD . Determine the couple M that must be applied to the crank AB to hold the mechanism in equilibrium when (a) $\alpha = 0$, (b) $\alpha = 30^\circ$.

6.134 Solve Prob. 6.133 when (a) $\alpha = 60^\circ$, (b) $\alpha = 90^\circ$.

6.135 and 6.136 Two rods are connected by a frictionless collar B . Knowing that the magnitude of the couple M_A is 500 lb-in., determine (a) the couple M_C required for equilibrium, (b) the corresponding components of the reaction at C .

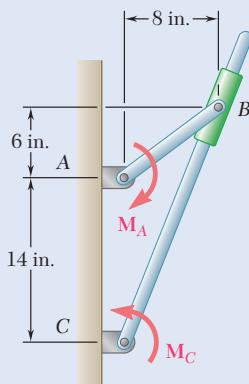


Fig. P6.135

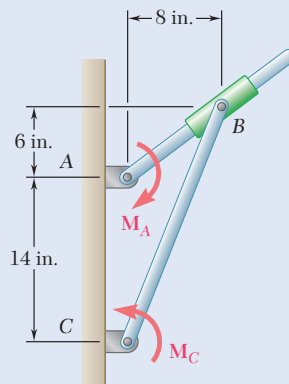


Fig. P6.136

6.137 and 6.138 Rod CD is attached to the collar D and passes through a collar welded to end B of lever AB . Neglecting the effect of friction, determine the couple M required to hold the system in equilibrium when $\theta = 30^\circ$.

6.139 Two hydraulic cylinders control the position of the robotic arm ABC . Knowing that in the position shown the cylinders are parallel, determine the force exerted by each cylinder when $P = 160 \text{ N}$ and $Q = 80 \text{ N}$.

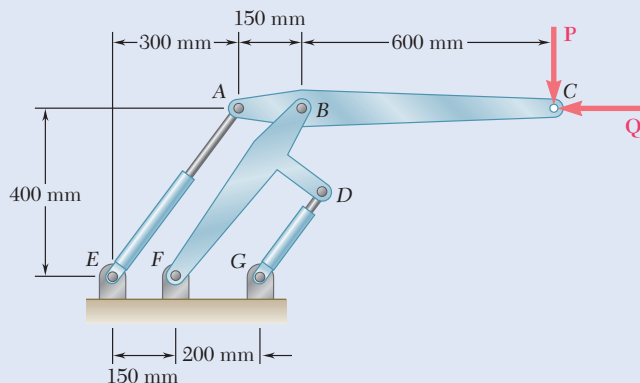


Fig. P6.139 and P6.140

6.140 Two hydraulic cylinders control the position of the robotic arm ABC . In the position shown, the cylinders are parallel and both are in tension. Knowing that $F_{AE} = 600 \text{ N}$ and $F_{DG} = 50 \text{ N}$, determine the forces P and Q applied at C to arm ABC .

6.141 A 39-ft length of railroad rail of weight 44 lb/ft is lifted by the tongs shown. Determine the forces exerted at D and F on tong BDF .

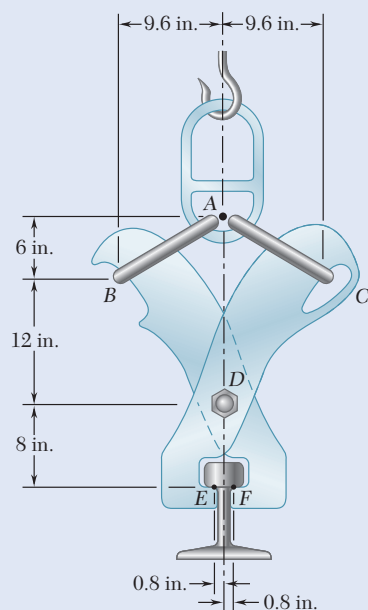


Fig. P6.141

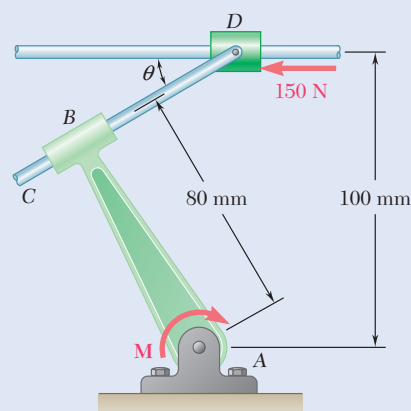


Fig. P6.137

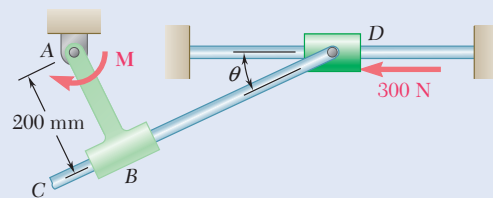


Fig. P6.138

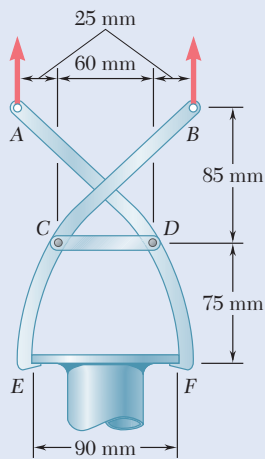


Fig. P6.143

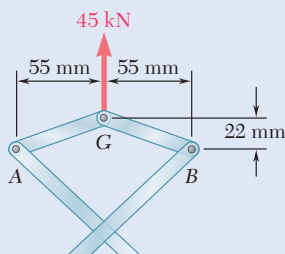


Fig. P6.144

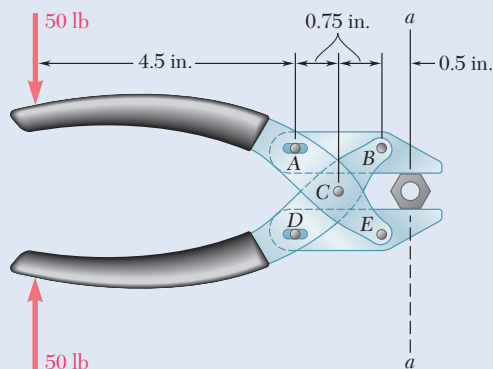


Fig. P6.146

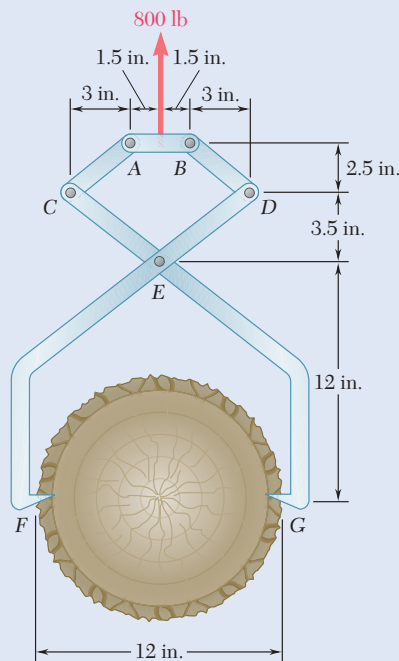


Fig. P6.142

- 6.142** A log weighing 800 lb is lifted by a pair of tongs as shown. Determine the forces exerted at E and F on tong DEF .
- 6.143** The tongs shown are used to apply a total upward force of 45 kN on a pipe cap. Determine the forces exerted at D and F on tong ADF .
- 6.144** If the toggle shown is added to the tongs of Prob. 6.143 and a single vertical force is applied at G , determine the forces exerted at D and F on tong ADF .
- 6.145** The pliers shown are used to grip a 0.3-in.-diameter rod. Knowing that two 60-lb forces are applied to the handles, determine (a) the magnitude of the forces exerted on the rod, (b) the force exerted by the pin at A on portion AB of the pliers.
- 6.146** Determine the magnitude of the gripping forces exerted along line aa on the nut when two 50-lb forces are applied to the handles as shown. Assume that pins A and D slide freely in slots cut in the jaws.

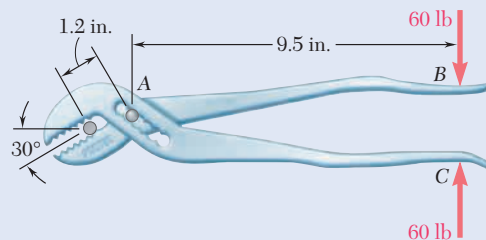


Fig. P6.145

6.147 In using the bolt cutter shown, a worker applies two 300-N forces to the handles. Determine the magnitude of the forces exerted by the cutter on the bolt.

6.148 Determine the magnitude of the gripping forces produced when two 300-N forces are applied as shown.

6.149 and 6.150 Determine the force $\mathbf{P}$ that must be applied to the toggle CDE to maintain bracket ABC in the position shown.

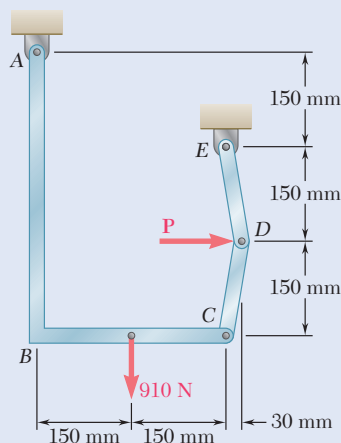


Fig. P6.149

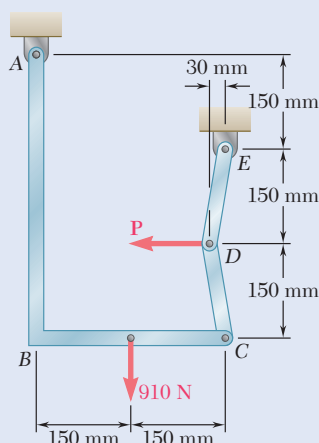


Fig. P6.150

6.151 Since the brace shown must remain in position even when the magnitude of $\mathbf{P}$ is very small, a single safety spring is attached at D and E . The spring DE has a constant of 50 lb/in. and an unstretched length of 7 in. Knowing that $l = 10$ in. and that the magnitude of $\mathbf{P}$ is 800 lb, determine the force $\mathbf{Q}$ required to release the brace.

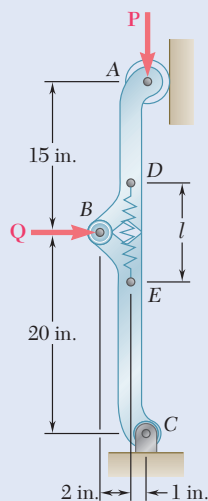


Fig. P6.151

6.152 The specialized plumbing wrench shown is used in confined areas (e.g., under a basin or sink). It consists essentially of a jaw BC pinned at B to a long rod. Knowing that the forces exerted on the nut are equivalent to a clockwise (when viewed from above) couple with a magnitude of 135 lb·in., determine (a) the magnitude of the force exerted by pin B on jaw BC , (b) the couple $\mathbf{M}_0$ that is applied to the wrench.

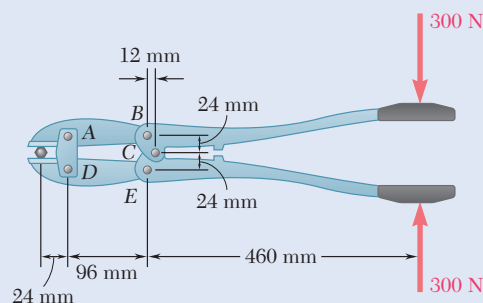


Fig. P6.147

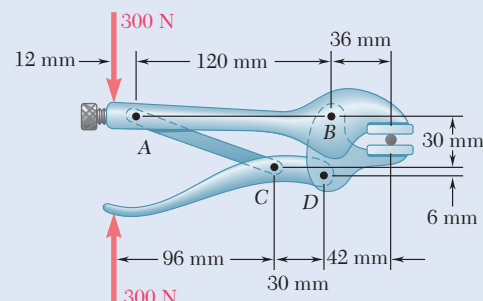


Fig. P6.148

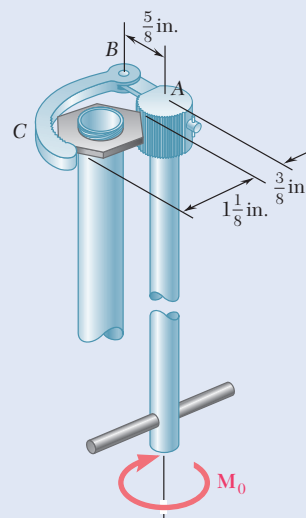


Fig. P6.152

- 6.153** The motion of the bucket of the front-end loader shown is controlled by two arms and a linkage that are pin-connected at D . The arms are located symmetrically with respect to the central, vertical, and longitudinal plane of the loader; one arm AFJ and its control cylinder EF are shown. The single linkage $GHDB$ and its control cylinder BC are located in the plane of symmetry. For the position and loading shown, determine the force exerted (a) by cylinder BC , (b) by cylinder EF .

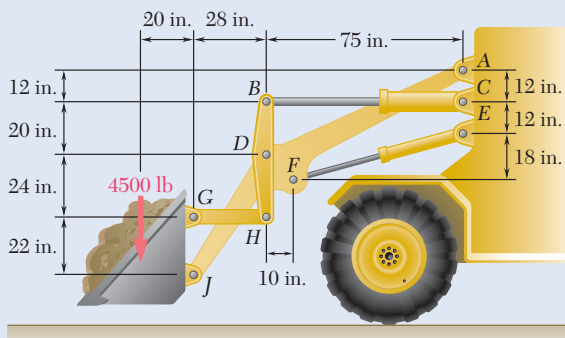


Fig. P6.153

- 6.154** The bucket of the front-end loader shown carries a 3200-lb load. The motion of the bucket is controlled by two identical mechanisms, only one of which is shown. Knowing that the mechanism shown supports one-half of the 3200-lb load, determine the force exerted (a) by cylinder CD , (b) by cylinder FH .

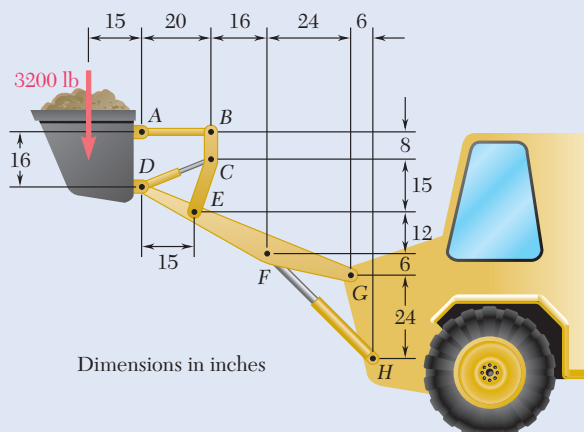


Fig. P6.154

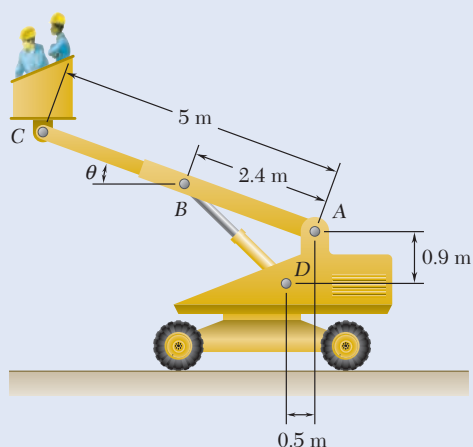


Fig. P6.155

- 6.155** The telescoping arm ABC is used to provide an elevated platform for construction workers. The workers and the platform together have a mass of 200 kg and have a combined center of gravity located directly above C . For the position when $\theta = 20^\circ$, determine (a) the force exerted at B by the single hydraulic cylinder BD , (b) the force exerted on the supporting carriage at A .

- 6.156** The telescoping arm ABC of Prob. 6.155 can be lowered until end C is close to the ground, so that workers can easily board the platform. For the position when $\theta = -20^\circ$, determine (a) the force exerted at B by the single hydraulic cylinder BD , (b) the force exerted on the supporting carriage at A .

- 6.157** The motion of the backhoe bucket shown is controlled by the hydraulic cylinders AD , CG , and EF . As a result of an attempt to dislodge a portion of a slab, a 2-kip force $\mathbf{P}$ is exerted on the bucket teeth at J . Knowing that $\theta = 45^\circ$, determine the force exerted by each cylinder.

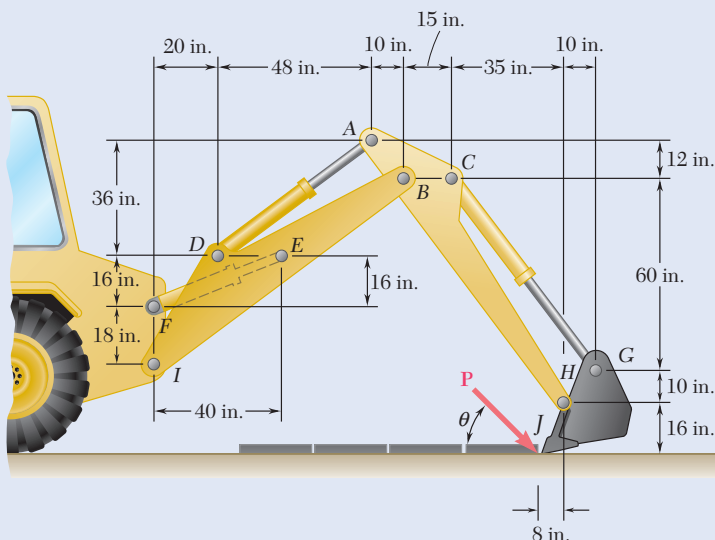


Fig. P6.157

- 6.158** Solve Prob. 6.157 assuming that the 2-kip force $\mathbf{P}$ acts horizontally to the right ($\theta = 0$).
- 6.159** The gears D and G are rigidly attached to shafts that are held by frictionless bearings. If $r_D = 90$ mm and $r_G = 30$ mm, determine (a) the couple $\mathbf{M}_0$ that must be applied for equilibrium, (b) the reactions at A and B .

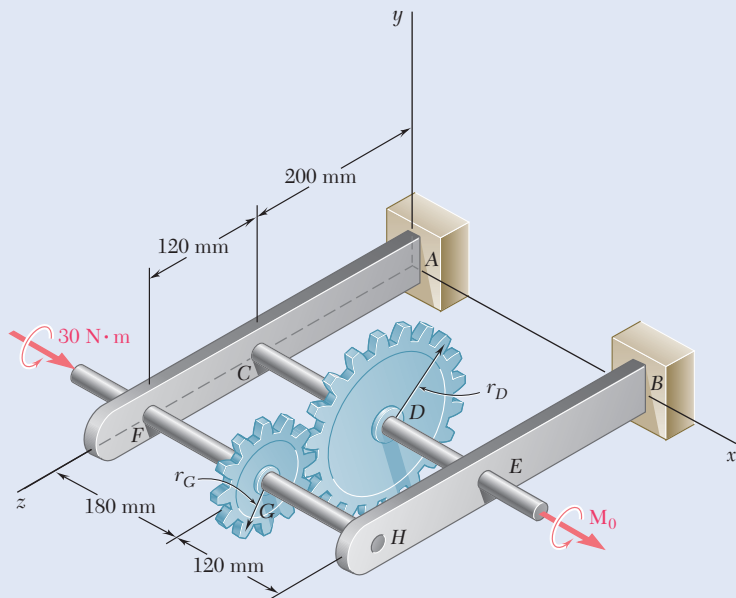


Fig. P6.159

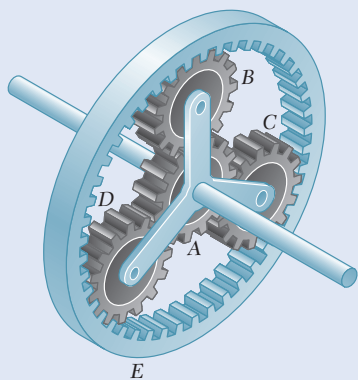


Fig. P6.160

6.160 In the planetary gear system shown, the radius of the central gear A is $a = 18$ mm, the radius of each planetary gear is b , and the radius of the outer gear E is $(a + 2b)$. A clockwise couple with a magnitude of $M_A = 10$ N·m is applied to the central gear A and a counterclockwise couple with a magnitude of $M_S = 50$ N·m is applied to the spider BCD. If the system is to be in equilibrium, determine (a) the required radius b of the planetary gears, (b) the magnitude M_E of the couple that must be applied to the outer gear E.

***6.161** Two shafts AC and CF, which lie in the vertical xy plane, are connected by a universal joint at C. The bearings at B and D do not exert any axial force. A couple with a magnitude of 500 lb·in. (clockwise when viewed from the positive x axis) is applied to shaft CF at F. At a time when the arm of the crosspiece attached to shaft CF is horizontal, determine (a) the magnitude of the couple that must be applied to shaft AC at A to maintain equilibrium, (b) the reactions at B, D, and E. (Hint: The sum of the couples exerted on the crosspiece must be zero.)

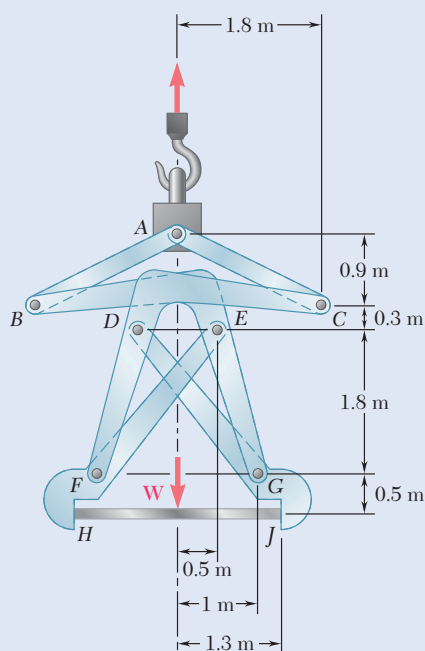


Fig. P6.163

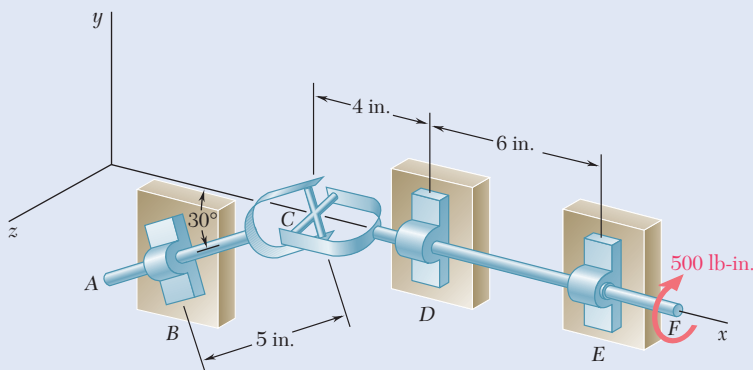


Fig. P6.161

***6.162** Solve Prob. 6.161 assuming that the arm of the crosspiece attached to shaft CF is vertical.

***6.163** The large mechanical tongs shown are used to grab and lift a thick 7500-kg steel slab HJ. Knowing that slipping does not occur between the tong grips and the slab at H and J, determine the components of all forces acting on member EFH. (Hint: Consider the symmetry of the tongs to establish relationships between the components of the force acting at E on EFH and the components of the force acting at D on DGJ.)

Review and Summary

In this chapter, you studied ways to determine the **internal forces** holding together the various parts of a structure.

Analysis of Trusses

The first half of the chapter presented the analysis of **trusses**, i.e., structures consisting of *straight members connected at their extremities only*. Because the members are slender and unable to support lateral loads, all of the loads must be applied at the joints; thus, we can assume that a truss consists of *pins and two-force members* [Sec. 6.1A].

Simple Trusses

A truss is **rigid** if it is designed in such a way that it does not greatly deform or collapse under a small load. A triangular truss consisting of three members connected at three joints is clearly a rigid truss (Fig. 6.24a). The truss obtained by adding two new members to the first one and connecting them at a new joint (Fig. 6.24b) is also rigid. Trusses obtained by repeating this procedure are called **simple trusses**. We may check that, in a simple truss, the total number of members is $m = 2n - 3$, where n is the total number of joints [Sec. 6.1A].

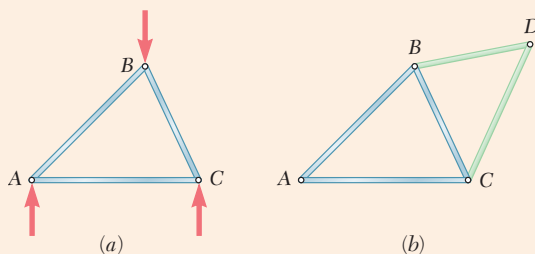


Fig. 6.24

Method of Joints

We can determine the forces in the various members of a simple truss by using the **method of joints** [Sec. 6.1B]. First, we obtain the reactions at the supports by considering the entire truss as a free body. Then we draw the free-body diagram of each pin, showing the forces exerted on the pin by the members or supports it connects. Since the members are straight two-force members, the force exerted by a member on the pin is directed along that member, and only the magnitude of the force is unknown. In the case of a simple truss, it is always possible to draw the free-body diagrams of the pins in such an order that only two unknown forces are included in each diagram. We obtain these forces from the corresponding two equilibrium equations or—if only three forces are involved—from the corresponding force triangle. If the force exerted by a member on a pin is directed toward

that pin, the member is in **compression**; if it is directed away from the pin, the member is in **tension** [Sample Prob. 6.1]. The analysis of a truss is sometimes expedited by first recognizing **joints under special loading conditions** [Sec. 6.1C]. The method of joints also can be extended for the analysis of three-dimensional or **space trusses** [Sec. 6.1D].

Method of Sections

The **method of sections** is usually preferable to the method of joints when we want to determine the force in only one member—or very few members—of a truss [Sec. 6.2A]. To determine the force in member BD of the truss of Fig. 6.25a, for example, we *pass a section* through members BD , BE , and CE ; remove these members; and use the portion ABC of the truss as a free body (Fig. 6.25b). Setting $\Sigma M_E = 0$, we determine the magnitude of force F_{BD} that represents the force in member BD . A positive sign indicates that the member is in *tension*; a negative sign indicates that it is in *compression* [Sample Probs. 6.2 and 6.3].

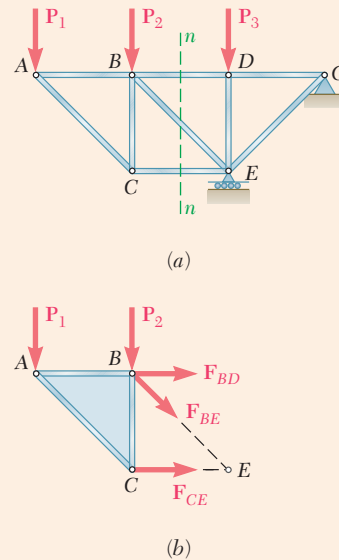


Fig. 6.25

Compound Trusses

The method of sections is particularly useful in the analysis of **compound trusses**, i.e., trusses that cannot be constructed from the basic triangular truss of Fig. 6.24a but are built by rigidly connecting several simple trusses [Sec. 6.2B]. If the component trusses are properly connected (e.g., one pin and one link, or three non-concurrent and unparallel links) and if the resulting structure is properly supported (e.g., one pin and one roller), the compound truss is **statically determinate, rigid, and completely constrained**. The following necessary—but not sufficient—condition is then satisfied: $m + r = 2n$, where m is the number of members, r is the number of unknowns representing the reactions at the supports, and n is the number of joints.

Frames and Machines

In the second part of the chapter, we analyzed **frames** and **machines**. These structures contain *multi-force members*, i.e., members acted upon by three or more forces. Frames are designed to support loads and are usually stationary, fully constrained structures. Machines are designed to transmit or modify forces and always contain moving parts [Sec. 6.3].

Analysis of a Frame

To analyze a frame, we first consider the entire frame to be a free body and write three equilibrium equations [Sec. 6.3A]. If the frame remains rigid when detached from its supports, the reactions involve only three unknowns and may be determined from these equations [Sample Probs. 6.4 and 6.5]. On the other hand, if the frame ceases to be rigid when detached from its supports, the reactions involve more than three unknowns, and we cannot determine them completely from the equilibrium equations of the frame [Sec. 6.3B; Sample Prob. 6.6].

Multi-force Members

We then dismember the frame and identify the various members as either two-force members or multi-force members; we assume pins form an integral part of one of the members they connect. We draw the free-body diagram of each of the multi-force members, noting that, when two multi-force members are connected to the same two-force member, they are acted upon by that member with *equal and opposite forces of unknown magnitude but known direction*. When two multi-force members are connected by a pin, they exert on each other *equal and opposite forces of unknown direction* that should be represented by *two unknown components*. We can then solve the equilibrium equations obtained from the free-body diagrams of the multi-force members for the various internal forces [Sample Probs. 6.4 and 6.5]. We also can use the equilibrium equations to complete the determination of the reactions at the supports [Sample Prob. 6.6]. Actually, if the frame is *statically determinate and rigid*, the free-body diagrams of the multi-force members could provide as many equations as there are unknown forces (including the reactions) [Sec. 6.3B]. However, as suggested previously, it is advisable to first consider the free-body diagram of the entire frame to minimize the number of equations that must be solved simultaneously.

Analysis of a Machine

To analyze a machine, we dismember it and, following the same procedure as for a frame, draw the free-body diagram of each multi-force member. The corresponding equilibrium equations yield the **output forces** exerted by the machine in terms of the **input forces** applied to it as well as the **internal forces** at the various connections [Sec. 6.4; Sample Prob. 6.7].

Review Problems

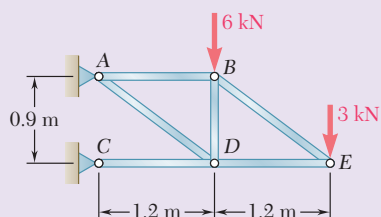


Fig. P6.164

6.164 Using the method of joints, determine the force in each member of the truss shown. State whether each member is in tension or compression.

6.165 Using the method of joints, determine the force in each member of the double-pitch roof truss shown. State whether each member is in tension or compression.

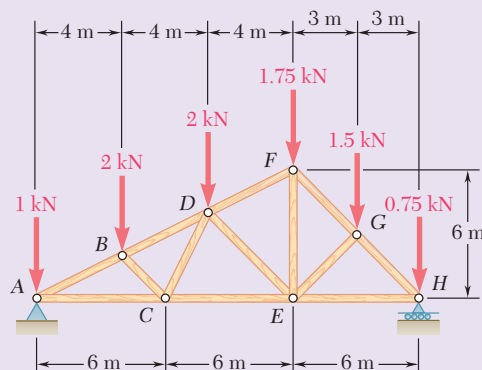


Fig. P6.165

6.166 A stadium roof truss is loaded as shown. Determine the force in members AB, AG, and FG.

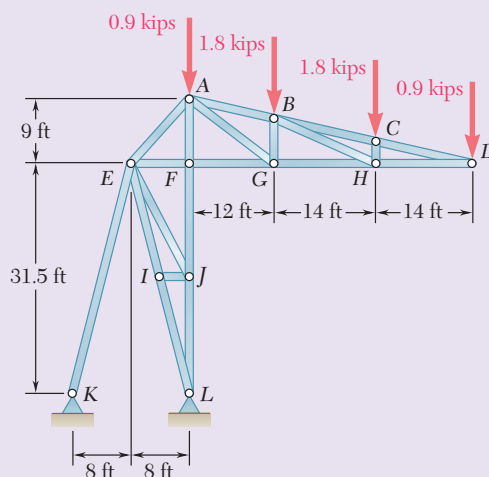


Fig. P6.166 and P6.167

6.167 A stadium roof truss is loaded as shown. Determine the force in members AE, EF, and FJ.

6.168 Determine the components of all forces acting on member ABD of the frame shown.

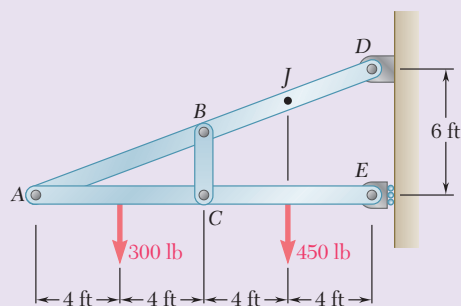


Fig. P6.168

6.169 Determine the components of the reactions at A and E if the frame is loaded by a clockwise couple of magnitude $36 \text{ N}\cdot\text{m}$ applied (a) at B , (b) at D .

6.170 Knowing that the pulley has a radius of 50 mm , determine the components of the reactions at B and E .

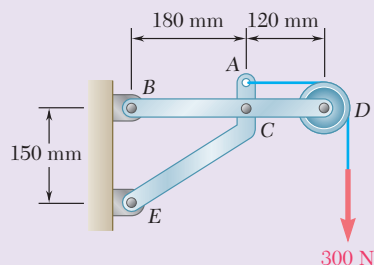


Fig. P6.170

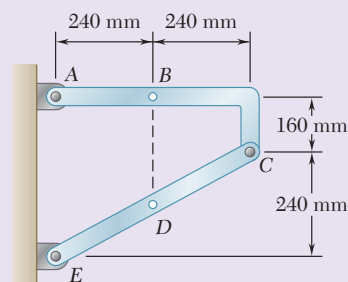


Fig. P6.169

6.171 For the frame and loading shown, determine the components of the forces acting on member CFE at C and F .

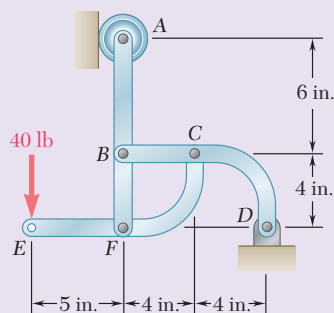


Fig. P6.171

6.172 For the frame and loading shown, determine the reactions at A , B , D , and E . Assume that the surface at each support is frictionless.

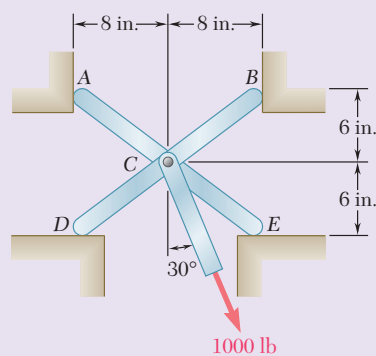


Fig. P6.172

- 6.173** Water pressure in the supply system exerts a downward force of 135 N on the vertical plug at A. Determine the tension in the fusible link DE and the force exerted on member BCE at B.

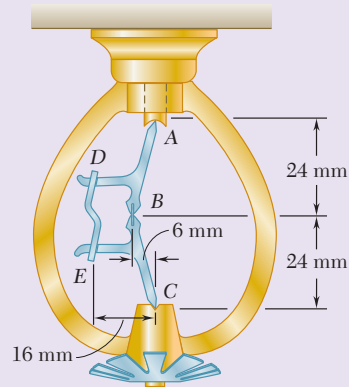


Fig. P6.173

- 6.174** A couple M with a magnitude of $1.5 \text{ kN}\cdot\text{m}$ is applied to the crank of the engine system shown. For each of the two positions shown, determine the force P required to hold the system in equilibrium.

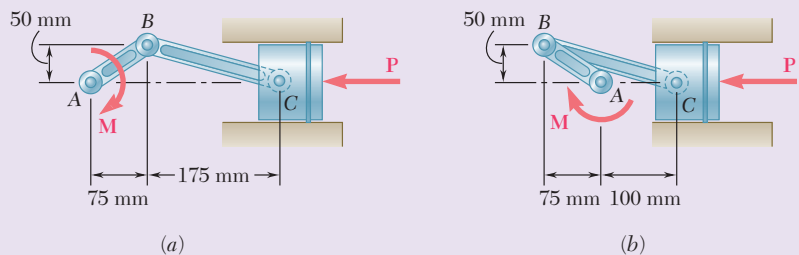


Fig. P6.174

- 6.175** The compound-lever pruning shears shown can be adjusted by placing pin A at various ratchet positions on blade ACE. Knowing that 300-lb vertical forces are required to complete the pruning of a small branch, determine the magnitude P of the forces that must be applied to the handles when the shears are adjusted as shown.

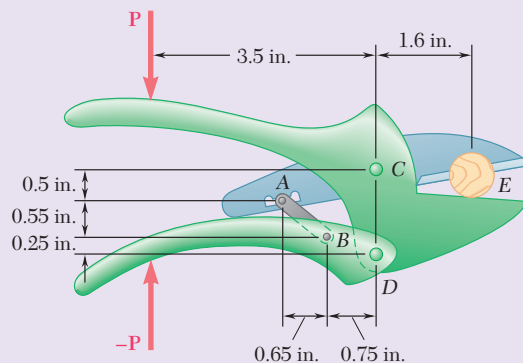


Fig. P6.175



7

Internal Forces and Moments

The Assut de l'Or Bridge in the City of Arts and Science in Valencia, Spain, is cable-stayed, where the bridge deck is supported by cables attached to the curved tower. The tower itself is partially supported by four anchor cables. The deck of the bridge consists of a system of beams that support the roadway.

Introduction

7.1 INTERNAL FORCES IN MEMBERS

7.2 BEAMS

7.2A Various Types of Loading and Support

7.2B Shear and Bending Moment in a Beam

7.2C Shear and Bending-Moment Diagrams

7.3 RELATIONS AMONG LOAD, SHEAR, AND BENDING MOMENT

*7.4 CABLES

7.4A Cables with Concentrated Loads

7.4B Cables with Distributed Loads

7.4C Parabolic Cables

*7.5 CATENARY CABLES

Objectives

- **Consider** the general state of internal member forces, which includes axial force, shearing force, and bending moment.
- **Apply** equilibrium analysis methods to obtain specific values, general expressions, and diagrams for shear and bending-moment in beams.
- **Examine** relations among load, shear, and bending-moment, and use these to obtain shear and bending-moment diagrams for beams.
- **Analyze** the tension forces in cables subjected to concentrated loads, loads uniformly distributed along the horizontal, and loads uniformly distributed along the cable itself.

Introduction

In previous chapters, we considered two basic problems involving structures: (1) determining the external forces acting on a structure (Chap. 4) and (2) determining the internal forces that hold together the various members forming a structure (Chap. 6). Now we consider the problem of determining the internal forces that hold together the parts of a given individual member.

We will first analyze the internal forces in the members of a frame, such as the crane considered in Fig. 6.1. Note that, whereas the internal forces in a straight two-force member can produce only **tension** or **compression** in that member, the internal forces in any other type of member usually produce **shear** and **bending** as well.

Most of this chapter is devoted to the analysis of the internal forces in two important types of engineering elements:

1. **Beams**, which are usually long, straight prismatic members designed to support loads applied at various points along it.
2. **Cables**, which are flexible members capable of withstanding only tension and are designed to support either concentrated or distributed loads. Cables are used in many engineering applications, such as suspension bridges and power transmission lines.

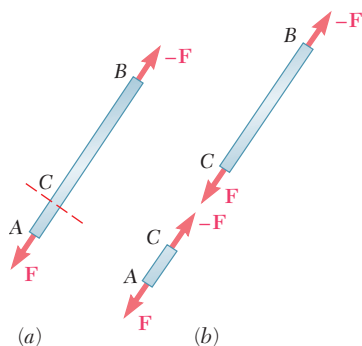


Fig. 7.1 A straight two-force member in tension. (a) External forces act at the ends of the member; (b) internal axial forces do not depend on the location of section C.

7.1 INTERNAL FORCES IN MEMBERS

Consider a straight two-force member AB (Fig. 7.1a). From Sec. 4.2A, we know that the forces F and $-F$ acting at A and B , respectively, must be directed along AB in opposite sense and have the same magnitude F . Suppose we cut the member at C . To maintain equilibrium of the resulting free bodies AC and CB , we must apply to AC a force $-F$ equal and

opposite to $\mathbf{F}$ and to CB a force $\mathbf{F}$ equal and opposite to $-\mathbf{F}$ (Fig. 7.1*b*). These new forces are directed along AB in opposite sense and have the same magnitude F . Since the two parts AC and CB were in equilibrium before the member was cut, **internal forces** equivalent to these new forces must have existed in the member itself. We conclude that, in the case of a straight two-force member, the internal forces that the two portions of the member exert on each other are equivalent to **axial forces**. The common magnitude F of these forces does not depend upon the location of the section C and is referred to as the *force in member AB* . In the case shown in Fig. 7.1, the member is in tension and elongates under the action of the internal forces. In the case represented in Fig. 7.2, the member is in compression and decreases in length under the action of the internal forces.

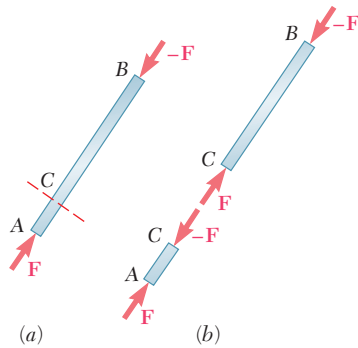


Fig. 7.2 A straight two-force member in compression. (a) External forces act at the ends; (b) internal axial forces are independent of the location of section C.

Next, consider a **multi-force member**. Take, for instance, member AD of the crane analyzed in Sec. 6.3A. This crane is shown again in Fig. 7.3*a*, and we drew the free-body diagram of member AD in Fig. 7.3*b*. Suppose we cut member AD at J and draw a free-body diagram for each of the portions JD and AJ (Fig. 7.3*c* and *d*). Considering the free body JD , we find that, to maintain its equilibrium, we need to apply at J a force $\mathbf{F}$ to balance the vertical component of $\mathbf{T}$; a force $\mathbf{V}$ to balance the

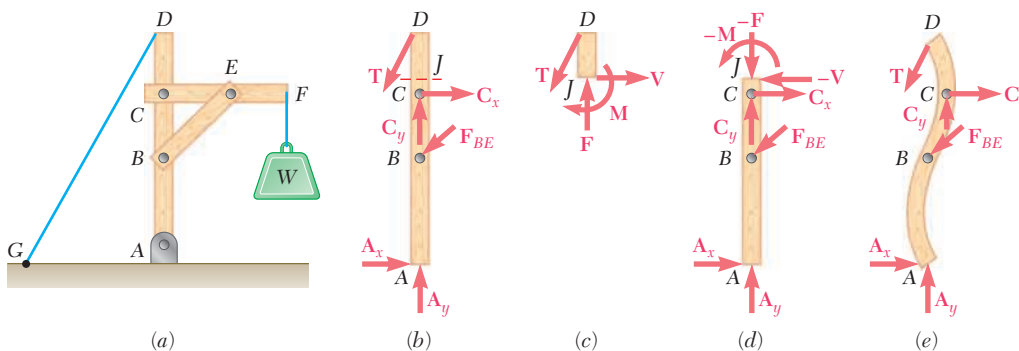


Fig. 7.3 (a) Crane from Chapter 6; (b) free-body diagram of multi-force member AD ; (c,d) free-body diagrams of sections of member AD showing internal force-couple systems; (e) deformation of member AD .



Photo 7.1 The design of the shaft of a circular saw must account for the internal forces resulting from the forces applied to the teeth of the blade. At a given point in the shaft, these internal forces are equivalent to a force-couple system consisting of axial and shearing forces and couples representing the bending and torsional moments.

horizontal component of $\mathbf{T}$; and a couple $\mathbf{M}$ to balance the moment of $\mathbf{T}$ about J . Again, we conclude that internal forces must have existed at J before member AD was cut, which is equivalent to the force-couple system shown in Fig. 7.3c.

According to Newton's third law, the internal forces acting on AJ must be equivalent to an equal and opposite force-couple system, as shown in Fig. 7.3d. Clearly, the action of the internal forces in member AD is *not limited to producing tension or compression*, as in the case of straight two-force members; the internal forces *also produce shear and bending*. The force $\mathbf{F}$ is an **axial force**; the force $\mathbf{V}$ is called a **shearing force**; and the moment $\mathbf{M}$ of the couple is known as the **bending moment at J** . Note that, when determining internal forces in a member, you should clearly indicate on which portion of the member the forces are supposed to act. The deformation that occurs in member AD is sketched in Fig. 7.3e. The actual analysis of such a deformation is part of the study of mechanics of materials.

Also note that, in a **two-force member that is not straight**, the internal forces are also equivalent to a force-couple system. This is shown in Fig. 7.4, where the two-force member ABC has been cut at D .

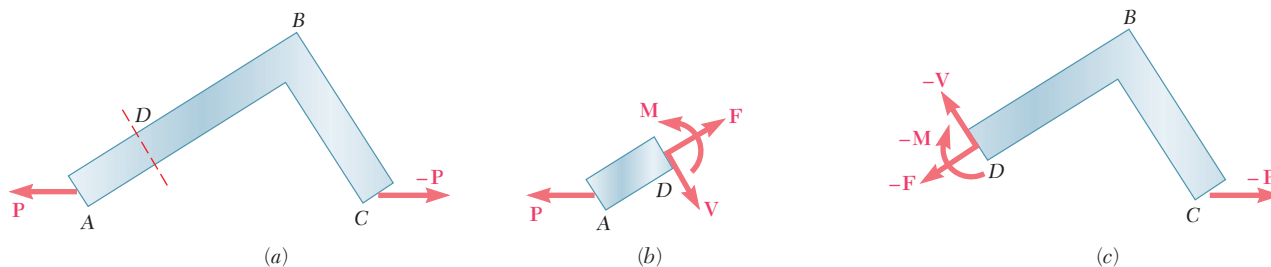
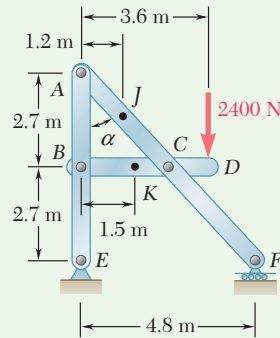


Fig. 7.4 (a) Free-body diagram of a two-force member that is not straight; (b, c) free-body diagrams of sections of member ABC showing internal force-couple systems.

Sample Problem 7.1

In the frame shown, determine the internal forces (a) in member ACF at point J , (b) in member BCD at point K . This frame was previously analyzed in Sample Prob. 6.5.



STRATEGY: After isolating each member, you can cut it at the given point and treat the resulting parts as objects in equilibrium. Analysis of the equilibrium equations, as we did before in Sample Problem 6.5, will determine the internal force-couple system.

MODELING: The reactions and the connection forces acting on each member of the frame were determined previously in Sample Prob. 6.5. The results are repeated in Fig. 1.

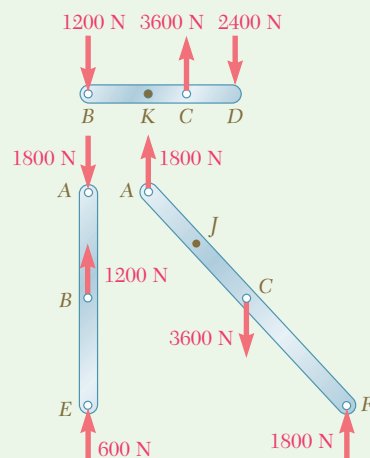


Fig. 1 Reactions and connection forces acting on each member of the frame.

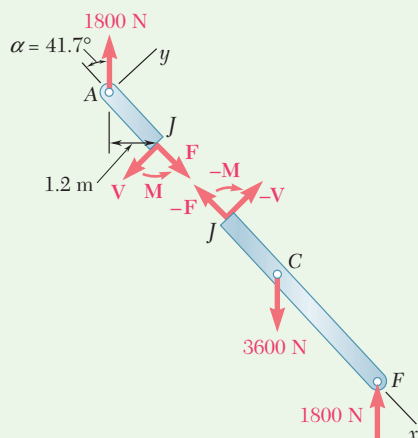


Fig. 2 Free-body diagrams of portion AJ and JF of member ACF .

ANALYSIS:

a. Internal Forces at J . Cut member ACF at point J , obtaining the two parts shown in Fig. 2. Represent the internal forces at J by an equivalent force-couple system, which can be determined by considering the equilibrium of either part. Considering the free body AJ , you have

$$\begin{aligned}
 +\curvearrowright \Sigma M_J &= 0: & -(1800 \text{ N})(1.2 \text{ m}) + M &= 0 \\
 & & M &= +2160 \text{ N}\cdot\text{m} & \mathbf{M = 2160 \text{ N}\cdot\text{m} \curvearrowright} \\
 +\searrow \Sigma F_x &= 0: & F - (1800 \text{ N}) \cos 41.7^\circ &= 0 \\
 & & F &= +1344 \text{ N} & \mathbf{F = 1344 \text{ N} \searrow} \\
 +\nearrow \Sigma F_y &= 0: & -V + (1800 \text{ N}) \sin 41.7^\circ &= 0 \\
 & & V &= +1197 \text{ N} & \mathbf{V = 1197 \text{ N} \nearrow}
 \end{aligned}$$

The internal forces at J are therefore equivalent to a couple $\mathbf{M}$, an axial force $\mathbf{F}$, and a shearing force $\mathbf{V}$. The internal force-couple system acting on part JCF is equal and opposite.

b. Internal Forces at K . Cut member BCD at K , obtaining the two parts shown in Fig. 3. Considering the free body BK , you obtain

$$\begin{aligned}
 +\curvearrowright \Sigma M_K &= 0: & (1200 \text{ N})(1.5 \text{ m}) + M &= 0 \\
 & & M &= -1800 \text{ N}\cdot\text{m} & \mathbf{M = 1800 \text{ N}\cdot\text{m} \curvearrowleft} \\
 +\rightarrow \Sigma F_x &= 0: & F &= 0 & \mathbf{F = 0} \\
 +\uparrow \Sigma F_y &= 0: & -1200 \text{ N} - V &= 0 \\
 & & V &= -1200 \text{ N} & \mathbf{V = 1200 \text{ N} \uparrow}
 \end{aligned}$$

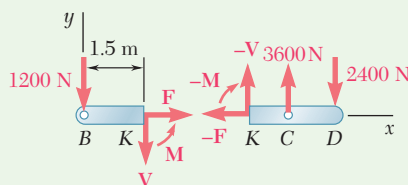


Fig. 3 Free-body diagrams of portion BK and DK of member BCD .

REFLECT and THINK: The mathematical techniques involved in solving a problem of this type are not new; they are simply applications of concepts presented in earlier chapters. However, the physical interpretation is new: we are now determining the internal forces and moments within a structural member. These are of central importance in the study of mechanics of materials.

SOLVING PROBLEMS ON YOUR OWN

In this section, we discussed how to determine the internal forces in the member of a frame. The internal forces at a given point in a **straight two-force member** reduce to an axial force, but in all other cases, they are equivalent to a **force-couple system** consisting of an **axial force F** , a **shearing force V** , and a couple **M** representing the **bending moment** at that point.

To determine the internal forces at a given point J of the member of a frame, you should take the following steps.

1. **Draw a free-body diagram of the entire frame**, and use it to determine as many of the reactions at the supports as you can.
2. **Dismember the frame and draw a free-body diagram of each of its members**. Write as many equilibrium equations as are necessary to find all of the forces acting on the member on which point J is located.
3. **Cut the member at point J and draw a free-body diagram of each resulting portion**. Apply to each portion at point J the force components and couple representing the internal forces exerted by the other portion. These force components and couples are equal in magnitude and opposite in sense.
4. **Select one of the two free-body diagrams** you have drawn and use it to write three equilibrium equations for the corresponding portion of the member.
 - a. **Summing moments about J** and equating them to zero yields the bending moment at point J .
 - b. **Summing components in directions parallel and perpendicular** to the member at J and equating them to zero yields, respectively, the axial and shearing forces.
5. **When recording your answers, be sure to specify the portion of the member** you have used, since the forces and couples acting on the two portions have opposite senses.

The solutions of the problems in this section require you to determine the forces exerted on each other by the various members of a frame, so be sure to review the methods used in Chap. 6 to solve this type of problem. When frames involve pulleys and cables, for instance, remember that the forces exerted by a pulley on the member of the frame to which it is attached have the same magnitude and direction as the forces exerted by the cable on the pulley [Prob. 6.90].

Problems

7.1 and 7.2 Determine the internal forces (axial force, shearing force, and bending moment) at point J of the structure indicated.

7.1 Frame and loading of Prob. 6.76

7.2 Frame and loading of Prob. 6.78

7.3 Determine the internal forces at point J when $\alpha = 90^\circ$.

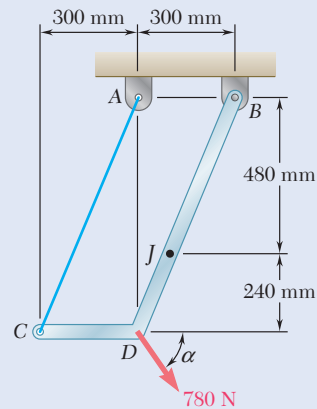


Fig. P7.3 and P7.4

7.4 Determine the internal forces at point J when $\alpha = 0$.

7.5 and 7.6 For the frame and loading shown, determine the internal forces at the point indicated:

7.5 Point J

7.6 Point K

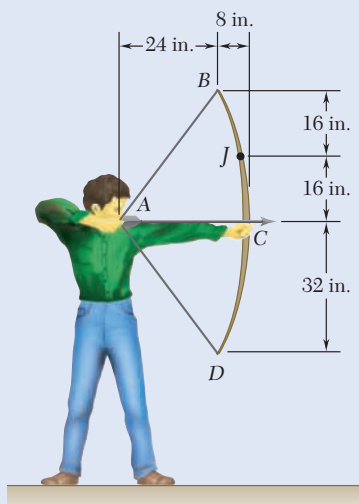


Fig. P7.7

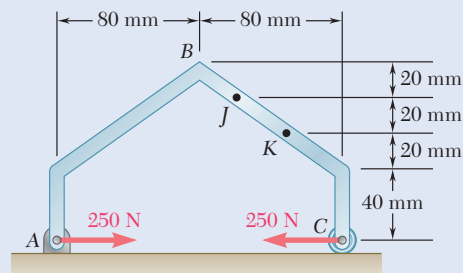


Fig. P7.5 and P7.6

7.7 An archer aiming at a target is pulling with a 45-lb force on the bowstring. Assuming that the shape of the bow can be approximated by a parabola, determine the internal forces at point J .

7.8 For the bow of Prob. 7.7, determine the magnitude and location of the maximum (a) axial force, (b) shearing force, (c) bending moment.

- 7.9** A semicircular rod is loaded as shown. Determine the internal forces at point J .

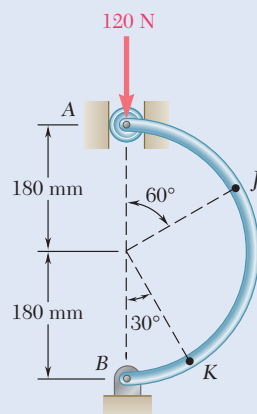


Fig. P7.9 and P7.10

- 7.10** A semicircular rod is loaded as shown. Determine the internal forces at point K .

- 7.11** A semicircular rod is loaded as shown. Determine the internal forces at point J knowing that $\theta = 30^\circ$.

- 7.12** A semicircular rod is loaded as shown. Determine the magnitude and location of the maximum bending moment in the rod.

- 7.13** The axis of the curved member AB is a parabola with vertex at A . If a vertical load $\mathbf{P}$ of magnitude 450 lb is applied at A , determine the internal forces at J when $h = 12$ in., $L = 40$ in., and $a = 24$ in.

- 7.14** Knowing that the axis of the curved member AB is a parabola with vertex at A , determine the magnitude and location of the maximum bending moment.

- 7.15** Knowing that the radius of each pulley is 200 mm and neglecting friction, determine the internal forces at point J of the frame shown.

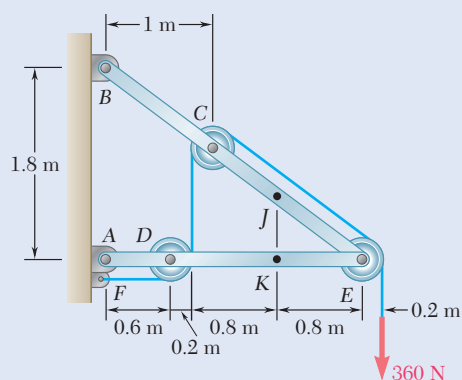


Fig. P7.15 and P7.16

- 7.16** Knowing that the radius of each pulley is 200 mm and neglecting friction, determine the internal forces at point K of the frame shown.

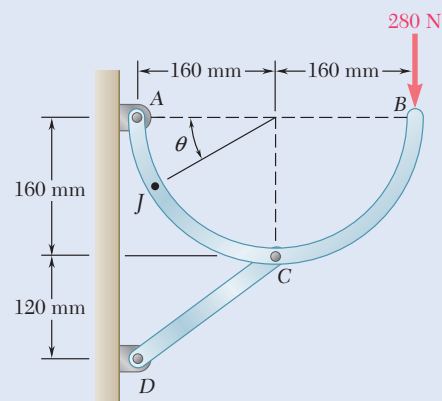


Fig. P7.11 and P7.12

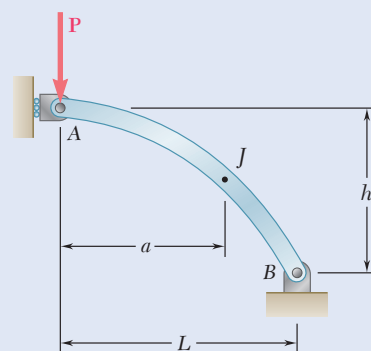
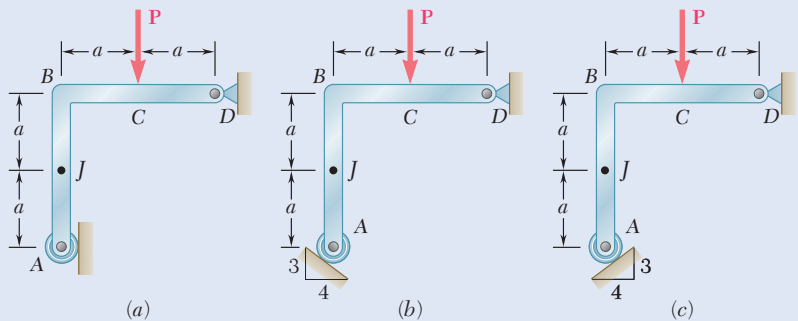


Fig. P7.13 and P7.14

-
- Diagram of a frame structure. A horizontal beam ADE is supported by a pin at A and a roller at E. A vertical cable is attached to the beam at D. A diagonal member BC is connected to the beam at B and has a roller support at C. A vertical force of 360 N is applied at E. Dimensions are given in meters.

Diagram of a mechanism with two bars, AB and BC , pivoted at A and B respectively. The bars meet at point C . A roller with radius $r = 2.5$ in. is in contact with the horizontal surface at point C . The horizontal distance from A to the vertical line through C is 12 in., and the horizontal distance from C to B is 6.75 in. The vertical height of B above the surface is 9 in.

7.21 and 7.22 A force $\mathbf{P}$ is applied to a bent rod that is supported by a roller and a pin and bracket. For each of the three cases shown, determine the internal forces at point J .



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- 7.23** A quarter-circular rod of weight W and uniform cross section is supported as shown. Determine the bending moment at point J when $\theta = 30^\circ$.

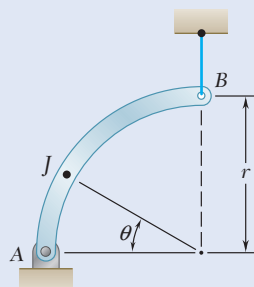


Fig. P7.23

- 7.24** For the rod of Prob. 7.23, determine the magnitude and location of the maximum bending moment.

- 7.25** A semicircular rod of weight W and uniform cross section is supported as shown. Determine the bending moment at point J when $\theta = 60^\circ$.

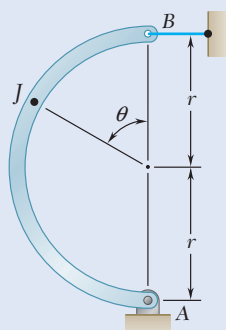


Fig. P7.25 and P7.26

- 7.26** A semicircular rod of weight W and uniform cross section is supported as shown. Determine the bending moment at point J when $\theta = 150^\circ$.

- 7.27 and 7.28** A half section of pipe rests on a frictionless horizontal surface as shown. If the half section of pipe has a mass of 9 kg and a diameter of 300 mm, determine the bending moment at point J when $\theta = 90^\circ$.

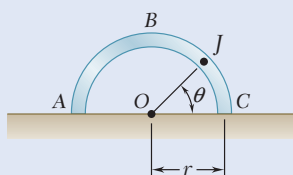


Fig. P7.27

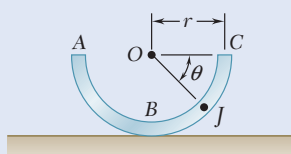


Fig. P7.28

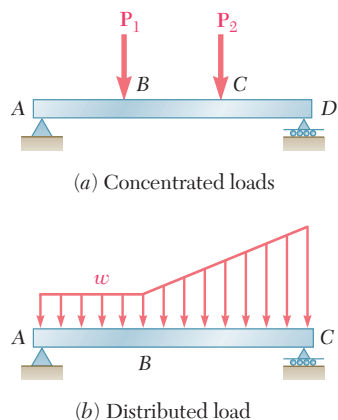


Fig. 7.5 A beam may be subjected to (a) concentrated loads or (b) distributed loads, or a combination of both.

7.2 BEAMS

A structural member designed to support loads applied at various points along the member is known as a **beam**. In most cases, the loads are perpendicular to the axis of the beam and cause only shear and bending in the beam. When the loads are not at a right angle to the beam, they also produce axial forces in the beam.

Beams are usually long, straight prismatic bars. Designing a beam for the most effective support of the applied loads is a two-part process: (1) determine the shearing forces and bending moments produced by the loads and (2) select the cross section best suited to resist these shearing forces and bending moments. Here we are concerned with the first part of the problem of beam design. The second part belongs to the study of mechanics of materials.

7.2A Various Types of Loading and Support

A beam can be subjected to **concentrated loads** $P_1, P_2, \dots$ that are expressed in newtons, pounds, or their multiples, kilonewtons and kips (Fig. 7.5a). We can also subject a beam to a **distributed load** w , expressed in N/m, kN/m, lb/ft, or kips/ft (Fig. 7.5b). In many cases, a beam is subjected to a combination of both types of load. When the load w per unit length has a constant value over part of the beam (as between A and B in Fig. 7.5b), the load is said to be **uniformly distributed** over that part of the beam. Determining the reactions at the supports is considerably simplified if we replace distributed loads by equivalent concentrated loads, as explained in Sec. 5.3A. However, you should not do this substitution, or at least perform it with care, when calculating internal forces (see Sample Prob. 7.3).

Beams are classified according to the way in which they are supported. Figure 7.6 shows several types of beams used frequently.

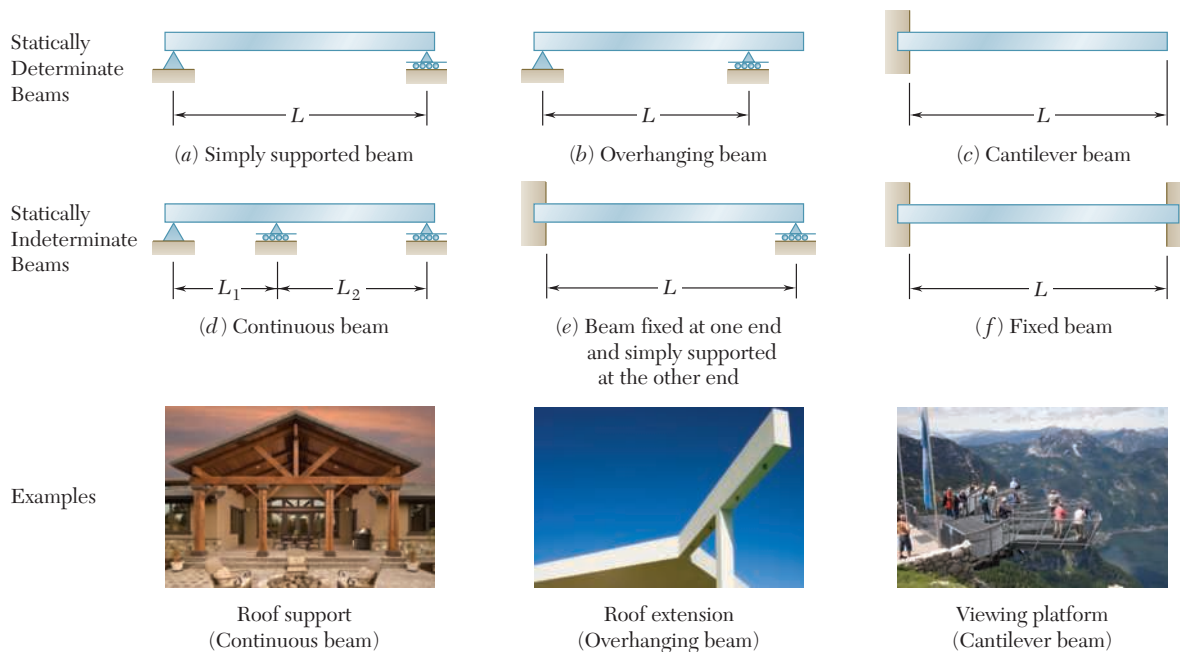


Fig. 7.6 Some common types of beams and their supports.

The distance L between supports is called the **span**. Note that the reactions are determinate if the supports involve only three unknowns. If more unknowns are involved, the reactions are statically indeterminate, and the methods of statics are not sufficient to determine the reactions. In such a case, we must take into account the properties of the beam with regard to its resistance to bending. Beams supported by only two rollers are not shown here; they are partially constrained and move under certain types of loadings.

Sometimes two or more beams are connected by hinges to form a single continuous structure. Two examples of beams hinged at a point H are shown in Fig. 7.7. Here the reactions at the supports involve four unknowns and cannot be determined from the free-body diagram of the two-beam system. However, we can determine the reactions by considering the free-body diagram of each beam separately. Analysis of this situation involves six unknowns (including two force components at the hinge), and six equations are available.

7.2B Shear and Bending Moment in a Beam

Consider a beam AB subjected to various concentrated and distributed loads (Fig. 7.8a). We propose to determine the shearing force and bending moment at any point of the beam. In the example considered here, the beam is simply supported, but the method used could be applied to any type of statically determinate beam.

First we determine the reactions at A and B by choosing the entire beam as a free body (Fig. 7.8b). Setting $\Sigma M_A = 0$ and $\Sigma M_B = 0$, we obtain, respectively, $\mathbf{R}_B$ and $\mathbf{R}_A$.

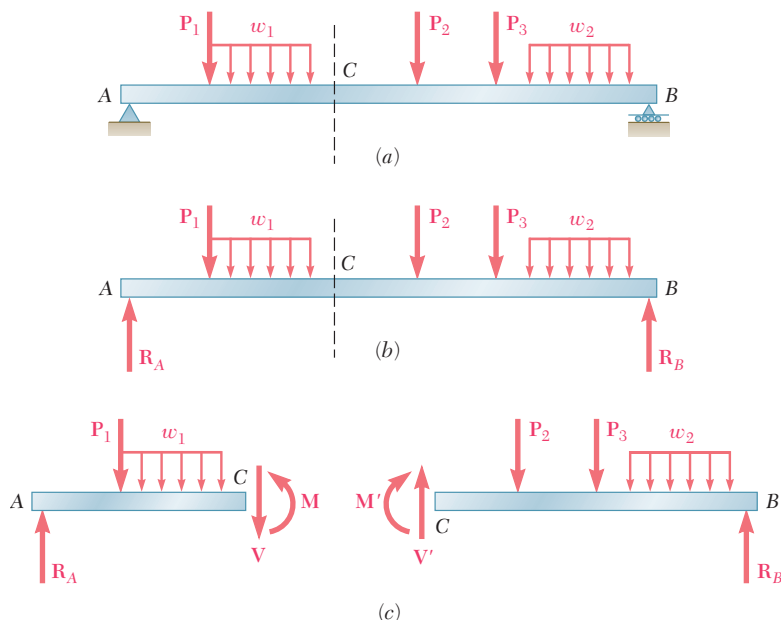


Fig. 7.8 (a) A simply supported beam AB ; (b) free-body diagram of the beam; (c) free-body diagrams of portions AC and CB of the beam, showing internal shearing forces and couples.

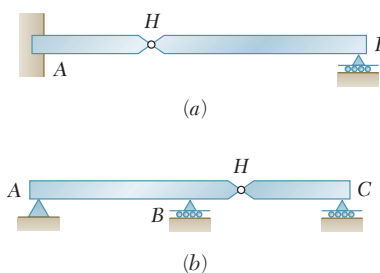


Fig. 7.7 Examples of two-beam systems connected by a hinge. In both cases, free-body diagrams of each individual beam enable you to determine the support reactions.



Photo 7.2 As a truck crosses a highway overpass, the internal forces vary in the beams of the overpass.

To determine the internal forces at an arbitrary point C , we cut the beam at C and draw the free-body diagrams of the portions AC and CB (Fig. 7.8c). Using the free-body diagram of AC , we can determine the shearing force $\mathbf{V}$ at C by equating the sum of the vertical components of all forces acting on AC to zero. Similarly, we can find the bending moment $\mathbf{M}$ at C by equating the sum of the moments about C of all forces and couples acting on AC to zero. Alternatively, we could use the free-body diagram of CB [†] and determine the shearing force $\mathbf{V}'$ and the bending moment $\mathbf{M}'$ by equating the sum of the vertical components and the sum of the moments about C of all forces and couples acting on CB to zero. Although this choice of free bodies may make the computation of the numerical values of the shearing force and bending moment easier, it requires us to indicate on which portion of the beam the internal forces considered are acting. If we want to calculate and efficiently record the shearing force and bending moment at every point of the beam, we must devise a way to avoid having to specify which portion of the beam is used as a free body every time. Therefore, we shall adopt the following conventions.

In determining the shearing force in a beam, we *always assume* that the internal forces $\mathbf{V}$ and $\mathbf{V}'$ are directed as shown in Fig. 7.8c. A positive value obtained for their common magnitude V indicates that this assumption is correct and that the shearing forces are actually directed as shown. A negative value obtained for V indicates that the assumption is wrong and the shearing forces are directed in the opposite way. Thus, to define completely the shearing forces at a given point of the beam, we only need to record the magnitude V , together with a plus or minus sign. The scalar V is commonly referred to as the **shear** at the given point of the beam.

Similarly, we *always assume* that the internal couples $\mathbf{M}$ and $\mathbf{M}'$ are directed as shown in Fig. 7.8c. A positive value obtained for their magnitude M , commonly referred to as the **bending moment**, indicates that this assumption is correct, whereas a negative value indicates that it is wrong.

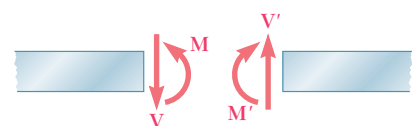
Summarizing these sign conventions, we state:

The shear V and the bending moment M at a given point of a beam are said to be positive when the internal forces and couples acting on each portion of the beam are directed as shown in Fig. 7.9a.

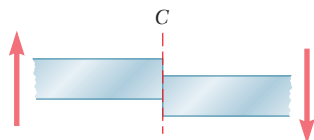
You may be able to remember these conventions more easily by noting that:

1. The shear at C is positive when the **external** forces (loads and reactions) acting on the beam tend to shear off the beam at C as indicated in Fig. 7.9b.
2. The bending moment at C is positive when the **external** forces acting on the beam tend to bend the beam at C in a concave-up fashion as indicated in Fig. 7.9c.

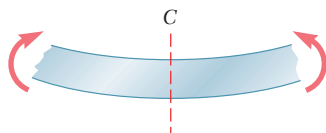
[†]We now designate the force and couple representing the internal forces acting on CB by $\mathbf{V}'$ and $\mathbf{M}'$, rather than by $-\mathbf{V}$ and $-\mathbf{M}$ as done earlier. The reason is to avoid confusion when applying the sign convention we are about to introduce.



(a) Internal forces at section
(positive shear and positive bending moment)



(b) Effect of external forces
(positive shear)



(c) Effect of external forces
(positive bending moment)

Fig. 7.9 Figure for remembering the signs of shear and bending moment.

It may also help to note that the situation described in Fig. 7.9, in which the values of both the shear and the bending moment are positive, is precisely the situation that occurs in the left half of a simply supported beam carrying a single concentrated load at its midpoint. This particular example is fully discussed in the following section.

7.2C Shear and Bending-Moment Diagrams

Now that we have clearly defined shear and bending moment in sense as well as in magnitude, we can easily record their values at any point along a beam by plotting these values against the distance x measured from one end of the beam. The graphs obtained in this way are called, respectively, the **shear diagram** and the **bending-moment diagram**.

As an example, consider a simply supported beam AB of span L subjected to a single concentrated load P applied at its midpoint D (Fig. 7.10a). We first determine the reactions at the supports from the free-body diagram of the entire beam (Fig. 7.10b); we find that the magnitude of each reaction is equal to $P/2$.

Next we cut the beam at a point C between A and D and draw the free-body diagrams of AC and CB (Fig. 7.10c). Assuming that shear and bending moment are positive, we direct the internal forces V and V' and the internal couples M and M' as indicated in Fig. 7.9a. Considering the free body AC , we set the sum of the vertical components and the sum of the moments about C of the forces acting on the free body to zero. From this, we find $V = +P/2$ and $M = +Px/2$. Therefore, both shear and bending moment are positive. (You can check this by observing that the reaction at A tends to shear off and to bend the beam at C as indicated in Fig. 7.9b and c.) Now let's plot V and M between A and D (Fig. 7.10e and f). The shear has a constant value $V = P/2$, whereas the bending moment increases linearly from $M = 0$ at $x = 0$ to $M = PL/4$ at $x = L/2$.

Proceeding along the beam, we cut it at a point E between D and B and consider the free body EB (Fig. 7.10d). As before, the sum of the vertical components and the sum of the moments about E of the forces acting on the free body are zero. We obtain $V = -P/2$ and $M = P(L - x)/2$. The shear is therefore negative and the bending moment is positive. (Again, you can check this by observing that the reaction at B bends the beam at E as indicated in Fig. 7.9c but tends to shear it off in a manner opposite to that shown in Fig. 7.9b.) We can now complete the shear and bending-moment diagrams of Fig. 7.10e and f. The shear has a constant value $V = -P/2$ between D and B , whereas the bending moment decreases linearly from $M = PL/4$ at $x = L/2$ to $M = 0$ at $x = L$.

Note that when a beam is subjected to concentrated loads only, the shear is of constant value between loads and the bending moment varies linearly between loads. However, when a beam is subjected to distributed loads, the shear and bending moment vary quite differently (see Sample Prob. 7.3).

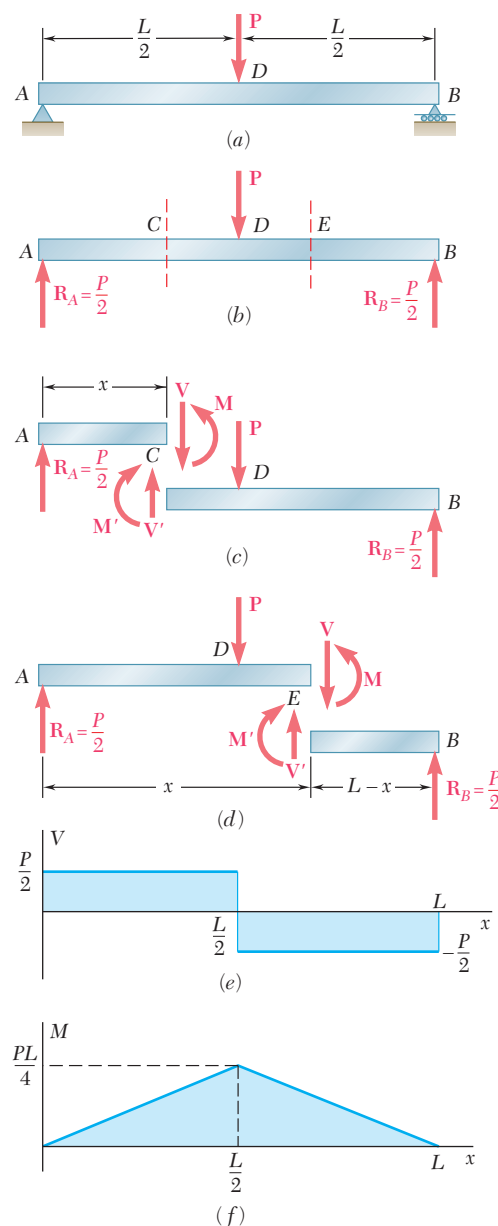


Fig. 7.10 (a) A beam supporting a single concentrated load at its midpoint; (b) free-body diagram of the beam; (c) free-body diagrams of parts of the beam after a cut at C ; (d) free-body diagrams of parts of the beam after a cut at E ; (e) shear diagram of the beam; (f) bending-moment diagram of the beam.

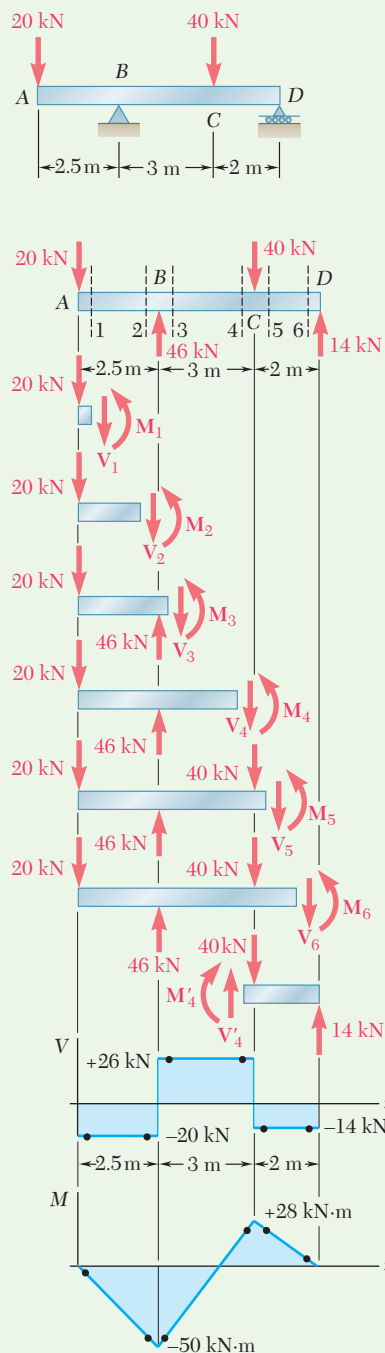


Fig. 1 Free-body diagrams of beam sections, and the resulting shear and bending-moment diagrams.

Sample Problem 7.2

Draw the shear and bending-moment diagrams for the beam and loading shown.

STRATEGY: Treat the entire beam as a free body to determine the reactions, then cut the beam just before and just after each external concentrated force (Fig. 1) to see how the shear and bending moment change along the length of the beam.

MODELING and ANALYSIS:

Free-Body, Entire Beam. From the free-body diagram of the entire beam, find the reactions at B and D:

$$\mathbf{R}_B = 46 \text{ kN } \uparrow \quad \mathbf{R}_D = 14 \text{ kN } \uparrow$$

Shear and Bending Moment. First, determine the internal forces just to the right of the 20-kN load at A. Consider the stub of beam to the left of point 1 as a free body, and assume V and M are positive (according to the standard convention). Then you have

$$\begin{aligned} +\uparrow \Sigma F_y = 0: & \quad -20 \text{ kN} - V_1 = 0 & \quad V_1 = -20 \text{ kN} \\ +\curvearrowright \Sigma M_1 = 0: & \quad (20 \text{ kN})(0 \text{ m}) + M_1 = 0 & \quad M_1 = 0 \end{aligned}$$

Next, consider the portion of the beam to the left of point 2 as a free body:

$$\begin{aligned} +\uparrow \Sigma F_y = 0: & \quad -20 \text{ kN} - V_2 = 0 & \quad V_2 = -20 \text{ kN} \\ +\curvearrowright \Sigma M_2 = 0: & \quad (20 \text{ kN})(2.5 \text{ m}) + M_2 = 0 & \quad M_2 = -50 \text{ kN}\cdot\text{m} \end{aligned}$$

Determine the shear and bending moment at sections 3, 4, 5, and 6 in a similar way from the free-body diagrams. The results are

$$\begin{aligned} V_3 &= +26 \text{ kN} & M_3 &= -50 \text{ kN}\cdot\text{m} \\ V_4 &= +26 \text{ kN} & M_4 &= +28 \text{ kN}\cdot\text{m} \\ V_5 &= -14 \text{ kN} & M_5 &= +28 \text{ kN}\cdot\text{m} \\ V_6 &= -14 \text{ kN} & M_6 &= 0 \end{aligned}$$

For several of the later cuts, the results are easier to obtain by considering as a free body the portion of the beam to the right of the cut. For example, consider the portion of the beam to the right of point 4. You have

$$\begin{aligned} +\uparrow \Sigma F_y = 0: & \quad V_4 - 40 \text{ kN} + 14 \text{ kN} = 0 & \quad V_4 = +26 \text{ kN} \\ +\curvearrowright \Sigma M_4 = 0: & \quad -M_4 + (14 \text{ kN})(2 \text{ m}) = 0 & \quad M_4 = +28 \text{ kN}\cdot\text{m} \end{aligned}$$

Shear and Bending-Moment Diagrams. Now plot the six points shown on the shear and bending-moment diagrams. As indicated in Sec. 7.2C, the shear is of constant value between concentrated loads, and the bending moment varies linearly. You therefore obtain the shear and bending-moment diagrams shown in Fig. 1.

REFLECT and THINK: The calculations are pretty similar for each new choice of free body. However, moving along the beam, the shear changes magnitude whenever you pass a transverse force and the graph of the bending moment changes slope at these points.

Sample Problem 7.3

Draw the shear and bending-moment diagrams for the beam AB . The distributed load of 40 lb/in. extends over 12 in. of the beam from A to C , and the 400-lb load is applied at E .

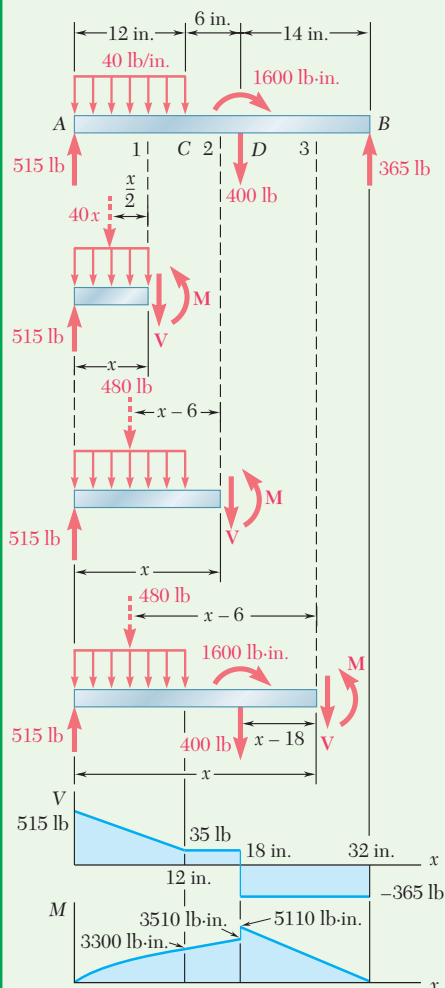
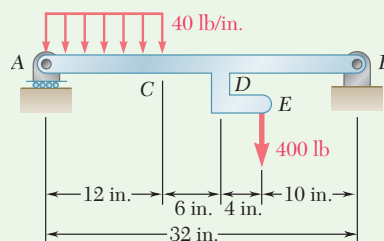


Fig. 2 Free-body diagrams of beam sections, and the resulting shear and bending-moment diagrams.

STRATEGY: Again, consider the entire beam as a free body to find the reactions. Then cut the beam within each region of continuous load. This will enable you to determine continuous functions for shear and bending moment, which you can then plot on a graph.

MODELING and ANALYSIS:

Free-Body, Entire Beam. Determine the reactions by considering the entire beam as a free body (Fig. 1).

$$\begin{aligned}
 +\uparrow \Sigma M_A &= 0: & B_y(32 \text{ in.}) - (480 \text{ lb})(6 \text{ in.}) - (400 \text{ lb})(22 \text{ in.}) &= 0 \\
 & & B_y &= +365 \text{ lb} & \mathbf{B_y = 365 \text{ lb} \uparrow} \\
 +\uparrow \Sigma M_B &= 0: & (480 \text{ lb})(26 \text{ in.}) + (400 \text{ lb})(10 \text{ in.}) - A(32 \text{ in.}) &= 0 \\
 & & A &= +515 \text{ lb} & \mathbf{A = 515 \text{ lb} \uparrow} \\
 \rightarrow \Sigma F_x &= 0: & B_x &= 0 & \mathbf{B_x = 0}
 \end{aligned}$$

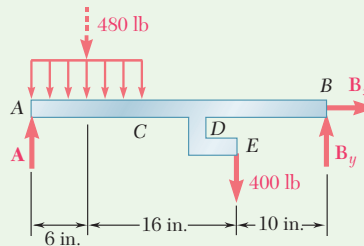


Fig. 1 Free-body diagram of entire beam.

Now, replace the 400-lb load by an equivalent force-couple system acting on the beam at point D and cut the beam at several points (Fig. 2).

Shear and Bending Moment. From A to C. Determine the internal forces at a distance x from point A by considering the portion of the beam to the left of point 1. Replace that part of the distributed load acting on the free body by its resultant. You get

$$\begin{aligned} +\uparrow \Sigma F_y = 0: & \quad 515 - 40x - V = 0 & \quad V = 515 - 40x \\ +\curvearrowright \Sigma M_1 = 0: & \quad -515x + 40x\left(\frac{1}{2}x\right) + M = 0 & \quad M = 515x - 20x^2 \end{aligned}$$

Note that V and M are not numerical values, but they are expressed as functions of x . The free-body diagram shown can be used for all values of x smaller than 12 in., so the expressions obtained for V and M are valid throughout the region $0 < x < 12$ in.

From C to D. Consider the portion of the beam to the left of point 2. Again replacing the distributed load by its resultant, you have

$$\begin{aligned} +\uparrow \Sigma F_y = 0: & \quad 515 - 480 - V = 0 & \quad V = 35 \text{ lb} \\ +\curvearrowright \Sigma M_2 = 0: & \quad -515x + 480(x - 6) + M = 0 & \quad M = (2880 + 35x) \text{ lb}\cdot\text{in.} \end{aligned}$$

These expressions are valid in the region $12 \text{ in.} < x < 18 \text{ in.}$

From D to B. Use the portion of the beam to the left of point 3 for the region $18 \text{ in.} < x < 32 \text{ in.}$ Thus,

$$\begin{aligned} +\uparrow \Sigma F_y = 0: & \quad 515 - 480 - 400 - V = 0 & \quad V = -365 \text{ lb} \\ +\curvearrowright \Sigma M_3 = 0: & \quad -515x + 480(x - 6) - 1600 + 400(x - 18) + M = 0 \\ & \quad \quad \quad M = (11,680 - 365x) \text{ lb}\cdot\text{in.} \end{aligned}$$

Shear and Bending-Moment Diagrams. Plot the shear and bending-moment diagrams for the entire beam. Note that the couple of moment 1600 lb·in. applied at point D introduces a discontinuity into the bending-moment diagram. Also note that the bending-moment diagram under the distributed load is not straight but is slightly curved.

REFLECT and THINK: Shear and bending-moment diagrams typically feature various kinds of curves and discontinuities. In such cases, it is often useful to express V and M as functions of location x as well as to determine certain numerical values.

SOLVING PROBLEMS ON YOUR OWN

In this section, you saw how to determine the **shear** V and the **bending moment** M at any point in a beam. You also learned to draw the **shear diagram** and the **bending-moment diagram** for the beam by plotting, respectively, V and M against the distance x measured along the beam.

A. Determining the shear and bending moment in a beam. To determine the shear V and the bending moment M at a given point C of a beam, take the following steps.

1. Draw a free-body diagram of the entire beam, and use it to determine the reactions at the beam supports.

2. Cut the beam at point C , and using the original loading, select one of the two resulting portions of the beam.

3. Draw the free-body diagram of the portion of the beam you have selected. Show:

a. The loads and the reactions exerted on that portion of the beam, replacing each distributed load by an equivalent concentrated load, as explained in Sec. 5.3A.

b. The shearing force and the bending moment representing the internal forces at C . To facilitate recording the shear V and the bending moment M after determining them, follow the convention indicated in Figs. 7.8 and 7.9. Thus, if you are using the portion of the beam located to the *left* of C , apply at C a *shearing force V directed downward* and a *bending moment M directed counterclockwise*. If you are using the portion of the beam located to the *right* of C , apply at C a *shearing force V' directed upward* and a *bending moment M' directed clockwise* [Sample Prob. 7.2].

4. Write the equilibrium equations for the portion of the beam you have selected. Solve the equation $\Sigma F_y = 0$ for V and the equation $\Sigma M_C = 0$ for M .

5. Record the values of V and M with the sign obtained for each of them. A positive sign for V means that the shearing forces exerted at C on each of the two portions of the beam are directed as shown in Figs. 7.8 and 7.9; a negative sign means they have the opposite sense. Similarly, a positive sign for M means that the bending couples at C are directed as shown in these figures, and a negative sign means that they have the opposite sense. In addition, a positive sign for M means that the concavity of the beam at C is directed upward, and a negative sign means that it is directed downward.

B. Drawing the shear and bending-moment diagrams for a beam. Obtain these diagrams by plotting, respectively, V and M against the distance x measured along the beam. However, in most cases, you need to compute the values of V and M at only a few points.

1. For a beam supporting only concentrated loads, note [Sample Prob. 7.2] that

a. The shear diagram consists of segments of horizontal lines. Thus, to draw the shear diagram of the beam, you need to compute V only just to the left or just to the right of the points where the loads or reactions are applied.

b. The bending-moment diagram consists of segments of oblique straight lines. Thus, to draw the bending-moment diagram of the beam, you need to compute M only at the points where the loads or reactions are applied.

2. For a beam supporting uniformly distributed loads, note [Sample Prob. 7.3] that under each of the distributed loads:

a. The shear diagram consists of a segment of an oblique straight line. Thus, you need to compute V only where the distributed load begins and where it ends.

b. The bending-moment diagram consists of an arc of parabola. In most cases, you need to compute M only where the distributed load begins and where it ends.

3. For a beam with a more complicated loading, you need to consider the free-body diagram of a portion of the beam of arbitrary length x and determine V and M as functions of x . This procedure may have to be repeated several times, since V and M are often represented by different functions in various parts of the beam [Sample Prob. 7.3].

4. When a couple is applied to a beam, the shear has the same value on both sides of the point of application of the couple, but the bending-moment diagram shows a discontinuity at that point, rising or falling by an amount equal to the magnitude of the couple. Note that a couple can either be applied directly to the beam or result from the application of a load on a member rigidly attached to the beam [Sample Prob. 7.3].

Problems

7.29 through 7.32 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

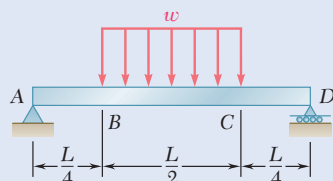


Fig. P7.29

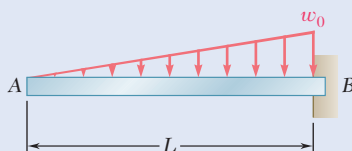


Fig. P7.30

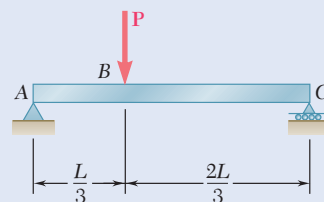


Fig. P7.31

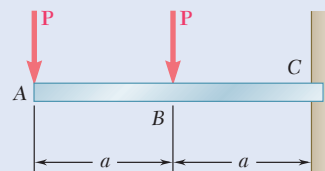


Fig. P7.32

7.33 and 7.34 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

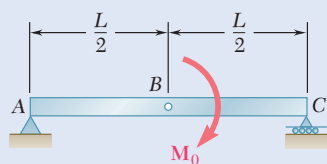


Fig. P7.33

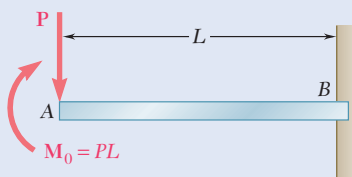


Fig. P7.34

7.35 and 7.36 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

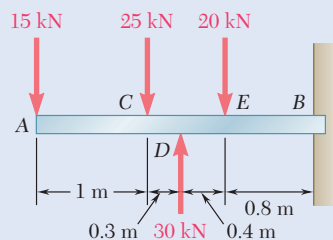


Fig. P7.35

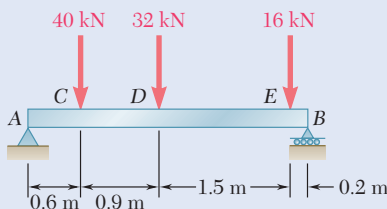


Fig. P7.36

7.37 and 7.38 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

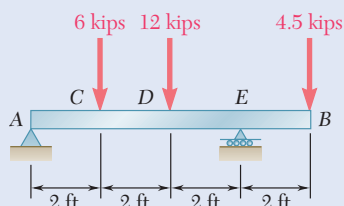


Fig. P7.37

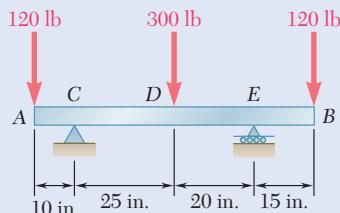


Fig. P7.38

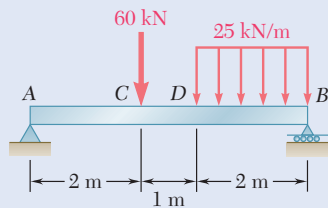


Fig. P7.39

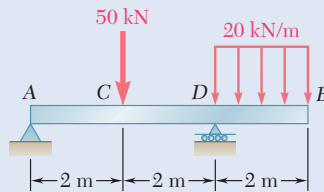


Fig. P7.40

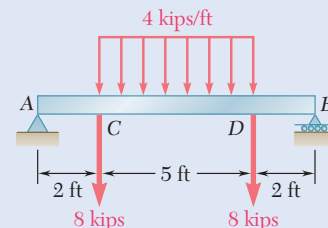


Fig. P7.41

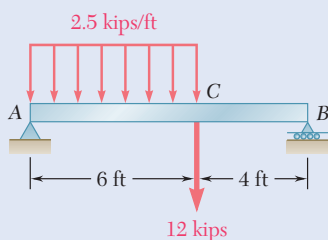


Fig. P7.42

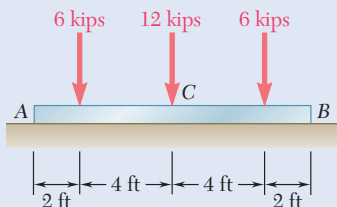


Fig. P7.45

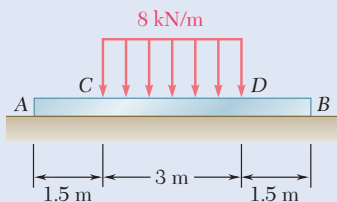


Fig. P7.47

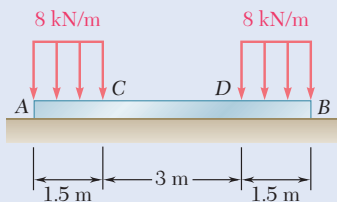


Fig. P7.48

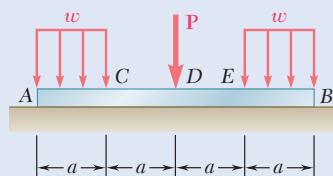


Fig. P7.43

7.39 through 7.42 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

7.43 Assuming the upward reaction of the ground on beam AB to be uniformly distributed and knowing that $P = wa$, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

7.44 Solve Prob. 7.43 knowing that $P = 3wa$.

7.45 Assuming the upward reaction of the ground on beam AB to be uniformly distributed, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

7.46 Solve Prob. 7.45 assuming that the 12-kip load has been removed.

7.47 and 7.48 Assuming the upward reaction of the ground on beam AB to be uniformly distributed, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

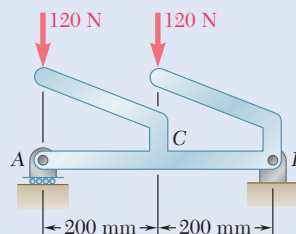


Fig. P7.49

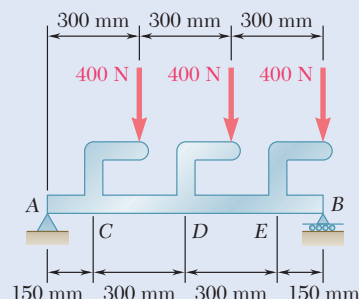


Fig. P7.50

7.49 and 7.50 Draw the shear and bending-moment diagrams for the beam AB, and determine the maximum absolute values of the shear and bending moment.

7.51 and 7.52 Draw the shear and bending-moment diagrams for the beam AB , and determine the maximum absolute values of the shear and bending moment.

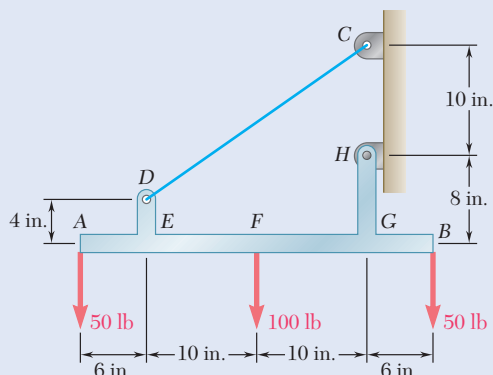


Fig. P7.51

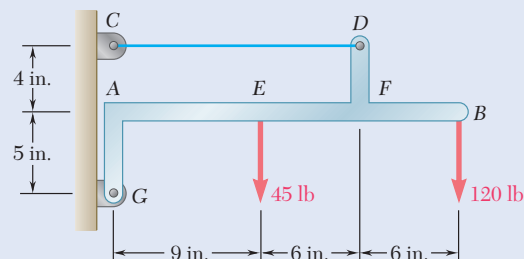


Fig. P7.52

7.53 Two small channel sections DF and EH have been welded to the uniform beam AB of weight $W = 3$ kN to form the rigid structural member shown. This member is being lifted by two cables attached at D and E . Knowing that $\theta = 30^\circ$ and neglecting the weight of the channel sections, (a) draw the shear and bending-moment diagrams for beam AB , (b) determine the maximum absolute values of the shear and bending moment in the beam.

7.54 Solve Prob. 7.53 when $\theta = 60^\circ$.

7.55 For the structural member of Prob. 7.53, determine (a) the angle θ for which the maximum absolute value of the bending moment in beam AB is as small as possible, (b) the corresponding value of $|M|_{\max}$. (Hint: Draw the bending-moment diagram and then equate the absolute values of the largest positive and negative bending moments obtained.)

7.56 For the beam of Prob. 7.43, determine (a) the ratio $k = P/wa$ for which the maximum absolute value of the bending moment in the beam is as small as possible, (b) the corresponding value of $|M|_{\max}$. (See hint for Prob. 7.55.)

7.57 Determine (a) the distance a for which the maximum absolute value of the bending moment in beam AB is as small as possible, (b) the corresponding value of $|M|_{\max}$. (See hint for Prob. 7.55.)

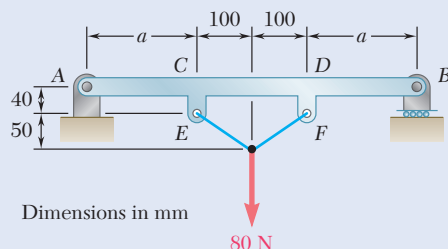


Fig. P7.57

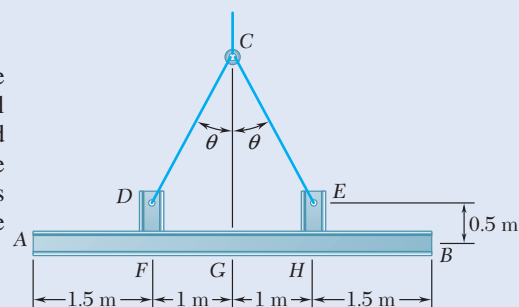


Fig. P7.53

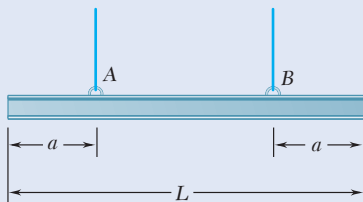


Fig. P7.59

- 7.58** For the beam and loading shown, determine (a) the distance a for which the maximum absolute value of the bending moment in the beam is as small as possible, (b) the corresponding value of $|M|_{\max}$. (See hint for Prob. 7.55.)

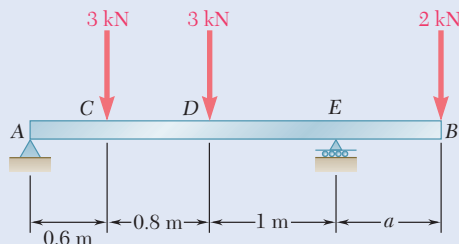


Fig. P7.58

- 7.59** A uniform beam is to be picked up by crane cables attached at A and B. Determine the distance a from the ends of the beam to the points where the cables should be attached if the maximum absolute value of the bending moment in the beam is to be as small as possible. (Hint: Draw the bending-moment diagram in terms of a , L , and the weight per unit length w , and then equate the absolute values of the largest positive and negative bending moments obtained.)

- 7.60** Knowing that $P = Q = 150$ lb, determine (a) the distance a for which the maximum absolute value of the bending moment in beam AB is as small as possible, (b) the corresponding value of $|M|_{\max}$. (See hint for Prob. 7.55.)

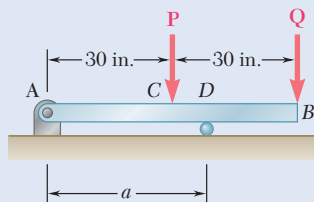


Fig. P7.60

- 7.61** Solve Prob. 7.60 assuming that $P = 300$ lb and $Q = 150$ lb.

- *7.62** In order to reduce the bending moment in the cantilever beam AB, a cable and counterweight are permanently attached at end B. Determine the magnitude of the counterweight for which the maximum absolute value of the bending moment in the beam is as small as possible and the corresponding value of $|M|_{\max}$. Consider (a) the case when the distributed load is permanently applied to the beam, (b) the more general case when the distributed load may either be applied or removed.

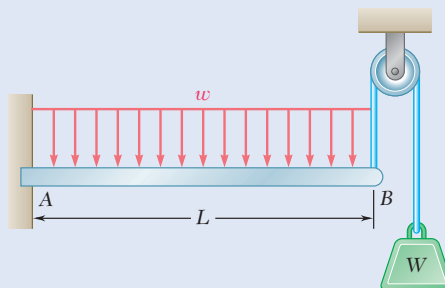


Fig. P7.62

7.3 RELATIONS AMONG LOAD, SHEAR, AND BENDING MOMENT

If a beam carries more than two or three concentrated loads or if it carries a distributed load, the method outlined in Sec. 7.2 for plotting shear and bending-moment diagrams is likely to be quite cumbersome. However, constructing a shear diagram and, especially, a bending-moment diagram, are much easier if we take into consideration some relations among load, shear, and bending moment.

Consider a simply supported beam AB carrying a distributed load w per unit length (Fig. 7.11a). Let C and C' be two points of the beam at a distance Δx from each other. We denote the shear and bending moment at C by V and M , respectively, and we assume they are positive. We denote the shear and bending moment at C' by $V + \Delta V$ and $M + \Delta M$.

Let us now detach the portion of beam CC' and draw its free-body diagram (Fig. 7.11b). The forces exerted on the free body include a load with a magnitude of $w \Delta x$ (indicated by a dashed arrow to distinguish it from the original distributed load from which it is derived) and internal forces and couples at C and C' . Since we assumed both shear and bending moment are positive, the forces and couples are directed as shown in the figure.

Relations Between Load and Shear. Because the free body CC' is in equilibrium, we set the sum of the vertical components of the forces acting on it to zero:

$$\begin{aligned} V - (V + \Delta V) - w \Delta x &= 0 \\ \Delta V &= -w \Delta x \end{aligned}$$

Dividing both sides of this equation by Δx and then letting Δx approach zero, we obtain

$$\frac{dV}{dx} = -w \quad (7.1)$$

Equation (7.1) indicates that, for a beam loaded as shown in Fig. 7.11a, the slope dV/dx of the shear curve is negative and the numerical value of the slope at any point is equal to the load per unit length at that point.

Integrating (7.1) between arbitrary points C and D , we have

$$V_D - V_C = - \int_{x_C}^{x_D} w dx \quad (7.2)$$

or

$$V_D - V_C = -(\text{area under load curve between } C \text{ and } D) \quad (7.2')$$

Note that we could also obtain this result by considering the equilibrium of the portion of beam CD , since the area under the load curve represents the total load applied between C and D .

Equation (7.1) is *not valid* at a point where a concentrated load is applied; the shear curve is discontinuous at such a point, as we saw in

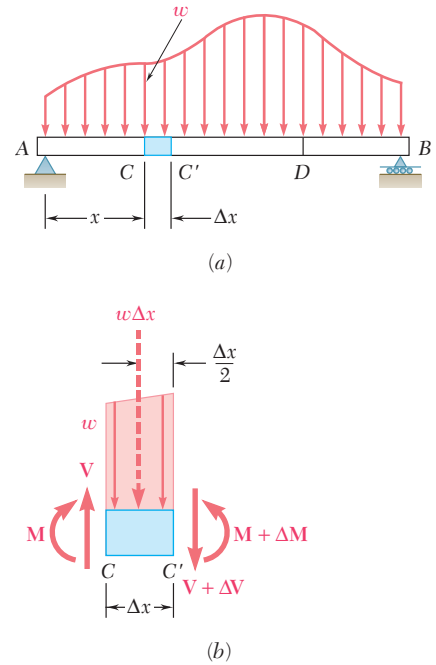


Fig. 7.11 (a) A simply supported beam carrying a distributed load; (b) free-body diagram of a portion CC' of the beam.

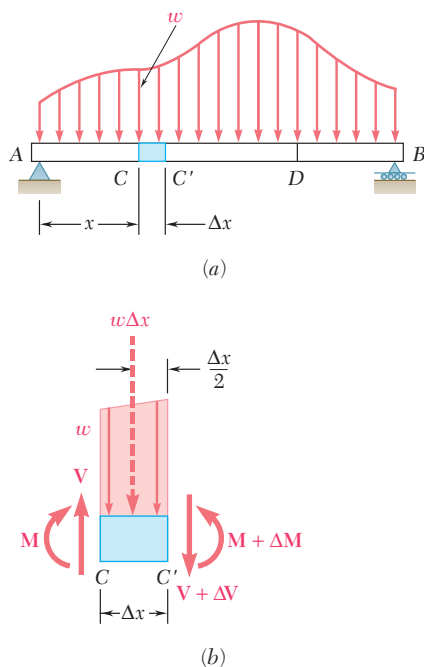


Fig. 7.11 (repeated)

Sec. 7.2. Similarly, formulas (7.2) and (7.2') cease to be valid when concentrated loads are applied between C and D , since they do not take into account the sudden change in shear caused by a concentrated load. Formulas (7.2) and (7.2'), therefore, should be applied only between successive concentrated loads.

Relations Between Shear and Bending Moment. Returning to the free-body diagram of Fig. 7.11b, we can set the sum of the moments about C' to be zero, obtaining

$$(M + \Delta M) - M - V \Delta x + w \Delta x \frac{\Delta x}{2} = 0$$

$$\Delta M = V \Delta x - \frac{1}{2} w (\Delta x)^2$$

Dividing both sides of this equation by Δx and then letting Δx approach zero, we have

$$\frac{dM}{dx} = V \quad (7.3)$$

Equation (7.3) indicates that the slope dM/dx of the bending-moment curve is equal to the value of the shear. This is true at any point where the shear has a well-defined value, i.e., at any point where no concentrated load is applied. Formula (7.3) also shows that the shear is zero at points where the bending moment is maximum. This property simplifies the determination of points where the beam is likely to fail under bending.

Integrating Eq. (7.3) between arbitrary points C and D , we obtain

$$M_D - M_C = \int_{x_C}^{x_D} V dx \quad (7.4)$$

$$M_D - M_C = \text{area under shear curve between } C \text{ and } D \quad (7.4')$$

Note that the area under the shear curve should be considered positive where the shear is positive and negative where the shear is negative. Formulas (7.4) and (7.4') are valid even when concentrated loads are applied between C and D , as long as the shear curve has been drawn correctly. The formulas cease to be valid, however, if a *couple* is applied at a point between C and D , since they do not take into account the sudden change in bending moment caused by a couple (see Sample Prob. 7.7).

In most engineering applications, you need to know the value of the bending moment at only a few specific points. Once you have drawn the shear diagram and determined M at one end of the beam, you can obtain the value of the bending moment at any given point by computing the area under the shear curve and using formula (7.4'). For instance, since $M_A = 0$ for the beam of Fig. 7.12, you can determine the maximum value of the bending moment for that beam simply by measuring the area of the shaded triangle in the shear diagram as

$$M_{\max} = \frac{1}{2} \frac{L}{2} \frac{wL}{2} = \frac{wL^2}{8}$$

In this example, the load curve is a horizontal straight line, the shear curve is an oblique straight line, and the bending-moment curve

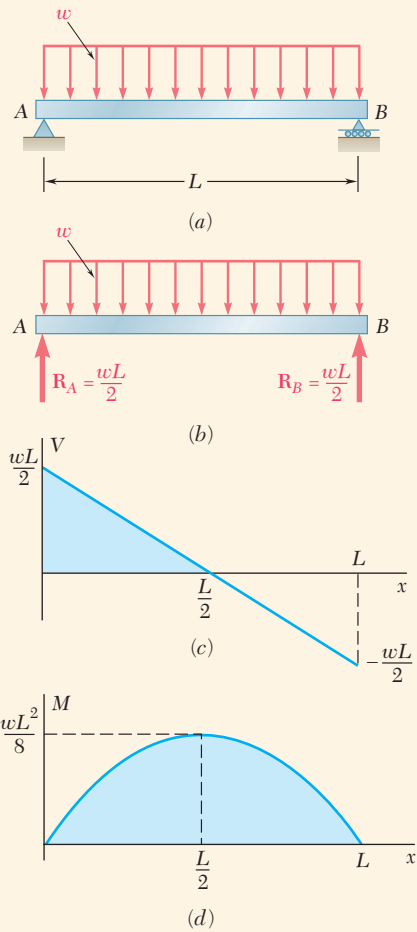


Fig. 7.12 (a) A simply supported beam carrying a uniformly distributed load; (b) free-body diagram of the beam to determine the reactions at the supports; (c) the shear curve is an oblique straight line; (d) the bending-moment diagram is a parabola.

Concept Application 7.1

Consider a simply supported beam AB with a span of L carrying a uniformly distributed load w (Fig. 7.12a). From the free-body diagram of the entire beam, we determine the magnitude of the reactions at the supports: $R_A = R_B = wL/2$ (Fig. 7.12b). Then we draw the shear diagram. Close to end A of the beam, the shear is equal to R_A ; that is, to $wL/2$, as we can check by considering a very small portion of the beam as a free body. Using formula (7.2), we can then determine the shear V at any distance x from A as

$$V - V_A = - \int_0^x w dx = -wx$$

$$V = V_A - wx = \frac{wL}{2} - wx = w \left(\frac{L}{2} - x \right)$$

The shear curve is thus an oblique straight line that crosses the x axis at $x = L/2$ (Fig. 7.12c). Now consider the bending moment. We first observe that $M_A = 0$. The value M of the bending moment at any distance x from A then can be obtained from Eq. (7.4), as

$$M - M_A = \int_0^x V dx$$

$$M = \int_0^x w \left(\frac{L}{2} - x \right) dx = \frac{w}{2} (Lx - x^2)$$

The bending-moment curve is a parabola. The maximum value of the bending moment occurs when $x = L/2$, since V (and thus dM/dx) is zero for that value of x . Substituting $x = L/2$ in the last equation, we obtain $M_{\max} = wL^2/8$.

is a parabola. If the load curve had been an oblique straight line (first degree), the shear curve would have been a parabola (second degree), and the bending-moment curve would have been a cubic (third degree). The equations of the shear and bending-moment curves are always, respectively, one and two degrees higher than the equation of the load curve. Thus, once you have computed a few values of the shear and bending moment, you should be able to sketch the shear and bending-moment diagrams without actually determining the functions $V(x)$ and $M(x)$. The sketches will be more accurate if you make use of the fact that, at any point where the curves are continuous, the slope of the shear curve is equal to $-w$ and the slope of the bending-moment curve is equal to V .

Sample Problem 7.4

Draw the shear and bending-moment diagrams for the beam and loading shown.

STRATEGY: The beam supports two concentrated loads and one distributed load. You can use the equations in this section between these loads and under the distributed load, but you should expect certain changes in the diagrams at the load points.

MODELING and ANALYSIS:

Free-Body, Entire Beam. Consider the entire beam as a free body and determine the reactions (Fig. 1):

$$+\uparrow \Sigma M_A = 0:$$

$$D(24 \text{ ft}) - (20 \text{ kips})(6 \text{ ft}) - (12 \text{ kips})(14 \text{ ft}) - (12 \text{ kips})(28 \text{ ft}) = 0$$

$$D = +26 \text{ kips} \quad \mathbf{D} = 26 \text{ kips } \uparrow$$

$$+\uparrow \Sigma F_y = 0: \quad A_y - 20 \text{ kips} - 12 \text{ kips} + 26 \text{ kips} - 12 \text{ kips} = 0$$

$$A_y = +18 \text{ kips} \quad \mathbf{A}_y = 18 \text{ kips } \uparrow$$

$$\rightarrow \Sigma F_x = 0: \quad A_x = 0 \quad \mathbf{A}_x = 0$$

Note that the bending moment is zero at both A and E ; thus, you know two points (indicated by small circles) on the bending-moment diagram.

Shear Diagram. Since $dV/dx = -w$, the slope of the shear diagram is zero (i.e., the shear is constant between concentrated loads and reactions). To find the shear at any point, divide the beam into two parts and consider either part as a free body. For example, using the portion of the beam to the left of point 1 (Fig. 1), you can obtain the shear between B and C :

$$+\uparrow \Sigma F_y = 0: \quad +18 \text{ kips} - 20 \text{ kips} - V = 0 \quad V = -2 \text{ kips}$$

You can also find that the shear is $+12$ kips just to the right of D and zero at end E . Since the slope $dV/dx = -w$ is constant between D and E , the shear diagram between these two points is a straight line.

Bending-Moment Diagram. Recall that the area under the shear curve between two points is equal to the change in bending moment between the same two points. For convenience, compute the area of each portion of the shear diagram and indicate it on the diagram (Fig. 1). Since you know the bending moment M_A at the left end is zero, you have

$$\begin{aligned} M_B - M_A &= +108 & M_B &= +108 \text{ kip}\cdot\text{ft} \\ M_C - M_B &= -16 & M_C &= +92 \text{ kip}\cdot\text{ft} \\ M_D - M_C &= -140 & M_D &= -48 \text{ kip}\cdot\text{ft} \\ M_E - M_D &= +48 & M_E &= 0 \end{aligned}$$

Since you know M_E is zero, this gives you a check of the calculations.

Between the concentrated loads and reactions, the shear is constant; thus, the slope dM/dx is constant. Therefore, you can draw the bending-moment diagram by connecting the known points with straight lines.

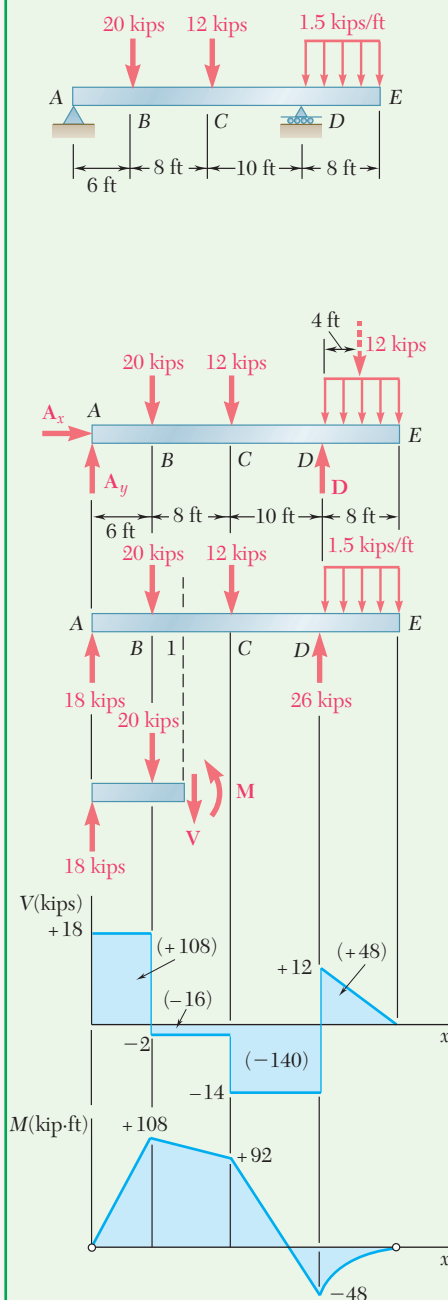


Fig. 1 Free-body diagrams of beam, free-body diagram of section to left of cut, shear diagram, bending-moment diagram.

Between D and E , where the shear diagram is an oblique straight line, the bending-moment diagram is a parabola.

From the V and M diagrams, note that $V_{\max} = 18$ kips and $M_{\max} = 108$ kip·ft.

REFLECT and THINK: As expected, the values of shear and slopes of the bending-moment curves show abrupt changes at the points where concentrated loads act. Useful for design, these diagrams make it easier to determine the maximum values of shear and bending moment for a beam and its loading.

Sample Problem 7.5

Draw the shear and bending-moment diagrams for the beam and loading shown and determine the location and magnitude of the maximum bending moment.

STRATEGY: The load is a distributed load over part of the beam with no concentrated loads. You can use the equations in this section in two parts: for the load and no load regions. From the discussion in this section, you can expect the shear diagram will show an oblique line under the load, followed by a horizontal line. The bending-moment diagram should show a parabola under the load and an oblique line under the rest of the beam.

MODELING and ANALYSIS:

Free-Body, Entire Beam. Consider the entire beam as a free body (Fig. 1) to obtain the reactions

$$\mathbf{R}_A = 80 \text{ kN } \uparrow \quad \mathbf{R}_C = 40 \text{ kN } \uparrow$$

Shear Diagram. The shear just to the right of A is $V_A = +80$ kN. Because the change in shear between two points is equal to *minus* the area under the load curve between these points, you can obtain V_B by writing

$$\begin{aligned} V_B - V_A &= -(20 \text{ kN/m})(6 \text{ m}) = -120 \text{ kN} \\ V_B &= -120 + V_A = -120 + 80 = -40 \text{ kN} \end{aligned}$$

Since the slope $dV/dx = -w$ is constant between A and B , the shear diagram between these two points is represented by a straight line. Between B and C , the area under the load curve is zero; therefore,

$$V_C - V_B = 0 \quad V_C = V_B = -40 \text{ kN}$$

and the shear is constant between B and C (Fig. 1).

Bending-Moment Diagram. The bending moment at each end of the beam is zero. In order to determine the maximum bending moment, you need to locate the section D of the beam where $V = 0$. You have

$$\begin{aligned} V_D - V_A &= -wx \\ 0 - 80 \text{ kN} &= -(20 \text{ kN/m})x \end{aligned}$$

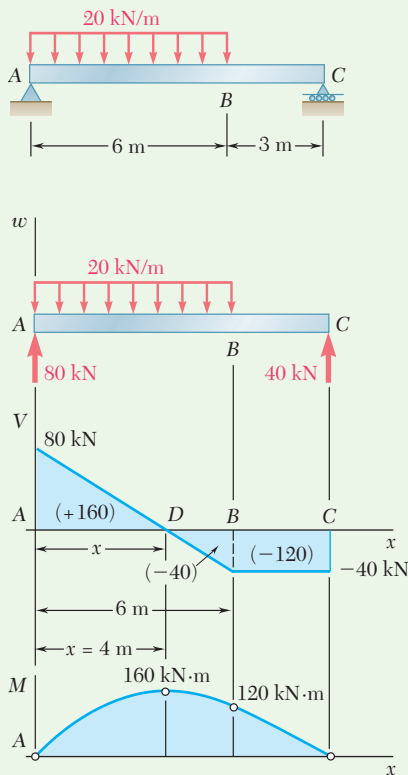


Fig. 1 Free-body diagram of beam, shear diagram, bending-moment diagram.

Solving for x :

$$x = 4 \text{ m} \quad \blacktriangleleft$$

The maximum bending moment occurs at point D , where we have $dM/dx = V = 0$. Calculate the areas of the various portions of the shear diagram and mark them (in parentheses) on the diagram (Fig. 1). Since the area of the shear diagram between two points is equal to the change in bending moment between those points, you can write

$$\begin{aligned} M_D - M_A &= +160 \text{ kN}\cdot\text{m} & M_D &= +160 \text{ kN}\cdot\text{m} \\ M_B - M_D &= -40 \text{ kN}\cdot\text{m} & M_B &= +120 \text{ kN}\cdot\text{m} \\ M_C - M_B &= -120 \text{ kN}\cdot\text{m} & M_C &= 0 \end{aligned}$$

The bending-moment diagram consists of an arc of parabola followed by a segment of straight line; the slope of the parabola at A is equal to the value of V at that point.

The maximum bending moment is

$$M_{\max} = M_D = +160 \text{ kN}\cdot\text{m} \quad \blacktriangleleft$$

REFLECT and THINK: The analysis conforms to our initial expectations. It is often useful to predict what the results of analysis will be as a way of checking against large-scale errors. However, final results can only depend on detailed modeling and analysis.

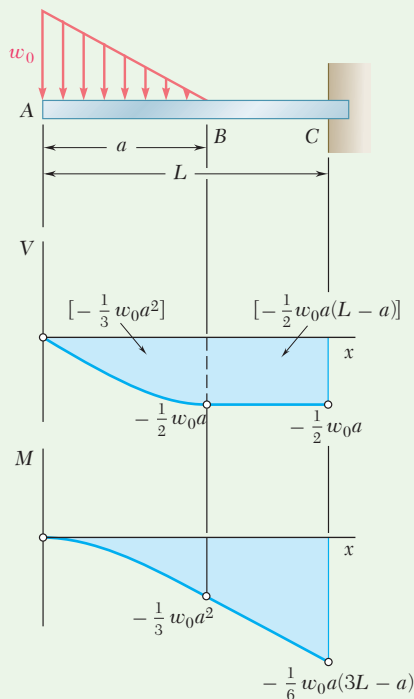


Fig. 1 Beam with load, shear diagram, bending-moment diagram.

Sample Problem 7.6

Sketch the shear and bending-moment diagrams for the cantilever beam shown.

STRATEGY: Because no support reactions appear until the right end of the beam, you can rely on the equations from this section without needing to use free-body diagrams and equilibrium equations. Due to the non-uniform load, you should expect the results to involve equations of higher degree with a parabolic curve in the shear diagram and a cubic curve in the bending-moment diagram.

MODELING and ANALYSIS:

Shear Diagram. At the free end of the beam, $V_A = 0$. Between A and B , the area under the load curve is $\frac{1}{2}w_0a$; we find V_B by writing

$$V_B - V_A = -\frac{1}{2}w_0a \quad V_B = -\frac{1}{2}w_0a$$

Between B and C , the beam is not loaded; thus, $V_C = V_B$. At A , we have $w = w_0$, and according to Eq. (7.1), the slope of the shear curve is $dV/dx = -w_0$. At B , the slope is $dV/dx = 0$. Between A and B , the loading decreases linearly, and the shear diagram is parabolic (Fig. 1). Between B and C , $w = 0$ and the shear diagram is a horizontal line.

Bending-Moment Diagram. Note that $M_A = 0$ at the free end of the beam. You can compute the area under the shear curve, obtaining

$$\begin{aligned} M_B - M_A &= -\frac{1}{3}w_0a^2 & M_B &= -\frac{1}{3}w_0a^2 \\ M_C - M_B &= -\frac{1}{2}w_0a(L - a) \\ M_C &= -\frac{1}{6}w_0a(3L - a) \end{aligned}$$

You can complete the sketch of the bending-moment diagram by recalling that $dM/dx = V$. The result is that between A and B the diagram is represented by a cubic curve with zero slope at A and between B and C the diagram is represented by a straight line.

REFLECT and THINK: Although not strictly required for the solution of this problem, determining the support reactions would serve as an excellent check of the final values of the shear and bending-moment diagrams.

Sample Problem 7.7

The simple beam AC is loaded by a couple of magnitude T applied at point B . Draw the shear and bending-moment diagrams for the beam.

STRATEGY: The load supported by the beam is a concentrated couple. Since the only vertical forces are those associated with the support reactions, you should expect the shear diagram to be of constant value. However, the bending-moment diagram will have a discontinuity at B due to the couple.

MODELING and ANALYSIS:

Free-Body, Entire Beam. Consider the entire beam as a free body and determine the reactions:

$$\mathbf{R}_A = \frac{T}{L}\uparrow \quad \mathbf{R}_B = \frac{T}{L}\downarrow$$

Shear and Bending-Moment Diagrams (Fig. 1). The shear at any section is constant and equal to T/L . Since a couple is applied at B , the bending-moment diagram is discontinuous at B ; because the couple is counterclockwise, the bending moment *decreases* suddenly by an amount equal to T . You can demonstrate this by taking a section to the immediate right of B and applying equilibrium to solve for the bending moment at this location.

REFLECT and THINK: You can generalize the effect of a couple applied to a beam. At the point where the couple is applied, the bending-moment diagram increases by the value of the couple if it is clockwise and decreases by the value of the couple if it is counterclockwise.

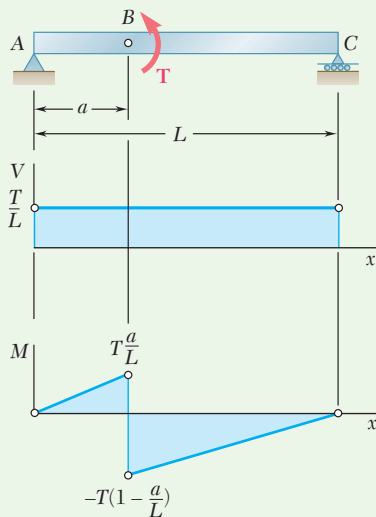


Fig. 1 Beam with load, shear diagram, bending-moment diagram.

SOLVING PROBLEMS ON YOUR OWN

In this section, we described how to use the relations among load, shear, and bending moment to simplify the drawing of shear and bending-moment diagrams. These relations are

$$\frac{dV}{dx} = -w \quad (7.1)$$

$$\frac{dM}{dx} = V \quad (7.3)$$

$$V_D - V_C = -(\text{area under load curve between } C \text{ and } D) \quad (7.2')$$

$$M_D - M_C = (\text{area under shear curve between } C \text{ and } D) \quad (7.4')$$

Taking these relations into account, you can use the following procedure to draw the shear and bending-moment diagrams for a beam.

1. Draw a free-body diagram of the entire beam, and use it to determine the reactions at the beam supports.

2. Draw the shear diagram. This can be done as in the preceding section by cutting the beam at various points and considering the free-body diagram of one of the two resulting portions of the beam [Sample Prob. 7.3]. You can, however, consider one of the following alternative procedures.

a. The shear V at any point of the beam is the sum of the reactions and loads to the left of that point; an upward force is counted as positive, and a downward force is counted as negative.

b. For a beam carrying a distributed load, you can start from a point where you know V and use Eq. (7.2') repeatedly to find V at all other points of interest.

3. Draw the bending-moment diagram, using the following procedure.

a. Compute the area under each portion of the shear curve, assigning a positive sign to areas above the x axis and a negative sign to areas below the x axis.

b. Apply Eq. (7.4') repeatedly [Sample Probs. 7.4 and 7.5], starting from the left end of the beam, where $M = 0$ (except if a couple is applied at that end, or if the beam is a cantilever beam with a fixed left end).

c. Where a couple is applied to the beam, be careful to show a discontinuity in the bending-moment diagram by *increasing* the value of M at that point by an amount equal to the magnitude of the couple if the couple is *clockwise*, or *decreasing* the value of M by that amount if the couple is *counterclockwise* [Sample Prob. 7.7].

4. Determine the location and magnitude of $|M|_{\max}$. The maximum absolute value of the bending moment occurs at one of the points where $dM/dx = 0$ [according to Eq. (7.3), that is at a point where V is equal to zero or changes sign]. You should

a. Determine from the shear diagram the value of $|M|$ where V changes sign; this will occur under a concentrated load [Sample Prob. 7.4].

b. Determine the points where $V = 0$ and the corresponding values of $|M|$; this will occur under a distributed load. To find the distance x between point C where the distributed load starts and point D where the shear is zero, use Eq. (7.2'). For V_C , use the known value of the shear at point C ; for V_D , use zero and express the area under the load curve as a function of x [Sample Prob. 7.5].

5. You can improve the quality of your drawings by keeping in mind that, at any given point according to Eqs. (7.1) and (7.3), the slope of the V curve is equal to $-w$ and the slope of the M curve is equal to V .

6. Finally, for beams supporting a distributed load expressed as a function $w(x)$, remember that you can obtain the shear V by integrating the function $-w(x)$, and you can obtain the bending moment M by integrating $V(x)$ [Eqs. (7.2) and (7.4)].

Problems

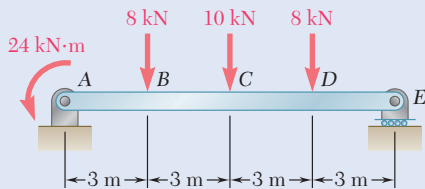


Fig. P7.69

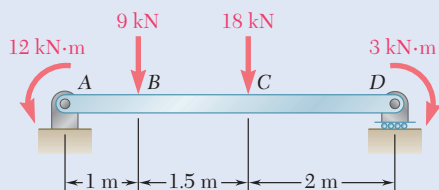


Fig. P7.70

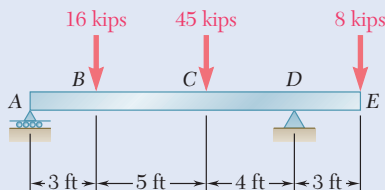


Fig. P7.75

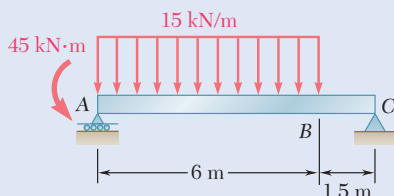


Fig. P7.77

7.63 Using the method of Sec. 7.3, solve Prob. 7.29.

7.64 Using the method of Sec. 7.3, solve Prob. 7.30.

7.65 Using the method of Sec. 7.3, solve Prob. 7.31.

7.66 Using the method of Sec. 7.3, solve Prob. 7.32.

7.67 Using the method of Sec. 7.3, solve Prob. 7.33.

7.68 Using the method of Sec. 7.3, solve Prob. 7.34.

7.69 and 7.70 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

7.71 Using the method of Sec. 7.3, solve Prob. 7.39.

7.72 Using the method of Sec. 7.3, solve Prob. 7.40.

7.73 Using the method of Sec. 7.3, solve Prob. 7.41.

7.74 Using the method of Sec. 7.3, solve Prob. 7.42.

7.75 and 7.76 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the maximum absolute values of the shear and bending moment.

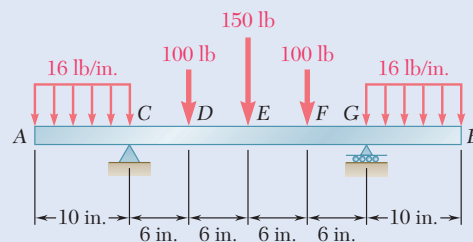


Fig. P7.76

7.77 and 7.78 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the magnitude and location of the maximum absolute value of the bending moment.

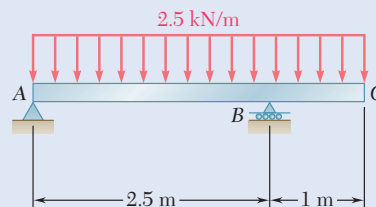


Fig. P7.78

7.79 and 7.80 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the magnitude and location of the maximum absolute value of the bending moment.

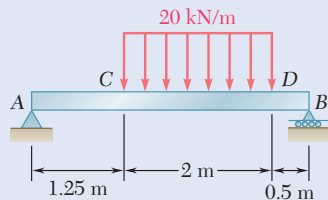


Fig. P7.79

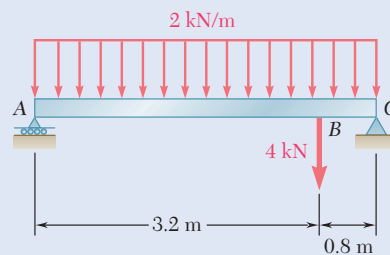


Fig. P7.80

7.81 and 7.82 For the beam and loading shown, (a) draw the shear and bending-moment diagrams, (b) determine the magnitude and location of the maximum absolute value of the bending moment.

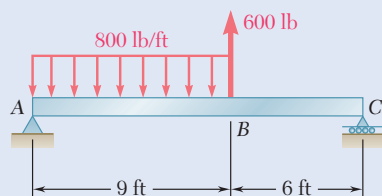


Fig. P7.81

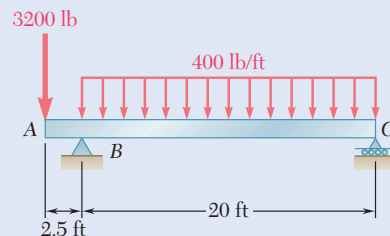


Fig. P7.82

7.83 (a) Draw the shear and bending-moment diagrams for beam AB, (b) determine the magnitude and location of the maximum absolute value of the bending moment.

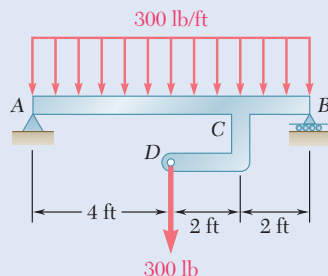


Fig. P7.83

7.84 Solve Prob. 7.83 assuming that the 300-lb force applied at D is directed upward.

7.85 and 7.86 For the beam and loading shown, (a) write the equations of the shear and bending-moment curves, (b) determine the magnitude and location of the maximum bending moment.

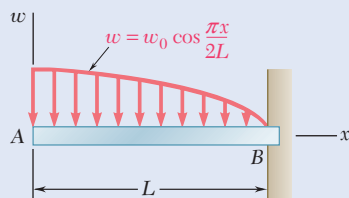


Fig. P7.85

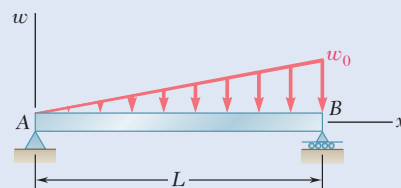


Fig. P7.86

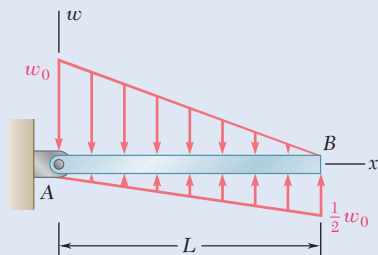


Fig. P7.87

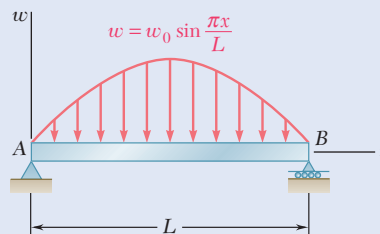


Fig. P7.88

7.87 and 7.88 For the beam and loading shown, (a) write the equations of the shear and bending-moment curves, (b) determine the magnitude and location of the maximum bending moment.

***7.89** The beam AB is subjected to the uniformly distributed load shown and to two unknown forces $\mathbf{P}$ and $\mathbf{Q}$. Knowing that it has been experimentally determined that the bending moment is $+800 \text{ N}\cdot\text{m}$ at D and $+1300 \text{ N}\cdot\text{m}$ at E , (a) determine $\mathbf{P}$ and $\mathbf{Q}$, (b) draw the shear and bending-moment diagrams for the beam.

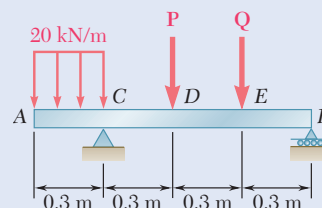


Fig. P7.89

***7.90** Solve Prob. 7.89 assuming that the bending moment was found to be $+650 \text{ N}\cdot\text{m}$ at D and $+1450 \text{ N}\cdot\text{m}$ at E .

***7.91** The beam AB is subjected to the uniformly distributed load shown and to two unknown forces $\mathbf{P}$ and $\mathbf{Q}$. Knowing that it has been experimentally determined that the bending moment is $+6.10 \text{ kip}\cdot\text{ft}$ at D and $+5.50 \text{ kip}\cdot\text{ft}$ at E , (a) determine $\mathbf{P}$ and $\mathbf{Q}$, (b) draw the shear and bending-moment diagrams for the beam.

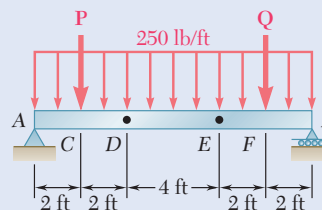


Fig. P7.91

***7.92** Solve Prob. 7.91 assuming that the bending moment was found to be $+5.96 \text{ kip}\cdot\text{ft}$ at D and $+6.84 \text{ kip}\cdot\text{ft}$ at E .

*7.4 CABLES

Cables are used in many engineering applications, such as suspension bridges, transmission lines, aerial tramways, guy wires for high towers, etc. Cables may be divided into two categories, according to their loading: (1) supporting concentrated loads and (2) supporting distributed loads.

7.4A Cables with Concentrated Loads

Consider a cable attached to two fixed points A and B and supporting n vertical concentrated loads $\mathbf{P}_1, \mathbf{P}_2, \dots, \mathbf{P}_n$ (Fig. 7.13a). We assume that the cable is *flexible*, i.e., that its resistance to bending is small and can be neglected. We further assume that the *weight of the cable is negligible* compared with the loads supported by the cable. We can therefore approximate any portion of cable between successive loads as a two-force member. Thus, the internal forces at any point in the cable reduce to a *force of tension directed along the cable*.

We assume that each of the loads lies in a given vertical line, i.e., that the horizontal distance from support A to each of the loads is known. We also assume that we know the horizontal and vertical distances between the supports. With these assumptions, we want to determine the shape of the cable (i.e., the vertical distance from support A to each of the points $C_1, C_2, \dots, C_n$) and also the tension T in each portion of the cable.

We first draw the free-body diagram of the entire cable (Fig. 7.13b). Since we do not know the slopes of the portions of cable attached at A and B , we represent the reactions at A and B by two components each. Thus, four unknowns are involved, and the three equations of equilibrium are not sufficient to determine the reactions. (Clearly, a cable is not a rigid body; thus, the equilibrium equations represent *necessary but not sufficient conditions*. See Sec. 6.3B.) We must therefore obtain an additional equation by considering the equilibrium of a portion of the cable. This is possible if we know the coordinates x and y of a point D of the cable.

We draw the free-body diagram of the portion of cable AD (Fig. 7.14a). From the equilibrium condition $\sum M_D = 0$, we obtain an additional relation between the scalar components A_x and A_y and can determine



Photo 7.3 The weight of the chairlift cables is negligible compared to the weights of the chairs and skiers, so we can use the methods of this section to determine the force at any point in the cable.

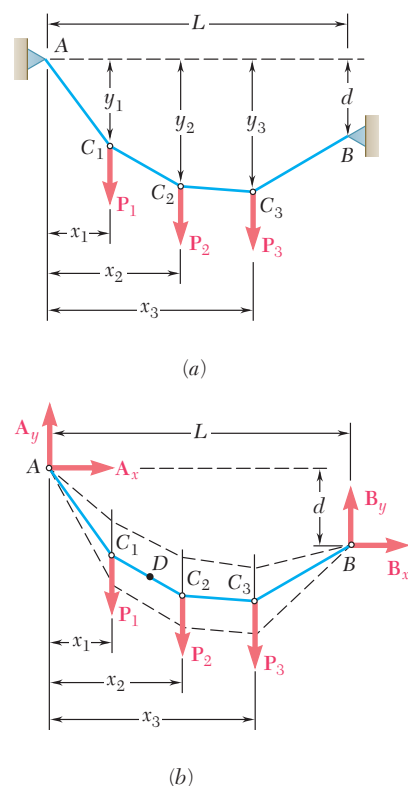


Fig. 7.13 (a) A cable supporting vertical concentrated loads; (b) free-body diagram of the entire cable.

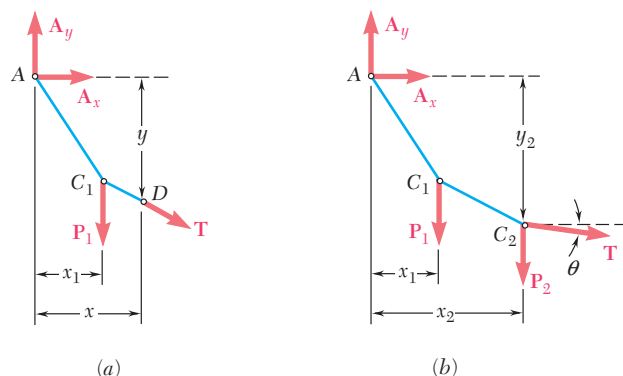


Fig. 7.14 (a) Free-body diagram of the portion of cable AD ; (b) free-body diagram of the portion of cable AC_2 .

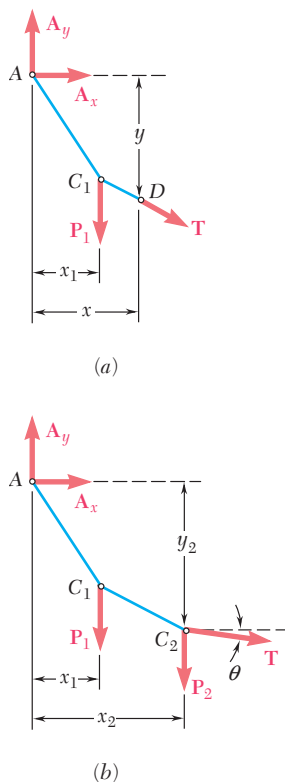


Fig. 7.14 (repeated)

the reactions at A and B . However, the problem remains indeterminate if we do not know the coordinates of D unless we are given some other relation between A_x and A_y (or between B_x and B_y). The cable might hang in any of various possible ways, as indicated by the dashed lines in Fig. 7.13b.

Once we have determined A_x and A_y , we can find the vertical distance from A to any point of the cable. Considering point C_2 , for example, we draw the free-body diagram of the portion of cable AC_2 (Fig. 7.14b). From $\Sigma F_{C_2} = 0$, we obtain an equation that we can solve for y_2 . From $\Sigma F_x = 0$ and $\Sigma F_y = 0$, we obtain the components of force $\mathbf{T}$ representing the tension in the portion of cable to the right of C_2 . Note that $T \cos \theta = -A_x$; that is, *the horizontal component of the tension force is the same at any point of the cable*. It follows that the tension T is maximum when $\cos \theta$ is minimum, i.e., in the portion of cable that has the largest angle of inclination θ . Clearly, this portion of cable must be adjacent to one of the two supports of the cable.

7.4B Cables with Distributed Loads

Consider a cable attached to two fixed points A and B and carrying a *distributed load* (Fig. 7.15a). We just saw that for a cable supporting concentrated loads, the internal force at any point is a force of tension directed along the cable. By contrast, in the case of a cable carrying a distributed load, the cable hangs in the shape of a curve, and the internal force at a point D is a force of tension $\mathbf{T}$ directed *along the tangent to the curve*. Here we examine how to determine the tension at any point of a cable supporting a given distributed load. In the following sections, we will determine the shape of the cable for two common types of distributed loads.

Considering the most general case of distributed load, we draw the free-body diagram of the portion of cable extending from the lowest point C to a given point D of the cable (Fig. 7.15b). The three forces acting on the free body are the tension force $\mathbf{T}_0$ at C , which is horizontal; the tension force $\mathbf{T}$ at D , which is directed along the tangent to the cable at D ; and the resultant $\mathbf{W}$ of the distributed load supported by the portion of cable CD . Drawing the corresponding force triangle (Fig. 7.15c), we obtain the relations

$$T \cos \theta = T_0 \quad T \sin \theta = W \quad (7.5)$$

$$T = \sqrt{T_0^2 + W^2} \quad \tan \theta = \frac{W}{T_0} \quad (7.6)$$

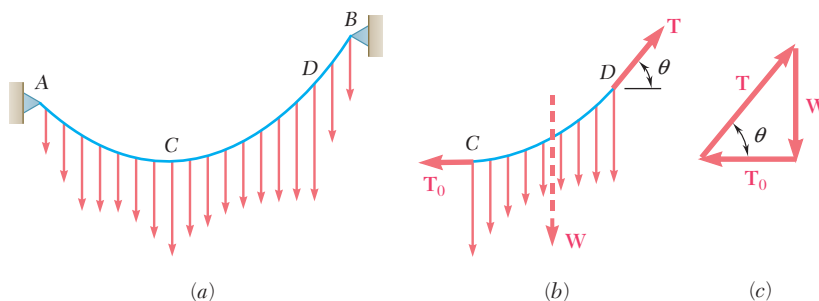


Fig. 7.15 (a) A cable carrying a distributed load; (b) free-body diagram of the portion of the cable CD ; (c) force triangle for the free-body diagram in part (b).

From the relations in Eqs. (7.5), we see that the horizontal component of the tension force $\mathbf{T}$ is the same at any point. Furthermore, the vertical component of $\mathbf{T}$ at any point is equal to the magnitude W of the load when measured from the lowest point (C) to the point in question (D). Relations in Eq. (7.6) show that the tension T is minimum at the lowest point and maximum at one of the two support points.

7.4C Parabolic Cables

Now suppose that cable AB carries a load *uniformly distributed along the horizontal* (Fig. 7.16a). We can approximate the load on the cables of a suspension bridge in this way, since the weight of the cables is small compared with the uniform weight of the roadway. We denote the load per unit length by w (*measured horizontally*) and express it in N/m or lb/ft. Choosing coordinate axes with the origin at the lowest point C of the cable, we find that the magnitude W of the total load carried by the portion of cable extending from C to the point D with coordinates x and y is $W = wx$. The relations in Eqs. (7.6) defining the magnitude and direction of the tension force at D become

$$T = \sqrt{T_0^2 + w^2 x^2} \quad \tan \theta = \frac{wx}{T_0} \quad (7.7)$$

Moreover, the distance from D to the line of action of the resultant $\mathbf{W}$ is equal to half of the horizontal distance from C to D (Fig. 7.16b). Summing moments about D , we have

$$+\circlearrowleft \sum M_D = 0: \quad wx \frac{x}{2} - T_0 y = 0$$

Solving for y , we have

Equation of parabolic cable

$$y = \frac{wx^2}{2T_0} \quad (7.8)$$

This is the equation of a *parabola* with a vertical axis and its vertex at the origin of coordinates. Thus, the curve formed by cables loaded uniformly along the horizontal is a parabola.[‡]

When the supports A and B of the cable have the same elevation, the distance L between the supports is called the **span** of the cable and the vertical distance h from the supports to the lowest point is called the **sag** of the cable (Fig. 7.17a). If you know the span and sag of a cable and if the load w per unit horizontal length is given, you can find the minimum tension T_0 by substituting $x = L/2$ and $y = h$ in Eq. (7.8). Equations (7.7) then yield the tension and the slope at any point of the cable and Eq. (7.8) defines the shape of the cable.

[‡]Cables hanging under their own weight are not loaded uniformly along the horizontal and do not form parabolas. However, the error introduced by assuming a parabolic shape for cables hanging under their own weight is small when the cable is sufficiently taut. In the next section, we give a complete discussion of cables hanging under their own weight.



Photo 7.4 The main cables of suspension bridges, like the Golden Gate Bridge above, may be assumed to carry a loading uniformly distributed along the horizontal.

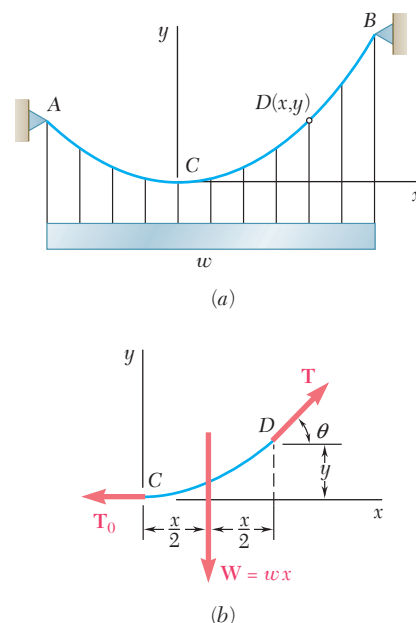


Fig. 7.16 (a) A cable carrying a uniformly distributed load along the horizontal; (b) free-body diagram of the portion of cable CD .

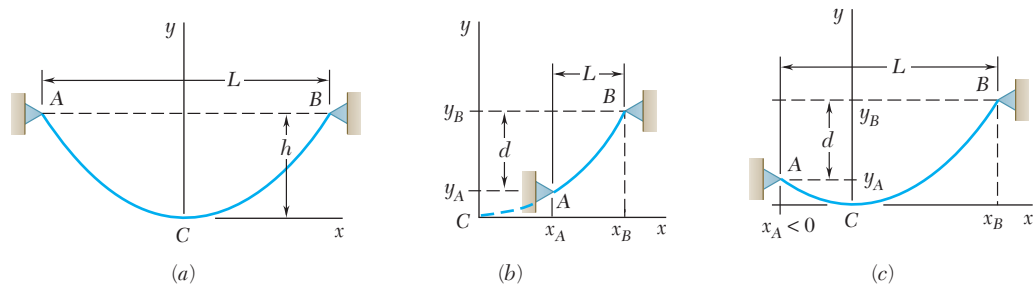


Fig. 7.17 (a) The shape of a parabolic cable is determined by its span L and sag h ; (b, c) span and vertical distance between supports for cables with supports at different elevations.

When the supports have different elevations, the position of the lowest point of the cable is not known, and we must determine the coordinates x_A , y_A and x_B , y_B of the supports. To do this, we note that the coordinates of A and B satisfy Eq. (7.8) and that

$$x_B - x_A = L \quad \text{and} \quad y_B - y_A = d$$

where L and d denote, respectively, the horizontal and vertical distances between the two supports (Fig. 7.17b and c).

We can obtain the length of the cable from its lowest point C to its support B from the formula

$$s_B = \int_0^{x_B} \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx \quad (7.9)$$

Differentiating Eq. (7.8), we obtain the derivative $dy/dx = wx/T_0$. Substituting this into Eq. (7.9) and using the binomial theorem to expand the radical in an infinite series, we have

$$\begin{aligned} s_B &= \int_0^{x_B} \sqrt{1 + \frac{w^2 x^2}{T_0^2}} dx = \int_0^{x_B} \left(1 + \frac{w^2 x^2}{2T_0^2} - \frac{w^4 x^4}{8T_0^4} + \cdots\right) dx \\ s_B &= x_B \left(1 + \frac{w^2 x_B^2}{6T_0^2} - \frac{w^4 x_B^4}{40T_0^4} + \cdots\right) \end{aligned}$$

Then, since $w x_B^2 / 2T_0 = y_B$, we obtain

$$s_B = x_B \left[1 + \frac{2}{3} \left(\frac{y_B}{x_B}\right)^2 - \frac{2}{5} \left(\frac{y_B}{x_B}\right)^4 + \cdots\right] \quad (7.10)$$

This series converges for values of the ratio y_B/x_B less than 0.5. In most cases, this ratio is much smaller, and only the first two terms of the series need be computed.

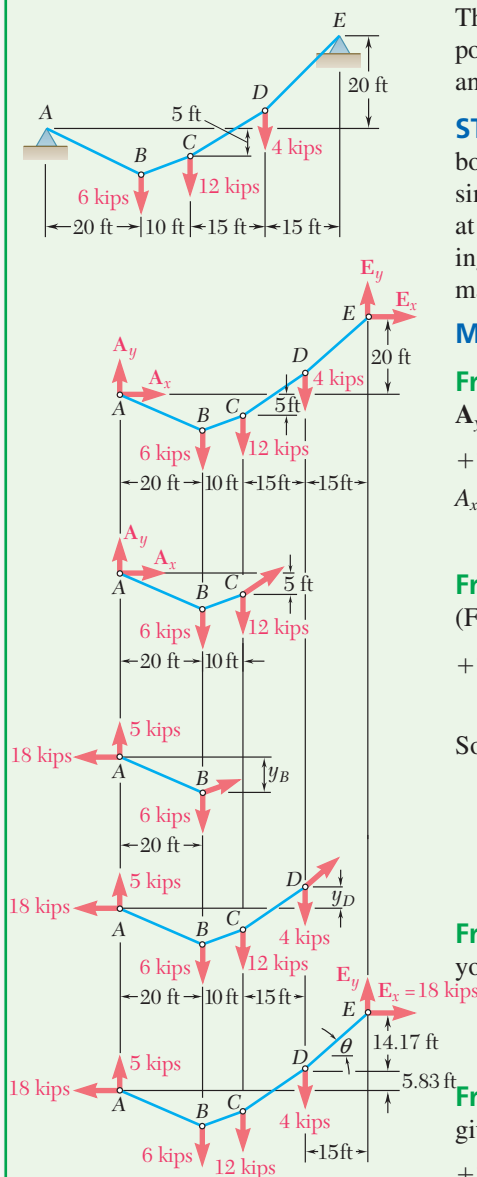


Fig. 1 Free-body diagrams of cable system.

Sample Problem 7.8

The cable AE supports three vertical loads from the points indicated. If point C is 5 ft below the left support, determine (a) the elevation of points B and D , (b) the maximum slope and the maximum tension in the cable.

STRATEGY: To solve for the support reactions at A , consider a free-body diagram of the entire cable as well as one that takes a section at C , since you know the coordinates of this point. Taking subsequent sections at B and D will then enable you to determine their elevations. The resulting cable geometry establishes the maximum slope, which is where the maximum tension in the cable occurs.

MODELING and ANALYSIS:

Free Body, Entire Cable. Determine the reaction components A_x and A_y as

$$+\circlearrowleft \sum M_E = 0:$$

$$A_x(20 \text{ ft}) - A_y(60 \text{ ft}) + (6 \text{ kips})(40 \text{ ft}) + (12 \text{ kips})(30 \text{ ft}) + (4 \text{ kips})(15 \text{ ft}) = 0$$

$$20A_x - 60A_y + 660 = 0$$

Free Body, ABC. Consider the portion ABC of the cable as a free body (Fig. 1). Then you have

$$+\circlearrowleft \sum M_C = 0: \quad -A_x(5 \text{ ft}) - A_y(30 \text{ ft}) + (6 \text{ kips})(10 \text{ ft}) = 0$$

$$-5A_x - 30A_y + 60 = 0$$

Solving the two equations simultaneously, you obtain

$$A_x = -18 \text{ kips} \quad A_x = 18 \text{ kips} \leftarrow$$

$$A_y = +5 \text{ kips} \quad A_y = 5 \text{ kips} \uparrow$$

a. Elevation of Points B and D :

Free Body, AB. Considering the portion of cable AB as a free body, you obtain

$$+\circlearrowleft \sum M_B = 0: \quad (18 \text{ kips})y_B - (5 \text{ kips})(20 \text{ ft}) = 0$$

$$y_B = 5.56 \text{ ft below } A \quad \blacktriangleleft$$

Free Body, ABCD. Using the portion of cable $ABCD$ as a free body gives you

$$+\circlearrowleft \sum M_D = 0:$$

$$-(18 \text{ kips})y_D - (5 \text{ kips})(45 \text{ ft}) + (6 \text{ kips})(25 \text{ ft}) + (12 \text{ kips})(15 \text{ ft}) = 0$$

$$y_D = 5.83 \text{ ft above } A \quad \blacktriangleleft$$

b. Maximum Slope and Maximum Tension. Note that the maximum slope occurs in portion DE . Since the horizontal component of the tension is constant and equal to 18 kips, you have

$$\tan \theta = \frac{14.17}{15 \text{ ft}} \quad \theta = 43.4^\circ \quad \blacktriangleleft$$

$$T_{\max} = \frac{18 \text{ kips}}{\cos \theta} \quad T_{\max} = 24.8 \text{ kips} \quad \blacktriangleleft$$